COLLABORATION HUMAN MACHINE INTERFACE (CHMI) ATFCM Reference Guide

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The following table identifies all management authorities who have successively approved the present issue of this document.

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Edition Change Record

This document does not require a change record, as it will only normally be used as a reference, when the latest reviewed and updated version will be provided, at which time changes are irrelevant to the application of the information held within this document.

With the integration of the NMP Flow Application in the CHMI environment, any information, and references to the NMP Flow Application are provided digitally via the Eurocontrol Learningzone; https://learningzone.eurocontrol.int – e-learning: Introduction to NMP Flow [NMO-FMP-NMP]

The NMP Flow training directly targets users with significant experience in ATFCM operational concepts and CHMI usage that are already, or will become, users of the NMP Flow application.

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1 Preamble

1.1 Introduction

1.1.1 Identification

(1) This document has a reference of "CHMI/SRD/ATFCM".

(2) This document has a title of "COLLABORATION HUMAN MACHINE INTERFACE (CHMI) ATFCM Reference Guide ".

1.1.2 Purpose

(1) The purpose of the Software Requirements Document (SRD) is to transpose the user's needs as expressed in the Functional Baseline from the User Requirements Phase into a well-structured set of software requirements.

(2) The SRD **shall** define the specific requirements for the Software Requirements Phase. The SRD is written to be as independent as possible of any implementation consideration.

1.1.3 Scope

(1) The intended audience of this document are the end-users, the user requirements staff and the system developers.

(2) This SRD is focussed on the ATFCM functionalities implemented in the frame of the CHMI System.

(3) This SRD will evolve in the sense that it **shall** reflect the changing user requirements and development needs for the maintenance of the ATFCM functions of the CHMI Application.

(4) This SRD sets the framework within which the Remote Client using a PC workstation accesses the Network Operations server systems (ETFMS, ENV, …).

(5) Signatures in the Header Pages of this SRD acknowledge acceptance by the aforementioned parties of all the requirements specified in this document.

1.1.4 Overview

(1) This SRD is subdivided into a number of parts.

(2) Each significant part is a separate Word Document.

(3) Each part can be retrieved, edited, stored, updated, and configuration managed independently from any other part.

(4) The list of all parts is contained in the "Table Of Document Parts".

1.2 References

1.2.1 External

(1) This Software Requirements Document makes reference to the following external documents, an external document being defined as a document not produced by the NM.

(2) Microsoft Windows XP Professional Version 2002 or later, User's Guide.

(3) The PANS-RAC (ICAO document 4444).

1.2.2 NM

(1) NM referenced documents **shall** take precedence over any referenced external documents wherever conflict arises between them. See the "Compound Table of Contents" of the Software Engineering Standards and Procedures for a complete list of Standards and Procedures.

(2) The Central Flow Management Unit (CFMU) Implementation Plan (EUROCONTROL DOC. No. 892005) dated June 1989.

(3) STD-CM/PRO/CCPUGChange Control Procedure Users Guide.

(4) STD-CM/PRO/SIMP NM Systems Incident Management Procedure

(5) "Tactical System Software Requirements", ed. 9.000 or later version.

(6) "ETFMS Interface Control Document for CFMU 10.0" or later version

(7) "Software Requirements Document of the CFMU Data Warehouse System", ed. 5.000 or later version.

(8) The "Software Requirements Document for the ATM Messages Exchange System" ed. 1.401 or later version.

(9) "Corp. SRD ENV" ed 1.800 or later version.

(10) "IFPS Software Requirements Document" ed. 3.502 or later version.

(11) "Geographical Display Specification" ed. 1.000 or later version.

1.3 System Introduction

1.3.1 Purpose of the System

(1) The aim of the CHMI System is to provide the local and remote PC clients with a Graphical User Interface to access the different Network Operations Server Systems.

(2) This document is focussed on the ATFCM functionalities available in the CHMI application.

(3) The main external ATFCM users of the CHMI System are Air Navigation Units (ANU) and Aircraft Operators. They wish to access the Network Operations Systems from a Personal Computer using a dedicated application.

1.3.2 Scope of the System

(1) The CHMI System shall communicate with the different Network Operations Server Systems (AME, DWH, ENV and ETFMS) to implement the desired ATFCM functionalities.

1.3.3 Overview of the System

(1) The CHMI System shall provide our clients (ANU, AO, …) with a GUI (Graphical User Interface) to the AME, DWH, ENV and ETFMS Systems (TACT / Predict / Simulation systems).

(2) The ATFCM functionalities of the CHMI System shall allow the clients:

a) To monitor the traffic

b) To display environment data

c) To apply pre-tactical and tactical updates to environment data definitions

d) To apply strategic updates to environment data definitions

e) To display AME data

f) To display counts

g) To display flight lists and individual flights

h) To send messages to the TACT / Predict / Simulation systems (not applicable in archive context).

(3) The GUI will provide the user with an application design based on the JDK (Java Development Kit) and Swing standard interface.

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2 General Requirements

2.1 System Perspective

(1) The ATFCM part of the CHMI system is subordinated to the implementation of:

a) The CHMI framework and front-end environment.

b) The ETFMS System release 5.0 or later version.

c) The DWH System release 5.0 or later version.

d) The ESC System release 7.0 or later version.

e) The IFPS System release 3.5 or later version.

f) The CUA System release 3.0 or later version.

2.2 System Functions

(1) The ATFCM component of the CHMI shall implement the following functionalities based on the AME data:

a) Maintain the list of latest released ATM Messages with automatic warning in case of new message issued.

b) Show the list of released AIM for one day.

c) Show the details for one released AIM.

d) Show ATFCM information message based on dedicated AIM.

e) Show the list of ANM release stamps for one day.

f) Show the latest ANM details for one day and eventually at a given stamp.

g) Show the details for one CRAM.

h) Show the ADP regulation list.

i) Show the current details for one ADP regulation.

j) Show the history log for one ADP regulation.

(2) The ATFCM component of the CHMI shall implement the following functionalities based on the ETFMS, Simulation and Prediction data:

a) Display traffic and delay states for selected FMPs or ATC configuration.

b) Display delay states for all FMPs.

c) Plot traffic and delays states on the map.

d) Display traffic counts and traffic listing for the different traffic types and a selected reference locations or Aircraft Operators.

e) Display flow counts for selected traffic volume.

f) Display the regulation list and detailed measures applied to FMPs.

g) Display the rerouting measures applied to FMPs.

h) Display environment data, i.e. capacities, aerodrome details, airspace configurations, route catalogue, Occupancy Traffic Monitoring Values (Permanent OTMV definitions retrieved from ENV, Daily applicability and ATC Airspace OTMV state list from ETFMS, Simulation or Prediction system).

i) Update pre-tactical and tactical environment data (i.e. Runway Configurations, ATC Sector Configuration Activation, ATC Capacity Figures, Temporary OTMV, Traffic Volume Activation).

j) Update ENV server data with strategic Permanent OTMV definitions.

k) Display delay figures for an aerodrome of departure or a regulation.

l) Allow to send CASA messages for a selected flight, i.e. REA, SAM, SMM, SRM, SRJ...

m) Implement functions to manage chaotic situations (CHAMAN)

n) Display detailed flight data including tactical and operational log entries for selected flights.

o) Plot flight profile details on horizontal and vertical view.

p) Allow to evaluate and apply alternate routes for selected flights.

q) Display automated tower departure flight list with the possibility to issue related update messages.

r) Implement functions to help identifying the optimized ATC configuration.

(3) The ATFCM component of the CHMI shall implement the following functionalities based on the DWH data:

a) Display traffic and delay states for selected FMPs.

b) Display traffic counts and traffic listing for the different traffic types and a selected reference locations or Aircraft Operators.

c) Display the regulation list and detailed measures applied to FMPs.

d) Display environment data, i.e. capacities, airspace configurations.

e) Display delay figures for a regulation.

f) Display detailed flight data including tactical and operational log entries for selected flights.

(4) List queries performed by a given user from the history log and allow displaying the corresponding reply image of the selected ATFCM traffic monitor function.

2.3 User Characteristics

(1) External users are mainly Air Navigation Units and Aircraft Operators accessing the system remotely.

(2) The user shall have a good ATC and ATFM knowledge.

(3) The user shall have a sufficient knowledge of the ICAO standard codification.

(4) To avoid confusion, unless differently described in specific requirements, in general it is recommended to apply the ICAO Doc 8400 to all interfaces artefacts; it saves space on the interface, and less information is required to explain abbreviations and acronyms that are not known in the aviation world.

(5) The user shall have a good knowledge of a Windows user interface.

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3 Data Requirements

3.1 Introduction

(1) The data requirements part shall describe the data used by the ATFCM component of the CHMI system.

(2) The data requirements shall be identified for each ATFCM sub-system.

(3) Internal and external data are defined separately.

(4) Internal data are defined for the ATFCM component of the CHMI system and kept in the CHMI system.

(5) External data are defined as data coming from other systems.

3.2 General Data Requirements

3.2.1 Overview

(1) As the CHMI system plays the role of interface between the remote users and the other Network Operations systems, the internal data requirements are limited.

(2) At the system level, the data requirements are restricted to the data needed to manage the interface.

(3) Data requirements defined in this document are related to the ATFCM functionalities of the CHMI system.

3.2.2 Internal Data Requirements

(1) The only use of internal data is for application customisation based on user preferences.

(2) The generic preference system implemented in the CHMI framework is used for this purpose.

3.2.3 External Data Requirements

3.2.3.1 Introduction

(1) Reference is made to the "ETFMS Interface Control Document".

3.2.3.2 Message List for the ATFCM functions

(1) The following messages related to ATFCM, Predict and Simulation functions are exchanged with ETFMS:

a) DBQ_MONITORING_VOL_ID

b) DBQ_COUNT_GENERAL

c) DBQ_DEPARTURE_AD_DELAYS

d) DBQ_REGULATION_DELAYS

e) DBQ_FLIGHT_GENERAL

f) DBQ_CHAOTIC_SITUATION

g) DBQ_FLIGHT_OPLOG

h) DBQ_AO_REROUTE

i) DBQ_CASA_MONITORING

j) DBQ_RG_DATA

k) DBQ_REROUTE_MONITORING

l) DBQ_LOCATION_CAPACITY

m) DBQ_MOD_LOCATION_CAPACITY

n) DBQ_ATC_CONFIGURATION

o) DBQ_MOD_ATC_CONFIGURATION

p) DBQ_AD_DATA

q) DBQ_MOD_AD_DATA

r) DBQ_TV_ACTIVATION

s) DBQ_MOD_TV_ACTIVATION

t) DBQ_SR

u) DBQ_VOL_SET

v) DBQ_CALCULATE_OPTICON

w) DBQ_CONVERT_TO_CRITERIA_FILE_FORMAT

x) DBQ_CONVERT_FROM_CRITERIA_FILE_FORMAT

y) DBQ_MERGE_OPTICON_CRITERIA

z) DBQ_APPLY_OPTICON_CHOICE

aa) DBQ_CDM_CONFIRMATION

bb) DBQ_LIST_PUBLISHED_DATA_SETS

cc) DBQ_TV_OTMV

dd) DBQ_MOD_TV_OTMV

ee) DBQ_AUA_OTMV

(2) The following messages related to ATFCM / AME functions are exchanged with ETFMS:

a) NDQ_AIM

b) NDQ_AIM_LIST

c) NDQ_CRAM

d) NDQ_ANM

e) NDQ_ANM_LIST

f) NDQ_ADP_DISP

(3) The following CASA messages are exchanged with ETFMS:

a) DBC_READY_TO_DEPART

b) DBC_READY_FOR_IMPROVEMENT

c) DBC_SIP_WANTED_MESSAGE

d) DBC_SLOT_PROPOSAL_ACCEPTANCE

e) DBC_SLOT_PROPOSAL_REJECTION

f) DBC_SLOT_MISSED

g) DBC_REROUTE_PROPOSAL_REJECTION

h) DBC_FLIGHT_CONFIRMATION

i) DBQ_TOWER_CTOT_MOD

j) DBQ_TOWER_TAXI_MOD

3.2.3.3 Message List for the Archive functions

(1) The following messages related to Archive functions are exchanged with DWH:

a) DBQ_MONITORING_VOL_ID

b) DBQ_COUNT_GENERAL

c) DBQ_REGULATION_DELAYS

d) DBQ_FLIGHT_GENERAL

e) DBQ_FLIGHT_OPLOG

f) DBQ_CASA_MONITORING

g) DBQ_RG_DATA

h) DBQ_LOCATION_CAPACITY

i) DBQ_ATC_CONFIGURATION

j) DBQ_VOL_SET

k) DBQ_TV_OTMV

(2) The following messages related to ATFCM / AME functions are exchanged with ETFMS:

a) NDQ_AIM

b) NDQ_AIM_LIST

c) NDQ_CRAM

d) NDQ_ANM

e) NDQ_ANM_LIST

f) NDQ_ADP_DISP

3.2.3.4 Services from the CUA system

(1) The following services related to CHMI ATFCM functions are requested from CUA:

a) Query_List_From_History_Log

b) Reply_Image_From_History_Log

3.2.3.5 Services from the ENV system

(1) The following service related to CHMI ATFCM function is requested from ENV:

a) Get_Permanent_OTMV_For_TV

b) Update_Permanent_OTMV_For_TV

4 ATFCM Collaboration Human Machine Interface application

4.1 Introduction

(1) This document contains:

a) The list of functionalities available in the ATFCM Collaboration Human Machine Interface application part with direct link to the detailed description.

b) The description of functionalities common to several ATFCM Collaboration Human Machine Interface application functions.

c) The identification of some specific behaviour related to the parallel Prediction, Archive, Query Replay and Simulation system access.

4.2 Specific behaviour

4.2.1 Archive Access

(1) Based on your user profile and selected role, the ATFCM Collaboration Human Machine Interface application menu may propose you to retrieve data from an alternative server: the **Archive** system.

(2) Read the **ATFCM Collaboration Human Machine Interface application** - **Archive Access** help page for a detailed description of the differences between the Archive and the ATFCM accesses.

4.2.2 Predict Access

(1) Based on your user profile and selected role, the ATFCM Collaboration Human Machine Interface application menu may propose you to retrieve data from an alternative server: the Prediction system (Predict).

(2) While the ATFCM option gives access to the Tactical (today) and the Pre-Tactical (tomorrow) operational data, the Predict gives access to prediction data.

(3) In order to better prepare ATFCM measures for the coming days, some similar archived traffic data are replayed, adapted and monitored.

(4) The proposed dates include today up to 6 days in the future (i.e. today, today+1 … today+6).

a) The days in the future are of interest for pre-tactical planning (preparation of the daily plan for each day)

b) The current day is of interest to the users to compare the current Operational situation with the Predicted situation. Keeping the predicted data of the current day available online allows the analysis of the plan (set

of measures: regulations, runway updates…) as foreseen during pre-tactical preparation and to compare it with the current tactical plan.

(5) Each windows' title bar includes the selected target application environment (ATFCM, Archive or Predict Access).

(6) When talking about dates, this help text refers to Tactical (today) and Pre-Tactical (tomorrow) dates. This is in fact the original purpose of this application.

(7) When a window is opened from the Predict Access menu, you must be aware that the proposed dates are for today, today+1 up to and including today+6. To avoid unnecessary complexity just for this specific usage, this has not been reflected in the detailed description text of each function.

4.2.3 Query Replay Access

(1) The ATFCM Collaboration Human Machine Interface application allows retrieving query list history data and displaying related reply images.

(2) This service is accessible from the Application / Query Replay menu option.

(3) This service is managed by the CUA system.

(4) This function is only displayed in the menu if your profile is authorising its usage.

4.2.4 Simulations Access

(1) Based on the user profile and selected role, the ATFCM Collaboration Human Machine Interface application menu could propose to retrieve data from alternative servers: the **Simulations** systems.

(2) See section Simulations Selection.

4.3 Main Menu

4.3.1 Description

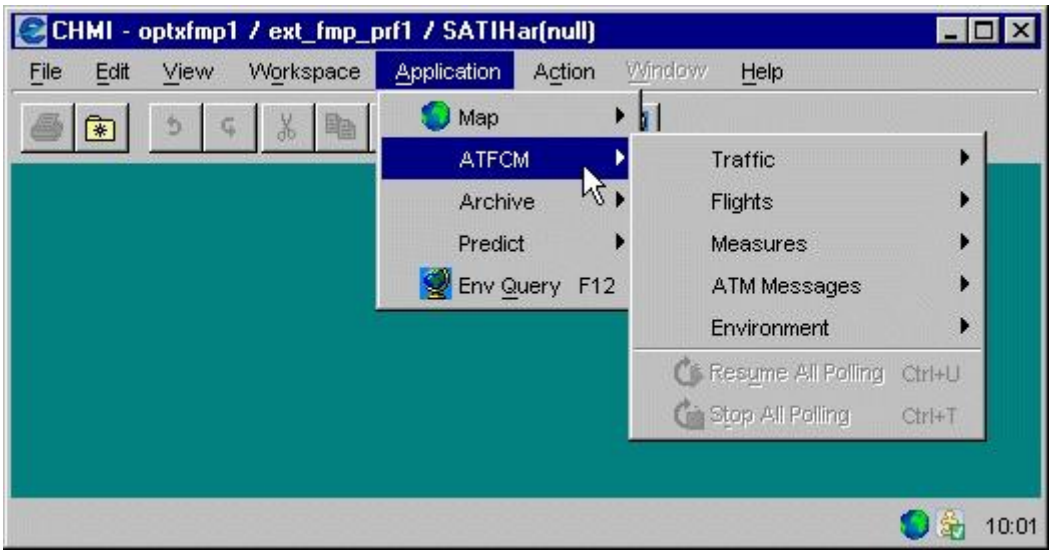


Figure Chmi - 1: Main Menu

(1) From the CHMI **Application / ATFCM** main menu, you can select the function you want to execute.

(2) Functions are only displayed in the menu if your profile is authorising its usage.

(3) It is only when a function has been selected that the corresponding dedicated Window appears.

4.3.2 Menu items

(1) **Traffic**

a) Monitor / FMP Monitor: displays a graph with the load states, regulation information and rerouting periods for the traffic volumes belonging to selected FMPs.

b) Monitor / ATC Airspace Monitor: displays a graph with the load states, regulation information and rerouting periods for the airspaces included in a selected configuration of an ATC Unit Airspace.

c) A separator line shall be added after the last Monitor option present in the Monitor group ("Monitor / FMP Monitor" or "Monitor / ATC Airspace Monitor") and before the following options in the Monitor group, if any is authorised.

d) Monitor / ATC Configuration: gives the ATC configuration and the activation table for one ATC unit airspace. This screen may give subsequent access to the maintenance function, if user is authorised.

e) **Monitor / ATC Configuration Optimizer:** Displays the optimal list of ATC configurations for one ATC Unit Airspace and for each interval of the selected period.

f) A separator line shall be added after the last ATC Configuration option present in the Monitor group ("Monitor / ATC Configuration" or "Monitor / ATC Configuration Optimizer") and before the following options in the Monitor group, if any is authorised.

g) **Monitor / Occupancy Traffic Monitoring Values:** Displays detailed Occupancy Traffic Monitor Values definition for a selected Traffic Volume and Duration. This screen may give subsequent access to the maintenance function, if user is authorised.

h) **Monitor / ATC Airspace OTMV List:** lists OTMV definition state on a selected date and time for all TV in the area of responsibility of an ATC Unit Airspace (A non clustered AUA or a cluster).

i) Traffic Counts: displays in a graph or a table the number of flights entering in a traffic volume or over a location. Related Regulated Rates are also available.

j) Flow Counts: displays in a graph or a table the number of flights per flow entering in a traffic volume.

k) Delays: displays in a graph or a table the delays for a Departure Aerodrome or a Regulation.

(2) **Flights**

a) Flight List: lists all the flights for a given traffic type and the associated selection criteria (traffic volume, aerodrome, point, ...).

b) ARCID List: list of flights corresponding to a specific aircraft identifier and associated optional selection criteria.

c) Suspended Flight List: lists all currently suspended flights based on the Traffic Demand for a combination of several query criteria.

d) Flight Data: gives the detailed information, route airspace profile and, if authorised, the restriction profile for one flight.

e) Point Profile: gives all details for one flight, pointing directly at the detailed route / point profile.

f) Airspace Profile: gives all details for one flight, pointing directly at the detailed airspace profile.

g) Flight History: displays Flight Event History list for one flight.

h) Flight Operational Log: lists Tactical Operational Log entries for one flight.

i) Rerouting Function: allows to evaluate and request the rerouting for the selected flight.

j) Or Alternate Routes, if the flight is not reroutable. It allows to list the different route alternative for the selected flight.

k) Tower Departure List: list of flights with related details and actions dedicated to a selected Aerodrome of departure.

(3) **Measures**

a) ATFCM Monitor: displays a graph with the regulation information for all FMPs.

b) CASA Regulations: lists the regulations for selected FMP(s) or all regulations.

c) Regulation Description: gives details for a selected Regulation including the Sub and Supplementary Periods.

d) Reroutings: lists the reroutings for selected or all FMP(s).

(4) **ATM Messages**

a) Release Notifier: lists the latest release stamps and notifies when some new ATM Messages have been released.

b) A separator line shall be added after the Release Notifier option, if present in the ATM Messages menu and before the following options, if any is authorised.

c) AIM: lists all AIMs valid for a selected date and displays additional details for the selected AIM.

d) ANM: lists the active ANM for a selected date and time.

e) ADP List: lists the latest ADP regulation list for a selected date.

f) ADP Current: gives details for a selected ADP Regulation including the Sub Periods and date.

g) ADP History: lists the history log for a selected ADP Regulation and date.

(5) **Environment**

a) **ATFCM Information:** displays the information message broadcasted by the server system.

b) A separator line shall be added after the ATFCM Information option, if present in the Environment menu and before the following options, if any is authorised.

c) **Traffic Capacity:** draws a graph with the capacity values for one location. This screen may give subsequent access to the maintenance function, if user is authorised.

d) **Aerodrome Details:** gives details for a selected Aerodrome. This screen may give subsequent access to the maintenance function, if user is authorised.

e) **Route Catalogue:** lists Standard Routes from the Tactical Route Catalogue.

f) **Traffic Volume Activation:** lists the activation status for a selected traffic volume. This screen may give subsequent access to the maintenance function, if user is authorised.

(6) **ACTION (CASA)**

a) Ready To Depart: sends an REA message for the selected flight.

b) Request For direct Improvement: sends an RFI message for the selected flight.

c) SIP Wanted Message: sends an SWM message for the selected flight.

d) Followed by a separator line to highlight the end of a logical group of menu options.

e) Slot Proposal Accept: sends an SPA message for the selected flight.

f) Slot Proposal Reject: sends an SRJ message for the selected flight.

g) Followed by a separator line to highlight the end of a logical group of menu options.

h) Rerouting Reject: sends an RJT message for the selected flight.

i) Followed by a separator line to highlight the end of a logical group of menu options.

j) Slot Missed Message: sends an SMM message for the selected flight.

k) Followed by a separator line to highlight the end of a logical group of menu options.

l) Flight Confirmation Message: sends an FCM message for the selected flight.

m) Airport Target TOT Confirmation: sends a confirmation message for a previously received Provisional Target TOT.

n) Followed by a separator line to highlight the end of a logical group of menu options.

o) Swap 1 with a free Slot: sends a request to exchange selected flight with an identified free slot.

p) Swap 2 Slots (new/reset): select/unselect a first flight to prepare the swapping of 2 allocated slots.

q) Swap Slot with: sends a request to exchange selected flight with the flight initially selected with the option Swap 2 Slots (new/reset). (only available from Regulation flight list)

4.4 Windows Title Structure

(1) To satisfy both occasional and expert user needs, the ATFCM Collaboration Human Machine Interface application shall implement two different title modes:

a) The normal window title mode uses with complete words to qualify the window title.

b) The abbreviated window title mode is intended for expert users. It uses many abbreviations to facilitate multi windowing specific requirements.

(2) The mode in use is identified by a user adjustable preference:

a) <CHMI / ATFCM application / General / Windows / Use abbreviated title >

i) Description= "Use abbreviated windows title to facilitate multi-windowing readability."

ii) *Default value= OFF (i.e. unchecked)*

(3) When the state of the *Use abbreviated title* preference is changed, the title of all opened window is immediately adjusted in line.

(4) When a window is minimized, a tooltip on the window title shall display the current window title as a reading help.

(5) Normal and abbreviated window titles shall respect the following element structure from left to right in sequence:

a) ***Main query criteria:***

b) After the first query/reply, the ***Main query criteria*** are displayed in front of the title string, when concise query description can be identified (see table "Abbreviated title details"):

i) For normal title mode, the query criteria are preceded by the NM type identifier (ex: "AD EBBR D").

ii) Abbreviated title mode only displays the query criteria (i.e. no type identifier in front) (ex: "EBBR D").

c) The ***Main query criteria*** string is appended with the optional keyword "***(Prop)"*** when the ***Normal/Proposal*** action is relevant to the function (see detailed specification) and when the displayed data is including flight "Proposals".

d) ***Function title:***

e) The ***function title*** is displayed after the Main query criteria when present, otherwise in front of the title string (see table "Abbreviated title details"):

i) Normal title uses full function text (ex: "Traffic Counts").

ii) Abbreviated title is based preferably on ICAO official acronym codes and then on Eurocontrol AIRIAL official abbreviations. (ex: "TFC CNT")

f) ***Query retrieval stamp:***

g) After the first query/reply, the ***query retrieval stamp*** is displayed next, in dd-hh:mm format:

i) For normal title mode, it is preceded by the keyword "at".

ii) Abbreviated title only displays the stamp without keyword.

h) When appropriate (see detailed specification), the ***Query retrieval stamp*** string is appended with the optional keyword **(Poll)** to indicate that the **Polling function** is active for this window.

i) When appropriate (see detailed specification), the ***Query retrieval stamp*** string is appended with the optional **>>** sign to indicate that the covered time period will shift on refresh

j) ***Target platform:***

k) As soon as the window is opened, the complete title made of above optional elements is appended with the ***target platform***

i) ATFCM, Archive or Predict

ii) Preceded by a "/" separator in normal title mode.

Function		
Normal title	**Abbreviated title (ICAO/AIRIAL)**	**Main query criteria** (- = none identified)
FMP Monitor	FMP MNT	-
ATC Airspace Monitor	ATC AS MNT	*"AUA aua_id Cfg cfg_id"*
ATC Configuration	ATC CFG	*"AUA aua_id"*
ATC Configuration Optimiser	ATC CFG OPT	*"AUA aua_id"*
Traffic Counts	TFC CNT	*"TV traffic_volume_id* or *"AS airspace_id"* or *"AD aerodrome_icao_id cat"* ** or *"AZ aerodrome_set_id cat"* ** or *"PT point_id"* or *"AO aircraft_operator_id"* + *"H/20"* if count type is "Entry Hour/20min" or *"H/10"* if count type is "Entry Hour/10min" or *"H"* if count type is "Entry Hour" or *"20"* if count type is "Entry 20min" or *"O/n"* if count type is "Occupancy" where *"n"* is the duration value (1-20). **=category (cat) only displayed if "D" or "A"
Flow Counts	FW CNT	*"TV traffic_volume_id"* + *"H/20"* if count type is "Entry Hour/20min" or *"H/10"* if count type is "Entry Hour/10min" or *"H"* if count type is "Entry Hour" or *"20"* if count type is "Entry 20min" or *"O/n"* if count type is "Occupancy". where *"n"* is the duration value (1-20).
Delays	DLA	*"AD aerodrome_icao_id D"* or *"RG regulation_id"*
Flight List	FLT LST	*"TV traffic_volume_id FW flow_id"* *(only if query on flow_id)* or *"AS airspace_id"* or *"AD aerodrome_icao_id cat"* ** or *"AZ aerodrome_set_id cat"* ** or *"PT point_id"* or *"AO aircraft_operator_id"* or *"RG regulation_id"* + *"O/n"* if count type is "Occupancy" where *"n"* is the duration value (1-20). **=category (cat) only displayed if "D" or "A"
ARCID List	ARCID LST	*"AC aircraft_id"*
Suspended Flight List	SUS FLT LST	*"TV traffic_volume_id FW flow_id"* *(only if query on flow_id)* or *"AS airspace_id"* or *"AD aerodrome_icao_id cat"* ** or *"AZ aerodrome_set_id cat"* ** or *"PT point_id"* or *"AO aircraft_operator_id"* or *"RG regulation_id"*

		**=category (cat) only displayed if “D” or “A”
Flight Data	FLT DATA	*“AC aircraft_id"*
Flight History	FLT HIST	*“AC aircraft_id"*
Operational Log	OPR LOG	*“AC aircraft_id"*
Alternate Routes	ALTN RT	*“AC aircraft_id"*
Tower Departure List	TWR DEP LST	*“AD aerodrome_icao_id"* or *“AZ aerodrome_set_id"*
Plotted Flight List	PLOT FLT LST	-
ATFCM Monitor	ATFCM MNT	-
CASA Regulations	CASA REGUL	-
Regulation Description	REGUL DESC	*“RG regulation_id"*
Reroutings	RER	-
Release Notifier	REL NOTIF	-
CRAM	CRAM	-
AIM List	AIM LST	-
ANM	ANM	-
ADP List	ADP LST	-
ADP Current	ADP CUR	*“RG regulation_id"*
ADP History	ADP HIST	*“RG regulation_id"*
ATFCM Information	ATFCM INFO	-
Traffic Capacity	TFC CAP	*“TV traffic_volume_id"* or *“AS airspace_id"* or *“AD aerodrome_icao_id cat"* ** or *“AZ aerodrome_set_id cat"* ** or *“PT point_id"* **=category (cat) only displayed if “D” or “A”
Aerodrome Details	AD DET	*“AD aerodrome_icao_id"*
Route Catalogue	RCAT	*“ADEP aerodrome_icao_id* *ADES aerodrome_icao_id"*
Traffic Volume Activation	TFC VOL ACT	*“TV traffic_volume_id"*
Query Replay	Query Replay	-
Occupancy Traffic Monitoring Values	OTMV	*"TV traffic_volume_id"*
ATC Airspace OTMV List	ATC AS OTMV LST	*"AUA aua_id"*

Table Chmi-1: Abbreviated title details

(6) Example with Traffic Counts window:

a) Normal title mode:

i) TV EDMDONU O/5 Traffic Counts at 20-09:58 / ATFCM

b) Same title in abbreviated title mode:

i) EDMDONU O/5 TFC CNT 20-09:58 ATFCM

(7) Each CHMI ATFCM window shall be associated with an identification parameter that can be used by other support software (ex: Test Director) to recognize the displayed window object.

(8) It is important that this identification parameter contains only static string information:

a) For example, the function normal title

b) But NO variable parts like the query data elements.

4.5 Common functionalities

4.5.1 Preferences dialog

(1) The **Preferences** dialog is started from the main menu option **File / Preferences...**.

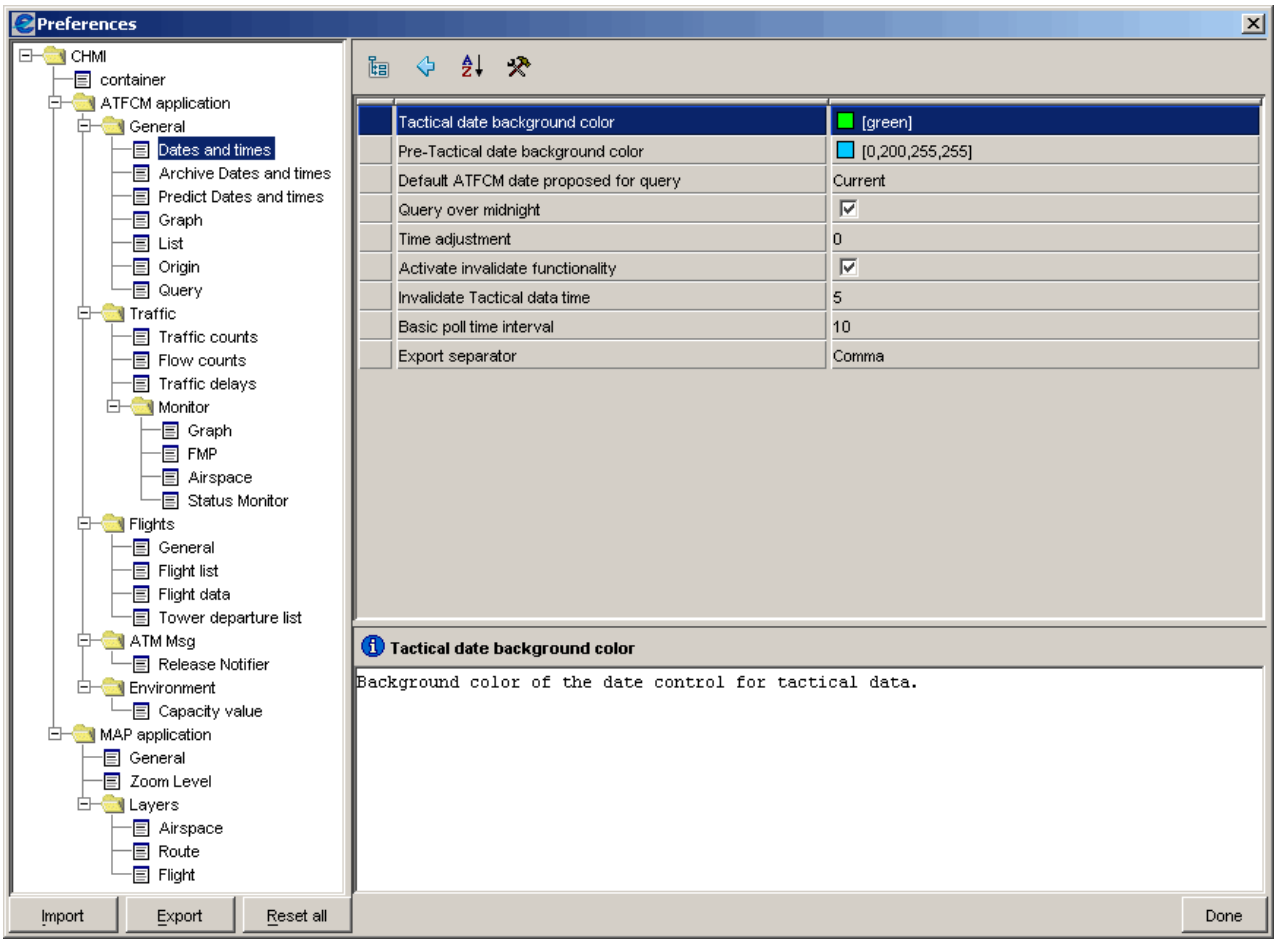


Figure Chmi - 2: Preferences dialog

(2) The Preferences dialog is composed of:

a) The preference **structure tree** on the left. The structure is based on the application menu.

b) A **button toolbar** at the top on the right side.

c) Below the toolbar on the right, the **list of preferences** related to the selected entry in the structure tree. Each preference is listed with the currently associated value. The string "<value not defined>" shall be displayed when there is no value associated with a preference.

d) At the bottom of the list, the **description text** of the currently selected preference.

(3) The ATFCM Collaboration Human Machine Interface application is customised based on

a) User adjustable preferences.

b) System properties.

(4) The Preferences dialog allows to display these properties and possibly maintain the preferences.

(5) When a preference is listed in grey, it cannot be updated by the user. It is there just for information.

(6) System properties are indicated by a picture of tools on the left of the preference name in the list.

(7) When a specific editor is available for updating one or more preferences, a button with three dots is displayed at the right when the corresponding edit box is selected.

(8) Each time a preference is mentioned in this document, the tree structure access path is provided in front of the preference name.

(9) The button toolbar allows the following actions:

a) The first button allows unfolding completely the preference tree.

b) The second button allows resetting the currently selected preference to the associated default value.

c) The third button allows swapping the sort sequence of the currently displayed preference list between ascending and descending alphabetical order of the identifiers. Initial display sequence can only be restored by closing and re-opening the preference dialog.

d) The fourth button allows showing or hiding the system properties in the list.

(10) Below the structure tree, three buttons allow to:

a) **Import** all preferences from a file created earlier with the Export button.

b) **Export** the current image of all your user adjusted preferences value into a non over-writable file.

c) **Reset all** preferences to the associated default value. A confirmation dialog will prompt you before the reset is performed.

(11) This functionality gives the possibility to save the user configured application at a given time and restore it to the identical configuration at any time later (for example: application customisation for a group of users).

(12) In practice, an Import command will:

a) First reset all user adjustable preferences to the initial default value.

b) Then possibly overwrite the default value with the user preference saved earlier by the Export.

(13) The **Export** / **Import** functionalities of the **Preferences** dialog shall save and restore all user defined **Show / Hide Columns** local system preferences.

(14) The application will refuse to overwrite an existing file to avoid undesired destruction of previous preferences set.

(15) This is a modal dialog which blocks access to all other functions in the application.

4.5.2 Tactical / Pre-Tactical dates

(1) The ATFCM Collaboration Human Machine Interface application gives access to:

a) The Tactical operational data (current day).

b) The Pre-Tactical operational data (tomorrow).

(2) Because of the possible misalignment of the clock on the local PC, when Tactical and Pre-Tactical dates shall be proposed in a query control, the ATFCM Collaboration Human Machine Interface application will always propose 4 dates:

a) Today - 1

b) Today

c) Today + 1

d) Today + 2

(3) On opening a new window, the system will propose a **default date** to be queried.

(4) Using the user preference **CHMI / ATFCM application / General / Dates and times / Default date**, there are two options to control the proposed default date:

a) **Current** will propose the current PC date as the default query date.

b) **Last used** will propose the same relative date as the one you have used for your last query. (for example: if you always query the Pre-Tactical day, you will always be prompted with the Pre-Tactical date)

(5) The Tactical and Pre-Tactical date selection is highlighted by different **background colours**.

(6) These colours shall be identified using the following user preferences:

a) CHMI / ATFCM application / General / Dates and times / Tactical date background colour

b) CHMI / ATFCM application / General / Dates and times / Pre-Tactical date background colour

4.5.3 Time adjustment

(1) Latest Windows technology is pretty safe in UTC and local time conversion.

(2) To allow for some flexibility in solving any remaining time alignment problems, the application is using a **Time adjustment** user preference.

(3) The **CHMI / ATFCM application / General / Dates and times / Time adjustment** preference shall be adjusted if time alignment problems arise.

(4) The property may contain a positive or negative integer expressing a number of minutes.

(5) This signed number will be added to the current UTC time before other default time calculation.

4.5.4 Default query period

(1) The calculation of the query period proposed by default is based on system properties and user adjustable preferences.

(2) When a default period is needed for the query, it is proposed based on the user preference **Default period type** related to the function used and the target server (ATFCM, Archive, Predict) (ex: CHMI / ATFCM application / Flights / Flight list / ATFCM default period type).

(3) This user adjustable preference identifies one of the following methods:

a) **0000 to 0000** for full day (WEF=00:00 UNT=00:00).

b) **WEF to 0000** for calculated WEF time to the end of the day (00:00).

c) **WEF to UNT** for calculated WEF and UNT times.

(4) Dependant on the **Maximum elapse** time allowed for the query period, the **Default period type** might be forced to the third method (Calculated WEF and UNT times).

(5) If the **Maximum elapse** is less than one day (1440 minutes), this third method is the only workable solution.

(6) This maximum elapse time for the query period is stored in the system property **Maximum elapse** related to the function used.

(7) This system property is not modifiable by the user. These system property values are expressed in minutes and have following values:

a) CHMI / ATFCM Application / Traffic / Traffic Counts / Entry / Entry maximum elapse: 1440

b) CHMI / ATFCM Application / Traffic / Traffic Counts / Occupancy / Occupancy maximum elapse: 720

c) CHMI / ATFCM Application / Traffic / Flow Counts / Entry / Entry maximum elapse: 1440

d) CHMI / ATFCM Application / Traffic / Flow Counts / Occupancy / Occupancy maximum elapse: 720

e) CHMI / ATFCM Application / Traffic / Traffic Delays / Maximum elapse: 1440

f) CHMI / ATFCM Application / Traffic / Monitor / FMP / Maximum elapse: 1440

g) CHMI / ATFCM Application / Traffic / Monitor / ATC Airspace / Maximum elapse: 1440

h) CHMI / ATFCM Application / Traffic / Monitor / ATFCM Monitor / Maximum elapse: 1440

i) CHMI / ATFCM Application / Flights / Flight List / Maximum elapse: 1440

j) CHMI / ATFCM Application / Flights / Flight List / Maximum elapse for AO list: 1440

(8) On opening a new window, the proposed default **WEF** is calculated using the following rules.

(9) If **Default period type** is **0000 to 0000**, the proposed default **WEF** is forced to 00:00.

(10) If **Default period type** is **WEF to 0000** or **WEF to UNT**, the proposed default **WEF** will be calculated as follow:

a) Basic time used is the current UTC time.

b) Corrected with the **Time adjustment** user preference as described in previous section (CHMI / ATFCM application / General / Dates and times / Time adjustment).

c) A lead time parameter is then subtracted from this adjusted UTC time. The **Query lead time** is a user adjustable preference related to each function and for each target server (ATFCM, Archive, Predict) (ex: CHMI / ATFCM application / Flights / Flight list / ATFCM query lead time).

d) The obtained time is then rounded to the previous 20 minutes step.

(11) The system property **Query over midnight** which authorises to query over two days, may possibly limit this time calculation process to the start of the day (00:00) (CHMI / ATFCM application / General / Dates and times / Query over midnight).

(12) On opening a new window, the proposed default **UNT** is calculated using the following rules.

(13) If **Default period type** is set to **0000 to 0000** or **WEF to 0000** method, and the **Maximum elapse** authorises to query on full day (1440 minutes), then the default **UNT** time will be set to the end of the selected day (00:00).

(14) If **Default period type** is set to **WEF to UNT** method, the **UNT** time will be calculated by adding the **Query default elapse** number of minutes to the WEF time provided in the related input control.

(15) The **Query default elapse** is a user adjustable preference related to each function and for each target server (ATFCM, Archive, Predict) (ex: CHMI / ATFCM application / Flights / Flight list / ATFCM query default elapse)

(16) The **Query default elapse** must be smaller or equal to the system property **Maximum elapse** corresponding to the function (ex: CHMI / ATFCM application / Flights / Flight list / Maximum elapse).

(17) The system property **Query over midnight** which authorises to query over two days, may possibly limit this time calculation process to the end of the day (00:00) (CHMI / ATFCM application / General / Dates and times / Query over midnight).

(18) On issuing a query with a blank **WEF** or **UNT** time for the period, the default value will be provided based on above user preference **Default period type**.

(19) When opening a new window from a contextual popup selection on another window, some query data elements are provided by the calling function.

(20) The **Force default period** is a user adjustable preference related to each function and for each target server (ATFCM, Archive, Predict)(ex: CHMI / ATFCM application / Flights / Flight list / ATFCM force default period).

(21) It allows to ignore the passed values and to use the defaults as if the window was opened from the main menu.

(22) When checked, the **Force default period** preference forces the use of calculated default values for WEF and UNT.

4.5.5 Send action

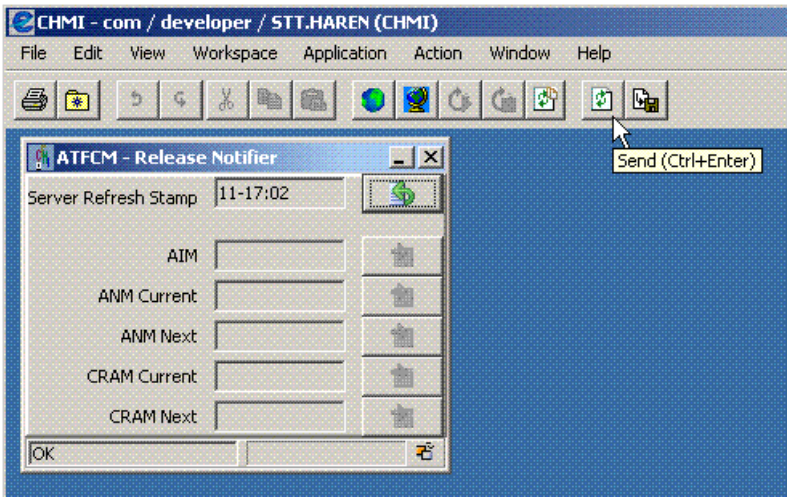


Figure Chmi - 3: Send Button

(1) Enabled: when the application is available for processing this action.

(2) Disabled: while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(3) Action: will issue the query to the Network Operations database if all necessary query elements are available and correct. Otherwise an error message will help in query preparation.

(4) Not applicable: never

(5) Shortcut key: **<Ctrl>** + **<ENTER>**

(6) **<ENTER>** key: the Send being the default action in most cases, the **<ENTER>** key will trigger it when no other default button is available on the active window.

(7) Menu option: **Action / Send**

4.5.6 Reset action

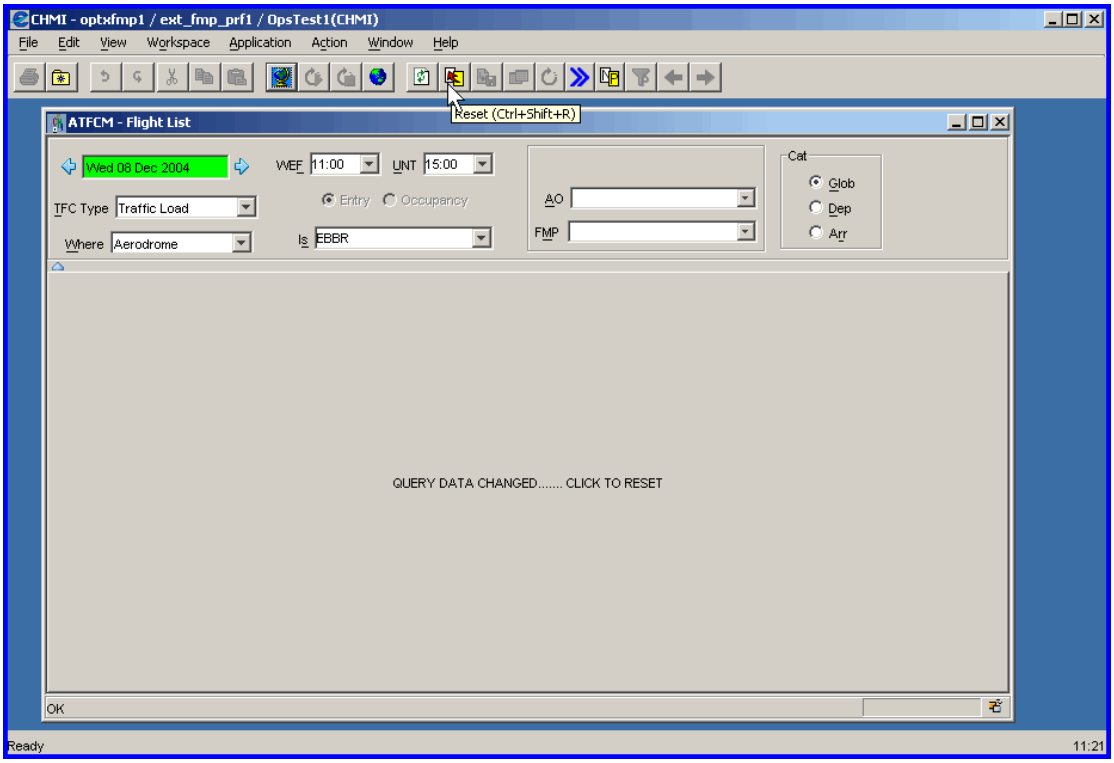


Figure Chmi - 4: Reset Button

(1) Enabled: after valid reply data are displayed, as soon as one of the initial query criteria is changed and the application is available for processing this action.

(2) Disabled: before initial query or while query criteria correspond to displayed reply data or while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(3) Applicable: to all windows where the criteria may be updated to issue an updated query.

(4) Large RESET button: in parallel with the activation of the Reset button from the Button Bar, a large RESET button covering all reply data will be displayed. This is implemented to avoid mismatch between currently retrieved reply data and modified query criteria.

(5) Action: will restore all query criteria corresponding to the currently retrieved reply data. The large RESET button hiding the reply data is consequently removed.

(6) Shortcut key: **<Ctrl> + <Shift> + <R>**

(7) Menu option: **Action / Reset**

4.5.7 Invalidate function

(1) Once some data have been retrieved from the Network Operations database, the ATFCM Collaboration Human Machine Interface application provides an automatic warning mechanism for out of date data.

(2) This mechanism is only implemented for volatile data related to flights and counts only.

(3) The elapse time before the display is marked as "**OBSOLETE DATA !**" may be adjusted using the following user preference:

a) CHMI / ATFCM application / General / Dates and times / Invalidate tactical data time

(4) The warning is displayed on a large button covering the data part of the window.

(5) Once the data has been marked as OBSOLETE, the user may:

a) Click the **Send** button or the "**OBSOLETE DATA... CLICK TO REFRESH**" large warning button to retrieve fresh data for the same query

b) Click the **Reset** button to keep the current data

c) Modify the query data elements to **Send** a modified query.

(6) The user may decide to deactivate this Invalidate mechanism by adjusting the following user preference:

a) CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality

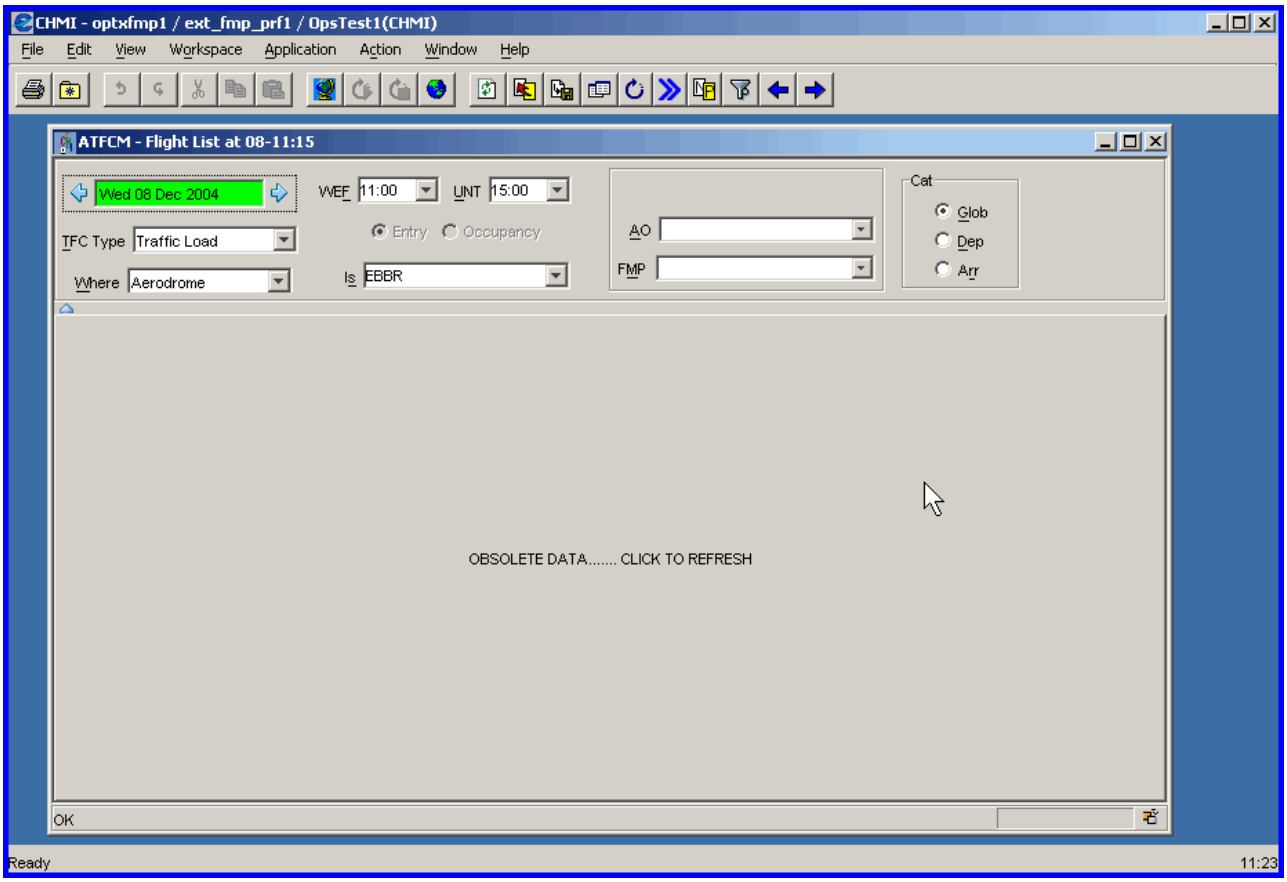


Figure Chmi - 5: OBSOLETE DATA warning

4.5.8 Sort action

(1) The sort action is available on most list display.

(2) It allows to alter the displayed sequence on retrieved data.

(3) The sorting is done locally and therefore doesn't need to retrieve additional data from the Network Operations database.

(4) To modify the sorting of a displayed list, click on column headers corresponding to your desired query criteria.

(5) A first click on an unsorted column will request to sort in ascending order on the selected column.

(6) Any subsequent click on an already sorted column will swap between ascending and descending order for the selected column.

(7) An arrow in front of the title of the column indicates the order of the currently applied sort for that column.

a) An arrow up indicates an ascending sequence.

b) An arrow down indicates a descending sequence.

(8) The number next to the arrow indicates the position of the column in combined sort sequences.

(9) Shift key combined with a click on the desired column allows the concatenation of several sort criteria.

(10) Default initial system sort sequence is provided in most of the cases.

(11) User is then free to modify the sort sequence used by the current active window.

(12) The modified sort sequence only applies to the window on which it was defined.

(13) The sort sequence may be changed before any data is retrieved.

(14) Modified sort sequence remains active even after a data refresh action.

4.5.9 Filter action

(1) The **Filter** action is only presented to the user when a window is displayed where a filter could be applied.

(2) Enabled: when the application is available for processing this action.

(3) Disabled: while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(4) Action: will prompt the **Filter table** dialog box proposing to implement a filter for each of the column of the main list. For each of the column, a filter may be created by selecting an operator and the value to compare to.

(5) Shortcut key: **<Ctrl>** + **<Shift>** + **<F>**

(6) Menu option: **Action / Filter**

(7) The **Action / Filter** menu option and the related button on the button bar shall have two possible states, when applicable for the currently selected window:

a) No filter active

i) No checkmark on menu option

ii) Normal background colour for the button

b) A filter is active on the currently selected window

i) Checkmark in front of menu option

ii) Specific background colour for the button - To be stored in the user adjustable preference **CHMI / container / Filter highlight colour** with default value "magenta" - Description: "Specific colour to be used for highlighting the presence of an active filter (i.e. filter button, column header …)"

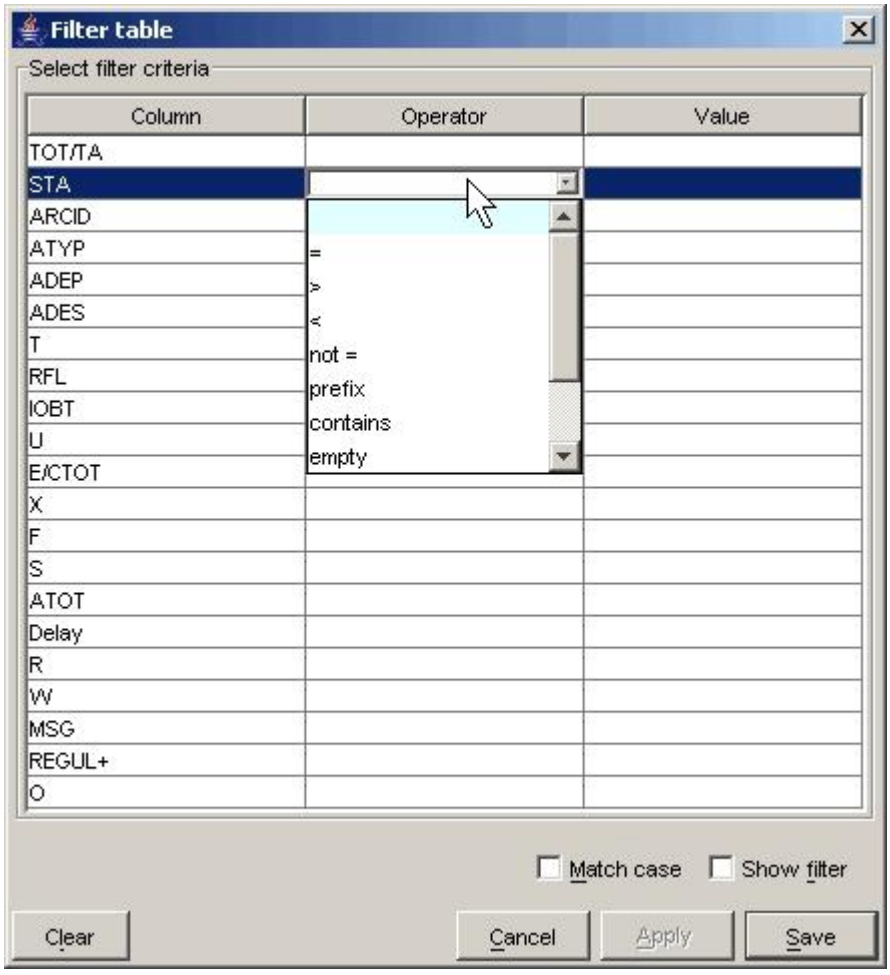


Figure Chmi - 6: Filter dialog box

(8) The Filter function allows to process sub-selection on retrieved data.

(9) The filtering is done locally and therefore doesn't need to retrieve additional data from the Network Operations database.

(10) The Filter function is then started for the currently active dialog, if this dialog contains a data list that could be filtered.

(11) This **Filter table** dialog is modal, so no other action is allowed while it is displayed.

(12) This dialog will allow to provide for each column in the table to be filtered:

a) The comparison operator that shall be used for filtering.

b) The value against which the selected column will be compared.

(13) The different comparison operators that can be used to Filter the detail lines are:

a) **none** (blank entry): when no filtering shall be applied

b) **=**: where the related column is equal to the entered value

c) **>**: where the related column is greater than the entered value

d) **<**: where the related column is smaller than the entered value

e) **not =**: where the related column is not equal to the entered value

f) **prefix (,)**: where the related column starts with one of the value entered in a comma separated list

g) **no prefix (,)**: where the related column does not start with any of the value entered in a comma separated list

h) **contains (,)**: where the related column contains one of the value entered in a comma separated list

i) **does not contain (,)**: where the related column does not contain any of the value entered in a comma separated list

j) **empty**: where the related column is empty

k) **not empty**: where the related column is not empty

l) **in (,)**: where the related column contains one of the values provided in a comma separated list

m) **not in (,)**: where the related column does not contain one of the values provided in a comma separated list

(14) For the "=", ">", "<" and "not =" comparison operator, if both the selected column and the entered value contains only numeric digits, a numeric comparison is performed. Otherwise an alphanumerical comparison is performed from left to right.

(15) The comparison value entered must include any special character like the ":" in case of a time comparison. The alphanumerical comparison is performed from left to right on the exact string of character you did provide.

(16) Leading and trailing blanks are ignored during comparison.

(17) When a filter is active on a selected window, the caption giving the number of rows in the list will contain:

a) The number of lines displayed after filtering

b) The total number of lines retrieved before filtering

c) The title of the column is displayed in italic using the specific **Filter highlight colour** (see requirement *1.5.9 Filter action (6) and (7)*) to indicate the presence of a filter on that column.

(18) Like for the sort, the filter only applies to one window.

(19) Any new window has no filter when it is created.

(20) The filter values may be entered before any data is retrieved.

(21) Clicking the **Apply** button on the dialog applies the filter to the active window without closing the dialog.

(22) Clicking the **Clear** button closes the filter dialog and removes any existing filter

(23) Clicking the **Cancel** button closes the filter dialog and keeps the situation as it was before invoking the Filter dialog.

(24) Clicking the **Save** button on the dialog applies the filter to the active window and closes the dialog.

(25) Checking the **Match case** checkbox forces all alphabetical comparisons to be case sensitive for the current dialog. The state of this checkbox is persistent for the currently defined filter.

(26) Checking the **Show filter** checkbox allows to temporarily limit the display to the columns on which a filter has been implemented.

4.5.10 Normal / Proposal Flights

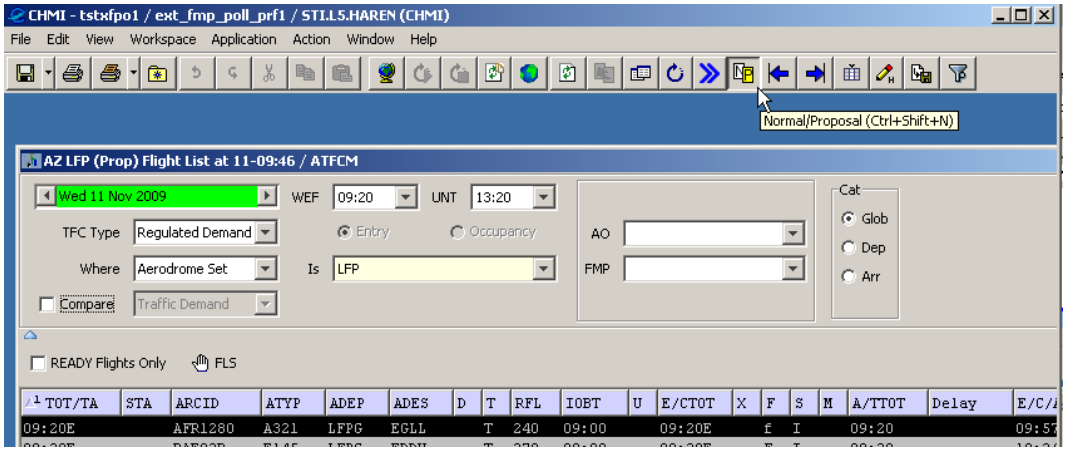


Figure Chmi - 7: Normal / Proposal Flights - Buttons and Title Bar

(1) The **Normal/Proposal** action is only presented to the user when it is used by the active window to prepare a query or to initiate a contextual popup request to another window that uses this indicator.

(2) Enabled: when the application is available for processing this action.

(3) Disabled: while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(4) Action: when issuing a new query or refreshing an existing window, where flights need to be retrieved or counted, this button is checked to prepare the request to the server. All replies that include flight data or counts will use this indicator to identify whether they shall take into account or ignore the flight Proposal issued by the Tactical system.

(5) Shortcut key: **<Ctrl>** + **<Shift>** + **<N>** to toggle between Normal and Proposal.

(6) Menu option: Action / Normal / Proposal

(7) The title bar of each window that contains flight counts or data, indicates the mode used for the data retrieval. The keyword **(Prop)** between the title and the stamp indicates that the displayed data are including the flight "Proposals". A refresh might alter the initial mode used.

(8) When a flight list might contain proposal flights, a special title background colour for the flight listbox is used. This colour may be adjusted using the following user preference:

a) **CHMI / ATFCM application / Flights / General / Proposal colour**

4.5.11 Print / Export function

(1) Enabled: as soon as some valid reply data are retrieved from the Network Operations database and the application is available for processing this action.

(2) Disabled: before any valid reply data are retrieved or when the query criteria are not in line with the available reply data or while the application is processing

some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(3) Applicable: all ATFCM Collaboration Human Machine Interface application functions.

(4) Action: will invoke the print or the export function for the currently active window.

(5) Shortcut key: **<Ctrl>** + **<P>** for Print and **<Ctrl>** + **<Shift>** + **<E>** for Export

(6) Menu option: **File / Print** or **Action / Export**

(7) The Print function allows to produce a hard copy of the retrieved data.

(8) Before invoking the print function, the displayed data may be sorted and filtered. The printout will reflect what is currently displayed on the window.

(9) The only additional information needed is the identification of the destination printer.

(10) A modal dialog will prompt for this information. Default printer is proposed.

(11) The Export function allows to save all the retrieved data into a text file.

(12) Before invoking the export function, the displayed data may be sorted and filtered. The exported data will reflect what is currently displayed on the window.

(13) Click the **Export** button or select the **Action / Export** menu option.

(14) The Export function is then started for the currently active dialog.

(15) This Export dialog is modal, so no other action is allowed while it is displayed.

(16) The following dialog box will ask you to provide the location and the file name in which the exported data will be stored.

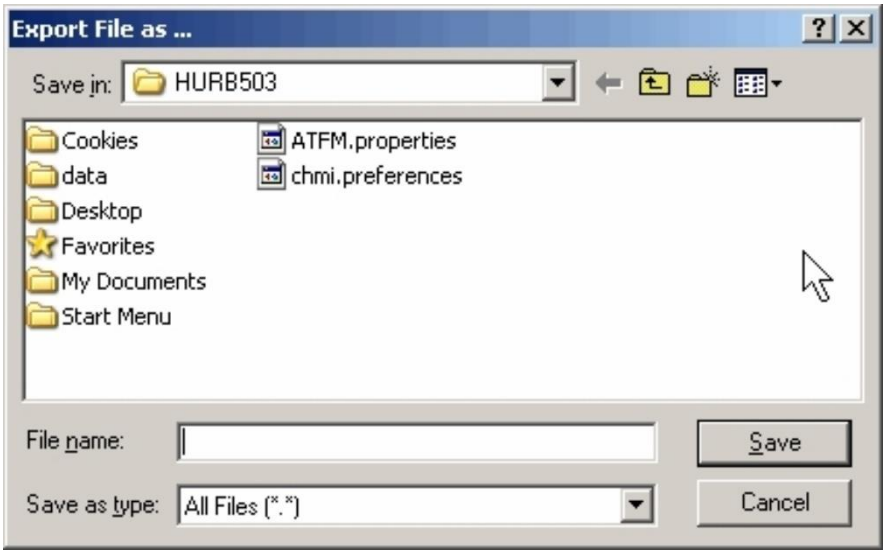


Figure Chmi - 8: Export File as dialog box

(17) The minimum input is the desired file name.

(18) Choose OK when you have selected the file reference.

(19) If the file already exists, you will be prompted with the following confirmation dialog.

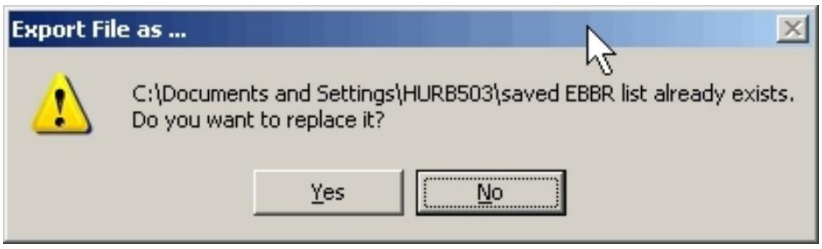


Figure Chmi - 9: Export - Confirmation dialog box

(20) The data is exported in a text file format, so it can be accessed with any editor.

(21) All query data elements are listed first, followed by the reply data.

(22) A special character separates each data element.

(23) **CHMI / ATFCM application / General / Dates and times / Export separator** user preference shall be adjusted to identify the separator character to be used.

(24) This feature allows to import the exported CHMI file into other standard software and to use the separator between the different data elements. (i.e. one data element per cell in a spreadsheet).

4.5.12 Period Shifting function

(1) Enabled: as soon as a window where period shifting is applicable becomes active and the application is available for processing this action.

(2) Disabled: before any valid reply data are retrieved or when the query criteria are not in line with the available reply data or while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(3) Applicable: all windows where the selection criteria include a time period that could be shifted.

(4) Action: Toggle button "**>>** " to switch on and off the shifting status of the currently active window.

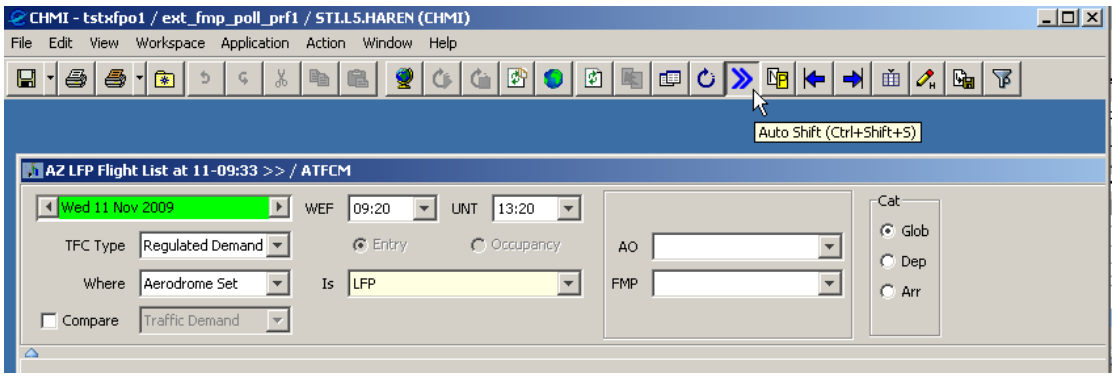


Figure Chmi - 10: Period shifting toggle button

(5) Shortcut key: **<Ctrl>** + **<Shift>** + **<S>**

(6) Menu option: **Action / Auto Shift**

(7) When Shifting is activated before a refresh, the query period is automatically shifted with the number of minutes elapsed since the last data retrieval from the server occurred.

(8) When Shifting is on for a window, it is indicated by the presence of **>>** in its title.

(9) By default, windows are in non-shifting state (fixed period), which implies that on refresh, the initial query period remains fixed.

(10) Shifting is activated either by clicking the **>>** button to toggle on, selecting the menu option **Action / Auto Shift** or when the automatic Polling is activated for a window.

(11) Shifting is de-activated either by clicking the **>>** button to toggle off, or by unchecking the menu option **Action / Auto Shift**.

(12) Shifting state is not reset when Polling is stopped for a window.

(13) Note: see also the **Polling function** paragraph in this document.

4.5.13 Partial Period Shifting function

(1) Enabled: when an earlier / later query period is available and the application is available for processing this action.

(2) This processing will take into account if the possibility to query over midnight is authorised by your user profile (CHMI / ATFCM application / General / Dates and times / Query over midnight).

(3) Disabled: **Shift-*x*** when WEF time is 00:00 or **Shift+*x*** when UNT time is 00:00 and query over midnight is not authorised or while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(4) Applicable: **Monitoring, Counts, Delays** and **Flight List** functions.

(5) Action: either button "**<< -x**" or "**>> +x**" will issue exactly the same query but for the previous or next period shifting by the number of minutes identified as described below.

(6) Depending on the type of currently active window, the shift value of ***x*** minutes may be assigned different values.

(7) **Traffic Monitoring, Counts and Delays functions** shall use following user adjustable preference values to identify the number of minutes to shift a query period related to:

a) **CHMI / ATFCM application / General / Dates and times / Entry period shift time**

i) Description: Time in minutes for partial shifting of query period on Entry Traffic.

ii) Default value: **60** minutes

b) **CHMI / ATFCM application / General / Dates and times / Occupancy period shift time**

i) Description: Time in minutes used for partial shifting of query period on Occupancy Traffic.

ii) Default value: **10** minutes

(8) To better answer ATFCM monitoring needs, **Flight List** shall use another ***x*** assignment method for partial period shifting.

a) When the **Flight List** is invoked from Traffic Counts or Flow Counts, ***x*** shall be assigned the **count step period** used by the invoking Traffic Counts window (i.e. 1 min for Occupancy, 20 or 60 minutes for Entry Counts).

b) When the **Flight List** is not invoked from Traffic Counts or Flow Counts (i.e. the count stepping cannot be identified), the **Partial Period Shifting function** is not proposed to the user (hidden buttons).

c) If any other parameter than the query period is changed, the **Partial Period Shifting function** is no more proposed to the user (hidden buttons) unless the RESET function is triggered.

(9) The shifted period is possibly truncated if we reach 00:00 and query over midnight is not authorised.

(10) Shortcut key: **<Ctrl>** + NumPad **<->** (minus sign) for **Shift-*x*** and **<Ctrl>** + NumPad **<+>** (plus sign) for **Shift+*x***

(11) Menu option: **Action / Shift-*x*** and **Action / Shift+*x*** where ***x*** is the number of minutes that will be used by the partial shifting action of the currently active window.

4.5.14 Polling function

(1) Enabled: as soon as some valid reply data are retrieved from the Network Operations database and the application is available for processing this action.

(2) Disabled: when the user security profile does not allow the use of the automatic polling facility or before any valid reply data are retrieved or while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(3) Applicable: all displays where non static traffic data are displayed (Flight Lists, Delays, Traffic and Flow Counts and Traffic Monitor functions).

(4) Action: Toggle button (circle shaped arrow) to switch on and off the polling function for the currently active window.

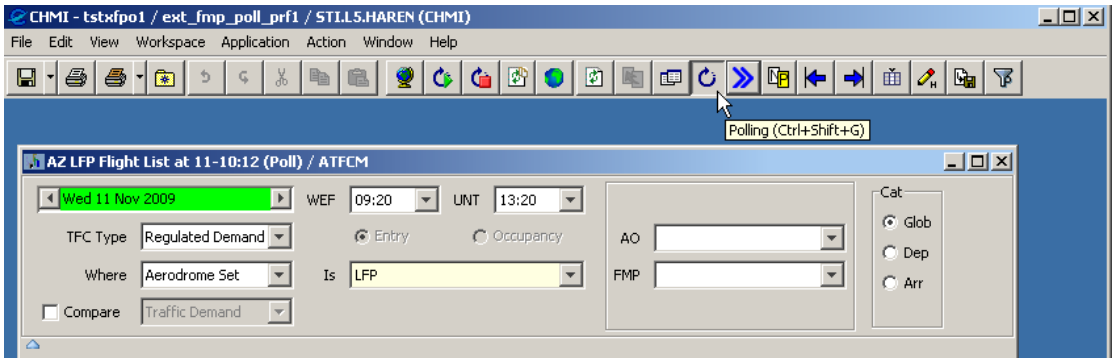


Figure Chmi - 11: Polling toggle button

(5) Shortcut key: **<Ctrl> + <Shift> + <G>**

(6) Menu option: **Action / Polling**

(7) Once some data have been retrieved from the Network Operations database, the ATFCM Collaboration Human Machine Interface application provides an automatic polling mechanism, which will keep the display up to date.

(8) This polling mechanism is only available for the functions where the situation is supposed to evolve over time and limited to the **Polling maximum elapse** system property associated to each function. This system property is not modifiable by the user. These system property values are expressed in minutes and have following values:

a) CHMI / ATFCM Application / Traffic / Traffic Counts / Entry / Entry polling maximum elapse: 600

b) CHMI / ATFCM Application / Traffic / Traffic Counts / Occupancy / Occupancy polling maximum elapse: 600

c) CHMI / ATFCM Application / Traffic / Flow Counts / Entry / Entry polling maximum elapse: 600

d) CHMI / ATFCM Application / Traffic / Flow Counts / Occupancy / Occupancy polling maximum elapse: 600

e) CHMI / ATFCM Application / Traffic / Traffic Delays / Polling maximum elapse: 300

f) CHMI / ATFCM Application / Traffic / Monitor / FMP / Polling maximum elapse: 1440

g) CHMI / ATFCM Application / Traffic / Monitor / ATC Airspace / Polling maximum elapse: 1440

h) CHMI / ATFCM Application / Traffic / Monitor / ATFCM Monitor / Polling maximum elapse: 1440

i) CHMI / ATFCM Application / Flights / Flight List / Polling maximum elapse: 240

(9) A message box will notify when the current query period of the window does not allow to start polling.

(10) To start / stop the polling mechanism for the current active window, the user has just to click the **Polling** toggle button or check the equivalent menu option **Action / Polling**.

(11) Activating the Polling mode for a window will force it at the same time to **Period Shifting** state.

(12) The application will notify when the maximum number of windows in poll mode is reached. This maximum is fixed by your user profile and is displayed with the following system property:

a) CHMI / ATFCM application / General / Dates and times / Maximum polling window.

(13) A message box will invite to stop first polling on another window.

(14) Following user preference shall be adjusted to modify basic time interval between two automatic refreshes of one window:

a) CHMI / ATFCM application / General / Dates and times / Basic poll time interval.

(15) This user preference may only be adjusted to a value greater or equal to the minimum authorised by your user profile and displayed with the following system property:

a) CHMI / ATFCM application / General / Dates and times / Enforced minimum polling interval.

(16) This basic interval is increased with a penalty, proportionally to the number of parallel windows started in poll mode. This penalty value is identified by your user profile and is displayed with the following system property:

a) CHMI / ATFCM application / General / Dates and times / Polling penalty.

(17) Once started, the automatic polling will process a pre-defined number of refreshes after each other. This pre-defined number is identified by your user profile and is displayed with the following system property:

a) CHMI / ATFCM application / General / Dates and times / Maximum consecutive poll.

(18) After this number of consecutive automatic refreshes is reached, the Polling function is stopped for that window. Automatic refresh then needs to be restarted by the user by clicking the **Polling** button again.

(19) The menu option **Application / CHMI / ATFCM application / Resume All Polling** (Shortcut key: **<Ctrl> + <U>**):

a) will reset the polling count for all windows currently active in auto refresh mode,

b) and restart polling for all windows that have reached the maximum number of consecutive poll.

(20) To stop automatic refresh for all windows currently in Poll mode, the user has just to select menu option **Application / CHMI / ATFCM application / Stop All Polling**. (Shortcut key: **<Ctrl>** + **<T>**)

(21) When the polling is started for a window, its title contains the indication **(Poll)** appended at the end. This allows an easy identification in the window list.

(22) The polling for one window is stopped:

a) By clicking the **Polling** button or menu option,

b) By closing the desired window,

c) When an error message is received from the Tactical server,

d) When the maximum number of consecutive poll is reached for that window,

e) By selecting the **Application / CHMI / ATFCM application / Stop All Polling** menu option.

(23) The **Polling** action is disabled during query processing for one window only. This action only applies to the currently selected window.

(24) The **Resume All Polling** and **Stop All Polling** actions are only enabled after at least one window has been started in Poll mode. They are not disabled while a window processes a query since they apply to all windows in poll mode (active or maximum consecutive poll reached).

(25) Due to the potential heavy impact on the network and the Network Operations server systems, the use of this facility will be monitored and limited by the security profile.

(26) The automatic refresh is stopped immediately if an error occurs during the communication with the Tactical system. It then needs to be restarted if necessary.

(27) Some polling behaviour is customisable by user profile. The user profile can be adjusted or newly composed at any moment by OSM. This change or creation is limited to a certain number of system properties:

a) **Maximum polling window**: The maximum number of concurrent active windows in polling mode.

b) **Enforced minimum polling interval**: The enforced minimum basic time interval between two automatic refreshes of the same window.

c) **Polling penalty**: Addition of a penalty in minutes to the base time per additional concurrent window in poll mode.

d) **Maximum consecutive poll:** Maximum consecutive auto-refresh per window after which the polling mode is automatically deactivated.

e) **Polling timer trigger:** Timer interval in second used to check if the next automatic refresh shall be triggered.

(28) For each of those system properties a choice can be made out of a pre-defined set of values defined in the system code. The following table provides the overview:

CHMI / ATFCM application / General / Dates and times / … *with **"Show System Properties"** option turned on*	**Number of Bits**	**Possible Values**			
		0	**1**	**2**	**3**
Maximum polling window	2	4 win	8 win	12 win	No Limit
Enforced minimum polling interval	2	10min	1min	5min	15min
Polling penalty	2	1min	2min	3min	None
Maximum consecutive poll	2	10	30	120	No Limit
Polling timer trigger	2	10sec	15 sec	30 sec	60 sec

Table Chmi-2: Polling System Properties

4.5.15 Previous / Next period

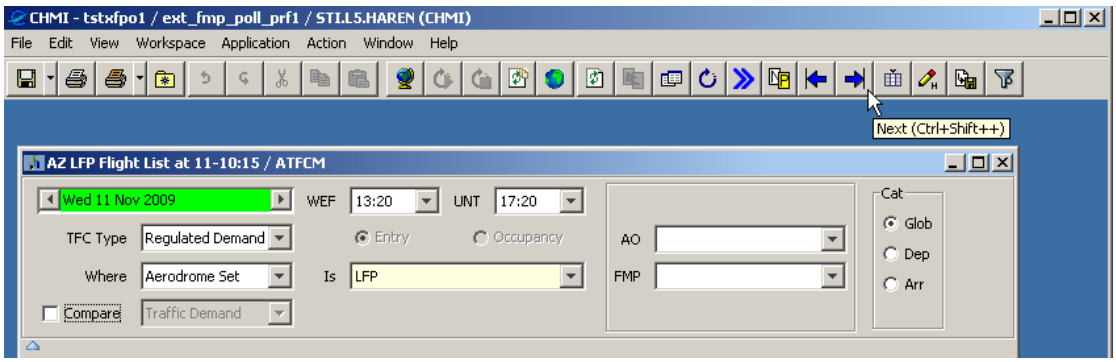


Figure Chmi - 12: Previous / Next Buttons

(1) Enabled: when an earlier / later query period is available and the application is available for processing this action.

(2) This processing will take into account if the possibility to query over midnight is authorised by your user profile (CHMI / ATFCM application / General / Dates and times / Query over midnight).

(3) Disabled: **Previous** when WEF time is 00:00 or **Next** when UNT time is 00:00 and query over midnight is not authorised or while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(4) Applicable: all windows where the selection criteria include a time period that could be shifted.

(5) Action: will issue a query identical to the previous one but for the previous / next period with the same elapse, possibly truncated if we reach 00:00 or 24:00 and query over midnight is not authorised.

(6) For example with the period 09:20 to 11:20, if you click the Next button, you will issue the same query for the period 11:20 to 13:20.

(7) Shortcut key: **<Ctrl>** + **<Shift>** + NumPad **<->** (minus sign) for Previous and **<Ctrl>** + **<Shift>** + NumPad **<+>** (plus sign) for Next

(8) Menu option: **Action / Previous** and **Action / Next**

4.5.16 Default Message Originator

(1) For some actions performed with the ATFCM Collaboration Human Machine Interface application, the user is invited to provide a SITA or an AFTN network address.

a) It is Optional for the CASA messages.

b) It is Mandatory for the Rerouting requests.

(2) This address will be included in the message issued to the Network Operations system.

(3) It will be used for later distribution of messages by the Network Operations system.

(4) To standardise the input of the addresses and the selection of the default one, a customised dialog may be invoked when updating the user preference **CHMI / ATFCM application / General / Origin / Default Address Id**. Just click on the little square at the right side of the Default Address ID input box on the Preferences dialog.

(5) Following system properties will be adjusted by this dialog:

a) **CHMI / ATFCM application / General / Origin / AFTN Address List:** a user pre-defined list of AFTN addresses to allow later choice.

b) **CHMI / ATFCM application / General / Origin / SITA Address List:** a user pre-defined list of SITA addresses to allow later choice.

c) **CHMI / ATFCM application / General / Origin / Default Address Type:** the type of the selected default address (AFTN or SITA).

d) **CHMI / ATFCM application / General / Origin / Default Address Id:** the selected default address.

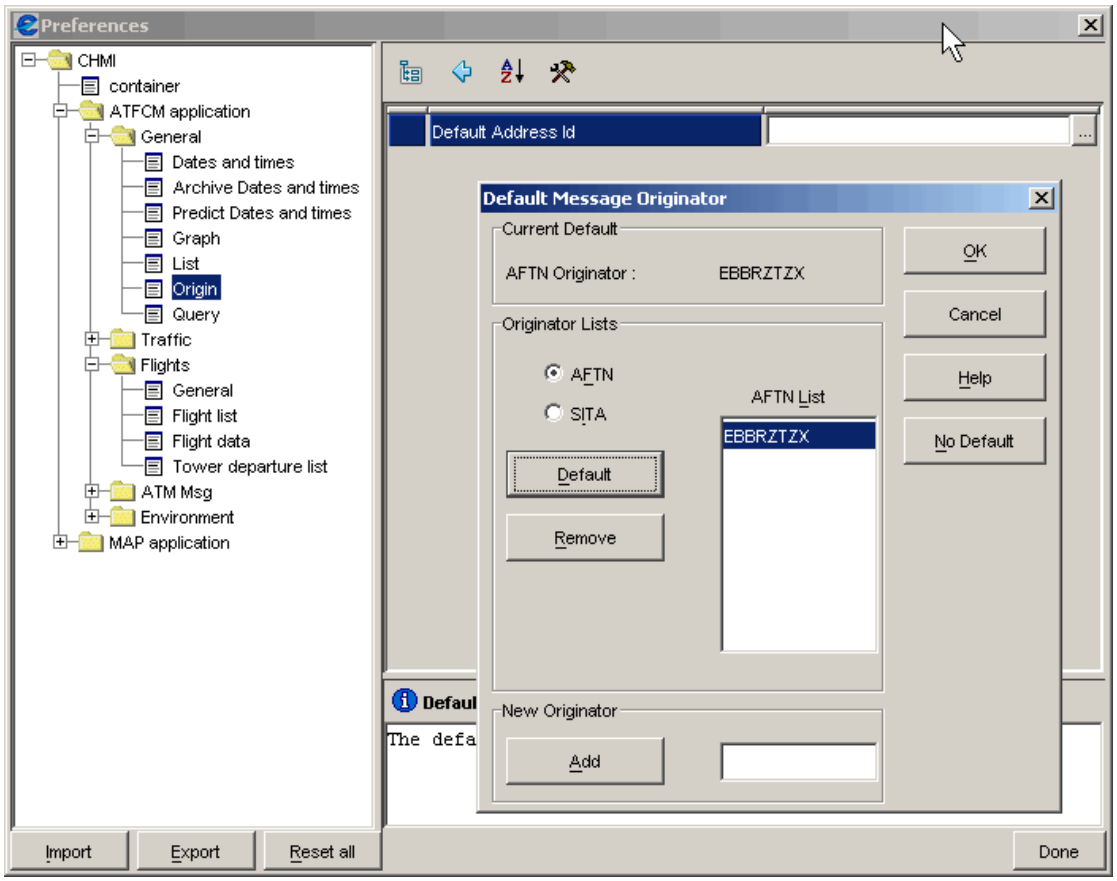


Figure Chmi - 13: Default Message Originator

(6) When such a window is opened, no other function may be invoked.

(7) The default address may be a SITA or an AFTN address.

(8) Several addresses may be defined but only one may be defined as default.

(9) To add a new address in the Originator List, you first select the list (SITA or AFTN), then introduce the address in the New Originator box and click on the **Add** button.

a) A valid AFTN address must have 8 alphabetics.

b) A valid SITA address must be composed as follow:

i) in position 1 .. 3: alphabetics

ii) in position 4 .. 5: any combination of alphabetic and numeric

iii) in position 6 .. 7: a combination of:

- 2 alphabetics
- one numeric and one alphabetic
- one alphabetic and one numeric

(10) To remove an address from a list, you first select the list (SITA or AFTN), then select the address in the list and click on the **Remove** button.

(11) To set the default, you first select the list (SITA or AFTN), then select the address in the list and click on the **Default** button.

(12) To change the default, you just set another originator address as **Default**.

(13) To remove the default, click on the **No Default** button.

(14) To store the default and the lists in your preferences, click on the **Ok** button.

(15) To ignore the changes to the default and to the lists, click on the **Cancel** button.

4.5.17 Workspace Management



Figure Chmi - 14: Workspace options

(1) The Collaboration Human Machine Interface application offers the use of **Workspaces** to help structuring the way of working and better manage the available display space.

(2) To preserve available space for displaying data windows, the Workspace tab is not displayed when there is a single workspace created.

(3) Basic management functions available in the **Workspace** menu are:

a) **Create** a new workspace next to the existing one(s).

b) **Remove** the workspace currently selected on display. A dialog will popup to ask for confirmation. This action will not affect any saved workspace configuration (see later).

c) **Rename ...** the workspace currently selected on display.

(4) On starting the Collaboration Human Machine Interface application, empty Workspaces and names are re-created as they existed on previous application close.

(5) In addition to the basic workspace management functions, some more sophisticated functions are implemented to save and restore workspace configuration with a given content.

(6) These enhanced Workspace management functions will not only **Create** the named Workspace, if it does not exist yet, but also restore the content as it was when the Save command was executed.

(7) Within a named Workspace, the content related to ATFCM windows is restored with the following rules:

a) Position and size of each window as it was on executing the Save command.

b) Query date and times are restored according to the user preference **Save Workspace with Absolute Time** as it was at the time of the Save workspace command:

i) Either relative to current time compare to the time of the Save command

ii) Or to absolute time (i.e. the same value as it was at the time of the Save command)

c) All other query criteria are restored to be identical to the saved value.

(8) A workspace can be defined as using relative or absolute query time when it is restored.

(9) The value of the user adjustable preference **Save Workspace with Absolute Time** is saved with the workspace configuration as it is at the time of the save command.

a) CHMI / ATFCM application / General / Dates and times / Save Workspace with Absolute Time

b) Description: With this option selected, when a workspace is saved the WEF and UNT times of applicable queries are saved as ABSOLUTE values (i.e. they are restored to the exact values they had when they were saved). When unchecked, WEF and UNT times are restored relative to current UTC time versus the time of the Save command.

c) Default value: Unchecked (i.e. OFF = restore relative to current time)

(10) Polling state and any filter are not restored automatically.

(11) Windows implementing the Open at startup feature are not taken into account (eg: Release Notifier).

(12) All other ATFCM windows are candidates to be restored but the query data and the initial send are only managed for functions monitoring a selected time period (eg: FMP Monitor).

(13) Following enhanced commands are available from the **Workspace** menu:

a) **Restore** to initiate the restore of All or one selected named Workspace. **All** will restore at once all Workspaces currently saved.

b) **Save** to save the currently selected Worskpace with its content. A dialog will ask to confirm the Save action and warn about overwrite if applicable.

c) **Unsave** to remove the saved configuration for All or one selected named Workspace. A dialog will ask to confirm the Unsave action.

(14) This restore process is done on request to allow selective workspace restore and avoid undesired massive restore on each application start.

4.5.18 Minimized Windows

(1) When minimized, the detailed ATFCM / Archive / Predict MDI windows have a Tool Tip which gives the full title of the window. The title of the window contains in a lot of cases essential data to facilitate the work of an operator.

(2) This Tool Tip avoids the necessity to re-open the windows to have access to this information.

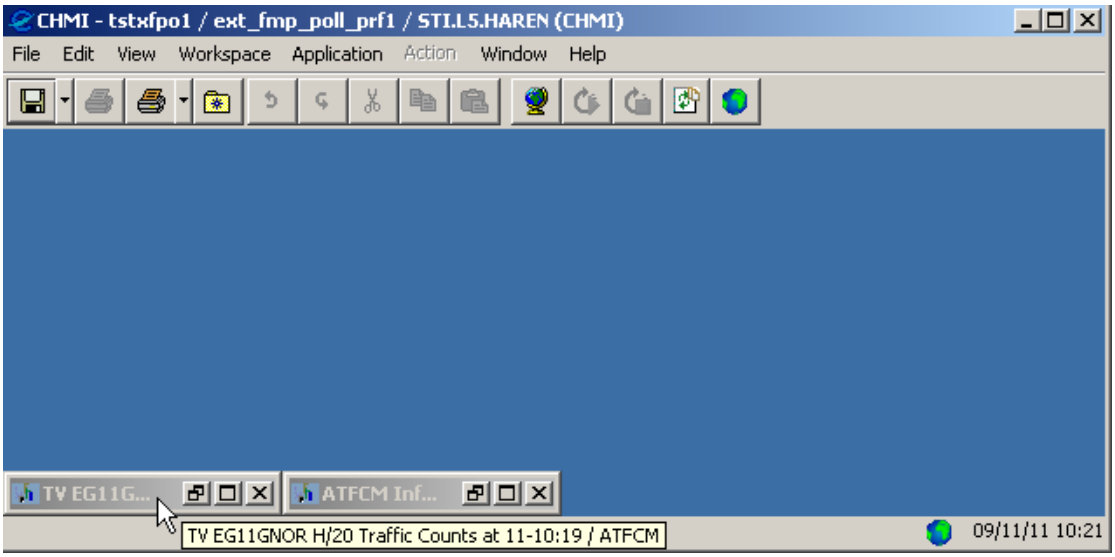


Figure Chmi - 15: Tool Tip on Minimized Windows

4.5.19 Global Refresh

(1) Enabled: when at least one window could be refreshed on the currently displayed workspace.

(2) Disabled: when no windows are available for refresh on the currently displayed workspace.

(3) Applicable: all displays where non static traffic data are displayed (Flight Lists, Delays, Traffic and Flow Counts and Traffic Monitor functions).

(4) This processing will do a refresh of all open windows for which the polling mechanism is available within the currently displayed workspace.

(5) A message box will notify when the current query period of a window does not allow automatic refresh (maximum elapse exceeded). Global refresh may continue for other valid windows on the workspace.

(6) Action: global refresh issues a query for all applicable windows on the workspace. This refresh mechanism is identical to the one triggered by the Polling mechanism eventually shifting the queired period.

(7) Menu option: **Action / Global Refresh**

(8) Shortcut key: **<Ctrl> + Shift + H**

(9) A button is implemented with a Tool Tip on the Main Tool Bar to ease the execution of this functionality

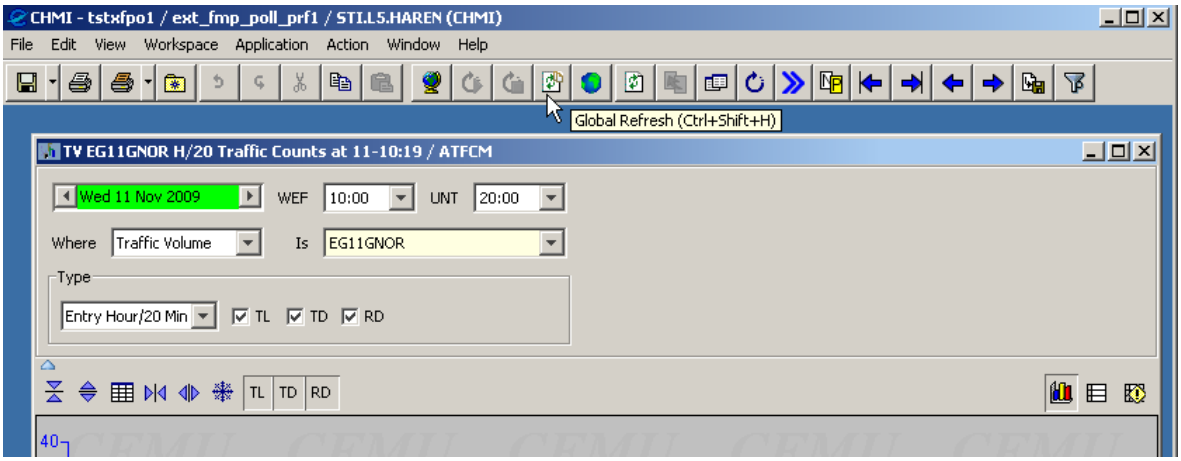


Figure Chmi - 16: Global Refresh button

4.5.20 Acronyms

(1) Tool tips provide the full names for acronyms used in the application. This is especially required on the ENV Query screen where a lot of sub-types of Environment Objects are displayed as acronyms. Also in the Map Configuration and Map Legend screen acronyms are widely used and foreseen with a Tool Tip.

(2) The Tool tips avoid browsing in help file's or documents to get the full name of used acronyms.

4.5.21 Map Management

(1) The functioning of the Map is described in his proper SRD. This SRD specifies the great set of common Map functionality for the Collaboration Human Machine Interface.

(2) This paragraph highlights some of the Map functional specificities for the ATFCM part of CHMI.

(3) A menu option "**ATFCM**" is available when right-clicking on a monitored and plotted Traffic Volume or Regulation in the Map.

(4) In case of a location (Traffic Volume) the sub-menu option **ATFCM - Load** gives direct access to following ATFCM functions:

a) **CASA Regulations:** shows the query results of all regulations impacting the FMP/Traffic Volume

b) **Reroutings:** shows the query results of all reroutings impacting the FMP/Traffic Volume

c) **Traffic Counts:** shows the query result Traffic Counts for the Traffic Volume

d) **Flow Counts:** shows the query result Flow Counts for the Traffic Volume

e) **Flight List:** shows the query result Flight List for the Traffic Volume

f) **Traffic Capacity** requests detailed capacity values for the Traffic Volume

(5) In case of a monitored and plotted Regulation the sub-menu option **ATFCM - Delay** gives direct access to following ATFCM functions:

a) **CASA Regulations:** shows the query results of all regulations impacting the FMP/Traffic Volume

b) **Regulation Description:** shows the detailed data of the Regulation

c) **Regulation Delays:** shows the query results Delay Counts for the Regulation

d) **Traffic Counts:** shows the query result Traffic Counts for the Traffic Volume

e) **Regulation Flight List:** shows the query result Flight List for the Traffic Volume

f) **Traffic Capacity** requests detailed capacity values for the Traffic Volume.

(6) Horizontal View:

a) When one or more flight(s) are plotted, the menu **Dossier - Flight** in the tab page **Layers** is enlarged with profile checkboxes: **Estimated**, **Calculated** and **Actual.**

b) The checkboxes allow the user to show/hide the different flight profile(s) present on the Map. The possibilities are **Estimated**, **Calculated** and **Actual**. All Flights plotted are impacted by this show/hide action.

(7) Vertical View:

a) A vertical view of a flight contains in the Tab page **Layers** the sub-menu **Flights.**

b) The sub-menu Flights contains max. 3 checkboxes: **Estimated**, **Calculated** and **Actual**.

c) Each of the checkbox facilitates the user to show/hide the specific flight profile on the vertical view.

(8) When a flight is plotted on the Horizontal Map in the tab Page **Layers,** the menu option **Dossier** has a sub-option **Flight**. Right-clicking opens the pop-up window **Customise for Flight**.

(9) This pop-up window **Customise for Flight** has two options that influence the display of regulations impacting the selected flight:

a) Display Regulation(s) impacting selected flight

b) Display label(s) of regulation(s) impacting the selected flight

(10) The label of the displayed regulation impacting the flight contains the reference location and the regulation id.

(11) On the vertical view by default the regulation(s) and its label(s) impacting the plotted flight are displayed.

4.5.22 Show / Hide Columns action

(1) The **Show / Hide Columns** action is only presented to the user when the focussed window has a list display implementing this **Show / Hide Columns** functionality.

(2) Enabled: when the application is available for processing this action, even before the first query is issued.

(3) Disabled: while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(4) Action: will prompt the **Show / Hide Columns** dialog box proposing to customise the list of columns to be shown by the corresponding window.

(5) Shortcut key: **<Ctrl> + <Shift> + <D>**

(6) Menu option: **Action / "Show / Hide Columns"**

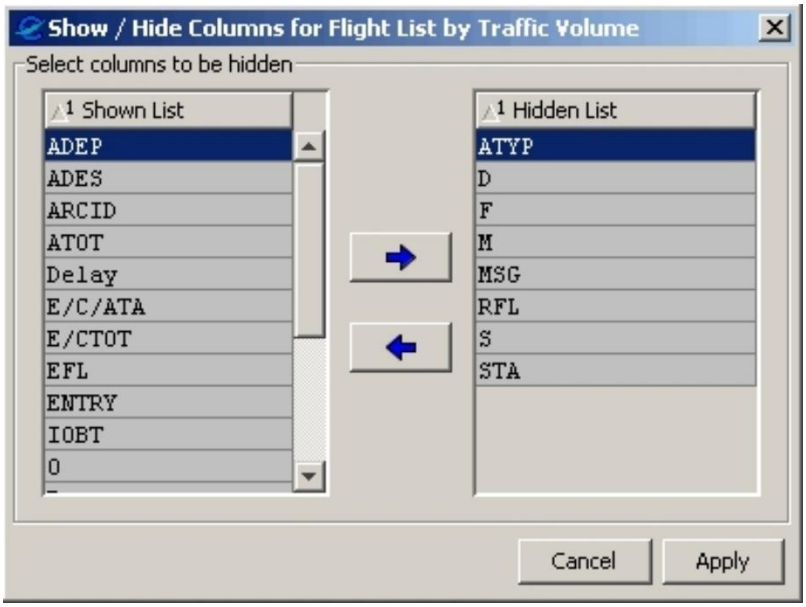


Figure Chmi - 17: Show / Hide Columns dialog box

(7) The **Show / Hide Columns** function allows identifying locally the columns to be displayed within the complete list of available columns.

(8) The column display filtering is done locally and therefore doesn't affect the data retrieval from the Network Operations database.

(9) When invoked, the **Show / Hide Columns** action opens the **Show / Hide Columns** dialog for the currently active window.

(10) This **Show / Hide Columns** dialog is modal, so no other action is allowed while it is displayed.

(11) The **Show / Hide Columns** dialog shall allow hiding any column defined for the corresponding elementary list type, but all columns cannot be hidden at the same time.

(12) This **Show / Hide Columns** dialog shall:

a) Identify clearly the scope of the list for which columns are going to be shown or hidden (ex: *Flight List by Aircraft Operator*).

b) **Hidden Columns:** List the name of the columns that are hidden

c) **Shown Column**: List the name of the columns that are shown

d) Implement action buttons ("→" and "←") to move selected column names from one list to the other. Multiple selections shall be possible on each list.

e) Implement action buttons to cancel, apply and save changes.

(13) Clicking the **Apply** button on the **Show / Hide Columns** dialog applies the currently displayed show / hide selection at least to the active window without closing the dialog.

(14) Clicking the **Cancel** button closes the **Show / Hide Columns** dialog and keeps the situation as it was before invoking the **Show / Hide Columns** dialog.

(15) Clicking the **Save** button on the **Show / Hide Columns** dialog updates the corresponding **Show / Hide Columns** local system preference and closes the dialog. Subsequent broadcast message will be sent to all window of the same elementary list type.

(16) If there is at least one column to be hidden, the customised list of columns to be shown shall be stored in a **Show / Hide Columns** local system preference.

(17) The presence of the **Show / Hide Columns** local system preference per elementary list type is optional.

(18) The **Show / Hide Columns** local system preference for a given elementary list type shall only exist if there is at least one column to be hidden.

(19) If the **Save** button is clicked on the **Show / Hide Columns** dialog while no column is displayed in the **Hidden Columns** list, the corresponding local system preference shall be deleted if it existed before opening the dialog.

(20) The **Show / Hide Columns** local system preference for a given elementary list type shall be used for any subsequent display of the same elementary list type (ex: *Flight List by Aircraft Operator*).

(21) Any maintenance action (create, update, delete) of the **Show / Hide Columns** local system preference for a given elementary list type shall be broadcasted to all opened window of the corresponding elementary list type for immediate update of the display.

(22) The **Show / Hide Columns** local system preference could be displayed but NOT updated via the **Preferences** dialog.

(23) Any window corresponding to a possible **Show / Hide Columns** local system preference of a given elementary list type shall implement the following behaviour on displaying newly retrieved reply data:

a) If the preference is defined:

i) The columns identified in the preference shall be respectively shown or hidden.

ii) The window shall implement a visual warning as an icon to the right of the counter textual indication at the top of the list with a tooltip explaining that some columns are hidden (same icon as for the **Show / Hide Columns** action button in the **Toolbar**). If applicable, this warning icon shall be displayed even when there is no counter information yet (i.e. before the first query is issued).

b) If the preference is NOT defined:

i) All columns are shown.

ii) The warning icon is not displayed.

(24) Any opened window corresponding to an updated **Show / Hide Columns** local system preference of a given elementary list type:

a) Shall show or hide the columns to reflect the newly saved preference.

b) Adjust the presence of the warning icon to the right of the counter, if needed.

4.5.23 Print Special / Save Special functions

(1) Enabled: as soon as the corresponding detail level (i.e. Container, Workspace, Component) is available for processing these actions.

(2) In practice, Container,and Workspace are always available, even when empty.

(3) Disabled: when the corresponding detail level (i.e. Container, Workspace, Component) is not available for processing these actions.

(4) In practice, only occurs when no Components are opened yet in the displayed Workspace.

(5) Applicable: **Print Special** and **Save Special** are global CHMI framework functionalities.

(6) Action: **Print Special** and **Save Special** will print or save the exact screen image (i.e. a snapshot) for the currently active components of the selected level of details:

a) **Container**: the complete application window including menu, button bar, workspace tabs and full content of the active workspace (including all active and non active windows).

b) **Workspace**: the name and full content of the active workspace (including all active and non active windows)

c) **Component**: the currently active window only

(7) Menu options and shortcut keys:

a) **File / Save Special / Container** **Ctrl+Shift+4**

b) **File / Save Special / Workspace** **Ctrl+Shift+5**

c) **File / Save Special / Component** **Ctrl+Shift+6**

d) **File / Print Special / Container** **Ctrl+Shift+7**

e) **File / Print Special / Workspace** **Ctrl+Shift+8**

f) **File / Print Special / Component** **Ctrl+Shift+9**

(8) The **Print Special** function allows to produce a hard copy for the selected HMI detail level (i.e. a WYSIWYG snapshot).

(9) A **Print Special** button on the button bar allows to trigger directly the action for the detail level identified by the user adjustable preference:

a) <CHMI / container / Default Print Special >.

i) Description= "Default level of details to be triggered by the Print Special button on the button bar."

ii) Three possible detail levels: **Container**, **Workspace** or **Component**

iii) Default value= **Component**

(10) The tooltip and the shortcut key associated to this button shall be the one of the corresponding menu option.

(11) A modal dialog will prompt to identify the destination printer.

(12) Default printer is proposed.

(13) The **Save Special** function allows to save a hard copy of the selected HMI detail level (i.e. a WYSIWYG snapshot).

(14) A **Save Special** button on the button bar allows to trigger directly the action for the detail level identified by the user adjustable preference

a) <CHMI / container / Default Save Special >

i) Description= "Default level of details to be triggered by the Save Special button on the button bar."

ii) Three possible detail levels: **Container**, **Workspace** or **Component**

iii) Default value= **Component**

(15) The tooltip and the shortcut key associated to this button shall be the one of the corresponding menu option.

(16) A modal dialog will prompt to identify the location and the file name in which the requested snapshot will be stored.

4.5.24 Plot All flights function

(1) To plot the Horizontal view of all flights displayed in a list:

a) Click on the **Plot All** toggle button

b) Or select the **Action / Plot All** from the menu.

(2) The **Plot All** action is accessible even if there is no flight displayed in the list (i.e. polling may justify such situation where subsequent refreshes may populate the list).

(3) The **Plot All** action is totally independent of the flight(s) currently selected in the list, if any.

(4) The **Plot All** action is stateful per window (i.e. each window remembers its last state).

(5) A warning message will inform the user when the flights to be plotted are having their off block time in two different AIRAC reference period:

"Warning

Some of the flights selected for plotting are referring to a different AIRAC data release which might result in changed or missing environment data definitions for these flights."

(6) When the **Plot All** action is triggered manually (i.e. not resulting from polling process), the originating list of flights shall first automatically refresh its content before querying for detailed plot data. This is to ensure flight data coherency between the displayed flight list details and the plotted flights.

(7) If a filter is currently active on the list, the **Plot All** action will only plot the flights currently visible in the list.

(8) Applying a different filter value will not be immediately reflected on the map. A refresh operation or a **Un-Plot All** (toggle the **Plot All** button while it is ON) / **Plot All** action sequence is required.

(9) The access to the **Plot All** functionality shall be controlled via a separate use case in the user security profile.

(10) See detailed software requirement per function for the requested presence of the **Plot All** function in the different list of flights.

(11) As a protective measure against potential system overload, the Network Operations systems shall implement an optional (i.e. this limitation could be turned on and off based on system monitoring) adjustable system parameter with the maximum number of flights that can be submitted in one plot request.

(12) A meaningful (i.e. non technical) error message shall be displayed when a plot request exceeds this limit.

4.5.25 Plot Selected flights function

(1) To plot the Horizontal view of one or more selected flights in a list:

a) Click on the **Plot Selected** button

b) Or select the **Action / Plot Selected** from the menu after selecting the desired flight(s) in the list.

(2) The **Plot Selected** action is only accessible when there is at least one flight selected in the list.

(3) A warning message will inform the user when the flights to be plotted are having their off block time in two different AIRAC reference period:

"Warning

Some of the flights selected for plotting are referring to a different AIRAC data release which might result in changed or missing environment data definitions for these flights."

(4) When the **Plot Selected** action is triggered, a new **Plotted Flight List** window will be created (see below). This is done to isolate the selected flights and allow for later refresh of the same list without freezing the originating Flight List.

(5) The **Plotted Flight List** window can only be created by invoking the **Plot Selected** action on a list of flights.

(6) The **Plotted Flight List** displays a set of flights identified for plotting on the map.

(7) For each flight in the list, the **Plotted Flight List** displays the following key criteria:

a) The Initial Estimated Off Block Date **(IOBD)**.

b) The Initial Estimated Off Block Time **(IOBT)**.

c) The Aircraft Identification **(ARCID)**.

d) The Aerodrome of Departure **(ADEP)**.

e) The Aerodrome of Destination **(ADES).**

(8) The title bar of the **Plotted Flight List** shall indicate the date and time stamp when the detailed flight data needed for plotting were retrieved.

(9) The **Plotted Flight List** window shall allow standard list sorting on any of the displayed columns.

(10) The initial display sequence of the flights in the **Plotted Flight List** is:

a) First in ascending sequence on the **IOBD**

b) And, for identical IOBD, in ascending sequence on the **IOBT**.

(11) The **Plotted Flight List** window shall allow:

a) Requesting a **refresh** of the flights on the map.

b) Starting the **polling** process to have automatic refresh of the flights on the map.

c) **Removing** an flight entry from the list (and consequently from the map) possibly via popup menu or contextual action on each flight.

d) Requesting **Flight Data** display possibly via popup menu or contextual action on each flight.

(12) The access to the **Plot Selected** functionality shall be controlled via a separate use case in the user security profile.

(13) See detailed software requirement per function for the requested presence of the **Plot Selected** function in the different list of flights.

4.5.26 Clone function

(1) Applicable:

a) Most justified for the windows where one or more criteria could be adjusted to compare results (i.e. compare different views on the same data or obtain very similar data selection).

b) Initially implemented for the **Traffic Counts**, **Flow Counts** and **Flight List** functionalities.

(2) Enabled: as soon as some valid reply data are retrieved from the Network Operations database and the application is available for processing this action.

(3) Disabled: before any valid reply data are retrieved or when the query criteria are not in line with the available reply data or while the application is processing some modal local actions (ex: filter, export, …) or during data retrieval for the same window.

(4) Action: based on the currently selected window, the Clone function is going to:

a) Open a new window of the same type (ex: Flight List),

b) Force the query criteria to the same value

c) And trigger immediately the Send action.

(5) Remark: any filtering or sorting criteria is not transferred to the new window.

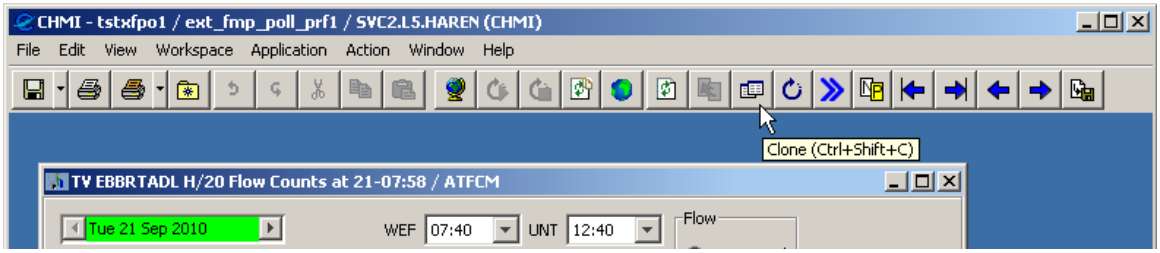


Figure Chmi - 18: Clone Button

(6) Menu option: **Action / Clone**

(7) Shortcut key: **<Ctrl> + <Shift> + <C>**

5 Monitor graph

5.1 Function presentation

(1) For the selected date and time period, the **Monitor** graph may display:

a) The load states.

b) The regulations period and delays states.

c) The rerouting periods.

(2) The monitoring may be requested:

a) For the Traffic Volumes belonging to a list of FMPs.

b) For one Configuration of a non clustered ATC Unit Airspace (AUA) or a Cluster.

c) For all FMPs. In this case Monitoring is limited to Regulation delay.

5.2 Opening the window

5.2.1 FMP Monitor

(1) From the Application / ATFCM menu, select the option **Traffic / Monitor / FMP Monitor**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl>** + **<Shift>** + **<F5>**.

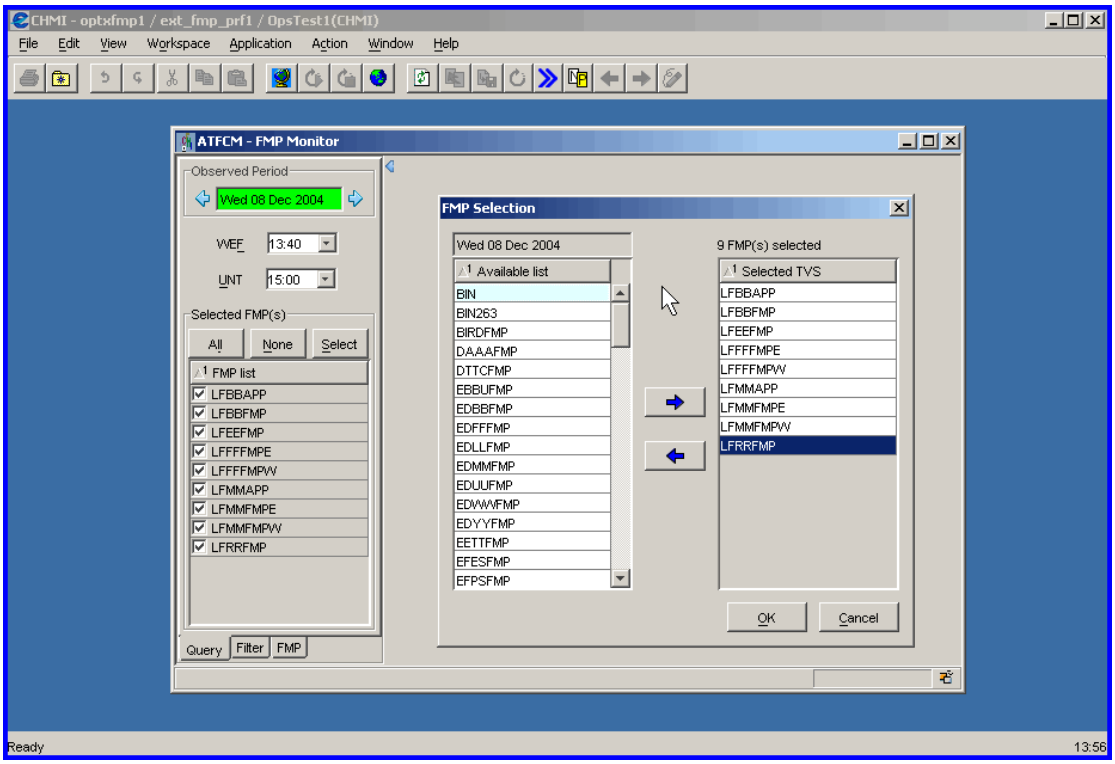


Figure Chmi - 19: Query FMP Monitor

(3) A new empty **FMP Monitor** window will be opened with the following defaults:

a) **Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** time: proposed based on the **Default query period** assignment using your user adjustable preferences.

c) **UNT** time: proposed based on the displayed WEF and the **Default query period** assignment.

d) **"Selected FMP(s)"** list is filled in with the last updated list maintained using the **FMP Selection** dialog.

e) FMPs are checked in the **"Selected FMP(s)"** list based on the last query issued that uses FMP selection.

(4) When no FMPs have been selected yet using the **FMP Selection** dialog (this is the case after application installation),

a) A message box will warn you and invite you to select first one or more FMPs.

b) On clicking OK, the **FMP Selection** dialog is opened automatically.

c) After adding one or more FMPs to the **"Selected list"**, click on the OK button.

d) This will close the dialog and update the "**Selected list**" on the calling window.

(5) On initial **FMP Monitor** query preparation, the focus is set on the date control.

(6) This window is designed with a standard tab traversal order; to move focus between the Observed Period controls, the WEF and UNT input fields, the Selected FMP(s) action buttons, the first item of the Selected list on the Query tabpage; the Tab keyboard key is used.

5.2.2 ATC Airspace Monitor

(1) From the Application / ATFCM menu, select the option **Traffic / Monitor / ATC Airspace Monitor**.

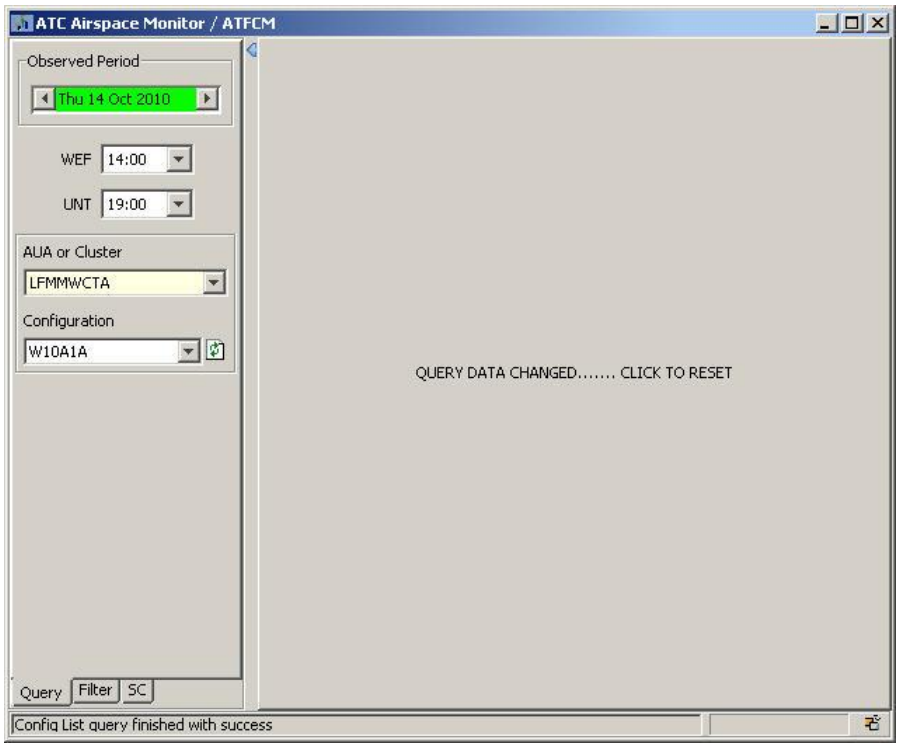


Figure Chmi - 20: Query ATC Airspace Monitor

(2) A new empty ATC **Airspace Monitor** window will be opened with the following defaults:

a) **Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** time: proposed based on the **Default query period** assignment using your user adjustable preferences.

c) **UNT** time: proposed based on the displayed WEF and the **Default query period** assignment.

d) **AUA or Cluster**: the input control is proposed empty. The drop-down list is filled in with the most recent IDs you have queried before with success.

e) **Configuration**: the input control is proposed empty. It is possible to enter the Configuration Id in the input control. The drop-down list is filled in with the most recent IDs you have queried before with success.

f) The icon next to the label **Configuration:** clicking it will query the associated Configuration Ids for the inputted ATC Unit Airspace Id and fill the drop-down list with the result to reflect the most up-to-date situation and facilitate the input for the ATC Airspace Monitor Query itself.

(3) On initial AUA or Cluster query preparation, the focus is set on the ATC Unit Airspace control.

(4) **ATC Airspace Monitor** query is accepted on Clusters and AUAs without Clusters. Querying a clustered AUA returns an Error.

(5) This window is designed with a standard tab traversal order; to move focus between the Observed Period controls, the WEF and UNT input fields, the ATC Unit Airspace and the Configuration input boxes on the Query tabpage; the Tab keyboard key is used.

5.2.3 ATFCM Monitor

(1) From the Application / ATFCM menu, select the option **Measures / ATFCM Monitor**.

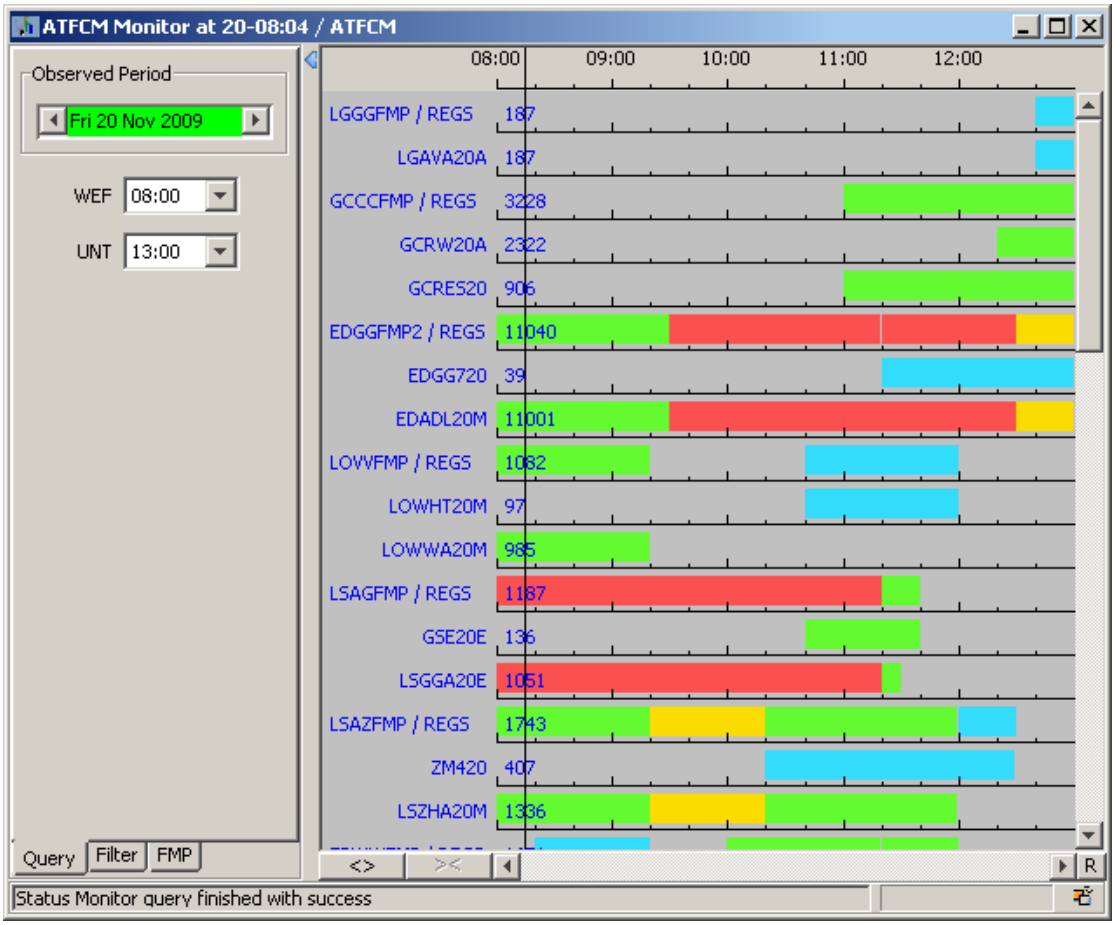


Figure Chmi - 21: Query ATFCM Monitor

(2) A new empty **ATFCM Monitor** window will be opened with the following defaults:

a) **Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** time: proposed based on the **Default query period** assignment using your user adjustable preferences.

c) **UNT** time: proposed based on the displayed WEF and the **Default query period** assignment.

(3) On initial **ATFCM Monitor** query preparation, the focus is set on the Date control.

(4) This window is designed with a standard tab traversal order; to move focus between the Observed Period controls, the WEF and UNT input fields on the Query tabpage; the Tab keyboard key is used.

5.3 Preparing to query

(1) The traffic **Observed period** may be adjusted as follow:

a) The traffic **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

b) The proposed default **WEF** and **UNT** times may be overwritten.

c) Any valid time value from 00:00 to 23:59 included may be entered in the **WEF** and/or the **UNT** entry boxes. This input must respect the HH:MM format.

d) Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(2) The period resulting from the input will be validated taking into account the system property **CHMI / ATFCM application / General / Dates and times / Query over midnight**.

(3) If query over two days is not authorised, the **WEF** time must be smaller than the **UNT** time, except if both times are set to 00:00, which corresponds to a full day query.

(4) The **Default query period** assignment will be used.

(5) For the **FMP Monitor**:

a) FMPs in the "**Selected list**" may then be checked or unchecked as desired.

b) Click on the "**Selected list**" title to adjust sort sequence (alternatively ascending and descending sequence). This shall not impact the list of selected FMPs and the fact that a given FMP is checked or not.

c) Click the "**All**" button to check all the FMPs present in the Selected list.

d) Click the "**None**" button to uncheck all the FMPs present in the Selected list.

e) To add or remove one or more FMPs in the Selected list, the **FMP selection** dialog shall be used. Click on the "**Select**" button to open this dialog.

f) When the Selected list is up to date on the **FMP selection** dialog, click on the OK button to save the list.

g) Rules for updating the **Selected list** on all opened windows using an FMP **Selected list** for the query is described in **FMP selection** dialog.

h) The FMPs checked in the "**Selected list**" before opening the **FMP selection** dialog shall remain checked.

i) Clicking the Cancel button of the **FMP selection** dialog stops editing the "**Selected list**" and come back to the previous screen without saving any modification to the list.

j) **FMP Monitor** window may be closed before issuing any query.

k) A maximum of 12 FMPs may be checked when issuing a query. Depending on your access rights, this limit may be extended.

(6) For the ATC Airspace Monitor:

a) An **ATC Unit Airspace** may be entered or selected from the list of previously queried IDs proposed in the drop-down list.

b) The related **Configuration** may be entered or selected from the list of previously queried IDs proposed in the drop-down list.

(7) For the **ATFCM Monitor** no other query citeria than the period is required.

(8) The **ATFCM Monitor** displays always all regulations for all FMPs for the selected period.

(9) Select the **Proposal** radio button on the button bar to take the proposed flights into account for the returned states. Otherwise the states will be based on the normal flights only.

(10) The **Send** button will send the query to the Network Operations system.

(11) The **Reset** action common functionality is implemented for this function.

(12) Following query data elements are considered for activating the Reset action:

a) Date.

b) WEF and UNT time controls.

c) Checked FMPs in the Selected list for the FMP Monitor.

d) ATC Unit Airspace and Configuration IDs for the ATC Airspace Monitor.

5.4 Displaying the query results

5.4.1 General

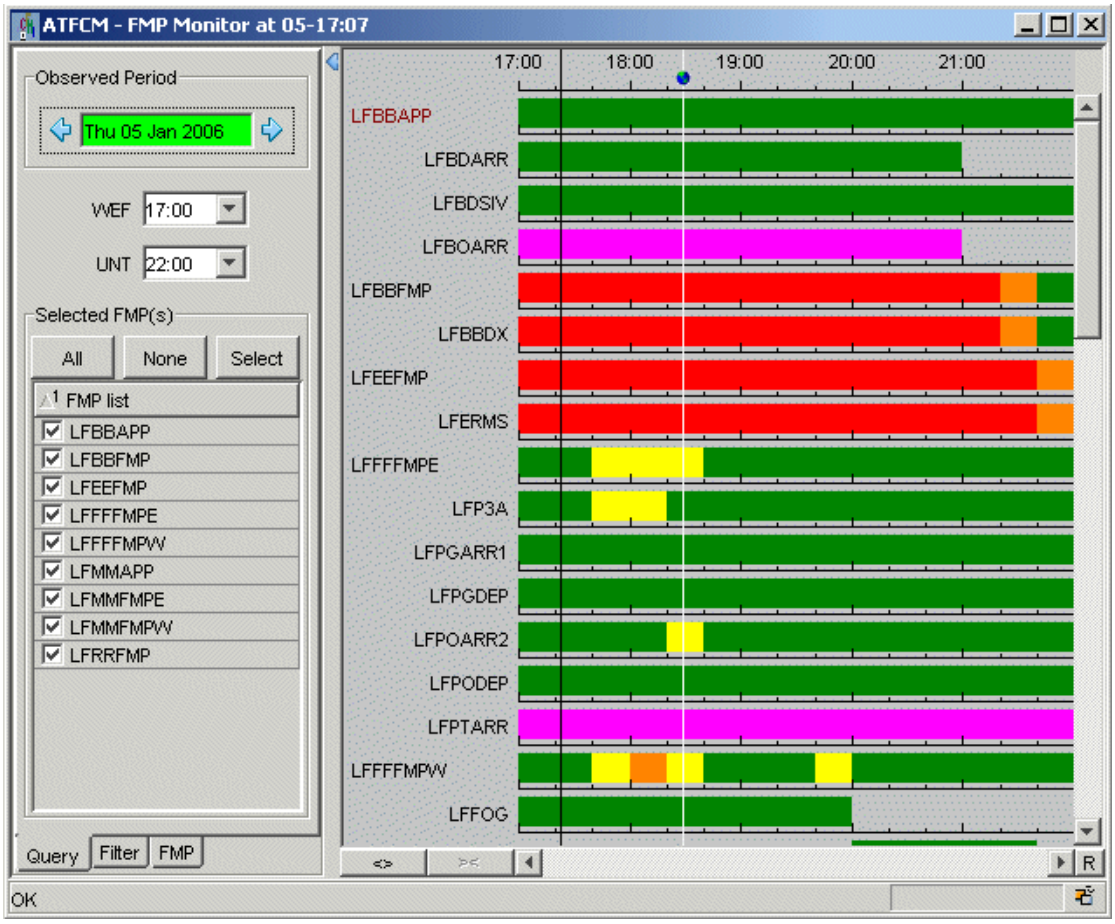


Figure Chmi - 22: FMP Monitor

(1) The main window caption indicates the type of Monitor display:

a) FMP Monitor

b) ATC Airspace Monitor

c) ATFCM Monitor

(2) In case of ATC Airspace Monitor, the **Configuration** ID is inserted between parentheses after the main window caption.

(3) On valid reply, the title contains the time stamp when the information was retrieved from the Network Operations system.

(4) The possible keyword **(Prop)** indicates that the displayed data (load and delay states) are including the flight "Proposals". So, if the keyword **(Prop)** is not there, the proposal flights are not included in the displayed monitor data.

(5) The optional **>>** sign indicates that the **Period Shifting** mechanism will be applied on refresh.

(6) The optional keyword **(Poll)** between parenthesis indicates that the **Polling function** is active for this window.

(7) The left part of the window contains the different **query data elements** used to retrieve the displayed monitor graph:

a) The observed **Period** (Date, WEF and UNT).

b) For the FMP Monitor, the list of **Selected FMPs** is displayed where the FMPs included in the query are checked.

c) For ATC Airspace Monitor, the **ATC Unit Airspace** and the **Configuration** identifiers.

(8) At the bottom of the left part of the window, following tabs are available:

a) The **Query** tab groups all the query data elements described earlier.

b) The **Filter** tab displays the different load categories with the currently associated colour. A check box next to each category allows to filter out the corresponding lines in the graph. The line will not be displayed if all the colours present on that line have their category unchecked. In other words, if at least one category present on a line is checked, the line remains on display.

c) The **FMP** tab (respectively SC tab) displays the list of Traffic Volumes and / or Regulations within each queried FMP (respectively Sector Configuration). A check box next to each FMP (respectively Sector Configuration), Traffic Volume and Regulation allows to filter out the corresponding lines in the graph. When an FMP is unchecked (respectively Sector Configuration), all Traffic Volume or Regulation lines for that FMP are (respectively Sector Configuration) hidden in the graph. When a Traffic Volume or a Regulation is unchecked, only that Traffic Volume or Regulation line is hidden in the graph.

(9) Just click on one of these tabs to display it.

(10) Each FMP (respectively Airspace Configuration) is represented by a group of lines structured as described below:

a) The group is composed of load lines and regulation delay lines.

b) If no regulation is related to the FMP (respectively Airspace Configuration), no regulation delay lines are displayed.

c) Load lines appear at the top of the group and regulation delay lines at the bottom of the group.

d) Load lines and regulation delay lines are described in details in the next paragraphs.

e) For the ATFCM Monitor, since the query is processed on all FMPs, only regulation delay lines are displayed. Load lines are not available.

(11) On query results display, the focus is set on the graph after selecting the first line in the list.

(12) A vertical line indicates the current UTC time on the monitor graph, if it is included within the displayed query period.

(13) The colour of this current UTC timeline is identified by the user adjustable preference **CHMI / ATFCM application / Traffic / Monitor / Graph / Current timeline colour**.

(14) When the Monitor content is plotted on the map, a vertical line with a globe at the top indicates the time currently displayed on the map. This timeline is only drawn if the map time is included within the displayed query period.

(15) The colour of this map timeline is identified by the user adjustable preference **CHMI / ATFCM application / Traffic / Monitor / Graph / Map timeline colour**.

(16) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

5.4.2 Load lines

(1) The graph shows the traffic load states during the query period for the selected FMPs (respectively Airspace Configuration) and the monitored Traffic Volumes.

(2) A summary Load line at the top of each group displays the worst load states, the regulated and the rerouted period per FMP (respectively airspace configuration).

(3) On initial display, the sequence of the Load lines per FMP (respectively Airspace Configuration), corresponds to the one received from the Network Operations server.

(4) This sequence of the Load lines within an FMP (respectively Airspace Configuration) group may have been adjusted by the user during a previous display of the same FMP (respectively Airspace Configuration). Once modified, the new sequence is stored locally on the PC and used for any subsequent display of the same FMP (respectively Airspace Configuration).

(5) A new Traffic Volume that was not in the saved sequence will remain at the end of the list until the user modifies again the display sequence.

(6) Each line begins with the corresponding FMP, ATC Configuration or Traffic Volume identifier.

(7) FMP and ATC Configuration identifiers are left aligned in front of the bar.

(8) Traffic Volume identifiers are right aligned in front of the bar.

(9) Each line contains a coloured horizontal bar using the following categories:

a) **Normal load colour**: load is maximun 90% of the hourly capacity (**Green**)

b) **Low Threshold colour**: load is between 90,1% and 100% of the hourly capacity (**Yellow**)

c) **High Threshold colour**: load is between 100,1% and 110% of the hourly capacity (**Orange**)

d) **Overloaded colour**: load is above 110,1% of the hourly capacity (**Red**)

e) **Undefined colour**: capacity not available (**Pink**).

(10) Each load category is associated to a user adjustable colour in the Preferences dialog (ex: CHMI / CHMI / ATFCM application / Traffic / Monitor / Graph / Normal load colour).

(11) Below the bar,

a) A stripe indicates that a non cancelled regulation (i.e. there should be no stripe after cancel time) has been implemented for this Traffic Volume. The colour of this stripe is identified by the user adjustable preference **CHMI / CHMI / ATFCM application / General / Graph / Regulation colour**.

b) Another stripe shows that a rerouting has been implemented for this traffic volume. The colour of this stripe is identified by the user adjustable preference **CHMI / CHMI / ATFCM application / General / Graph / Reroute colour**.

5.4.3 Regulation delay lines

(1) The first regulation delay line displays a summary of the "worst" average delay of all regulations related to the FMP (respectively airspace configuration).

(2) This line begins with the FMP identifier followed by the keyword **/ REGS** (for example: "EGTTFMP / REGS").

(3) Under this summary line, one line per regulation is displayed.

(4) The line related to a regulation begins with the Regulation's identifier. The identifier of the regulation is right justified.

(5) Each line contains a horizontal bar coloured according to the average delay (this is the "CASA Regulation worse Slice<Delayed average> Delay on the segment" defined in the TACT SRD, Part HMI-1, section 3.1, paragraph (5), Note 10) generated by the regulation. The different colour categories used are:

a) **Delay less than 15' colour**: average delay < 15 minutes

b) **Delay less than 30' colour**: average delay < 30 minutes

c) **Delay less than 45' colour**: average delay < 45 minutes

d) **Delay greater equal to 45' colour**: average delay >= 45 minutes

(6) Each delay category is associated to a user adjustable colour in the Preferences dialog (ex: CHMI / CHMI / ATFCM application / Traffic / Monitor / Graph / Delay less than 15' colour).

(7) Following default colour preference values shall be used:

a) **Delay less than 15' colour**: RGB 98, 184, 255 (blue)

b) **Delay less than 30' colour**: RGB 255, 255, 0 (yellow)

c) **Delay less than 45' colour**: RGB 255, 125, 0 (orange)

d) **Delay greater equal to 45' colour**: RGB 255, 0, 0 (red)

(8) When a regulation has been cancelled during its activation period (i.e. after start time and before planned end time), the average delay state shall still be drawn on the horizontal bar with the same colour codification after the cancel time, but using a shaded pattern (versus solid for NON cancelled periods).

(9) The coloured horizontal bar of the regulation summary line shall use the same shaded pattern, if the displayed "worst" average delay colour for a given time slice is ONLY based on the average delay of a regulation after its cancel time.

(10) When the Monitor content is plotted on the map, regulated locations are drawn using the colour category based on the average delay value for the current map time.

(11) Cancelled regulations shall NOT be visible after cancel time.

(12) The **total delay** generated by the regulation is displayed at the beginning of the coloured bar.

(13) The delay displayed on the summary line is the total delay for all regulations of the FMP.

5.5 User interactions

(1) Just click on one of the tabs (Query / Filter / FMP) at the bottom of the left part to display its content.

(2) Using the **Query tab**, this window allows to modify a previous query selection and to issue a new request.

(3) On query modification, the **Reset action** is enabled.

(4) To display / hide the query part, click respectively the Right / Left arrow at the top next to the split bar.

(5) On the **Filter tab**, check or uncheck the different category boxes to display or not the corresponding bars on the graph.

(6) A state bar will not be displayed on the graph if all the colours present on that bar have their categories unchecked.

(7) When one or more colour categories are unchecked, the keyword **Filter** is displayed in red at the top of the graph, next to the time scale.

(8) On the **FMP tab**, check or uncheck the different FMP, Traffic Volume or Regulation boxes to display or not the corresponding bars on the graph.

(9) The state bar will not be displayed on the graph for the Traffic Volumes or Regulations unchecked on the FMP tab.

(10) The state bars for all Traffic Volumes of the unchecked FMP will not be displayed on the graph.

(11) When one or more FMPs, Traffic Volumes or Regulations are unchecked on the FMP tab, the keyword **FMP** is displayed in red at the top of the graph, next to the time scale.

(12) To adjust the displayed **map timeline**, if present, just click on the globe at the top of the vertical line. Then hold the mouse button down while moving the mouse to the desired time. On release, the map will adjust automatically to the newly selected time. The initial click must happen on the globe, but to ease the selection, you may release the mouse button anywhere on the monitor graph.

(13) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(14) The **Normal / Proposal** button may be adjusted before issuing a new query.

(15) Filtering done on the Filter and the FMP tabs is kept over a data refresh.

(16) An horizontal cursor enables the change of the displayed period when the full queried period is not visible at once.

(17) Click on the "**<>**" button to reduce the scope of the shown period.

(18) Click on the "**><**" button to enlarge the scope of the shown period.

(19) Click on the "**R**" button at the bottom right of the graph to Reset to the original scope of the shown period.

(20) A vertical cursor gives the possibility to scroll in the list of FMPs, Traffic Volumes and Regulations.

(21) The list allows only single selection.

(22) The display sequence of the Load lines may be adjusted using the following method:

a) Select the Traffic Volume line to move in the displayed sequence.

b) Click on the **Move Up** sub-menu option or **<Ctrl>** + **<Up>** shortcut key to move the line before the previous one.

c) Click on the **Move Down** sub-menu option or **<Ctrl>** + **<Down>** shortcut key to move the line after the next one.

d) Repeat this operation as many times as needed.

(23) The sequence of the Regulation delay lines is automatically adjusted to respect the same requested sequence as for the Traffic Volume Load lines.

(24) No mixture is allowed between Traffic Volumes of different FMPs or AUA configurations.

(25) Once modified, the new sequence is stored locally on the PC.

(26) This new sequence will be used for any user on the same PC if a Monitor display is requested for the same FMP or AUA configuration.

(27) A new Traffic Volume that was not in the saved sequence will remain at the end of the list until the user modifies again the display sequence.

(28) Click on the **Reset Sequence** option of the FMP / Airspace Configuration popup menu to restore the corresponding original Traffic Volume sequence. This action will trigger a refresh of the window.

(29) The **Previous / Next period** function may be used to issue the same query but for the previous or the next period in time.

(30) The **Polling function** may be activated to start automatic refresh processing on this window.

(31) The **Period Shifting function** may be activated to start automatic period shifting on subsequent refresh of this window.

(32) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(33) To print the Monitor graph, first retrieve, arrange Traffic Volume sequence and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(34) To display the FMP / Airspace Configuration sub menu, click on one FMP or Airspace Configuration to select it, then click on the right button of the mouse.

(35) To display the Traffic Volume sub menu, click on one Traffic Volume to select it, then click on the right button of the mouse.

(36) To display the Regulations summary sub menu, click on one REGS line to select it, then click on the right button of the mouse.

(37) To display the Regulation sub menu, click on one Regulation to select it, then click on the right button of the mouse.

(38) Increasing / reducing the window height will impact the size of the FMP Selected list and the Monitor graph controls.

(39) Increasing / reducing the window width will impact the size of the Monitor graph control.

(40) The **Partial Period Shifting** function may be used to issue the same query but for the previous or the next period with time overlap (see **Partial Period Shifting** function detailed description).

5.6 Contextual actions

5.6.1 FMP / Airspace Configuration sub menu

(1) When an FMP or Configuration line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected FMP or Airspace Configuration and may contain the following options:

a) **CASA Regulations** requests all regulations for the ATC Airspace Monitor or the regulations linked to the FMP attached to the selected bar in the FMP Monitor.

b) **Reroutings** requests all reroutings for the ATC Airspace Monitor or the reroutings linked to the FMP attached to the selected bar in the FMP Monitor.

c) **ATC Configuration** requests ATC configuration details and the activation table for the ATC Unit Airspace.

d) **ATC Configuration Optimizer** requests optimal ATC Configuration list for the ATC Unit Airspace and for default interval of the selected period (Only applicable for ATC Airspace Monitor).

e) **Reset sequence** only available when a user preferred sequence is in use. A confirmation dialog will be displayed before the current user sequence is deleted.

f) **Show Traffic Volume Set Description** to display the CHMI ENV detailed description screen for the selected FMP line in the FMP Monitor (Lines are named FMP for the external user understanding but in practice, they correspond to Traffic Volume Set identifiers).

g) **Show Airspace Description** to display the CHMI ENV detailed description screen for the ATC Unit Airspace or Cluster selected in the ATC Airspace Monitor.

5.6.2 Traffic volume sub menu

(1) When a Traffic Volume line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Traffic Volume and may contain the following options:

a) **CASA Regulations** requests all regulations for the ATC Airspace Monitor or the regulations linked to the FMP attached to the selected bar in the FMP Monitor.

b) **Reroutings** requests all reroutings for the ATC Airspace Monitor or the reroutings linked to the FMP attached to the selected bar in the FMP Monitor.

c) **Traffic Capacity** requests detailed capacity values for the selected Traffic Volume.

d) **Traffic Volume Activations** requests all (de)activations of the selected Traffic Volume.

e) **ATC Configuration** requests ATC configuration details and the activation table for the ATC Unit Airspace.

f) **ATC Configuration Optimizer** requests

i) To open the ATC Configuration Optimizer without an initial query.

ii) Following query criteria are forced: **ATC Unit Airspace** and **Observed Period, WEF and UNT** from selected period.

iii) The **Configuration Order Criteria** dialog is opened for the selected period and the selected Airspace identifier is automatically inserted in the **Forbidden Sectors** input box.

iv) This menu option is only available for the ATC Airspace Monitor if the selected Traffic Volume line is based on a Reference Location of type Airspace (AS).

g) **Traffic Counts** to query counts for traffic type previously queried and the selected Traffic Volume.

h) **Flow Counts** to query Traffic Load counts per Flow for the selected Traffic Volume.

i) **Flight List** to query Traffic Load list of flights for the selected Traffic Volume and period.

j) **Move Up** to modify the display sequence with the previous Load line inside the FMP or the AUA.

k) **Move Down** to modify the display sequence with the next Load line inside the FMP or the AUA.

l) **Show Traffic Volume Description** to display the CHMI ENV detailed description screen for the selected Traffic Volume.

(3) For the Traffic Counts, Flow Counts and Flight List options, the query period used is the same as the one used for the Monitor query.

(4) The transferred period may be affected by user preferences related to **Default query period identification**.

5.6.3 Regulations summary sub menu

(1) When a Regulations summary line (REGS) is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected FMP / Regulation group and may contain the following option:

a) **CASA Regulations** requests all regulations for the ATC Airspace Monitor or the regulations linked to the FMP attached to the selected bar in the FMP Monitor and the ATFCM Monitor.

5.6.4 Regulation sub menu

(1) When a Regulation line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Regulation and may contain the following options:

a) **CASA Regulations** requests all regulations for the ATC Airspace Monitor or the regulations linked to the FMP attached to the selected bar in the FMP Monitor and the ATFCM Monitor.

b) **Regulation Description**: displays detailed information for the selected Regulation.

c) **Regulation Delays**: displays the delays for the selected regulation and its activation period.

d) **Regulation Flight List**: displays the Provisional Slot Allocation List (Traffic Load) for the selected regulation and its activation period starting either with the current time or the start time of the regulation.

e) **Traffic counts**: displays the count graph for the regulated traffic volume for the selected day.

(3) For the Regulation Delays, Regulation Flight List and Traffic Counts options, the query period used is the same as the one used for the Monitor query.

(4) The transferred period may be affected by user preferences related to **Default query period identification**.

6 FMP Selection

6.1 Function presentation

(1) This display gives the list of available FMPs and allows maintaining a user-preferred list.

(2) This user preferred list will be proposed by default for the different queries when a **Selected list** of FMPs has to be presented.

(3) This user preferred list is stored in a property file for later references.

6.2 Opening the window

(1) This modal dialog may be opened from the different functions where the query to the server system includes a list of FMPs.

(2) Currently these functions are:

a) **CASA Regulations**

b) **Reroutings**

c) **ADP list**

d) **FMP Monitor**

(3) A click on the **Select** button in the **Selected FMP(s)** frame in one of the functions listed above, will open the **FMP Selection** dialog.

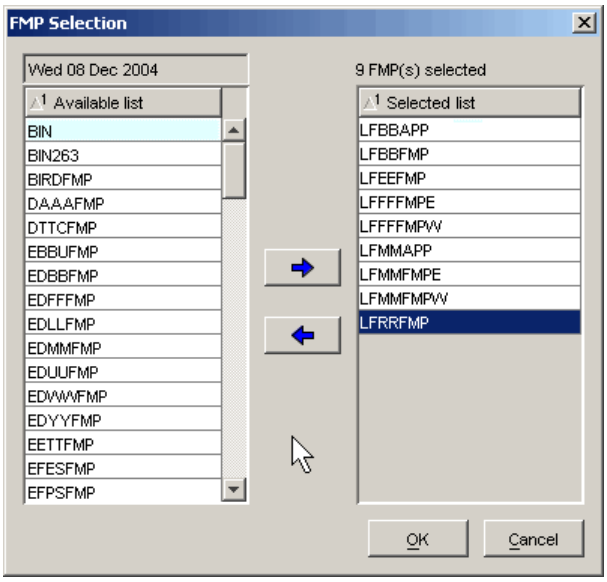


Figure Chmi - 23: FMP Selection

(4) On opening the dialog, the list of available FMPs is retrieved from the server.

(5) The date to be used for the retrieval is the one selected on the invoking window. It is displayed in the top left corner of the dialog.

(6) This **Available list** of FMPs is displayed on the left side of the dialog.

(7) The FMP **Selected list** is stored in the user preference **CHMI / ATFCM application / General / Query / Selected FMP Identifiers**.

(8) This FMP **Selected list**, if any, is displayed on the right side of the dialog.

(9) The focus sequence between the different controls starts from the Available list and the attached scroll bar, then it reaches the Selected TVS list and the attached scroll bar. Finally it goes on the different buttons: Right and Left arrows and then OK and Cancel.

(10) There is no resizing allowed on this dialog.

6.3 User interactions

(1) This dialog allows adding and removing FMPs from the user **Selected list**, based on the retrieved FMP **Available list**.

(2) FMPs stored previously in the **Selected list** may no more exist in the newly retrieved **Available list**. In such a case, a message box will prompt the user to ask if these erroneous FMP entries may be removed from the **Selected list** because this might lead to unsuccessful queries.

(3) Select one or more FMPs from the **Available list** on the left side and click on the "**>>** " button to add them to the **Selected list**.

(4) Multiple selection is possible by combining the mouse click with the <Ctrl> or the <Shift> keys.

(5) Eventual duplicates are automatically suppressed.

(6) Select one or more FMPs from the **Selected list** on the right side and click on the "**<<** " button to remove them from the current **Selected list**.

(7) A double click on an FMP in either list will immediately request the transfer action.

(8) When the **Selected list** is up to date, click on the **OK** button to save the list and come back to the previous window.

(9) This will save the new preferred list in the user preference **CHMI / ATFCM application / General / Query / Selected FMP Identifiers**.

(10) Following update rules will be applied on all opened windows using a **Selected FMP(s)** list for the query:

a) If the opened window has not yet processed a valid query / reply, **newly added FMPs** are inserted and checked in the Selected FMP(s) list. **Deleted**

entries are removed from the Selected FMP(s) list independently from the current check state.

b) If a valid query / reply has already been processed, **newly added FMPs** are inserted unchecked in the Selected FMP(s) list. **Deleted entries** are removed from the Selected FMP(s) list only if they were unchecked before opening the dialog.

c) All other existing FMPs remain in the Selected FMP(s) list with the previous check state.

(11) New windows opened after this list update will use the new user preferred **Selected list**.

(12) To stop editing and leave the FMP Selection without saving the modifications to the list, click on the **Cancel** button.

(13) If the user clicks the close dialog button ("X" on top right) and the **Selected list** was changed, a confirmation dialog will pop up asking if the modifications to the list shall be saved or ignored.

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7 Traffic Counts

7.1 Function presentation

(1) This display gives the number of flights for a given Traffic Type.

(2) Flights may be selected on different query criteria for given time period and interval.

(3) The flights are selected on:

a) The time over the reference location for a traffic Volume and a Point.

b) The take off time for the Aircraft Operator, the Departure Aerodrome and Aerodrome Set.

c) The arrival time for the Arrival Aerodrome and Aerodrome Set.

d) The entry time for the Airspace.

7.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Traffic / Traffic Counts**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl>** + **<Shift>** + **<F6>**.

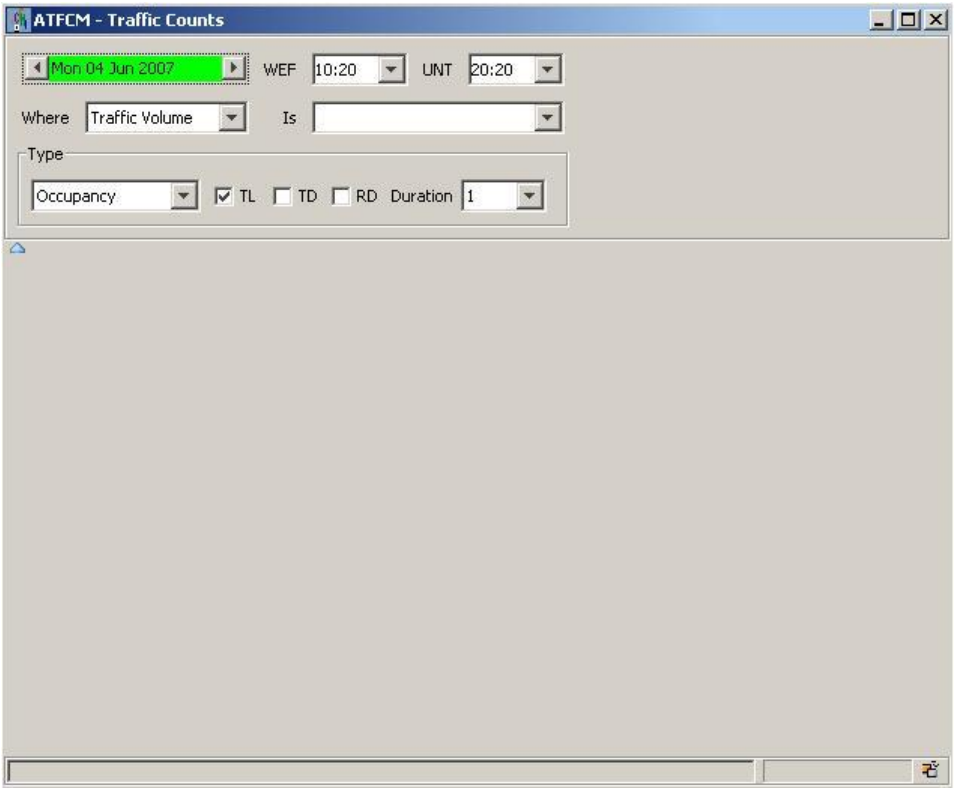


Figure Chmi - 24: Query Traffic Counts

(3) A new empty **Traffic Counts** window will be opened with the following defaults:

a) Traffic Date: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** time: proposed based on the **Default query period** assignment using your user adjustable preferences (see **Important remark** below).

c) **UNT** time: proposed based on the displayed WEF and the **Default query period** assignment (see **Important remark** below).

d) **Where**: last element type used with this query. This selection is based on the system property **CHMI / ATFCM application / Traffic / Traffic counts / General / Last Used Reference Location Type**.

e) **Is**: based on the "Where" selection, the drop-down box will propose the most recent query data elements you have queried before with success.

f) Count **Type**: last selection used with your last similar query. This selection is based on the system property **CHMI / ATFCM application / Traffic / Traffic counts / General / Last Used Count Rate**.

g) Traffic Type (**TL - TD - RD**): last selection used with your last similar query. This selection is based on the system property **CHMI / ATFCM application / Traffic / Traffic counts / General / Last Used Type**. See next section **Preparing to query** for an explanation of these codes.

(4) **Important remark**: when referring to common requirements for **Default query period** assignment, the Traffic Counts function shall use typed preferences based on selected Entry or Occupancy count type. For example:

a) <CHMI / ATFCM application / Traffic / Traffic counts / Entry / ATFCM entry query lead time> or <CHMI / ATFCM application / Traffic / Traffic counts / Occupancy / ATFCM occupancy query lead time>

b) <CHMI / ATFCM application / Traffic / Traffic counts / Entry / ATFCM entry query default elapse> or <CHMI / ATFCM application / Traffic / Traffic counts / Occupancy / ATFCM occupancy query default elapse>

(5) When **Occupancy** Count **Type** is selected, the **Duration** parameter must be provided.

(6) For occupancy counts a snap-shot is taken of all flights present in the sector during a time period identified by the **Duration** parameter.

a) For a radar controller this is the flights present in the sector during one minute.

b) For a planner controller it can be the flights present in the sector during up to twenty minutes.

(7) See section "*Preparing to query*" for initialisation details of the **Duration** control.

(8) When Aerodrome or Aerodrome Set is selected, the related category controls (**Cat**: Global (**Glob**), Departure (**Dep**) or Arrival (**Arr**)) are proposed. Global button comes preselected.

(9) When Point is selected, the Maximum (**MAX**) and Minimum (**MNM**) Flight Level (**FL**) controls are proposed. They are initialised with the extreme values (990 for MAX and 000 for MNM).

(10) On initial display, the focus is set on the first mandatory query control with no default value: the **Is** input control.

7.3 Preparing to query

(1) The observed period may be adjusted as follow:

a) The traffic **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

b) The proposed default **WEF** and **UNT** times may be overwritten.

c) Any valid time value from 00:00 to 23:59 included may be entered in the **WEF** and/or the **UNT** entry boxes. This input must respect the HH:MM format.

d) Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(2) The period resulting from the input will be validated taking into account the system property **CHMI / ATFCM application / General / Dates and times / Query over midnight**.

(3) If query over two days is not authorised, the **WEF** time must be smaller than the **UNT** time, except if both times are set to 00:00, which corresponds to a full day query.

(4) The **Default query period** assignment will be used.

(5) Every time the count type query control is changed from Entry to Occupancy and vice versa, the query period shall immediately be adjusted based on the following rules:

a) If current UTC time is within the displayed query period (i.e. "UTC >= WEF" and "UTC <= UNT"), then WEF and UNT input controls shall be adjusted based on the **Default query period** preferences setting corresponding to the newly selected count type (i.e. like when a new window is opened).

b) If current UTC time is outside the displayed query period (i.e. "UTC < WEF" or "UTC > UNT"), then WEF value remains untouched and the relevant default query elapse preference from the **Default query period** assignment is added to WEF to identify the new UNT value.

(6) The **Where** parameter may be adjusted to one of the following values:

a) Traffic Volume

b) Aerodrome

c) Aerodrome Set

d) Airspace

e) Point

f) Aircraft Operator

(7) The identifier corresponding to the selected **Where** shall be keyed in the associated **Is** entry box or selected from the drop-down list (when a previous query was successful).

(8) When Aerodrome or Aerodrome Set is selected, the related category (**Cat**) must be checked (Global, Departure or Arrival).

(9) When Point is selected, the Maximum (**MAX**) and Minimum (**MNM**) Flight Levels (**FL**) may be overwritten to reduce or enlarge selected altitude range.

(10) A valid integer value (maximum of 3 digits) entered in the Maximum (**MAX**) box, will be rounded as follow:

a) 0 remains 0.

b) Input between 1 and 30 is rounded up to the nearest multiple of 10.

c) Input between 31 and 290 is rounded up to the nearest multiple of 5.

d) Input between 291 and 990 is rounded up to the nearest multiple of 10.

e) Above 990 is forced to 990.

(11) A valid integer value (maximum of 3 digits) entered in the Minimum (**MNM**) box, will be rounded as follow:

a) Input between 0 and 34 is rounded down to the nearest multiple of 10.

b) Input between 35 and 294 is rounded down to the nearest multiple of 5.

c) Input between 295 and 999 is rounded down to the nearest multiple of 10.

(12) Following Count **Types** may be available depending on the selected Where criteria:

a) **Entry Hour/20 Min**: Entry hourly counts are given every 20 minutes step.

b) **Entry Hour/10 Min**: Entry hourly counts are given every 10 minutes step.

c) **Entry Hour**: Entry hourly counts are given every 60 minutes step.

d) **Entry 20 Min**: Entry 20 minutes counts are given every 20 minutes step.

e) **Occupancy**

(13) **Entry** counts are available for all possible **Where** entries.

(14) Access to **Entry** counts will be restricted by security profile.

(15) **Occupancy** Type is only available for Airspace, Traffic Volume based on an Airspace and Aircraft Operator.

(16) In the case of Occupancy Type, the returned counts contain:

a) either all flights being in the selected airspace at the given time,

b) or all flights being on fly at the given time.

(17) The **Duration** parameter is automatically adjusted by the application when **Occupancy** type is selected and the **Where / Is** identifier is adjusted. The application will use:

a) Either the **Duration** value that was last queried with the specified **Where / Is** identifier

b) Or the default value stored in the user adjustable preference **CHMI / ATFCM application / Traffic / Traffic counts / Occupancy / Default Occupancy Duration** (initially set to 1 minute) when occupancy counts for the specified **Where / Is** identifier is queried for the first time. Possible values for the user preference are from 1 to 20 minutes by 1 minute increment.

(18) On a restore of a saved Workspace the value is set as it was stored by the save workspace action.

(19) After adjustment by the application, the user can modify the **Duration** query criteria.

a) The drop-down list proposes one minute increment values from 1 to 20 minutes.

b) **Duration** values for which OTMV parameters are defined shall be highlighted in the list.

(20) The **Occupancy** counts are given every one minute step.

(21) This step of one minute is defined in system property **CHMI / ATFCM Application / Traffic / Traffic counts / Occupancy / Occupancy Step** and cannot be changed by the user**.**

(22) When **Occupancy** Type is selected and when the user adjustable preference **CHMI / ATFCM application / General / Graph / Minutes on time scale** is enabled,

a) At the minimum, the time scale is always displaying hourly, 20 and 40 minutes labels.

b) If current scale and window size allows correct readability, the time scale labels are displayed every 10 minutes.

c) Furthermore, if current scale and window size allows correct readability, the time scale labels are displayed every 1 minute.

(23) The different Traffic Types can be checked or unchecked:

a) **TL**: Traffic Load

b) **TD**: Traffic Demand

c) **RD**: Regulated Demand

(24) Any combination of these Traffic Types may be requested.

(25) At least one Traffic Type must be checked on issuing the query.

(26) Select the **Proposal** radio button on the button bar to include the proposed flights in the retrieved counts. Otherwise the counts will be based on the normal flights only.

(27) The **Send** button will send the query to the Network Operations system.

(28) This window may be closed before any query is issued.

(29) The **Reset** action common functionality is implemented for this function. All query criteria except Normal/Proposal selection are considered for activating the Reset action.

(30) A checkbox "**Show VFR/OAT at Aerodrome**" shall be visible and editable if (and only if) the "Where" dropdown is Aerodrome, Aerodrome Set or Traffic Volume.

(31) When "Show VFR/OAT at Aerodrome" is selected on a traffic volume it has effect only if the reference location is an aerodrome or an aerodrome set.

7.4 Displaying the query results

7.4.1 Common Display

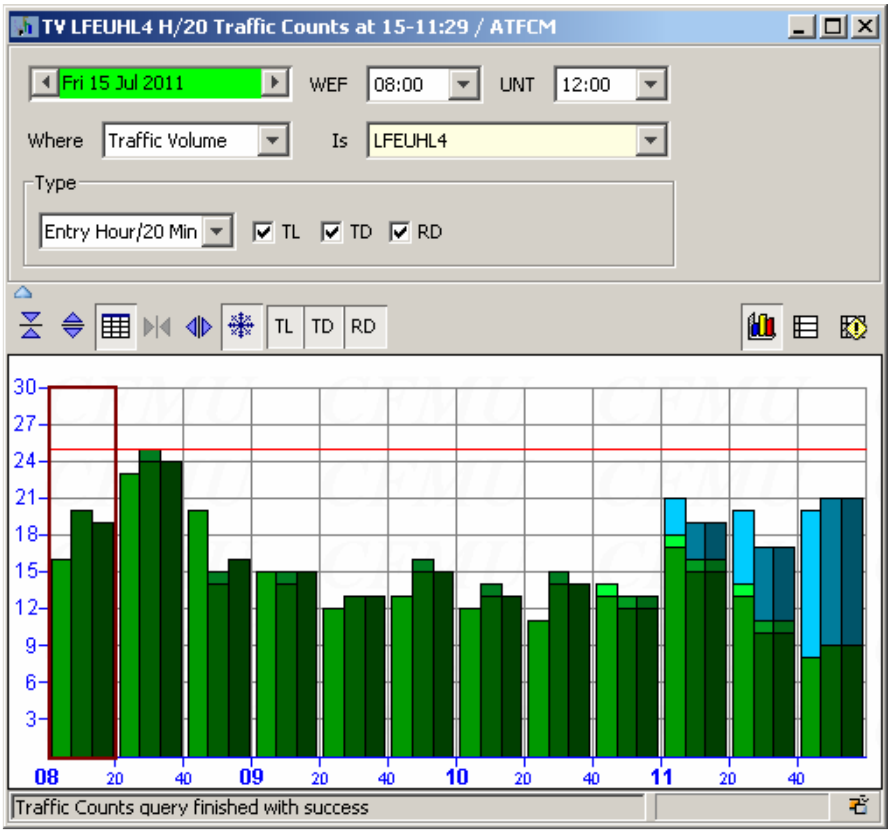


Figure Chmi - 25: Traffic Counts graph

(1) The header box contains the selected query criteria:

a) The period (Traffic date, **WEF** and **UNT** times)

b) The main **Where** selection criteria (Traffic Volume, Aerodrome or Set with the associated **Category**, Airspace, Point with associated Flight Level Range (**FL**), Aircraft Operator),

c) The selected Count **Type** (Entry / Occupancy)

i) With the associated Count rate (elapse / step) in case of Entry counts.

ii) With the **Duration** additional box in case of Occupancy counts.

d) The combination of Traffic **Types** retrieved from the server (**TL, TD, RD**).

(2) A split bar separates the query fields from the query results. It includes on the left, a small arrow button (Up/Down) to Hide/Show the query part.

(3) The query part may be hidden automatically on displaying the query results, depending on the user adjustable preference **CHMI / ATFCM application / Traffic / Traffic counts / General / Hide query on reply**.

(4) Underneath the split bar that separates the query fields from the query results there are three data presentation buttons; they allow the user to select one of the following display types:

a) Graph: bar chart representation of the counts.

b) Table: tabular presentation of the different count values.

c) Regulation: detailed list of all regulations affecting these counts.

(5) The Graph display is shown initially.

(6) When the user chooses to change the display type then the active window will use the data previously retrieved.

(7) When the query results are displayed, the focus is set on the Graph, the Table or the Regulation list, depending on the selected type.

(8) The first bar of the Graph, the first cell of the Table or the first line of the list is then selected.

(9) A **High Counts** warning sign is displayed at the top of the graph, the table or the list, ONLY if at least a part of a high occupancy counts OTMV alert is reported for the future (i.e. after current UTC time).

(10) This warning shall remain active until the reply data are refreshed and future high occupancy counts situation is back to "normal" values (i.e. High occupancy counts OTMV alerts are no more reported for future time).

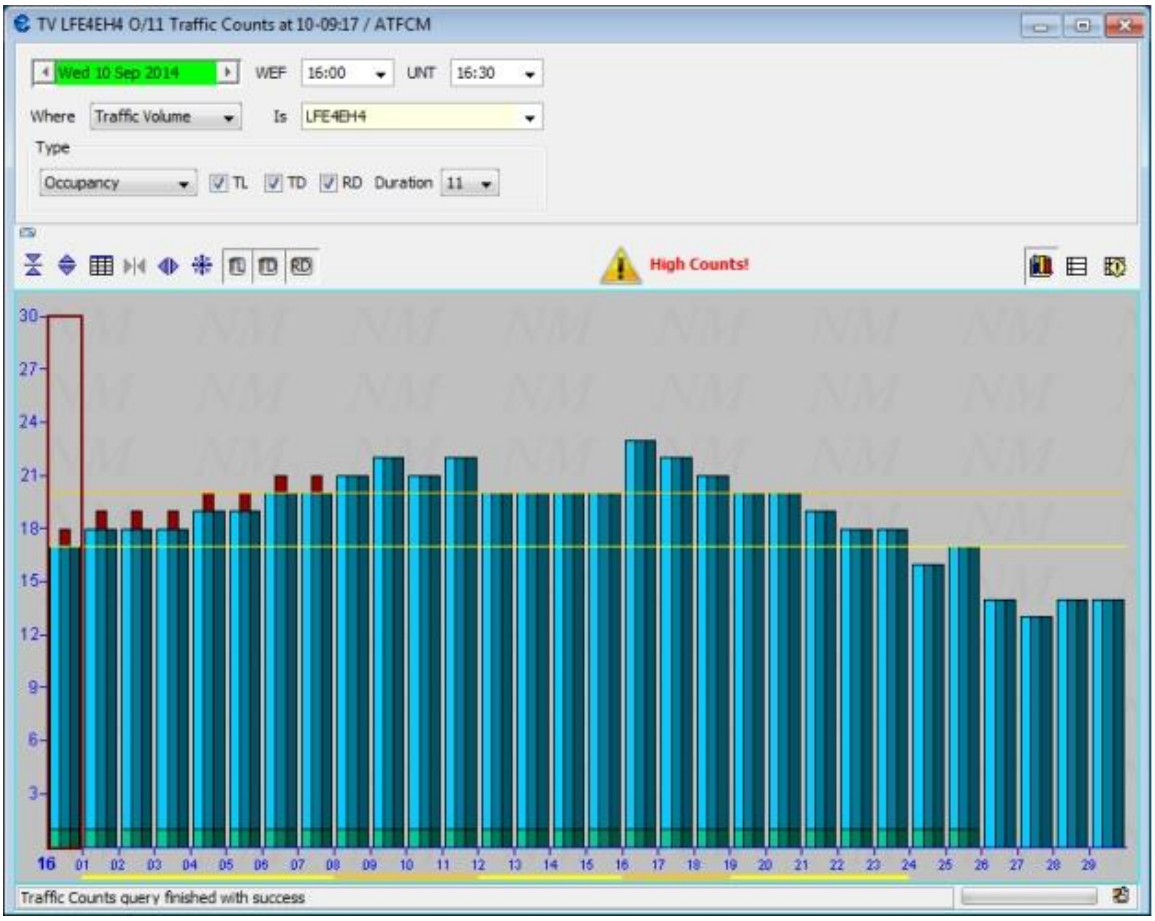


Figure Chmi - 26: High Occupancy counts display

7.4.2 Graph display

(1) On the top of the graph, a group of three toggle buttons allows to select the traffic type(s) to be included in the graph:

a) **TL**: Traffic Load.

b) **TD**: Traffic Demand.

c) **RD**: Regulated Demand.

(2) Each button is activated when the corresponding Traffic type has been successfully retrieved from the Network Operations systems.

(3) All retrieved Traffic types are displayed initially.

(4) Each traffic type is rendered differently on the graph. From the first Traffic Type displayed to the last, the brightness of the different colour used is reduced. This is done to ease the recognition of each bar based on the Traffic Type.

(5) The graph contains a series of vertical bars representing the number of flights according to the count **Type** requested.

(6) Each vertical bar is composed of a series of stacked coloured blocs representing the following flight sub-totals:

a) ATC activated

b) TACT activated with no FSA expected

c) RPL

d) PFD

e) IFPL

f) TACT activated with FSA expected

g) currently Suspended

(7) The above sequence is used by default from bottom to the top of the bar.

(8) Some of these blocs might be absent if the corresponding sub-total is 0.

(9) Dedicated user preferences may be adjusted for the piling up sequence (**CHMI / ATFCM application / Traffic / Traffic counts / General / Count piling up order**) and the colour associated to each sub-total (i.e. **CHMI / ATFCM application / Traffic / Traffic counts / General / Suspended count colour**).

(10) The "RPL + PFD + FPL" sub-total is split into individual sub-totals and so the Count piling up order now defines the order of the 7 sub-totals. Furthermore, the user preference shall be able to adjust the colour of each of the 7 sub-totals.

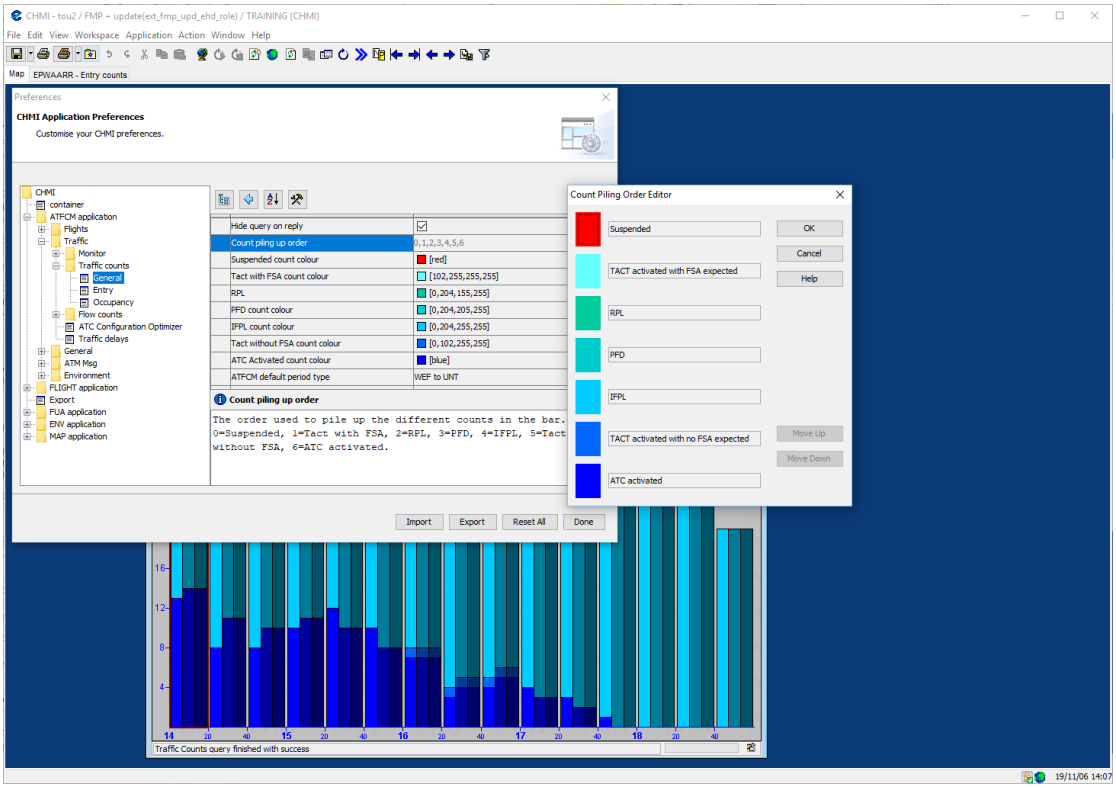


Figure Chmi - 27: Count Colour Editor

(11) The count value represents:

a) The number of flights over the next 20 or 60 minutes for Entry counts depending on the count **Type** selected during query preparation.

b) For Occupancy counts it is depending on the **Duration** value either the number of flights during the next x minutes where x stands for the **Duration** value or during each minute if **Duration** value is equal to one minute.

(12) The count value is shown every 20 or 60 minutes for Entry counts or every 1 minute for Occupancy counts depending on the initial choice made during the query preparation.

(13) The capacity value is drawn as line on the graph, if 60 minutes counts have been requested.

(14) To avoid confusion, hourly capacity is not shown on the graph when 20 minutes counts are displayed.

(15) The colour of the capacity line is identified by the user adjustable preference **CHMI / ATFCM application / General / Graph / Capacity colour**.

(16) When the user adjustable parameter **Peak OTMV** (Occupancy Traffic Monitoring Value) is defined for the queried **Where** identifier and the associated **Duration**, this **Peak OTMV** value, taking possible temporal OTMV overwrite into account, is drawn as horizontal lines on the graph if **Occupancy** Type is selected.

(17) The colour of the **Peak OTMV** line is identified by the user adjustable preference **CHMI / ATFCM application / Traffic / Traffic counts / Occupancy / Peak OTMV colour**

(18) When the user adjustable parameter **Sustained OTMV** (Occupancy Traffic Monitoring Value) is defined for the queried **Where** identifier and the associated **Duration**, this **Sustained OTMV** value, taking possible temporal OTMV overwrite into account, is drawn as horizontal lines on the graph if **Occupancy** Type is selected.

(19) The colour of the **Sustained OTMV** line is identified by the user adjustable preference **CHMI / ATFCM application / Traffic / Traffic counts / Occupancy / Sustained OTMV colour**

(20) If at least a part of a high occupancy counts OTMV alert is reported for the future (i.e. after current UTC time), a warning modal dialog shall highlight:

a) The fact a **High Occupancy** counts situation is detected.

b) The queried **Where** identifier and the associated **Duration.**

c) The number of different periods detected per severity level for reporting this **High Occupancy** counts situation.

(21) The condition for displaying this additional warning modal dialog in case of **High Occupancy** counts situation is based on the following user adjustable preference:

a) **CHMI / ATFCM application / Traffic / Traffic counts / Occupancy / High OTMV Warning**

b) Info text: **Popup a warning dialog when High Occupancy situation is detected?**

c) Possible values are:

i) **Inactive**

ii) **First Detection**

iii) **Always**

d) Default value: **First Detection**

(22) If the preference **High OTMV Warning** is set to **First Detection**, the warning dialog shall only popup on initial detection of a given **High Occupancy** counts situation. Further consecutive detection on the same window is no more reported by a modal dialog until the situation is back to "normal" values for at least one refresh (i.e. **High Occupancy** counts situation is no more detected).

(23) If the preference **High OTMV Warning** is set to **Always**, the warning dialog shall popup on any detection of a given **High Occupancy** counts situation (even on subsequent data refresh of the same display).

(24) If a **High Occupancy** counts situation is detected and the preference **High OTMV Warning** is set to **Inactive**, the warning dialog shall NOT popup, even in case of a **High Occupancy** counts situation.

(25) This is a modal dialog which blocks access to all other functions in the application until it is acknowledged (i.e. closed).



Figure Chmi - 28: High Occupancy counts warning dialog

(26) If a **High Occupancy** counts situation is reported, a stripe is drawn below the graph and aligned on the time line to indicate past and future periods when High Counts are reported based on the defined OTMV:

a) Using the user adjustable **Peak OTMV Colour** for Peak traffic periods

b) Using the user adjustable **Sustained OTMV Colour** for Sustained traffic periods

(27) The alerts displayed using stripes are the ones detected on the highest Traffic Type currently displayed on the graph (Traffic Load is higher than Regulated Demand, which is higher than Traffic Demand).

(28) The applicability of the **OTMV** (Occupancy Traffic Monitoring Value) display and related alert processing in the Occupancy Traffic Counts is subordinated to the authorisation via security profile

(29) Below the graph,

a) A stripe indicates that a regulation has been implemented for this Traffic Volume. The colour of this stripe is identified by the user adjustable preference **CHMI / CHMI / ATFCM application / General / Graph / Regulation colour**.

b) Another stripe shows that a rerouting has been implemented for this traffic volume. The colour of this stripe is identified by the user adjustable preference **CHMI / CHMI / ATFCM application / General / Graph / Reroute colour**.

(30) Whenever the mouse is over one of the stripes drawn at the bottom of the graph, the corresponding text legend giving details over the stripe (stripe type, state, period) is displayed below the graph area (*ex: "From 17:12 to 17:18 Sustained OTMV Alert"*).

(31) On the graph itself, a thin line shows the period when a regulation is active and the corresponding flight rate. The colour of this line is identified by the user adjustable preference **CHMI / CHMI / ATFCM application / General / Graph / Regulation colour**.

(32) The **vertical axis** indicates the number of flights.

(33) On initial display of newly retrieved data, if the **Zoom Freeze** function is OFF:

a) The graph shall be "pre-"zoomed vertically so that the highest count to be displayed in the complete graph is visible (i.e. taking into account all counts for the complete query period).

b) When displaying Hourly Entry Counts, the initial vertical zooming shall take the hourly capacity values into account to make sure they are initially visible.

c) When displaying Occupancy Counts where OTMV parameters are defined, the initial vertical zooming shall take these values into account to make sure they are initially visible.

(34) The **horizontal axis** indicates the time periods.

(35) On initial display of newly retrieved data, if the **Zoom Freeze** function is OFF:

a) The graph shall be "pre-"zoomed horizontally so that the elapse time displayed in one view is equal or just above (i.e. next multiple of time that corresponds to a complete vertical bar) the number of minutes identified in the **Initial Display Elapse** preference corresponding to the selected

count type (ex: *CHMI / ATFCM application / Traffic / Traffic counts / Entry / Entry initial display elapse*).

b) The graph shall be "pre-"scrolled horizontally so that the first visible vertical bar contains the time obtained by summing WEF with the number of minutes identified in the **Initial Display Shift** preference corresponding to the selected count type (ex: *CHMI / ATFCM application / Traffic / Traffic counts / Entry / Entry initial display shift*).

(36) At the top of the graph, next to the "graph related buttons" (grid and scope controls), a **Zoom Freeze ON / OFF** toggle button allows freezing the graph scaling (i.e. Vertical and Horizontal zooming) and time scrolling in its current level over a data refresh operation.

a) When the **Zoom Freeze** toggle button is **OFF** during a refresh operation, the new graph is displayed with the initial default zoom and period alignment settings.

b) When the **Zoom Freeze** toggle button is **ON** during a refresh operation with or without **Period Shifting**, the graph shall keep current level of zooming (i.e. same time elapse on horizontal axis and same count numbers on vertical axis) and time scrolling.

c) In case of refresh while the **Period Shifting** and **Zoom Freeze** are **ON**, the same time shifting value shall also be applied to the period displayed horizontally.

d) Any manual modification (i.e. automatic **Period Shifting** is not considered as a manual modification) of any query criteria implies a RESET of the current zoom levels when the new reply data is displayed.

e) On opening a new window or on RESET of the current zoom level, the **Zoom Freeze** toggle button is forced to **OFF.**

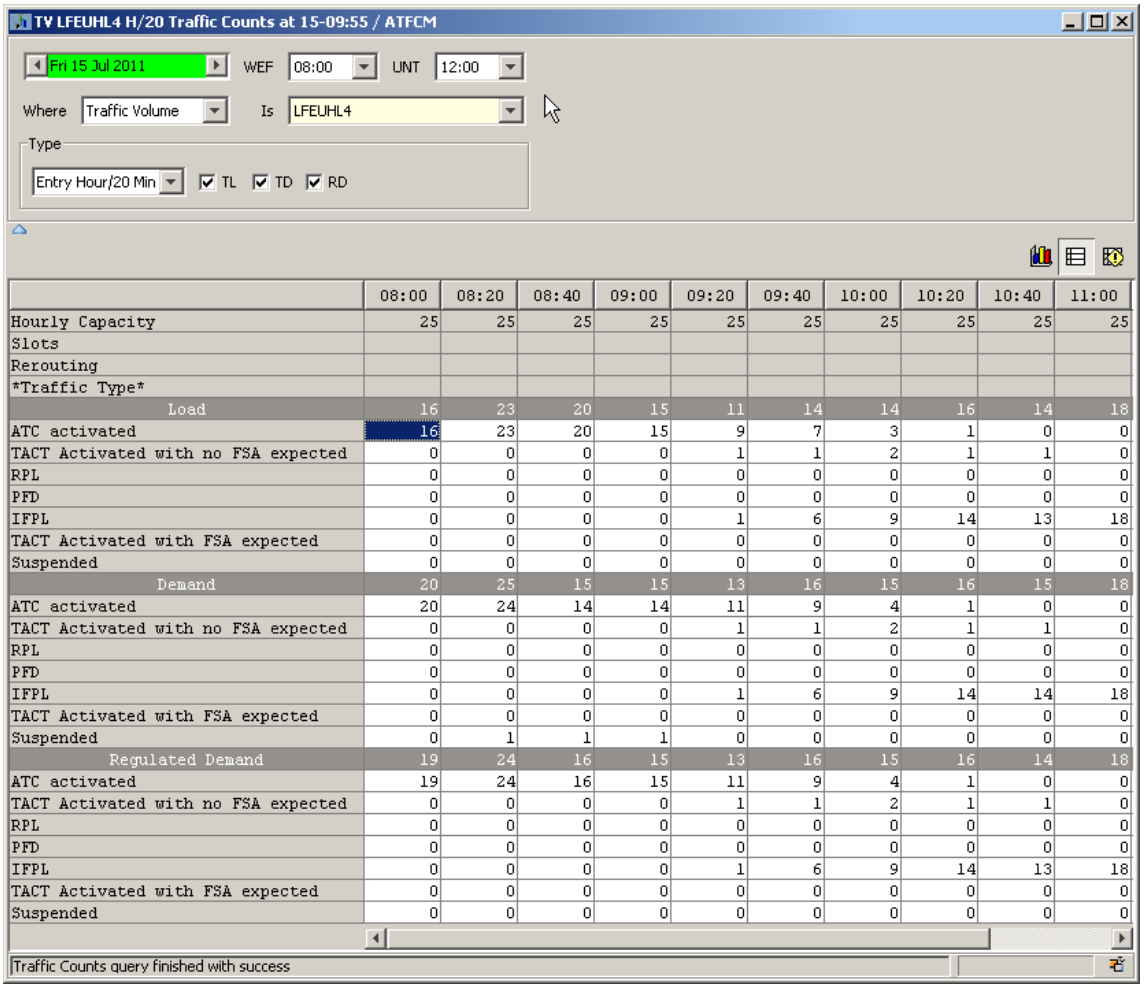
f) As soon as the user adjusts manually the zooming levels or the time scrolling, the **Zoom Freeze** toggle button is forced to **ON**.

g) The user can adjust at any time the state of the **Zoom Freeze** toggle button by clicking on it.

h) When the **Zoom Freeze** toggle button is forced manually to **OFF**, the graph is immediately displayed with the initial default zoom and period alignment settings (even before any refresh is performed).

(37) A warning sign **Zoom Hiding High Values** shall be displayed at the top of the graph vertical axis to indicate when the current vertical zooming level does not allow displaying the highest values of the complete queried period.

7.4.3 Table display



	08:00	08:20	08:40	09:00	09:20	09:40	10:00	10:20	10:40	11:00
Hourly Capacity	25	25	25	25	25	25	25	25	25	25
Slots										
Rerouting										
Traffic Type										
Load	16	23	20	15	11	14	14	16	14	18
ATC activated	16	23	20	15	9	7	3	1	0	0
TACT Activated with no FSA expected	0	0	0	0	1	1	2	1	1	0
RPL	0	0	0	0	0	0	0	0	0	0
PFD	0	0	0	0	0	0	0	0	0	0
IFPL	0	0	0	0	1	6	9	14	13	18
TACT Activated with FSA expected	0	0	0	0	0	0	0	0	0	0
Suspended	0	0	0	0	0	0	0	0	0	0
Demand	20	25	15	15	13	16	15	16	15	18
ATC activated	20	24	14	14	11	9	4	1	0	0
TACT Activated with no FSA expected	0	0	0	0	1	1	2	1	1	0
RPL	0	0	0	0	0	0	0	0	0	0
PFD	0	0	0	0	0	0	0	0	0	0
IFPL	0	0	0	0	1	6	9	14	14	18
TACT Activated with FSA expected	0	0	0	0	0	0	0	0	0	0
Suspended	0	1	1	1	0	0	0	0	0	0
Regulated Demand	19	24	16	15	13	16	15	16	14	18
ATC activated	19	24	16	15	11	9	4	1	0	0
TACT Activated with no FSA expected	0	0	0	0	1	1	2	1	1	0
RPL	0	0	0	0	0	0	0	0	0	0
PFD	0	0	0	0	0	0	0	0	0	0
IFPL	0	0	0	0	1	6	9	14	13	18
TACT Activated with FSA expected	0	0	0	0	0	0	0	0	0	0
Suspended	0	0	0	0	0	0	0	0	0	0

Figure Chmi - 29: Traffic Counts table

(1) The tabular representation provides:

a) The different **time slices** covered by the count query.

b) Depending on the displayed count type:

i) For Entry Counts, the **Hourly Capacity** for the corresponding time slice.

ii) For Occupancy Counts, the **Peak OTMV** and the **Sustained OTMV** applicable for the corresponding time slice, if defined.

c) An indicator when the slice is subject to a **Regulation** (CASA **Slots** allocation).

d) An indicator when the slice is subject to a **Rerouting** measure.

e) The different **count values**.

(2) Following breakdown is given for the **count values**:

a) The total count values for the different requested **Traffic Types**. Whenever a Peak or Sustained alert is reported for a past or future time slice of a Traffic Type, the cell background of the corresponding total count is coloured using the respective user adjustable **Peak OTMV Colour** or **Sustained OTMV Colour.**

b) For each Traffic Type, the sub-total count values are listed for the different count categories.

(3) These count categories are listed in following sequence:

a) ATC activated

b) TACT activated with no FSA expected

c) RPL

d) PFD

e) IFPL

f) TACT activated with FSA expected

g) currently Suspended

7.4.4 Regulation display

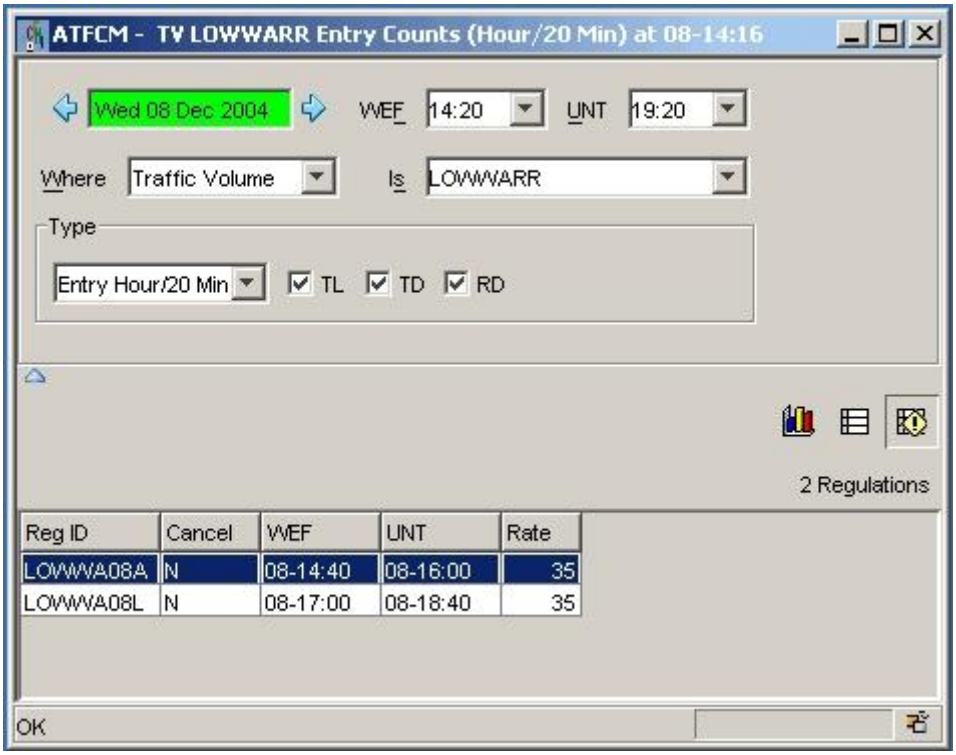


Figure Chmi - 30: Traffic Counts Regulations

(1) A counter identifies the number of Regulation entries.

(2) The main listbox of the window shows the list of Regulations affecting the selected count data.

(3) One Regulation may appear on several detail lines, if several periods and rates are applicable.

(4) Each detailed Regulation line contains:

a) The Regulation identifier. (**REG ID**)

b) An indicator for regulations cancelled before their termination (**Cancel**).

c) The With Effect From day and time (**WEF**).

d) The Until day and time (**UNT**).

e) The corresponding rate applied (**Rate**).

7.5 User interactions

7.5.1 Common

(1) The **Normal / Proposal** button may be adjusted before issuing a new query.

(2) This window allows to modify a previous query selection and to issue a new request.

(3) On query modification, the **Reset action** is enabled.

(4) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(5) The user adjustable preference **CHMI / ATFCM application / Traffic / Traffic counts / General / Hide query on reply** allows to hide automatically or not the query data part on display of valid reply count data.

(6) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(7) The **Previous / Next period** function may be used to issue the same query but for the previous or the next period in time.

(8) The **Polling function** may be activated to start automatic refresh processing on this window.

(9) The **Period Shifting function** may be activated to start automatic period shifting on subsequent refresh of this window.

(10) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(11) To print Traffic Counts data, first retrieve the data in the window and select the desired display type (Graph, table, regulation). Adjust possibly the displayed

period, scale and other details. When ready, click on the **Print** button or select the **File / Print** menu option.

(12) To export the retrieved count details, click on the **Export** button or select the **Action / Export** menu option.

(13) Increasing / reducing the window height will impact the size of the main display control (graph, table or list).

(14) Increasing / reducing the window width will impact the size of the main display control (graph, table or list). Regulation list counter moves with the edge.

(15) The **Partial Period Shifting** function may be used to issue the same query but for the previous or the next period with time overlap (see **Partial Period Shifting** function detailed description).

7.5.2 Graph display

(1) It is possible to toggle on/off any combination of the **Traffic Type** buttons to display the corresponding count bars on the graph.

(2) Following dedicated user preferences may be adjusted using the **Preferences** dialog for the **piling up sequence** and the **colour** associated to each sub-total:

a) CHMI / ATFCM application / Traffic / Traffic counts / General / Count piling up order

b) CHMI / ATFCM application / Traffic / Traffic counts / General / Suspended count colour

c) CHMI / ATFCM application / Traffic / Traffic counts / General / Tact with FSA count colour

d) CHMI / ATFCM application / Traffic / Traffic counts / General / RPL count colour

e) CHMI / ATFCM application / Traffic / Traffic counts / General / PFD count colour

f) CHMI / ATFCM application / Traffic / Traffic counts / General / IFPL count colour

g) CHMI / ATFCM application / Traffic / Traffic counts / General / Tact without FSA count colour

h) CHMI / ATFCM application / Traffic / Traffic counts / General / ATC Activated count colour

(3) Clicking on a bar puts the focus on the graph and sets the bar as selected item.

(4) When the focus is on the graph, the left and right arrows on the keyboard moves selection from one bar to the other.

(5) The right mouse button pops up a **contextual menu** for the selected bar.

(6) The **bottom scroll bar** within the graph pane is available only when the scope of the shown period is reduced or enlarged.

(7) Another scroll bar may appear at the bottom of the Traffic Counts component when the window is too small to see all contents; it will disappear again when all of the windows contents are visible.

(8) Underneath the split bar that separates the Query fields from the query results several buttons may be displayed. When the graph presentation button on the right side of the pane is selected, then graph related buttons appear on the left of the pane to allow the user to adjust the following graph characteristics:

a) **To enlarge the scope of the vertical axis**: click on the two vertical arrows pointing to each other with a horizontal line in the middle.

b) **To reduce the scope of the vertical axis**: click on the two vertical arrows pointing away from each other with a horizontal line in the middle.

c) **To display / hide a grid on the graph**: click on the toggle button with a table representation.

d) **To enlarge the scope of the shown period**: click on the two horizontal arrows pointing to each other with a vertical line in the middle.

e) **To reduce the scope of the shown period**: click on the two horizontal arrows pointing away from each other with a vertical line in the middle.

f) **To freeze / unfreeze the current level of vertical and horizontal zooming and the current time scrolling**: click on the **Zoom Freeze ON / OFF** toggle button.

7.5.3 Table display

(1) Clicking on a cell puts the focus on the table and select the clicked bar.

(2) When the focus is on a cell with a count, the left/right and up/down arrows on the keyboard moves selection from one cell to the other.

(3) The right mouse button pops up a **contextual menu** for the selected cell.

7.5.4 Regulation display

(1) To display a sub-selection of the retrieved Regulation list, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(2) To modify the **Sort sequence** of the Regulation list, click on column headers corresponding to your desired sort criteria.

(3) When a line is selected in the list, the right mouse button pops up a **contextual menu** for the selected Regulation.

7.6 Contextual actions

7.6.1 Graph display

(1) When a bar is selected on the graph, the right mouse button pops up a **contextual menu** for the corresponding period.

(2) This popup menu offers sub-menu parts related to the selected item in the higher-level popup menu.

(3) The higher-level popup menu displays:

a) The **time slice** corresponding to the selected bar

b) A separator line.

c) For **Entry Counts**:

i) The **Hourly Capacity** for the period of the selected bar

ii) **Traffic Capacity** to display detailed capacity values for the selected date and reference location (i.e. not applicable when querying AO counts). May give subsequent access to the related maintenance function, if the user is authorised.

iii) **Occupancy Counts** allows focussing on the Occupancy Counts for a selected Entry Counts elementary period (i.e. a bar of the graph). Selecting this option will issue the same query in another window where only the following query parameters are adjusted:

i. The count type is forced to **Occupancy**

ii. The WEF time is set to the start time of the selected bar minus the corresponding "Occupancy query lead time" preference from the **Default query period** assignment.

iii. The UNT time is set to the calculated WEF time plus the corresponding "Occupancy query default elapse" preference from the **Default query period** assignment.

d) For **Occupancy Counts**:

i) The **Peak OTMV** parameter value for the period of the selected bar, if defined and if the user is authorised.

ii) The **Sustained OTMV** parameter value for the period of the selected bar, if defined and if the user is authorised.

iii) **Occupancy Traffic Monitoring Values** to display detailed Occupancy Traffic Monitoring Values for the selected date and Traffic Volume (i.e. only applicable when querying counts for a TV defined on an Airspace). May give subsequent access to the related maintenance function, if the user is authorised.

e) A separator line.

f) The different **Traffic Types** retrieved with the count value corresponding to the selected period. Each Traffic Type line gives access to a popup sub-menu.

(4) Each popup sub-menu may contain the following entries:

a) The breakdown of the count value per sub-category.

b) A separator line.

c) **Flight List** to query the list of flights for the selected Traffic Type with the same Reference Location as the initial Count query. Requested period is the one identified by the selected bar. The "Show VFR/OAT at Aerodrome" and "Show Forecast" (when access is authorized) checkboxes are selected/deselected accordingly.

d) **Flow Counts** to query the same Traffic Type counts per Flow for the selected Traffic Volume. This option is only available from **Traffic Volume Counts** display. The query period used is the same as the one displayed on top of the window. The "Show VFR/OAT at Aerodrome" and "Show Forecast" (when access is authorized) checkboxes are selected/deselected accordingly.

(5) The transferred period may be affected by user preferences related to **Default query period identification**.

7.6.2 Table display

(1) When a count cell is selected on the table, the right mouse button pops up a **contextual menu** for the corresponding period.

(2) This popup menu lists the possible actions for the selected cell.

(3) Following options may be available in the popup menu:

a) The **time slice** corresponding to the selected cell is displayed.

b) A separator line.

c) For **Entry Counts**:

i) **Traffic Capacity** to display detailed capacity values for the selected date and reference location (i.e. not applicable when querying AO counts). May give subsequent access to the related maintenance function, if the user is authorised.

ii) **Occupancy Counts** allows focussing on the Occupancy Counts for a selected Entry Counts elementary period (i.e. a column of the table). Selecting this option will issue the same query in another window where only the following query parameters are adjusted:

- The count type is forced to **Occupancy**

- The WEF time is set to the start time of the selected bar minus the corresponding "Occupancy query lead time" preference from the **Default query period** assignment.
- The UNT time is set to the calculated WEF time plus the corresponding "Occupancy query default elapse" preference from the **Default query period** assignment.

d) For **Occupancy Counts**:

i) **Occupancy Traffic Monitoring Values** to display detailed Occupancy Traffic Monitoring Values for the selected date and Traffic Volume (i.e. only applicable when querying counts for a TV defined on an Airspace). May give subsequent access to the related maintenance function, if the user is authorised.

e) A separator line.

f) **Flight List** to query list of flights for the selected Traffic Type with the same Reference Location as the initial Count query. Requested period is the one identified by the selected cell. The "Show VFR/OAT at Aerodrome" and "Show Forecast" (when access is authorized) checkboxes are selected/deselected accordingly.

g) **Flow Counts** to query the same Traffic type counts per Flow for the selected Traffic Volume. This option is only available from **Traffic Volume Counts** display. The query period used is the same as the one displayed on top of the window. The "Show VFR/OAT at Aerodrome" and "Show Forecast" (when access is authorized) checkboxes are selected/deselected accordingly.

(4) The transferred period may be affected by user preferences related to **Default query period identification**.

7.6.3 Regulation display

(1) When a regulation line is selected on the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected regulation.

(3) Following options may be available in the popup menu:

a) **Flight List**: displays the Provisional Slot Allocation List (Traffic Load) for the selected regulation and its activation period starting either with the current time or the start time of the regulation.

b) **Regulation Description**: displays detailed information for the selected regulation.

c) **Regulation Delays**: displays the delays for the selected regulation and its activation period.

(4) For the Regulation Delays and the Regulation Flight List the transferred period may be affected by user preferences related to **Default query period identification**.

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8 Flow Counts

8.1 Function presentation

(1) This display gives the number of flights entering in a traffic volume for a given Traffic Type.

(2) The counts are given for each flow of the Traffic Volume for a given time period and interval.

8.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Traffic / Flow Counts**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl> + <Shift> + <F7>**.

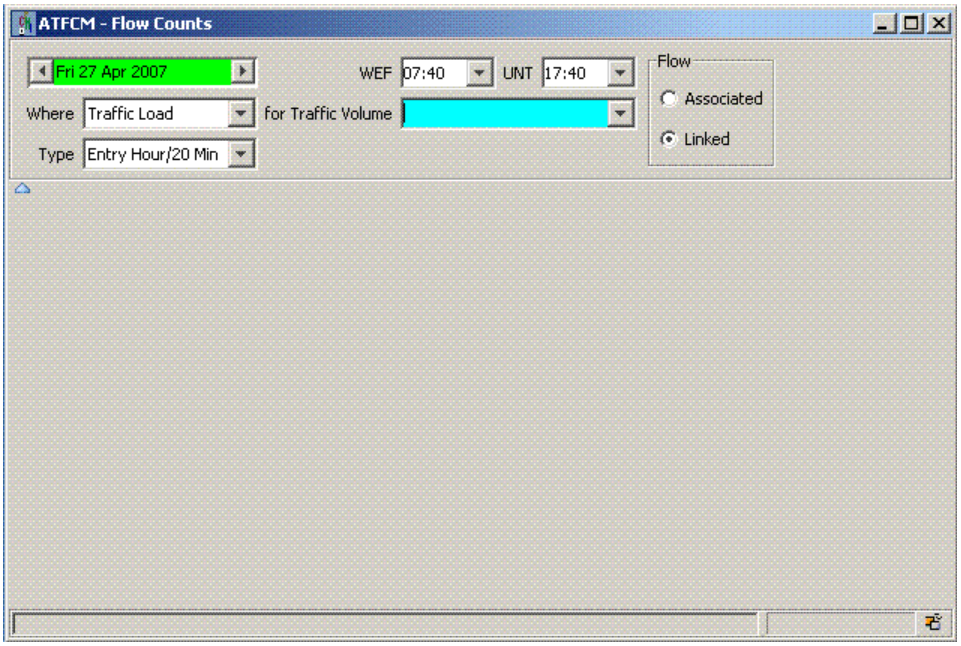


Figure Chmi - 31: Query Flow Counts

(3) A new empty **Flow Counts** window will be opened with the following defaults:

a) Traffic Date: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** time: proposed based on the **Default query period** assignment using your user adjustable preferences (see **Important remark** below).

c) **UNT** time: proposed based on the displayed WEF and the **Default query period** assignment (see **Important remark** below).

d) **Where**: last Traffic Type used with this query. This selection is based on the system property **CHMI / ATFCM application / Traffic / Flow counts / General / Last Used Traffic Type**.

e) **For Traffic Volume**: the drop-down box will propose the most recent Traffic Volumes you have queried before with success.

f) Count **Type**: last selection used with your last similar query. This selection is based on the system property **CHMI / ATFCM application / Traffic / Flow counts / General / Last Used Count Rate.** On a restore of a saved Workspace the value of this preference property is not used but the value which was stored by the save workspace action.

g) If Count **Type** is Occupancy **Duration** is filled in with last selection used with your last similar query. This selection is based on the system property **CHMI / ATFCM application / Traffic / Flow counts / Occupancy / Last Used Occupancy Duration.** On a restore of a saved Workspace the value of this preference property is not used but the value which was stored by the save workspace action.

h) The group box titled **"Flow"** contains two radio buttons: "**Associated"** and "**Linked"**. On initial display the radio button **"Associated"** is selected.

(4) **Important remark**: when referring to common requirements for **Default query period** assignment, the Flow Counts function shall use typed preferences based on selected Entry or Occupancy count type. For example:

a) <CHMI / ATFCM application / Traffic / Flow counts / Entry / ATFCM entry query lead time> or <CHMI / ATFCM application / Traffic / Flow counts / Occupancy / ATFCM occupancy query lead time>

b) <CHMI / ATFCM application / Traffic / Flow counts / Entry / ATFCM entry query default elapse> or <CHMI / ATFCM application / Traffic / Flow counts / Occupancy / ATFCM occupancy query default elapse>

(5) On initial display, the focus is set on the first mandatory query control with no default value: the **Traffic Volume** input control.

8.3 Preparing to query

(1) The observed period may be adjusted as follow:

a) The traffic **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

b) The proposed default **WEF** and **UNT** times may be overwritten.

c) Any valid time value from 00:00 to 23:59 included may be entered in the **WEF** and/or the **UNT** entry boxes. This input must respect the HH:MM format.

d) Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(2) The period resulting from the input will be validated taking into account the system property **CHMI / ATFCM application / General / Dates and times / Query over midnight**.

(3) If query over two days is not authorised, the **WEF** time must be smaller than the **UNT** time, except if both times are set to 00:00, which corresponds to a full day query.

(4) The **Default query period** assignment will be used.

(5) Every time the count type query control is changed from Entry to Occupancy and vice versa, the query period shall immediately be adjusted based on the following rules:

a) If current UTC time is within the displayed query period (i.e. "UTC >= WEF" and "UTC <= UNT"), then WEF and UNT input controls shall be adjusted based on the **Default query period** preferences setting corresponding to the newly selected count type (i.e. like when a new window is opened).

b) If current UTC time is outside the displayed query period (i.e. "UTC < WEF" or "UTC > UNT"), then WEF value remains untouched and the relevant default query elapse preference from the **Default query period** assignment is added to WEF to identify the new UNT value.

(6) The **Where** Traffic Type parameter may be adjusted to one of the following values:

a) Traffic load,

b) Traffic demand,

c) Regulated demand.

(7) The desired Traffic Volume identifier shall be keyed or selected from the drop-down list (when a previous query was successful).

(8) Select the **Proposal** radio button on the button bar to include the proposed flights in the retrieved counts. Otherwise the counts will be based on the normal flights only.

(9) The Count **Type** parameter may be adjusted to one of the following values:

a) **Entry Hour/20 Min:** Entry hourly counts are given every 20 minutes step.

b) **Entry Hour/10 Min**: Entry hourly counts are given every 10 minutes step.

c) **Entry Hour:** Entry hourly counts are given every 60 minutes step.

d) **Entry 20 Min:** Entry 20 minutes counts are given every 20 minutes step.

e) **Occupancy:** Occupancy counts are given every minute. In case of Occupancy counts the **Duration** may be filled in. A drop-down box facilitates the input from 1 to 20 minutes. The step for occupancy counts is a system property fixed at value 1 minute. It is defined in system property

CHMI / ATFCM Application / Traffic / Flow counts / Occupancy / Occupancy Step and **cannot** be changed by the user.

(10) Access to **Entry** flow counts will be restricted by security profile.

(11) Indicate in the group box titled **"Flow"** a way how to display traffic flow counts distribution

a) Select the **"Associated"** radio button**:** The counts correspond to the traffic of each one of the "Associated" flows and the "OTHERS" flow specified on the reference location that intersects the traffic inside the traffic volume excluding the traffic of the "Excluded" flows. The display of the counts is per "Associated" and "OTHERS" flow. Following possibilities exist:

i) TV without flows: The system shows traffic counts for the traffic of each "Associated" flow.

ii) TV with included flows: The system shows traffic counts for the traffic of each "Associated" flow intersecting with the traffic of the "Included" flows. The traffic counts are equal to the counts of the traffic intersection (common traffic) of each "Associated" flow with the "Included" flows.

iii) TV with excluded flows: The system shows the traffic counts for the traffic of each "Associated" flow and the "OTHERS" flow, excluding the traffic of the excluded flows. The traffic counts are equal to the counts of the traffic of each "Associated" and "OTHERS" flow minus the traffic of the excluded flows.

iv) TV with included and excluded flows: The system shows traffic counts for the traffic of each "Associated" flow with traffic intersecting the traffic of the "Included" flows, excluding the traffic of the excluded flows. The traffic counts are equal to the counts of the traffic intersection (common traffic) of each "Associated" flow with the "Included" flows minus the traffic of the excluded flows.

b) Select the **"Linked"** radio button**:** The counts correspond to the traffic inside each one of the "Included", "Exempted" and "Included/Exempted" flows that are specified on the traffic volume. The display of the counts is per "Included", "Exempted" and "Included/Exempted" flow.

(12) The **Send** button will send the query to the Network Operations system.

(13) This window may be closed before any query is issued.

(14) The **Reset** action common functionality is implemented for this function. All query criteria except Normal/Proposal selection are considered for activating the Reset action.

(15) A checkbox "**Show VFR/OAT at Aerodrome**" shall allow including/excluding flights that are in a non-extracted portion of the route (VFR/OAT) at the Aerodrome.

(16) The behaviour shall be the same for Predict and Simulations but slightly different for Archive.

8.4 Displaying the query results

8.4.1 Common Display

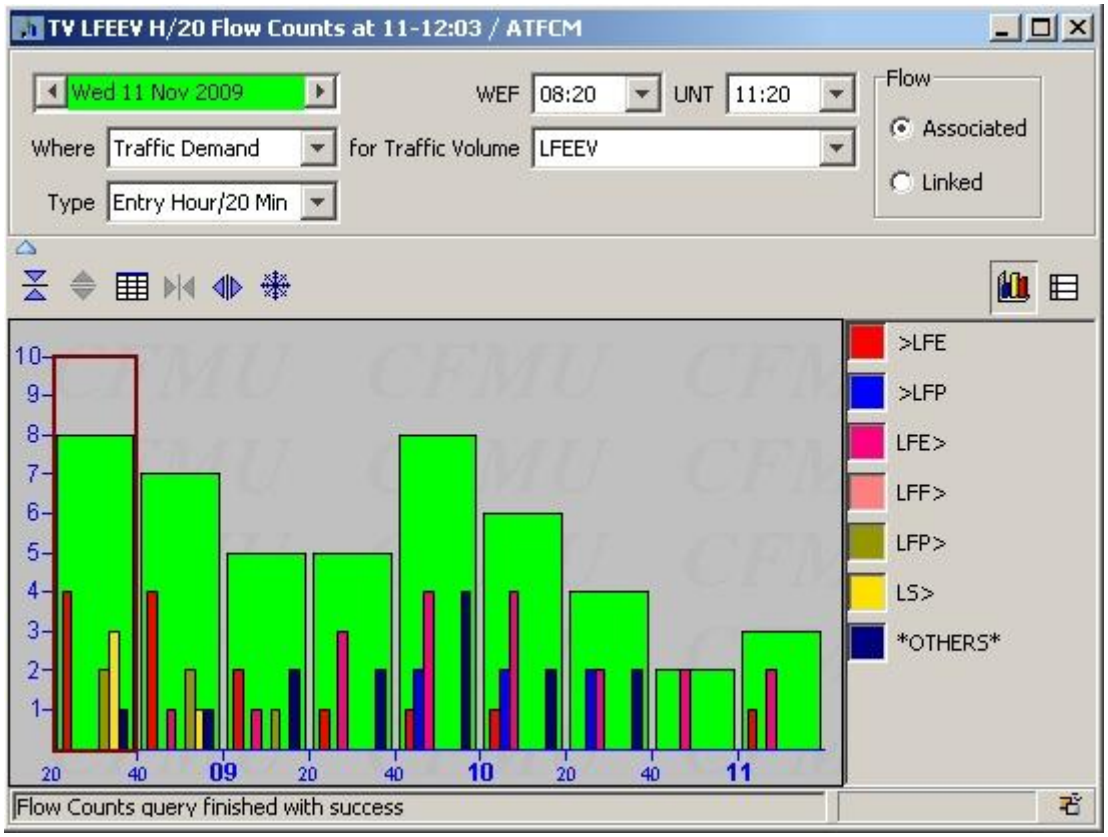


Figure Chmi - 32: Flow Counts graph

(1) The header box contains the selected query criteria:

a) The observed period (Traffic date, **WEF** and **UNT** times)

b) The queried Traffic Type.

c) The **Traffic Volume** identifier.

d) The selected Count **Type** (Entry / Occupancy) and count rate (elapse/step)

e) In case of Occupancy count type the **Duration**

f) In the group box "**Flows**" the selected radio-button either:

i) **Associated**

ii) **Linked**

(2) A split bar separates the Query fields from the query results. It includes on the left, a small arrow button (Up/Down) to Hide/Show the query part.

(3) Underneath the split bar that separates the Query fields from the query results there are two data presentation buttons; they allow the user to select one of the following display types:

a) Graph: bar chart representation of the counts.

b) Table: tabular presentation of the different count values.

(4) The Graph display is shown initially.

(5) When the user chooses to change the display type then the active window will use the data previously retrieved.

(6) When the query results are displayed, the focus is set on the Graph or the Table, depending on the selected type.

(7) The first bar of the Graph or the first cell of the Table is then selected.

8.4.2 Graph display

(1) The graph shows:

a) The total number of flights for the queried **Traffic Volume**. This count is represented by a large surrounding bar for each time slice. The colour of the large surrounding Traffic Volume bar is identified by the user adjustable preference **CHMI / ATFCM application / General / Graph / Bar default colour**.

b) The sub-total number of flights for the selected **Flow(s)**. These sub-counts are represented by thinner bars inside each large surrounding Traffic Volume bar for each time slice. Flow colours are assigned in sequence from a pool of ten user adjustable preferences **CHMI / ATFCM application / Traffic / Flow counts / General / Flow colour nn** where nn is a number between 1 and 10.

(2) The list box on the right side of the graph shows the different Flows related to the selected Traffic Volume.

(3) Each Flow in this list is associated to a check box.

(4) When activated, the colour of the check box indicates the colour of the corresponding bars in the graph.

(5) The capacity value is drawn as line on the graph.

(6) The colour of the capacity line is identified by the user adjustable preference **CHMI / ATFCM application / General / Graph / Capacity colour**. The **vertical axis** indicates the number of flights per hour.

(7) The **horizontal axis** indicates the time periods.

(8) On initial display of newly retrieved data, if the **Zoom Freeze** function is OFF:

a) The graph shall be "pre-"zoomed horizontally so that the elapse time displayed in one view is equal or just above (i.e. next multiple of time that corresponds to a complete vertical bar) the number of minutes identified in the **Initial Display Elapse** preference corresponding to the selected count type (ex: *CHMI / ATFCM application / Traffic / Traffic counts / Entry / Entry initial display elapse*).

b) The graph shall be "pre-"scrolled horizontally so that the first visible vertical bar contains the time obtained by summing WEF with the number of minutes identified in the **Initial Display Shift** preference corresponding to the selected count type (ex: *CHMI / ATFCM application / Traffic / Traffic counts / Entry / Entry initial display shift*).

(9) At the top of the graph, next to the "graph related buttons" (grid and scope controls), a **Zoom Freeze ON / OFF** toggle button allows freezing the graph scaling (i.e. Vertical and Horizontal zooming) and time scrolling in its current level over a data refresh operation.

a) When the **Zoom Freeze** toggle button is **OFF** during a refresh operation, the new graph is displayed with the initial default zoom and period alignment settings.

b) When the **Zoom Freeze** toggle button is **ON** during a refresh operation with or without **Period Shifting**, the graph shall keep current level of zooming (i.e. same time elapse on horizontal axis and same count numbers on vertical axis) and time scrolling.

c) In case of refresh while the **Period Shifting** and **Zoom Freeze** are **ON**, the same time shifting value shall also be applied to the period displayed horizontally.

d) Any manual modification (i.e. automatic **Period Shifting** is not considered as a manual modification) of any query criteria implies a RESET of the current zoom levels when the new reply data is displayed.

e) On opening a new window or on RESET of the current zoom level, the **Zoom Freeze** toggle button is forced to **OFF**

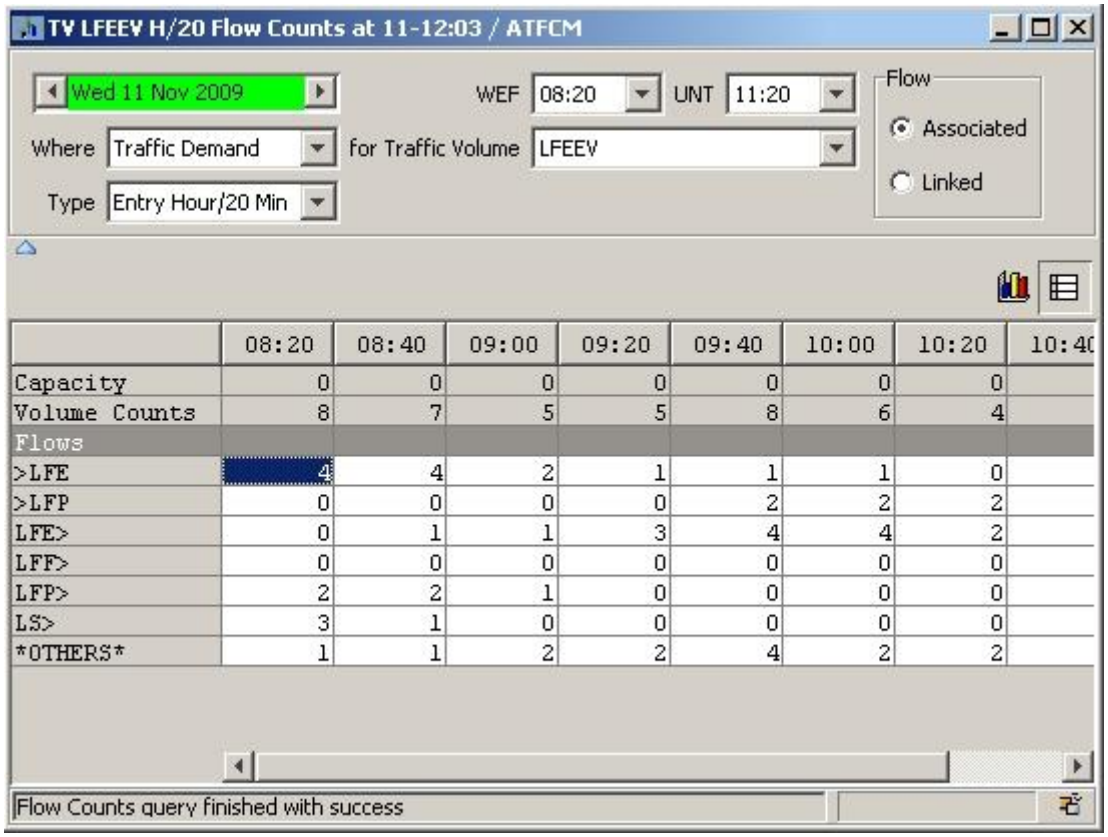
f) As soon as the user adjusts manually the zooming levels or the time scrolling, the **Zoom Freeze** toggle button is forced to **ON**

g) The user can adjust at any time the state of the **Zoom Freeze** toggle button by clicking on it.

h) When the **Zoom Freeze** toggle button is forced manually to **OFF**, the graph is immediately displayed with the initial default zoom and period alignment settings (even before any refresh is performed).

(10) A warning sign **Zoom Hiding High Values** shall be displayed at the top of the graph vertical axis to indicate when the current vertical zooming level does not allow displaying the highest values of the complete queried period.

8.4.3 Table display



	08:20	08:40	09:00	09:20	09:40	10:00	10:20	10:40
Capacity	0	0	0	0	0	0	0	
Volume Counts	8	7	5	5	8	6	4	
Flows								
>LFE	4	4	2	1	1	1	0	
>LFP	0	0	0	0	2	2	2	
LFE>	0	1	1	3	4	4	2	
LFF>	0	0	0	0	0	0	0	
LFP>	2	2	1	0	0	0	0	
LS>	3	1	0	0	0	0	0	
OTHERS	1	1	2	2	4	2	2	

Figure Chmi - 33: Flow Counts table

(1) The tabular representation provides:

a) The different **time slices** covered by the count query.

b) The hourly **Capacity** for the corresponding time slice.

c) The Traffic **Volume Counts** values for the selected Traffic Type.

d) The different **Flow count** values related to each flow of this Traffic Volume.

8.5 User interactions

8.5.1 Common

(1) The **Normal / Proposal** button may be adjusted before issuing a new query.

(2) This window allows to modify a previous query selection and to issue a new request.

(3) On query modification, the **Reset action** is enabled.

(4) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(5) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(6) The **Previous / Next period** function may be used to issue the same query but for the previous or the next period in time.

(7) The **Polling function** may be activated to start automatic refresh processing on this window.

(8) The **Period Shifting function** may be activated to start automatic period shifting on subsequent refresh of this window.

(9) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(10) To print Flow Counts data, first retrieve the data in the window and select the desired display type (Graph or table). Adjust possibly the displayed period, scale and other details. When ready, click on the **Print** button or select the **File / Print** menu option.

(11) To export the retrieved count details, click on the **Export** button or select the **Action / Export** menu option.

(12) Increasing / reducing the window height will impact the size of the main display control (graph or table).

(13) Increasing / reducing the window width will impact the size of the main display control (graph or table).

(14) The **Partial Period Shifting** function may be used to issue the same query but for the previous or the next period with time overlap (see **Partial Period Shifting** function detailed description).

8.5.2 Graph display

(1) It is possible to toggle on/off any combination of the **Flow** buttons in the list at the right of the graph.

(2) When selected, these toggle buttons display the colour assigned for the drawing of the corresponding bar on the graph.

(3) Colours are assigned on a first come first serve base in a pool of ten colours.

(4) For readability reasons, a maximum of ten flows may be displayed at once. A message box will warn the user when this limit is reached. To display another flow on the graph, you first have to unselect at least one of the ten.

(5) To adjust the colour of the large surrounding Traffic Volume bar, you have to modify **CHMI / ATFCM application / General / Graph / Bar default colour** in the Preferences dialog.

(6) To adjust the pool of colours assigned to the selected flows, you have to modify one of the following preference: **CHMI / ATFCM application / Traffic / Flow counts / General / Flow colour nn** where nn is a number between 1 and 10.

(7) To adjust the colour of capacity line, you have to modify **CHMI / ATFCM application / General / Graph / Capacity colour** in the Preferences dialog. This preference is common with other functions in the application.

(8) When clicking on a bar in the graph, then focus and selection is set on this particular bar.

(9) When the focus is on the graph, the left and right arrows on the keyboard moves selection from one bar to the other.

(10) The right mouse button pops up a **contextual menu** for the selected bar.

(11) The **bottom scroll bar** within the graph pane is available only when the scope of the shown period is reduced or enlarged.

(12) Another scroll bar may appear at the bottom of the Flow Counts component when the window is too small to see all contents; it will disappear again when all of the windows contents are visible.

(13) Underneath the split bar that separates the Query fields from the query results several buttons may be displayed. When the graph presentation button on the right side of the pane is selected, then graph related buttons appear on the left of the pane to allow the user to adjust the following graph characteristics:

a) **To enlarge the scope of the vertical axis**: click on the two vertical arrows pointing to each other with an horizontal line in the middle.

b) **To reduce the scope of the vertical axis**: click on the two vertical arrows pointing away from each other with an horizontal line in the middle.

c) **To display / hide a grid on the graph**: click on the toggle button with a table representation.

d) **To enlarge the scope of the shown period**: click on the two horizontal arrows pointing to each other with an vertical line in the middle.

e) **To reduce the scope of the shown period**: click on the two horizontal arrows pointing away from each other with an vertical line in the middle.

f) **To freeze / unfreeze the current level of vertical and horizontal zooming and the current time scrolling**: click on the **Zoom Freeze ON / OFF** toggle button.

8.5.3 Table display

(1) When clicking on a cell in the table, then focus is set on this particular cell.

(2) When clicking on the Flow Id (left column), then the row is selected but previous column selection remains.

(3) When the focus is on a cell with a count, the left/right and up/down arrows on the keyboard moves selection from one cell to the other.

(4) When the focus is on a Flow Id (left column), the up/down arrows on the keyboard moves selection from one cell to the other.

(5) The right mouse button pops up a **contextual menu** for the selected cell.

8.6 Contextual actions

8.6.1 Graph display

(1) When a bar is selected on the graph, the right mouse button pops up a **contextual menu** for the corresponding period.

(2) The higher-level popup menu displays:

a) The **time slice** corresponding to the selected bar.

b) The **Hourly Capacity** for the period of the selected bar.

c) The total count value for the queried **Traffic Volume** within this time slice.

d) The different selected **Flows** with the sub-total value corresponding to the displayed time slice. Each Flow line gives access to a popup sub-menu.

e) **Flight List**: to query the list of flights for the queried Traffic Volume and selected Type. Requested period is the one identified by the selected bar.

f) **Occupancy Counts**, if currently displaying Entry Counts, allows focussing on the Occupancy Counts for a selected Entry Counts elementary period (i.e. a bar of the graph). Selecting this option will issue the same query in another window where only the following query parameters are adjusted:

i) The count type is forced to **Occupancy**

ii) The WEF time is set to the start time of the selected bar minus the corresponding "Occupancy query lead time" preference from the **Default query period** assignment.

iii) The UNT time is set to the calculated WEF time plus the corresponding "Occupancy query default elapse" preference from the **Default query period** assignment.

(3) This contextual menu offers sub-menu parts related to the selected Flow in the higher-level popup menu.

(4) Each Flow popup sub-menu may contain the following entry:

a) The menu option **Flight List** for the displayed Traffic Volume associated to the selected Flow. Requested period is the one identified by the selected bar. The "Show VFR/OAT at Aerodrome" and "Show Forecast" (when access is authorized) checkboxes are selected/deselected accordingly.

b) The menu option **Show Flow Description**: to query the detailed data associated to the selected Flow. This menu option allows a direct access to the relevant Flow data instead of querying the data via the Environment Query screen.

(5) The transferred period may be affected by user preferences related to **Default query period identification**.

8.6.2 Table display

(1) When a count cell is selected on the table, the right mouse button pops up a **contextual menu** for the corresponding period.

(2) The higher-level popup menu displays:

a) The **time slice** corresponding to the selected cell is displayed.

b) **Flight List**: to query the list of flights for the queried Traffic Volume and selected Type. Requested period is the one identified by the selected cell.

c) **Occupancy Counts**, if currently displaying Entry Counts, allows focussing on the Occupancy Counts for a selected Entry Counts elementary period (i.e. a column of the table). Selecting this option will issue the same query in another window where only the following query parameters are adjusted:

i) The count type is forced to **Occupancy**

ii) The WEF time is set to the start time of the selected column minus the corresponding "Occupancy query lead time" preference from the **Default query period** assignment.

iii) The UNT time is set to the calculated WEF time plus the corresponding "Occupancy query default elapse" preference from the **Default query period** assignment.

d) The **Flow** corresponding to the selected cell. This Flow line gives access to a popup sub-menu.

(3) This contextual menu offers sub-menu parts related to the selected Flow in the higher-level popup menu.

(4) Each Flow popup sub-menu may contain the following entry:

a) The menu option **Flight List** for the displayed Traffic Volume associated to the selected Flow and Type. Requested period is the one identified by the selected cell. The "Show VFR/OAT at Aerodrome" and "Show Forecast" (when access is authorized) checkboxes are selected/deselected accordingly.

b) The menu option **Show Flow Description**: to query the detailed data associated to the selected Flow. This menu option allows a direct access to the relevant Flow data instead of querying the data via the Environment Query screen.

(5) The transferred period may be affected by user preferences related to **Default query period identification**.

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9 Departure and Regulation Delays

9.1 Function presentation

(1) This function displays the Delays figures for a Departure Aerodrome or a Regulation associated to a selected date and time period.

9.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Traffic / Delays**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl>** + **<Shift>** + **<F8>**.

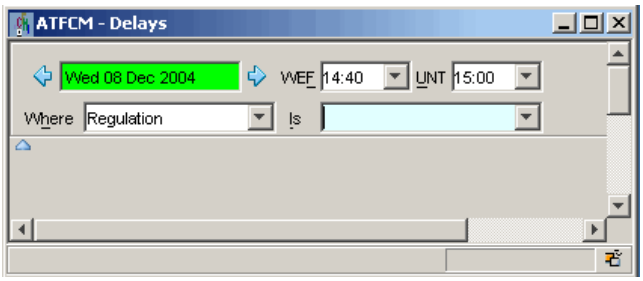


Figure Chmi - 34: Query Delays

(3) A new empty **Delays** window will be opened with the following defaults:

a) Traffic Date: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** time: proposed based on the **Default query period** assignment using your user adjustable preferences.

c) **UNT** time: proposed based on the displayed WEF and the **Default query period** assignment.

d) **Where**: last element used with this query This selection is based on the system property **CHMI / ATFCM application / Traffic / Traffic delays / Last Used Reference Location Type**.

e) **Is**: based on the "Where" selection, the drop-down box will propose the most recent query data elements you have queried before with success.

(4) The **Where** parameter can be:

a) Regulation,

b) Departure Aerodrome.

(5) On initial display, the focus is set on the first mandatory query control with no default value: the **Is** input control.

9.3 Preparing to query

(1) The observed period may be adjusted as follow:

a) The traffic **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

b) The proposed default **WEF** and **UNT** times may be overwritten.

c) Any valid time value from 00:00 to 23:59 included may be entered in the **WEF** and/or the **UNT** entry boxes. This input must respect the HH:MM format.

d) Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(2) The period resulting from the input will be validated taking into account the system property **CHMI / ATFCM application / General / Dates and times / Query over midnight**.

(3) If query over two days is not authorised, the **WEF** time must be smaller than the **UNT** time, except if both times are set to 00:00, which corresponds to a full day query.

(4) The **Default query period** assignment will be used.

(5) The **Where** parameter may be adjusted to one of the following values:

a) Regulation.

b) Departure Aerodrome.

(6) The identifier corresponding to the selected **Where** shall be keyed in the associated **Is** entry box or selected from the drop-down list (when a previous query was successful).

(7) Select the **Proposal** radio button on the button bar to include the proposed flights in the delays figures. Otherwise the figures will be based on the normal flights only.

(8) The **Send** button will send the query to the Network Operations system.

(9) This window may be closed before any query is issued.

(10) The **Reset** action common functionality is implemented for this function. All query criteria except Normal/Proposal selection are considered for activating the Reset action.

9.4 Displaying the query results

9.4.1 Common Display

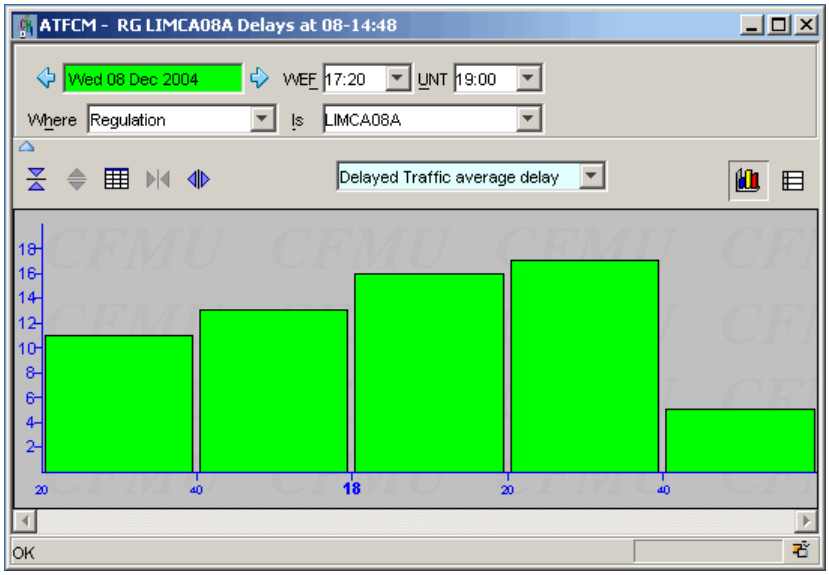


Figure Chmi - 35: Departure Delays graph

(1) The header box contains the selected query criteria:

a) The observed period (Traffic date, **WEF** and **UNT** times).

b) The main **Where** selection criteria (Departure Aerodrome or Regulation).

c) The identifier associated to the selected Where.

(2) A split bar separates the Query fields from the query results. It includes on the left, a small arrow button (Up/Down) to Hide/Show the query part.

(3) Underneath the split bar that separates the Query fields from the query results there are two data presentation buttons; they allow the user to select one of the following display types:

a) Graph: bar chart representation of the delay figures.

b) Table: tabular presentation of the different delay figures.

(4) The Graph display is shown initially.

(5) When the user chooses to change the display type then the active window will use the data previously retrieved.

(6) When the query results are displayed, the focus is set on the Graph or the Table, depending on the selected type.

(7) The first bar of the Graph or the first cell of the Table is then selected.

9.4.2 Graph display

(1) Underneath the splitter bar and when the graph is displayed, a drop-down menu is available from which the user can choose to display another category of delay figures.

(2) These categories are either the amount of delay in minutes for:

a) Delayed Traffic average delay

b) Delayed Traffic maximum delay

c) Delayed Traffic total delay

(3) Or the number of flights that are impacted:

a) CASA Regulated Traffic Demand

b) Delayed Traffic Load

(4) The counts are made every 20 minutes. The value associated is always an hourly value (e.g. 12 flights delayed between 11h20 and 12h20).

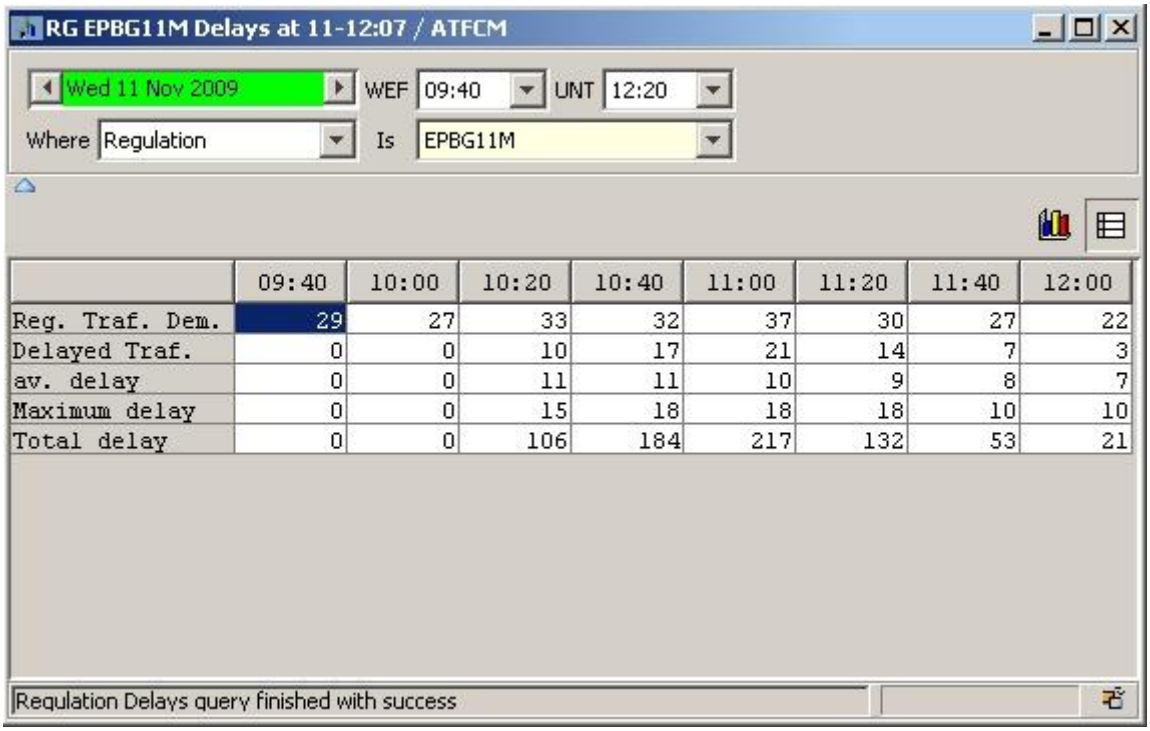
(5) Each bar on the graph represents an hourly count for a time slice of the selected delay category.

(6) The colour of the delay bars is identified by the user adjustable preference **CHMI / ATFCM application / General / Graph / Bar default colour**.

(7) The **vertical axis** indicates either the number of flights per hour or the number of minutes per hour.

(8) The **horizontal axis** indicates the time periods.

9.4.3 Table display



	09:40	10:00	10:20	10:40	11:00	11:20	11:40	12:00
Reg. Traf. Dem.	29	27	33	32	37	30	27	22
Delayed Traf.	0	0	10	17	21	14	7	3
av. delay	0	0	11	11	10	9	8	7
Maximum delay	0	0	15	18	18	18	10	10
Total delay	0	0	106	184	217	132	53	21

Figure Chmi - 36: Departure Aerodrome Delays table

(1) The tabular representation provides:

a) The different **time slices** covered by the delays query.

b) The Traffic **Delay** values for the different categories.

(2) These delay values are either in minutes for:

a) Delayed Traffic average delay (**av. delay**).

b) Delayed Traffic maximum delay (**Maximum delay**).

c) Delayed Traffic total delay (**Total delay**).

(3) Or in number of impacted flights for:

a) CASA Regulated Traffic Demand (**Reg. Traf. Dem.**).

b) Delayed Traffic Load (**Delayed Traf.**).

9.5 User interactions

9.5.1 Common

(1) The **Normal / Proposal** button may be adjusted before issuing a new query.

(2) This window allows to modify a previous query selection and to issue a new request.

(3) On query modification, the **Reset action** is enabled.

(4) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(5) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(6) The **Previous / Next period** function may be used to issue the same query but for the previous or the next period in time.

(7) The **Polling function** may be activated to start automatic refresh processing on this window.

(8) The **Period Shifting function** may be activated to start automatic period shifting on subsequent refresh of this window.

(9) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(10) To print Delays data, first retrieve the data in the window and select the desired display type (Graph or table). Adjust possibly the displayed period, scale and other details. When ready, click on the **Print** button or select the **File / Print** menu option.

(11) To export the retrieved Delays details, click on the **Export** button or select the **Action / Export** menu option.

(12) Increasing / reducing the window height will impact the size of the main display control (graph or table).

(13) Increasing / reducing the window width will impact the size of the main display control (graph or table).

(14) The **Partial Period Shifting** function may be used to issue the same query but for the previous or the next period with time overlap (see **Partial Period Shifting** function detailed description).

9.5.2 Graph display

(1) To adjust the colour of the bars, you have to modify **CHMI / ATFCM application / General / Graph / Bar default colour** in the Preferences dialog.

(2) When clicking on a bar in the graph, then focus and selection is set on this particular bar.

(3) When the focus is on the graph, the left and right arrows on the keyboard moves selection from one bar to the other.

(4) The right mouse button pops up a **contextual menu** for the selected bar.

(5) The **bottom scroll bar** within the graph pane is available only when the scope of the shown period is reduced or enlarged.

(6) Another scroll bar may appear at the bottom of the Delays component when the window is too small to see all contents; it will disappear again when all of the windows contents are visible.

(7) The display another category of delay figures, change the selection of the drop-down list box at the top of the graph.

(8) Underneath the split bar that separates the Query fields from the query results several buttons may be displayed. When the graph presentation button on the right side of the pane is selected, then graph related buttons appear on the left of the pane to allow the user to adjust the following graph characteristics:

a) **To enlarge the scope of the vertical axis**: click on the two vertical arrows pointing to each other with an horizontal line in the middle.

b) **To reduce the scope of the vertical axis**: click on the two vertical arrows pointing away from each other with an horizontal line in the middle.

c) **To display / hide a grid on the graph**: click on the toggle button with a table representation.

d) **To enlarge the scope of the shown period**: click on the two horizontal arrows pointing to each other with an vertical line in the middle.

e) **To reduce the scope of the shown period**: click on the two horizontal arrows pointing away from each other with an vertical line in the middle.

f) **To freeze / unfreeze the current level of vertical and horizontal zooming and the current time scrolling**: click on the **Zoom Freeze ON / OFF** toggle button.

9.5.3 Table display

(1) When clicking on a cell in the table, then focus is set on this particular cell.

(2) When the focus is on a cell with a count, the left/right and up/down arrows on the keyboard moves selection from one cell to the other.

(3) When the focus is on a Delay Category (left column), the up/down arrows on the keyboard moves selection from one cell to the other.

(4) The right mouse button pops up a **contextual menu** for the selected cell.

9.6 Contextual actions

9.6.1 Graph display

(1) When a bar is selected on the graph, the right mouse button pops up a **contextual menu** for the corresponding period.

(2) This popup menu lists the possible actions for the selected bar.

(3) This popup menu displays:

a) The **time slice** corresponding to the selected bar.

b) The **Delays value** next to the text of the currently selected Delays category.

c) **Flight List** or **Departure List**: to query list of flights for the same Reference Location as the initial Delays query. Requested period is the one identified by the selected bar.

(4) The transferred period may be affected by user preferences related to **Default query period identification**.

9.6.2 Table display

(1) When a delay cell is selected on the table, the right mouse button pops up a **contextual menu** for the corresponding period.

(2) This popup menu lists the possible actions for the selected cell.

(3) This popup menu displays:

a) The **time slice** corresponding to the selected cell.

b) **Flight List** or **Departure List**: to query list of flights for the same Reference Location as the initial Delays query. Requested period is the one identified by the selected cell.

(4) The transferred period may be affected by user preferences related to **Default query period identification**.

10 Flight List

10.1 Function presentation

(1) This function lists all the flights for a combination of several query criteria.

(2) The main query criteria include a period, a traffic type and a main selection criterion with related sub-criteria.

(3) In addition, there are two optional criteria: an Aircraft Operator list and a most penalizing FMP.

(4) The flights are selected on:

a) The time over the reference location for a Traffic Volume, a Regulation and a Point.

b) The take off time for the Aircraft Operator, the Departure Aerodrome and Aerodrome Set.

c) The arrival time for the Arrival Aerodrome and Aerodrome Set.

d) The entry time for the Airspace.

(5) A count type (Entry/Occupancy) is requested when flights are selected on a Traffic Volume, Aircraft Operator or Airspace.

(6) This function also allows to compare the flight list against a flight list of a different traffic type. In such case, the differences in flight's entry time, entry flight level and position are shown.

10.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Flights / Flight List**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl>** + **<Shift>** + **<F9>**.

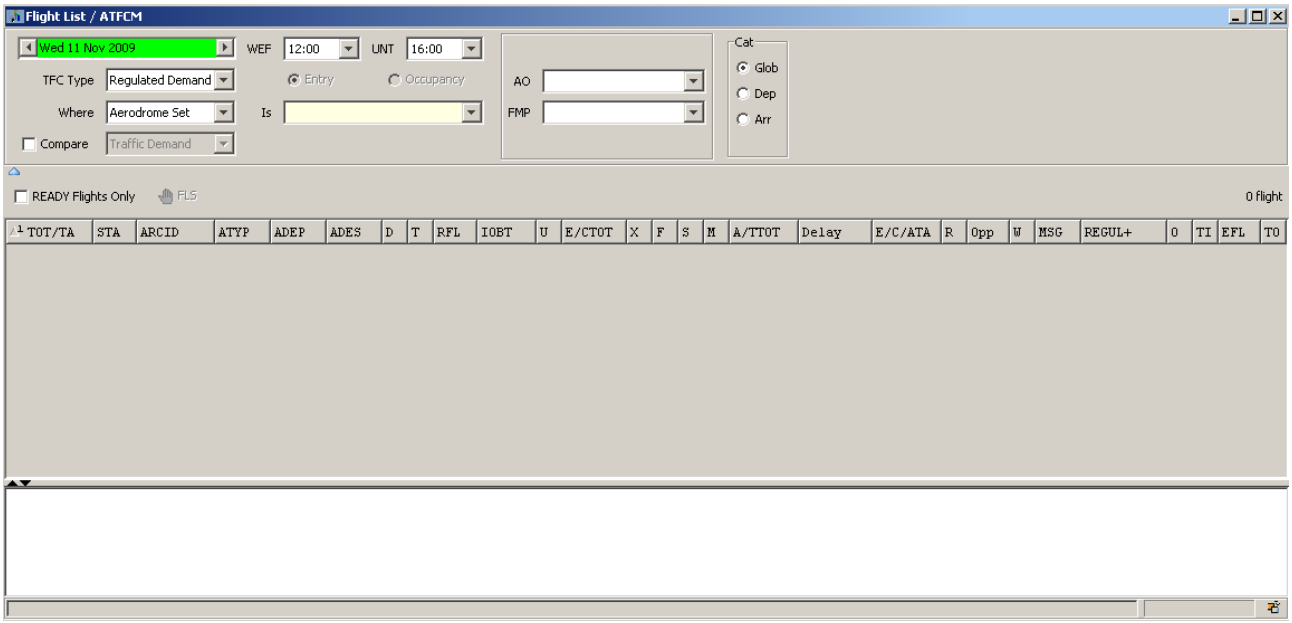


Figure Chmi - 37: Query Flight List

(3) A new empty **Flight List** window will be opened with the following defaults:

a) Observed period: current or last queried traffic date is proposed (relative to current UTC date) dependant on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** time: proposed based on the **Default query period** assignment using your user adjustable preferences. Replaced by **At** time when Occupancy count type is selected.

c) **UNT** time: proposed based on the displayed WEF and the **Default query period** assignment. Replaced by **Duration** time when Occupancy count type is selected. Default **Duration** value is based on the system **property CHMI / ATFCM application / Flights / Flight list / Last Used Occupancy Duration**.

d) Traffic Type (**TFC Type**): last traffic type used with this query. This selection is based on the system property **CHMI / ATFCM application / Flights / Flight list / Last Used Traffic Type**.

e) **Compare** checkbox: If the **Flight List** window was invoked on **Traffic Load** from the contextual popup menu of either the **Traffic Counts** window or the **Flow Counts** window, the **Compare** checkbox is checked otherwise it is unchecked.

f) With dropdown: the default setting of this depends on the value selected for the Traffic Type (**TFC Type**) as follows:

selected **TFC Type**	Default With
Traffic Load	Regulated Demand (**RD**)
Traffic Demand	Traffic Load (**TL**)
Regulated Demand	Traffic Demand (**TD**)

Table Chmi-3: Default With dropdown

g) Count type (**Entry/Occupancy**): last selection used with your last similar query. This selection is based on the system property **CHMI / ATFCM application / Flights / Flight list / Last Used count calculation type**.

h) **Where**: last element used with this query.

i) **Is**: based on the "Where" selection, the drop-down box will propose the most recent query data elements you have queried before with success.

(4) When Traffic Volume is selected, the **Flow** control is proposed empty. The drop-down list is filled in with the most recent IDs you have queried before with success.

(5) When Aerodrome or Aerodrome Set is selected, the related category controls (**Cat**) are proposed. Global button comes preselected.

(6) When Point is selected, the Maximum (**MAX**) and Minimum (**MNM**) Flight Level (**FL**) controls are proposed. They are initialised with the extreme values (990 for MAX and 000 for MNM).

(7) The **FMP** input box is proposed empty. The drop-down list is filled in with the most recent IDs you have queried before with success.

(8) When **Where** is not set to Aircraft Operator(s), the secondary **AO(s)** input box is proposed empty. The drop-down list is filled in with the most recent IDs you have queried before with success.

(9) All these **Where** query elements may be associated to the ID of one or more Aircraft Operators, except when the main query criterion is already one or more Aircraft Operators.

(10) On initial display, the focus is set on the first mandatory query control with no default value: the **Is** input control.

10.3 Preparing to query

(1) The traffic **Observed period** may be adjusted as follows:

a) The traffic **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

b) The proposed default **WEF** and **UNT** times (respectively **At** and **Duration** for Occupancy**)** may be overwritten.

c) Any valid time value from 00:00 to 23:59 included may be entered in the **WEF** and/or the **UNT** entry boxes. This input must respect the HH:MM format.

d) Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(2) The period resulting from the input will be validated taking into account the system property **CHMI / ATFCM application / General / Dates and times / Query over midnight**.

(3) If query over two days is not authorised, the **WEF** time must be smaller than the **UNT** time, except if both times are set to 00:00, which corresponds to a full day query.

(4) The Traffic Type (**TFC Type**) parameter may be adjusted to one of the following values:

a) Traffic load,

b) Traffic demand,

c) Regulated demand.

(5) If a flight list comparison is required with another Traffic Type, the **Compare** checkbox must be checked.

(6) When the **Compare** checkbox is unchecked, the With dropdown is disabled.

(7) When the **Compare** checkbox is checked:

a) the With dropdown becomes accessible,

b) the content of the With dropdown is automatically adjusted every time the **TFC Type** is adjusted as follows:

selected **TFC Type**	contents of With dropdown
Traffic Load	Traffic Demand (**TD**) Regulated Demand (**RD**)
Traffic Demand	Traffic Load (**TL**) Regulated Demand (**RD**)
Regulated Demand	Traffic Load (**TL**) Traffic Demand (**TD**)

Table Chmi-4: Contents of With dropdown

c) the default With value is as defined in Table Chmi-4.

(8) The **Where** drop-down list provides the user with following selection:

a) Traffic Volume

b) Airspace

c) Aerodrome

d) Aerodrome Set

e) Point

f) Aircraft Operator(s)

g) Regulation

(9) A tooltip on the **Is** entry box shall describe:

a) The expected input

b) With a sample format when the user can enter a list of identifiers.

(10) Example: “Enter one or more Aircraft Operator identifiers (ex: ‘AFR,FIN’)”

(11) The identifier corresponding to the selected **Where** shall be keyed in the associated **Is** entry box or selected from the drop-down list (when a previous query was successful).

(12) When Traffic Volume is selected, a **Flow** may be entered or selected from the list of previously queried IDs proposed in the drop-down list.

(13) When Aerodrome or Aerodrome Set is selected, the related category (**Cat**) must be checked (Global, Departure or Arrival).

(14) When Point is selected, the Maximum (**MAX**) and Minimum (**MNM**) Flight Levels (**FL**) may be overwritten to reduce or enlarge selected altitude range.

(15) To issue a valid query, the Maximum (**MAX**) value shall be greater or equal to the Minimum (**MNM**) Flight Level value.

(16) A valid integer value (maximum of 3 digits) entered in the Maximum (**MAX**) box, will be rounded as follow:

a) 0 remains 0.

b) Input between 1 and 30 is rounded up to the nearest multiple of 10.

c) Input between 31 and 290 is rounded up to the nearest multiple of 5.

d) Input between 291 and 990 is rounded up to the nearest multiple of 10.

e) Above 990 is forced to 990.

(17) A valid integer value (maximum of 3 digits) entered in the Minimum (MNM) box, will be rounded as follow:

a) Input between 0 and 34 is rounded down to the nearest multiple of 10.

b) Input between 35 and 294 is rounded down to the nearest multiple of 5.

c) Input between 295 and 999 is rounded down to the nearest multiple of 10.

(18) All these **Where** query elements may be associated to the ID of a most penalising FMP, where the selected flights must have the most penalising Regulation from a given FMP.

(19) All these **Where** query elements may be associated to the ID of one or more Aircraft Operators (**AO(s)**), except when the main query criterion is already one or more Aircraft Operators.

(20) When **Aircraft Operator(s)** is selected from the **Where** drop-down list, the associated **Is** entry box shall accept a comma separated list of 1 to 10 Aircraft Operator Identifiers.

(21) When accessible, the secondary **AO(s)** entry box shall accept a comma separated list of 1 to 10 Aircraft Operator Identifiers.

(22) The comma is the official standard separator character. Other characters which are invalid in AO identifier (ex: semicolon, blank …) could be accepted by the HMI.

(23) After a successful query, valid AO comma separated list shall be proposed in the AO related drop down list when preparing a new query.

(24) Changing the **Where** value could change the settings of the other query criteria except the period and the traffic type.

(25) The **Count type** may be set to either:

a) **Entry**: Entry type is available for all possible Where parameters.

b) **Occupancy**: Occupancy type is available for Airspace, Traffic Volume defined on an Airspace and Aircraft Operator Flight Lists.

(26) When only **Entry** is available, both options are disabled.

(27) When **Occupancy** Type is selected, the time of the snapshot and the **Duration** parameter must be provided:

a) The **WEF** label is replaced by the keyword **At**.

b) The **UNT** label is replaced by the keyword **Duration**.

c) Default **Duration** is last selection used with your last similar query. This selection is based on the system **property CHMI / ATFCM application / Flights / Flight list / Last Used Occupancy Duration**. On a restore of a saved Workspace the value of this preference property is not used but the value which was stored by the save workspace action.

d) A drop-down box allows a choice up to twenty minutes.

e) For occupancy counts a snap-shot is taken of all flights present in the sector during a time period. For a radar controller this is the flights present in the sector during one minute, for a planner controller it can be the flights present in the sector during up to twenty minutes.

(28) In the case of **Entry** counts, the returned list contains:

a) Either the flights entering the reference location during the queried period,

b) Or the flights taking off during the queried period (AO selection).

(29) In the case of **Occupancy** counts, the returned list contains:

a) Either all flights being in the selected airspace or traffic volume at the given time,

b) Or all flights being on fly at the given time (AO selection).

(30) The main listbox header line will be adjusted when receiving the reply to reflect the kind of query you have issued.

(31) Select the **Proposal** radio button on the button bar to include the proposed flights in the retrieved list. Otherwise the list will be based on the normal flights only.

(32) The **Send** button will send the query to the Network Operations system.

(33) If an Aircraft Operator code is part of the selection criteria, the query period will be limited to the value of the system property **CHMI / ATFCM application / Flights / Flight list / Maximum elapse for AO list**.

(34) Otherwise, the normal **Default query period** assignment will be used. This window may be closed before any query is issued.

(35) The **Reset** action common functionality is implemented for this function. All query criteria except Normal/Proposal selection are considered for activating the Reset action.

(36) A checkbox **"Show VFR/OAT at Aerodrome"** shall be visible and editable if (and only if) the "Where" dropdown is Aerodrome, Aerodrome Set or Traffic Volume.

(37) When "Show VFR/OAT at Aerodrome" is selected on a traffic volume it has effect only if the reference location is an aerodrome or an aerodrome set.

(38) The behaviour shall be the same for Predict and Simulations and slightly different from Archive.

10.4 Displaying the query results

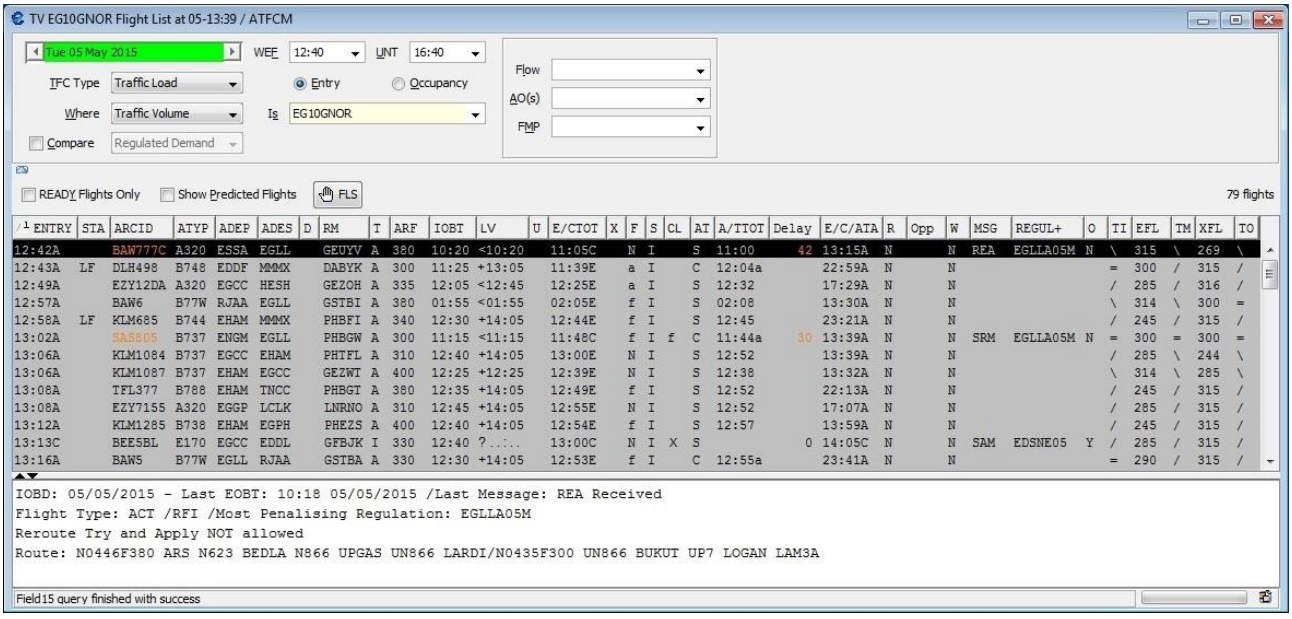


Figure Chmi - 38: Flight List

(1) The header box of the window contains the different **query data elements** used to retrieve the displayed flight data:

a) The observed period (Traffic date, **WEF** and **UNT** times) (respectively **At** and **Duration** for Occupancy**)**

b) The selected Traffic Type (**TFC Type**),

c) The **Compare** checkbox status and the With selection,

d) The selected Count type (**Entry** / **Occupancy**),

e) The main **Where** selection criteria (Traffic Volume, Airspace, Aerodrome or Set with the associated Category, Point with associated Flight Level Range (**FL**), Regulation or Aircraft Operator(s)),

f) The optional selection criteria for the list (Most Penalising **FMP**, Aircraft Operator (**AO(s)**) or **Flow** for a Traffic Volume).

(2) At the top of the Flight List, a counter indicates the **number of flights** retrieved.

(3) This counter may be affected by **filtering** (fff / rrr Flights where fff is the filtered number and rrr is the retrieved number).

(4) When the flights are selected on a Regulation identifier, following related information is displayed at the top of the list:

a) The **Traffic Volume Identifier (ID)** on which the regulation applies,

b) The related reference **Location**.

(5) On query results display, the focus is set on the Flight List after selecting the first flight in the list.

(6) The **READY Flights only** checkbox allows to display only the flights being in READY state. This sub-selection is done locally by the application, no query to the server is issued for this request. The counter is adjusted to reflect the number of displayed flights.

(7) For each flight the **Flight List** display contains following columns:

a) The Time of Entry into the Location (see later).

b) The Status of the flight **(STA)** (see later).

c) The Aircraft Identification **(ARCID)**.

d) The ICAO identifier of the Aircraft Type **(ATYP)**.

e) The Aerodrome of Departure **(ADEP)**.

f) The Aerodrome of Destination **(ADES)** of the highest available requested model.

i) For a diverting flight, when the requested traffic kind selects the Estimated model or the Calculated model, the originally filed ADES will be displayed in the column with header ADES.

ii) For a diverting flight, when the requested traffic kind selects the Actual model, the new ADES will be displayed in the column with header ADES.

g) Diversion indicator **(D),** indicated with **>** symbol. This symbol is present for a diverted flight, independent of the requested model.

h) The most recent Registration Mark **(RM)** of the flight. The order, from the most recent to the least is:

i) CDM Registration Mark

ii) IFPS Registration Mark (filed)

If the CDM Registration Mark of the flight is marked as discrepancy, the registration mark is displayed with the discrepancy colour.

i) The Type of flight data **(T)** (see later).

j) The Requested Flight Level applicable on the portion of the flight route that penetrates the sector. If the reference location is an aerodrome, set of aerodromes or aircraft operator, this is the highest Requested Flight Level of the flight.

i) When the Flight List is based on the Traffic Demand or the Regulated Demand, the displayed Requested Flight Level (**RFL**) corresponds to the Flight Level filed by the AO.

ii) When the Flight List is based on the Traffic Load, the displayed Actual Requested Flight Level (**ARF**) corresponds to the Flight Level possibly adapted by profile recalculation for the active flights, otherwise it is the Flight Level filed by the AO.

k) The Initial Estimated Off Block Time **(IOBT)**.

l) Just after the IOBT, the presence of the letter **P**. When a flight is still a Proposal that needs to be confirmed, then the letter P will appear along with the IOBT value.

m) The Last Valid (**LV**) EOBT acceptable for the flight before triggering IFPS errors.

ValidityPeriodEnd HHMM is prefixed with character "<", "+" or "?" depending respectively on ProfileValidityKind value E, M or N.

When ValidityPeriodEnd is not available (ProfileValidityKind=N), an empty time is displayed as "..:.." after the "?" prefix character (i.e. ?..:..).

Tooltip on column title: "**Last Validity to remain IFPS compliant**" followed on subsequent lines:

"**<HH:MM** flight profile becomes invalid after HH:MM"

"**+HH:MM** flight profile valid until HH:MM (check time limit)"

"**? . . : . .** flight profile validity has not been checked"

n) The Suspension status (**U**) if the flight is currently suspended or the last suspension status if the flight had been previously suspended (see later).

o) The Estimated (**E**) or Calculated (**C**) Take Off Time (**E/CTOT**) (only shown when flight is published).

p) The number of regulations from which the flight is Excluded **(X)**, possibly preceded by a "**!**" when the flight is Excluded from one or more regulations defined on the Traffic Volume used for the Flight List selection. If the flight was previously excluded from one or more regulations but is no longer, a 'W' will be displayed.

q) The Flight Activation Monitoring (FAM) status **(F)** (see later).

r) The Ready status **(S)** (see later).

s) The CTOT Limit (**CL**) reason: Possible exceptional reason that may affect the CTOT allocation of the flight, if any. Possible values are as described in the tooltip below.

Tooltip on column title build online to reflect ONLY the letter code present in the currently displayed list:

"**CTOT Limit**" followed on subsequent lines for each letter code present in the list:

- **t** = Forced by Tower
- **f** = Forced by NMOC
- **w** = Was Forced by NMOC
- **e** = Slot Extension

- **c** = Forced by CHAMAN
- **s** = Forced by STAM
- **V** = By Violation
- **X** = By Violation / Zero Rate / RVR

t) The Take Off Time information from the Aerodrome of Departure **(A/TTOT)** (see later).

i) Following tooltip shall be implemented on the column header: Take Off Time info from Aerodrome of Departure

u) The Airport Type (**AT**) associated to the Aerodrome of Departure.

i) Following tooltip shall be implemented on the column header: Departure Airport Type (S=Standard, T=Advanced ATC/TWR, C=CDM)

v) The CDM DPI Target Off-Block Time (hh:mm), if any (**TOBT**)

i) Following tooltip shall be implemented on the column header: Target Off-Block Time

w) The CDM DPI Target Start-Up Approval Time (hh:mm), if any (**TSAT**)

i) Following tooltip shall be implemented on the column header: Target Start-Up Approval Time

x) The latest Taxi Time in minutes identified by the Network Operations system, possibly based on CDM DPI message, to calculate the Actual flight profile (**TT**)

i) Following tooltip shall be implemented on the column header: Actual Taxi Time

y) The Slot Tolerance Window for a regulated flight or the Departure Tolerance Window for a non regulated flight (**TW**).

i) The **TW** column shall only be displayed when authorized by the user profile. A single common use case shall grant the display of the Tolerance Window data in all relevant functionalities (Flight List, Flight Data &).

ii) The Lower and Upper time limit of the Tolerance Window are displayed using the following format: [*low-hh:mm*,*upper-hh:mm*] (ex: "*[10:20,10:40]*").

iii) The complete string is displayed in *italic* when the values are defined as an extended tolerance window.

iv) Following tooltip shall be implemented on the column header: Slot Tolerance Window for Regulated Flights or Departure Tolerance Window for Non-Regulated Flights.

v) Each time newly retrieved Tolerance Window values are displayed, the flight list shall be resorted if the **TW** column is part of the selected sort criteria for ordering the list.

z) The **Delay** in minutes.

i) A case sensitive letter may possibly be appended to the delay value to indicate following specific delay characteristics:

- "D" when the delay value calculated for the flight is exceeding the delay confirmation threshold of a regulation affecting the flight
- "a" when the delay value of the flight has been adjusted to the clock

ii) Provisional delays (slot not yet published) are indicated by an asterisk on either side of the value including the possible delay characteristic letter (e.g. *12D*).

aa) The Estimated (**E**), Calculated (**C**) or Actual (**A**) Time of Arrival (**E/C/ATA**) (see later).

bb) The Rerouting indicator **(R)** (see later).

cc) Opportunity indicator (**Opp**). Set to (**Y**) if an opportunity for the flight exists, **blank** otherwise.

dd) The Yoyo indicator **(YY).** This column is by default hidden.

Set to N if no yoyo RFL detected or yoyo never computed, otherwise (Y, C, S, E, A, O) except blank is shown as N and NN is still shown as blank.

ee) The "AO What If Reroute" indicator **(W)** (see later).

ff) The last Message sent or received **(MSG)** (see later).

gg) The identifier of the most penalising Regulation **(REGUL+)**.

hh) If Other Regulations apply **(O)**.

ii) The Trend In **(TI)**: the vertical movement indicator at the entry of the location (see later).

jj) The Entry Flight Level **(EFL)**, the flights flight level over the entry point of the associated countable location. If the reference location is an aerodrome or set of aerodromes or an aircraft operator, the entry flight level (EFL) value is the calculated highest flight level.

kk) The Trend Middle **(TM)**: the overall vertical movement indicator inside the location obtained from the comparison between EFL and XFL (see later). Not displayed when the reference location is a point, an aerodrome, a set of aerodromes or an aircraft operator.

ll) The Exit Flight Level **(XFL)**, the flights flight level over the exit point of the associated countable location. Not displayed when the reference location is a point, an aerodrome, a set of aerodromes or an aircraft operator.

mm) The Trend Out **(TO)**: the vertical movement indicator at the exit of the location (see later)

nn) Following columns are only displayed if the Compare checkbox was checked in the query:

i) The intruder indicator

Following tooltip shall be implemented on the column header:

"**Intruder indicator**
(**H**=Horizontal Intruder,
V=Vertical Intruder,
M=Mixed (Vertical + Horizontal) Intruder,
blank=Not an Intruder)"

ii) Airspace **Origin** of the deviation for an intruder, if applicable and identified.

iii) The delta time (**D.Time**), if the **Compare** checkbox in the query was checked.

iv) The delta flight level (**D.Level**), if the **Compare** checkbox in the query was checked.

v) The delta position (**D.Pos**), if the **Compare** checkbox in the query was checked

oo) **CCAMS**: SSR code assigned by CCAMS to a flight. Only showed while the code is active for that flight (status selected).

(8) In the first column, the Time of Entry into the location can be:

a) the Time of **ENTRY** into the Airspace, Traffic Volume or reference location on which the Regulation is defined,

b) the Time **OVER** the selected Point,

c) the Take Off Time (**TOT**) for the Departure Aerodrome, Departure Set of Aerodromes or Aircraft Operator,

d) the Time of Arrival (**TA**) for the Aerodrome of Arrival or Set of Aerodromes of Arrival

(9) Time of Entry may be of three sources:

a) Estimated (**E**) when based on the Estimated flight profile.

b) Calculated (**C**) when based on the Calculated flight profile (allocated slot).

c) Actual (**A**) when based on the Actual flight profile (after departure).

(10) The Status of the flight may be:

a) **LF** for Late Filer

b) **LU** for Late Updater

c) **LFU** for Late Filer and Updater

d) **blank** when not applicable

(11) The Type of flight data may be:

a) **A** for ATC Activated flight

b) **I** for IFPL

c) **P** for PFD (Predicted Flight Data)

d) **R** for RPL

e) **M** for MFD (Mini Flight Data)

f) **T** for TACT Activated flight when the ATOT is before or equal to the current UTC time

g) **t** for TACT Activated flight when the ATOT is in the future of the current UTC time

h) **E** for Terminated

(12) A tooltip shall be implemented on the title of the Type (**T**) column. It shall indicate "*Type of Data*" appended on each new line with each type text and the corresponding number of occurrences currently present in the displayed list (Example: *"ATC Activated: 5", "TACT Activated (Past ATOT): 2"*).

(13) The flight might be suspended or might have been suspended with the following status:

a) **ST** for SIT Time Out

b) **SM** for Slot Missed

c) **RC** for FCM required

d) **DC** for Delay Threshold exceeded, Flight Confirmation Message (FCM) required

e) **TV** for RVR

f) **NR** for Not Reported as Airborne

g) **RV** for FP Revalidation

h) **MS** for Manual Suspension

i) **AS** for Airport Suspension

(14) The column remains blank if the flight is not suspended and has never been suspended.

(15) The Flight Activation Monitoring (FAM) status may be (case sensitive !):

a) **N** not under FAM yet or never under FAM,

b) **F** currently subject to FAM, if no data received, flight will be shifted soon,

c) **f** was subject to FAM but new data received before first shift (airborne, FPL …),

d) **S** currently shifted by FAM,

e) **s** was shifted by FAM, FPL data received when shifted,

f) **a** was shifted by FAM, airborne data received when shifted,

g) **U** currently suspended by FAM,

h) **u** was suspended by FAM, FPL data received when suspended,

i) **A** was suspended by FAM, airborne data received when suspended.

(16) The Ready status may be:

a) **I** when the flight is in Request For direct Improvement mode (RFI) and the flight is not yet Ready to Depart,

b) **S** when the SIP Wanted Message mode is on (SWM), per opposition to the RFI mode and the flight is not yet Ready to Depart.

c) **R** when the flight is Ready to Depart (REA message received or set by DPI pre-departure processing (T-DPI_s)).

(17) The Ready to Depart state has priority (overwrites) the RFI/SWM state.

(18) The **A/TTOT** column comprises of the following elements in sequence:

a) A take off time (TOT) in hh:mm format, if applicable:

i) Effective Actual Take Off Time (ATOT), if available.

ii) If effective ATOT is not available, dependant on the current CDM Flight Status, the Target Take Off Time (TTOT) provided in the last received DPI message is displayed as follows:

- If CDM Flight Status is **e** or **t**, the latest **Turnaround TTOT** received within a DPI message;

- If the CDM Flight Status is **s**, the latest **Consolidated TTOT** if present, else the latest **Earliest TTOT** if present, else the latest **Turnaround TTOT** received within a DPI message;
- If the CDM Flight Status is **a**, the latest **ATC TTOT** received within an A-DPI message;
- Otherwise left blank.

b) Possibly the CDM Flight Status.

i) Only displayed if CDM Flight Status is either **e**, **t**, **s** or **a**

ii) The CDM Flight Status value is always appended right aligned to the ATOT column, even when no time value is displayed.

c) A CDM Discrepancy indicator, if applicable.

i) The letter **d** is appended if CDM Discrepancies are reported.

d) When the **Departure Status** is not Ok for a flight and the flight state is not yet ATC activated (A) or Terminated (E):

i) The string displayed in the **A/TTOT** cell is appended with an exclamation mark (**!**).

ii) The background of the **A/TTOT** cell is coloured using the user adjustable preference:

- **CHMI / ATFCM application / Flights / General / Departure Status colour**
- Default value is RGB 0, 225, 225 (aqua)
- Description is **Colour used for the display of Departure Status**

(19) If the take off time displayed in the **A/TTOT** column is the effective ATOT and its value is in the future of the current UTC time and the flight type is TACT activated, the entire entry in the **A/TTOT** column is displayed in italic.

(20) If there are CDM Discrepancies, the entire entry in the **A/TTOT** column is displayed in **CDM Discrepancy colour**.

(21) The colour used may be adjusted in the Preferences dialog: **CHMI / ATFCM application / Flights / General / CDM Discrepancy colour**.

(22) When the flight is selected, the CDM Discrepancy details are displayed in the details pane of the Flight List.

(23) The time in the **E/C/ATA** column may be of different sources. It reflects the highest available requested model depending on the requested Traffic Type:

a) For **Traffic Demand**

i) The **Estimated Time of Arrival** is displayed.

b) For **Regulated Demand,**

i) The **Calculated Time of Arrival** is displayed when available.

ii) Otherwise the **Estimated Time of Arrival** is displayed.

c) For **Traffic Load**

i) The **Actual Time of Arrival** is displayed when already available.

ii) If the flight has no actual model yet, the **Calculated Time of Arrival** is displayed when available.

iii) Otherwise the **Estimated Time of Arrival** is displayed.

(24) The appended letter indicates if the Actual (**A**), the Calculated (**C**) or the Estimated (**E**) Time of Arrival is displayed.

(25) Values in this **E/C/ATA** column shall only be displayed when the flight is published (i.e. when the calculated slot has been made public).

(26) The rerouting indicator is composed of a reason and optionally a state.

(27) The rerouting reason code may be:

a) **N** for not rerouted

b) **M** for ATFCM rerouting

c) **C** for NM rerouting (CWIR) (i.e. former “CFMU” or “CEU” rerouting)

d) **A** for Auto rerouting (AWIR)

e) **O** for AO rerouting (AOWIR)

(28) If the rerouting reason is **N**, there is no state.

(29) Otherwise, the rerouting state code may be:

a) **P** for Produced. There is a valid rerouting going on, waiting to be realised by either an FPL or a CHG.

b) **E** for Executed. Rerouting has been done.

c) **T** for Timed Out. No FPL/CHG received on time.

d) **R** for Rejected. Rerouting proposal has been rejected.

e) **V** for Revoked. Rerouting proposal has been revoked.

f) **N** for No match. Message received did not match the proposal. Rerouting has been invalidated.

(30) The "AO What If Reroute" indicator code may be:

a) **N** for TRY and APPLY are NOT possible.

b) **T** for TRY is allowed.

c) **A** for TRY and APPLY are allowed.

(31) The reroutable indicator grants only access to the initial display of the **Rerouting Function**.

(32) Further detailed analysis of the flight in the **Rerouting Function** might disable some rerouting possibilities.

(33) The types of NM ATFCM Messages that may appear in the column **MSG** can be consulted in the ATFCM Handbook part ATFCM Message Examples.

(34) When displaying the flight list for a cherry picked measure, the HMI will clearly identify the flights which are impacted (i.e. non-exempted) by this cherry picked measure.

(35) In such case, the font of the **ARCID** and **REGUL+** cells is using a colour from the user adjustable preference:

a) CHMI / ATFCM application / Flights / General / Cherry Picked Flight colour

b) Default value is RGB 255, 0, 204

c) Description is “Colour used for highlighting Cherry Picked flights”

(36) Cherry picking colouring is taking precedence (i.e. possibly overwriting) over the possible colouring based on delay values.

(37) The Trend In **(TI)**, Trend Middle **(TM)** and Trend Out **(TO)** are attached before and after the entry **(EFL)** and the exit **(XFL)** flight levels and are indications of the flight’s trend at the reference location’s entry, inside and exit point as follows:

a) Cruise. This is indicated as “**=**”. A cruise indication is shown when:

i) The flight vector that includes the entry point as an end point for **TI** is a cruise vector

ii) The comparison between entry **(EFL)** and exit **(XFL)** flight levels indicate an overall cruise for **TM**

iii) The flight vector that includes the exit point as a start point for **TO** is a cruise vector

b) Climb. This is indicated as “**/**”. A climb indication is shown for each trend as detailed here above for **TI**, **TM** and **TO**.

c) Descent. This is indicated as “****”. A descent indication is shown for each trend as detailed here above for **TI**, **TM** and **TO**.

(38) If the reference location is an aerodrome or a set of aerodromes or an aircraft operator, the flight trend indicators have an empty value.

(39) If the **Compare** checkbox in the query was checked, the system creates a comparison flight list from the same query parameters as those used to create the displayed flight list except for:

a) the query period: this is set to an infinite period, and

b) **TFC Type**: this is set to the With.

(40) The comparison consists of searching for each flight in the displayed flight list, the matching flight entry (same flight) in the system's comparison flight list.

(41) **D.Time** is the time difference in minutes (max 4 digits) between the flight's entry time at the reference location in the displayed flight list and in the system's comparison flight list.

(42) The value is preceded by a "+" or a "-" indicating if the entry time in the displayed flight list is later than or before respectively the matched flight in the system's comparison flight list. If the difference is zero, the value is not displayed (i.e. blank).

(43) **D.Time** value is only shown if it is equal or greater than **Absolute D.Time threshold**.

(44) **Absolute D.Time threshold** may be adjusted in Preferences dialog **CHMI / ATFCM application / Flights / Flight list / Absolute D.Time threshold.**

(45) **D.Level** is the flight level difference (3 digits with leading zeros if necessary) between the flight's entry flight level at the reference location in the displayed flight list and in the system's comparison flight list.

(46) The value is preceded by a "+" or a "-" indicating if the entry flight level in the displayed flight list is higher than or lower than respectively the matched flight in the system's comparison flight list. If the difference is zero, the value is not displayed (i.e. blank).

(47) **D.Level** value is only shown if it is equal or greater than **Absolute D.Level threshold**.

(48) **Absolute D.Level threshold** may be adjusted in Preferences dialog **CHMI / ATFCM application / Flights / Flight list / Absolute D.Level threshold**.

(49) **D.Pos** is the distance in nautical miles between the geographical position of the flights entry point into the reference location in the displayed flight list and in the systems comparison flight list.

(50) If the difference is zero, the value is not displayed (i.e. blank).

(51) **D.Pos** value is only shown if it is equal or greater than **D.Pos threshold**.

(52) **D.Pos threshold** may be adjusted in Preferences dialog **CHMI / ATFCM application / Flights / Flight list / D.Pos threshold**.

(53) A value "999" in **D.Pos** indicates a difference in distance greater or equal to 999 nautical miles.

(54) If a flight in the displayed flight list is not matched with a flight in the systems comparison flight list (i.e. Intruder indicator is NOT blank), the **I**, **Origin**, **D.Time**, **D.Level** and **D.Pos** values are displayed in **Intruder colour**.

(55) **Intruder colour** may be adjusted in Preferences dialog **CHMI / ATFCM application / Flights / General / Intruder colour**.

(56) The initial display sort sequence for the Flight List is **Ascending** order on the **Time of Entry into the Location** (first column).

(57) The Flight List allows single or multiple flight selection:

a) Single selection to display additional details or to invoke contextual actions for one flight.

b) Multiple selections to trigger an action on a group of flights (ex: Plot Selected).

(58) As soon as more than one flight is selected, additional details and contextual actions for a single flight are no more available.

(59) When a single flight is selected in the list, the note box at the bottom is filled in with more details:

a) The Initial Estimated Off Block Date (**IOBD**).

b) The last received Estimated Off Block Date and Time (**Last EOBT**).

c) The **Type** of flight, only present if it is a **Proposal**.

d) The title of the **Last Message** sent or received for the flight.

e) The full text corresponding to the state in the list (**Late Filer**, **Late Updater**, **Late Filer and Updater**). Blank if not applicable.

f) The type of flight data (**Flight Type**) (**ATC Activated**, **IFPL**, **PFD**, **MFD**, **RPL**, **TACT Activated**, **Terminated**).

g) The ready state displaying either:

i) **Ready to Depart** when the *readyToDepart* state is true

ii) Or **RFI** when not *readyToDepart* but *readyForImprovement*

iii) Or **SWM** when not readyToDepart and not readyForImprovement

h) The originally **Filed ADES**: only visible in case the selected flight diverted and the flight list contains the diversion ADES.

i) The name of the **Most Penalising Regulation,** if any,

j) and if **Other Regulations Apply** to the same flight.

k) The Proposed Calculated Take Off Time (**Proposed CTOT**) if present for the flight,

l) with the related limit in time for the response (**Response by**).

m) The **CTOT Limit** reason, if non blank. Possible reasons are:

i) **Forced by Tower**

ii) **Forced by FMD**

iii) **Forced by CHAMAN**

iv) **Forced by STAM**

v) **By Violation**

vi) **By Violation / Zero Rate / RVR**

n) The suspension status if the flight is currently suspended. Blank if not currently suspended. Possible reasons for suspension are:

i) **Flight suspended - SIT Time Out**

ii) **Flight suspended - Slot Missed**

iii) **Flight suspended - FCM Required**

iv) **Flight suspended - Delay Threshold exceeded, FCM Required**

v) **Flight suspended - RVR**

vi) **Flight suspended - Not Reported as Airborne**

vii) **Flight suspended - FP Revalidation**

viii) **Flight suspended - Manual Suspension**

ix) **Flight suspended - Airport Suspension**

o) The time limit for the Confirmation message (FCM) after the issuance of a Flight Suspension (**FLS Response by**), if present for the flight.

p) If the flight is subject to a rerouting, the reason and the current state in full text.

i) If the flight has been rerouted, the displayed rerouting reason may be:

- **ATFCM rerouting,**
- **NM rerouting (CWIR) (i.e. former CFMU or CEU rerouting),**
- **Auto rerouting (AWIR),**
- **AO rerouting (AOWIR).**

ii) When there is a rerouting reason, it is followed by one of these rerouting states:

- **Produced** for a valid rerouting going on, waiting to be realised by either an FPL or a CHG.
- **Executed** when the rerouting has been processed.
- **Timed-out** when no FPL/CHG was received on time.
- **Rejected** for a rerouting proposal that has been rejected.
- **Revoked** for a rerouting proposal has been revoked.
- **No-match** when the message received did not match the proposal. Rerouting has been invalidated.

q) The "AO What If Reroute" possibility in full text.

i) **Reroute Try and Apply NOT allowed.**

ii) **Reroute Try possible.**

iii) **Reroute Try and Apply possible.**

r) The CDM Discrepancy details are displayed if any. (see later)

s) The complete **Route (**ICAO Field 15) information comprising of initial requested speed, flight level and route details.

(60) CDM Discrepancy details might include the following:

a) **"IFPS-Airport Discrepancy with Target Off-Block Time"**

b) **"IFPS-Airport Discrepancy with Aircraft Type=xxxx"** where xxxx is the Aircraft Type received in the CDM info.

c) **"IFPS-Airport Discrepancy with Registration Mark=xxxxxxx"** where xxxxxxx is the Registration Mark received in the CDM info.

(61) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

10.5 User interactions

(1) Click on a line to **select the flight** and have more detailed information displayed for the flight.

(2) If the **Route** and the Tolerance Window (**TW**) information are already available locally on the CHMI workstation, it is displayed immediately.

(3) If not yet available locally and if the user is authorised by security profile and has activated the user preference **Auto Display Extra Info**, the **Display Extra Infos** contextual action is automatically triggered after a small “selection elapse time” (i.e. order of 500 milliseconds) to ensure the user is not just scrolling quickly through the list and the focus remains on the list.

(4) The user preference **Auto Display Extra Info** has the following characteristics:

a) Path name: **CHMI / ATFCM application / Flights / General / Auto Display Extra Info**

b) Default value: **ON** (i.e. automatically trigger the retrieval of extra info)

c) Description: "Allows enabling / disabling the automatic retrieval and display of extra flight information (Route, Tolerance Window …) in list of flights".

(5) A double click will **select the flight** as described above and it will immediately trigger the default **Display Extra Infos** contextual action if the **Route** and the Tolerance Window (**TW**) information are not yet available locally.

(6) The standard "Ctrl" or "Shift" key combination shall allow the selection of multiple flights.

a) The bottom details pane shall be blanked out when multiple flights are selected in the list.

b) When multiple flights are selected, the contextual popup menu shall only display the actions where the identification of a single flight is not mandatory, if any.

(7) The **Flight List** shall implement the global **Plot All flights function**.

(8) The **Flight List** shall implement the global **Plot Selected flights function**.

(9) The **Normal / Proposal** button may be adjusted before issuing a new query.

(10) Following user preferences may be adjusted to associate colours with one of the pre-defined delay categories.

(11) Both ARCID and Delay values are coloured using the selected preferences.

a) **CHMI / ATFCM application / Flights / General / Delay less than 1' colour:** Colour used for the display of delays less than 1 minute.

b) **CHMI / ATFCM application / Flights / General / Delay less than 15' colour:** delays between 1 and 15 minutes.

c) **CHMI / ATFCM application / Flights / General / Delay less than 30' colour:** delays between 15 and 30 minutes.

d) **CHMI / ATFCM application / Flights / General / Delay less than 45' colour:** delays between 30 and 45 minutes.

e) **CHMI / ATFCM application / Flights / General / Delay greater or equal 45' colour:** delays greater or equal to 45 minutes.

(12) Following default colour preference values shall be used:

a) **Delay less than 1' colour:** RGB 0, 0, 0 (black)

b) **Delay less than 15' colour:** RGB 51, 51, 255 (blue)

c) **Delay less than 30' colour**: RGB 255, 255, 0 (yellow)

d) **Delay less than 45' colour**: RGB 255, 125, 0 (orange)

e) **Delay greater equal to 45' colour**: RGB 255, 0, 0 (red)

(13) Following user preference may be adjusted to associate a colour with the first flight selected for Swap 2 operation: **CHMI / ATFCM application / Flights / General / Swap colour**.

(14) When the line is selected, colour is used in reverse video (background/foreground).

(15) This window allows to modify a previous query selection and to issue a new request.

(16) On query modification, the **Reset action** is enabled.

(17) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(18) To display the **Suspended Flight List** for the same query criteria as the current flight list, click on the **FLS** button at the top of the list.

(19) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(20) The **Previous / Next period** function may be used to issue the same query but for the previous or the next period in time.

(21) The **Polling function** may be activated to start automatic refresh processing on this window.

(22) The **Period Shifting function** may be activated to start automatic period shifting on subsequent refresh of this window.

(23) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(24) To print the Flight list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(25) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(26) To modify the **Sort sequence** of the flights, click on column headers corresponding to your desired sort criteria.

(27) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(28) To adjust the display selection within the columns available in a Flight List (Flight List + associated where criteria), click on the **Show / Hide Columns** button on the button bar or select the **Action / Show / Hide Columns** menu option and adjust the desired display selection.

(29) The user preference **Show flight details** allows to automatically show / hide the detail box at the bottom of the window on opening a new Flight List window. It may be adjusted for each target server (ATFCM, Archive, Predict) (ex: CHMI / ATFCM application / Flights / Flight list / ATFCM Show flight details).

(30) Increasing / reducing the window height will impact the size of the main Flight list control.

(31) Increasing / reducing the window width will impact the size of the main Flight list and the bottom note box controls.

(32) The **Partial Period Shifting** function may be used to issue the same query but for the previous or the next period with time overlap (see **Partial Period Shifting** function detailed description).

10.6 Contextual actions

(1) When a line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Flight and may contain the following options:

a) **Flight Data**: displays the details for the selected flight (profiles ...).

b) **Point Profile**: same as Flight Data but activates immediately the Point Profile display when flight details are retrieved.

c) **Airspace Profile**: same as Flight Data but activates immediately the Airspace Profile display when flight details are retrieved.

d) **Flight History**: displays Flight Event History list for one flight.

e) **Operational Log**: lists TACT Operational Log entries for one flight.

f) **Departure Aerodrome Delays**: displays the delays for the departure aerodrome of the flight.

g) **Rerouting Function**: allows to evaluate and request the rerouting for the selected flight.

h) If the flight is not reroutable, the option **Alternate Routes**: allows to list the different route alternative for the selected flight.

i) **Display Extra Infos**: (default action which would be triggered by a double click)

i) Retrieves the ICAO Field 15 format and the Tolerance Window time information for the selected flight, if not yet available locally.

ii) Insert the Tolerance Window time information in the list (**TW**) for the selected flight, if authorised by the user profile.

iii) And append the flight route details as described in the ICAO Field 15 format to the current details displayed in the box at the bottom of the list.

j) **Regulation Delays**: displays the delays due to the regulation, if the selected flight is regulated.

k) **Ready To Depart**: sends an REA message for the selected flight(see conditions in related dialog section).

l) **Request For direct Improvement**: sends an RFI message for the selected flight (see conditions in related dialog section).

m) **SIP Wanted Message**: sends an SWM message for the selected flight (see conditions in related dialog section).

n) **Slot Proposal Accept**: sends an SPA message for the selected flight (see conditions in related dialog section).

o) **Slot Proposal Reject**: sends an SRJ message for the selected flight (see conditions in related dialog section).

p) **Slot Missed Message**: sends an SMM message for the selected flight (see conditions in related dialog section).

q) **Flight Confirmation Message**: sends an FCM message for the flight (see conditions in related dialog section).

r) **Swap 1 with a free Slot**: sends a request to exchange selected flight with an identified free slot. (only available from Regulation Flight List)

s) **Swap 2 Slots (new/reset)**: select/unselect a first flight to prepare the swapping of 2 allocated slots. (only available from Regulation Flight List)

t) **Swap Slot with**: sends a request to exchange selected flight with the flight initially selected with the option Swap 2 Slots (new/reset). (only available from Regulation Flight List)

11 ARCID List

11.1 Function presentation

(1) The ARCID flight list gives the list of flights corresponding to a specific aircraft identifier.

(2) The query may optionally include a given day, a selected Aerodrome of Departure and/or an Initial Off Block Time.

11.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Flights / ARCID List**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl>** + **<Shift>** + **<F10>**.

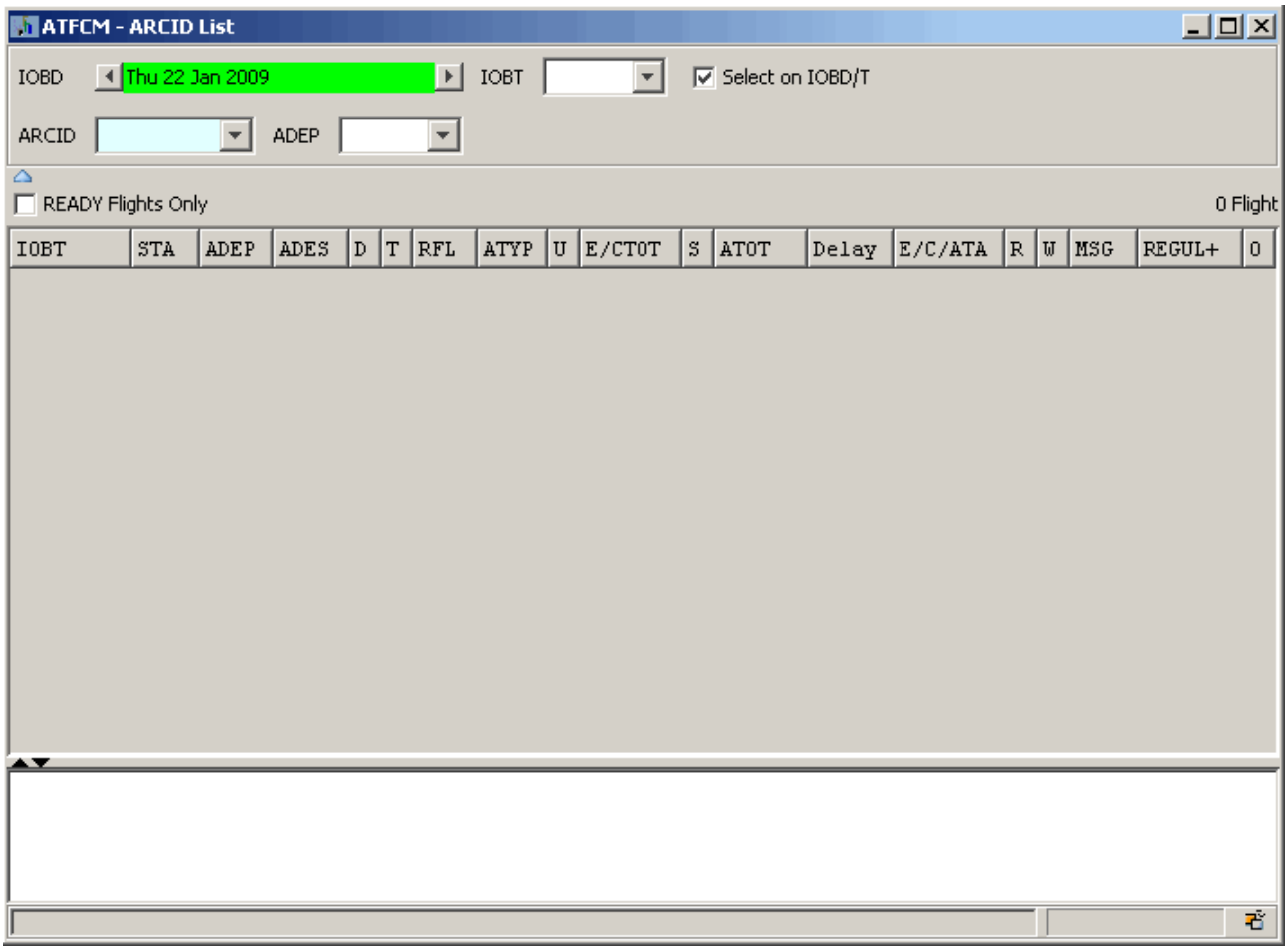


Figure Chmi - 39: Query ARCID List

(3) A new empty **ARCID List** window will be opened with the following defaults:

a) IOBD: current or last queried traffic date is proposed (relative to current UTC date) dependant on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) IOBT: blank. Drop-down list will propose pre-formatted time on a 20 minutes step base.

c) ARCID: the drop-down box will propose the most recent query data elements you have queried before with success.

d) ADEP: the drop-down box will propose the most recent query data elements you have queried before with success.

(4) The **Select on IOBD/T** check box is always initially pre-checked.

(5) On initial display, the focus is set on the first mandatory query control with no default value: the **ARCID** input control.

11.3 Preparing to query

(1) The Initial Estimated Off Block Date (**IOBD**) may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The Initial Estimated Off Block Time (**IOBT**) input box may be adjusted to a valid time value from 00:00 to 23:59 included. This time does not have to be exact. The system will allow some flexibility to identify the precise flight the user wants to query. This is an optional query data element.

(3) By unchecking the **Select on IOBD/T** box, the user indicates he does not want any selection of the flights on date and time criteria. Consequently, the IOBD and IOBT input controls are disabled.

(4) The Aircraft Identifier (**ARCID**) shall be filled in or selected from the drop-down list (when a previous query was successful). It is the only mandatory query data element.

(5) The Aerodrome of Departure (**ADEP**) may be filled in with an Aerodrome ICAO code or selected from the drop-down list (when a previous query was successful). This is an optional query data element.

(6) Select the **Proposal** radio button on the button bar to include the proposed flights in the retrieved list. Otherwise the list will be based on the normal flights only.

(7) The **Send** button will send the query to the Network Operations system.

(8) This window may be closed before any query is issued.

(9) The **Reset** action common functionality is implemented for this function. All query criteria except Normal/Proposal selection are considered for activating the Reset action.

11.4 Displaying the query results

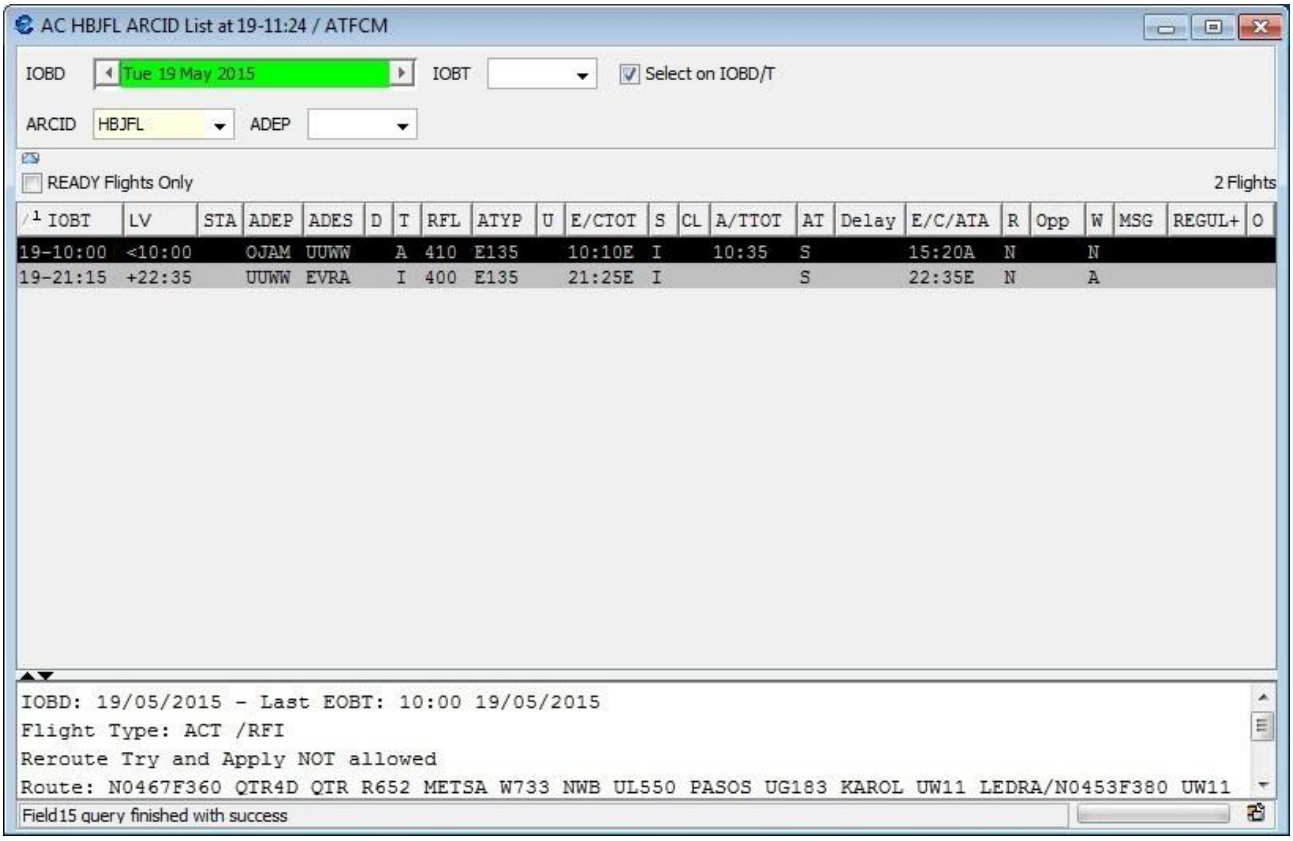


Figure Chmi - 40: ARCID list

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) Presence of the keyword **(Prop)** in the title indicates that the displayed data are including the flight "Proposals".

(3) The header box of the window contains the different **query data elements** used to retrieve the displayed flight data:

a) IOBD: Initial Estimated Off Block Date

b) IOBT: Initial Estimated Off Block Time

c) ARCID: Aircraft Identifier

d) ADEP: Aerodrome of Departure

(4) IOBD and IOBT controls may be disabled if **Select on IOBD/T** is unchecked.

(5) At the top of the flight list, a counter indicates the **number of flights** retrieved.

(6) This counter may be affected by **filtering** (fff / rrr Flights where fff is the filtered number and rrr is the retrieved number).

(7) On query results display, the focus is set on the flight list after selecting the first flight in the list.

(8) The **READY Flights only** checkbox allows to display only the flights being in READY state. This sub-selection is done locally by the application, no query to the server is issued for this request. The counter is adjusted to reflect the number of displayed flights.

(9) For each flight the **ARCID List** display contains following columns:

a) The Initial Estimated Off Block Time **(IOBT)**.

b) Just after the IOBT, the presence of the letter **P**. When a flight is still a Proposal that needs to be confirmed, then the letter P will appear along with the IOBT value.

c) The Last Valid (**LV**) EOBT acceptable for the flight before triggering IFPS errors

See **LV** column in Flight List for detailed display format, tooltip and ICD source data.

d) The Status of the flight **(STA)** (see later).

e) The Aerodrome of Departure **(ADEP)**.

f) The Aerodrome of Destination **(ADES)** of the highest available model. For a diverting flight, the new ADES will be displayed in the column with header ADES.

g) Diversion indicator **(D),** indicated with **>** symbol. This symbol is present for a diverted flight, independent of the requested model.

h) The Type of flight data **(T)** (see later).

i) The highest Requested Flight Level **(RFL)**.

j) The ICAO identifier of the Aircraft Type **(ATYP)**.

k) The Suspension status **(U)** if the flight is currently suspended or the last suspension status if the flight had been previously suspended (see later).

l) The Estimated (**E**) or Calculated (**C**) Take Off Time (**E/CTOT**) (only shown when flight is published).

m) The Ready status **(S)** (see later).

n) The CTOT Limit (**CL**) reason: Possible exceptional reason that may affect the CTOT allocation of the flight, if any. Possible values are as described in the tooltip below.

See **CL** column in Flight List for detailed display format, tooltip and ICD source data.

o) The Take Off Time information from the Aerodrome of Departure **(A/TTOT)** (see later).

i) Following tooltip shall be implemented on the column header: **Take Off Time info from Aerodrome of Departure.**

p) The Airport Type (**AT**) associated to the Aerodrome of Departure.

   i) Following tooltip shall be implemented on the column header: **Departure Airport Type (S=Standard, T=Advanced ATC/TWR, C=CDM).**

q) The Airport Type (**AT**) associated to the Aerodrome of Departure.

r) The CDM DPI Target Off-Block Time (hh:mm), if any (**TOBT**)

   i) Following tooltip shall be implemented on the column header: Target Off-Block Time

s) The CDM DPI Target Start-Up Approval Time (hh:mm), if any (**TSAT**)

   i) Following tooltip shall be implemented on the column header: Target Start-Up Approval Time

t) The latest Taxi Time in minutes identified by the Network Operations system, possibly based on CDM DPI message, to calculate the Actual flight profile (**TT**)

   i) Following tooltip shall be implemented on the column header: Actual Taxi Time

u) The Slot Tolerance Window for a regulated flight or the Departure Tolerance Window for a non-regulated flight (**TW**).

   i) The **TW** column shall only be displayed when authorized by the user profile. A single common use case shall grant the display of the Tolerance Window data in all relevant functionalities (Flight List, Flight Data &).

   ii) The Lower and Upper time limit of the Tolerance Window are displayed using the following format: [*low-hh:mm*,*upper-hh:mm*] (ex: "*TW [10:20,10:40]*").

   iii) The complete string is displayed in *italic* when the values are defined as an extended tolerance window.

   iv) Following tooltip shall be implemented on the column header: Slot Tolerance Window for Regulated Flights or Departure Tolerance Window for Non-Regulated Flights.

   v) Each time newly retrieved Tolerance Window values are displayed, the flight list shall be resorted if the **TW** column is part of the selected sort criteria for ordering the list.

v) The **Delay** in minutes.

   i) A case sensitive letter may possibly be appended to the delay value to indicate following specific delay characteristics:

- "D" when the delay value calculated for the flight is exceeding the delay confirmation threshold of a regulation affecting the flight
- "a" when the delay value of the flight has been adjusted to the clock.

ii) Provisional delays (slot not yet published) are indicated by an asterisk on either side of the value including the possible delay characteristic letter (e.g. *12D*).

w) The Estimated (**E**), Calculated (**C**) or Actual (**A**) Time of Arrival (**E/C/ATA**) (see later).

x) The Rerouting indicator **(R)** (see later).

y) Opportunity indicator (**Opp**). Set to (**Y**) if an opportunity for the flight exists, **blank** otherwise.

z) The Yoyo indicator **(YY)**. This column is by default hidden.

aa) The "AO What If Reroute" indicator **(W)** (see later).

bb) The last Message sent or received **(MSG)** (see later).

cc) The identifier of the most penalising Regulation **(REGUL+)**.

dd) If Other Regulations apply **(O)**.

ee) **CCAMS**: SSR code assigned by CCAMS to a flight. Only showed while the code is active for that flight (status selected).
This column should only be displayed if the user profile permits it.

(10) The Status of the flight may be:

a) **LF** for Late Filer

b) **LU** for Late Updater

c) **LFU** for Late Filer and Updater

d) **blank** when not applicable

(11) The Type of flight data may be :

a) **A** for ATC Activated flight

b) **I** for IFPL

c) **P** for PFD (Predicted Flight Data)

d) **R** for RPL

e) **M** for MFD (Mini Flight Data)

f) **T** for TACT Activated flight when the ATOT is before or equal to the current UTC time

g) **t** for TACT Activated flight when the ATOT is in the future of the current UTC time

h) **E** for Terminated

(12) The flight might be suspended or might have been suspended with the following status:

a) **ST** for SIT Time Out

b) **SM** for Slot Missed

c) **RC** for FCM required

d) **DC** for Delay Threshold exceeded, Flight Confirmation Message (FCM) required

e) **TV** for RVR

f) **NR** for Not Reported as Airborne

g) **RV** for FP Revalidation

h) **MS** for Manual Suspension

i) **AS** for Airport Suspension

(13) The column remains blank if the flight is not suspended and has never been suspended.

(14) The Ready status may be:

a) **I** when the flight is in Request For direct Improvement mode (RFI) and the flight is not yet Ready to Depart,

b) **S** when the SIP Wanted Message mode is on (SWM), per opposition to the RFI mode and the flight is not yet Ready to Depart.

c) **R** when the flight is Ready to Depart (REA message received or set by DPI pre-departure processing (T-DPI_s)).

(15) The Ready to Depart state has priority (overwrites) the RFI/SWM state.

(16) If the take off time displayed in the **A/TTOT** column is the effective ATOT and its value is in the future of the current UTC time and the flight type is TACT activated, the entire entry in the **A/TTOT** column is displayed in italic.

(17) If there are CDM Discrepancies, the entire entry in the **A/TTOT** column is displayed in **CDM Discrepancy colour**.

(18) The colour used may be adjusted in the Preferences dialog: **CHMI / ATFCM application / Flights / General / CDM Discrepancy colour**.

(19) When the flight is selected, the CDM Discrepancy details are displayed in the details pane of the Flight List.

(20) The time in the **E/C/ATA** column may be of different sources. It reflects the highest available model for the flight:

a) The **Actual Time of Arrival** is displayed when available.

b) If the flight has no actual model yet, the **Calculated Time of Arrival** is displayed when available.

c) Otherwise the **Estimated Time of Arrival** is displayed.

(21) The appended letter indicates if the Actual (**A**), the Calculated (**C**) or the Estimated (**E**) Time of Arrival is displayed.

(22) Values in this **E/C/ATA** column shall only be displayed when the flight is published (i.e. when the calculated slot has been made public).

(23) The rerouting indicator is composed of a reason and optionally a state.

(24) The rerouting reason code may be:

a) **N** for not rerouted

b) **M** for ATFCM rerouting

c) **C** for NM rerouting (CWIR) (i.e. former “CFMU” or “CEU” rerouting)

d) **A** for Auto rerouting (AWIR)

e) **O** for AO rerouting (AOWIR)

(25) If the rerouting reason is **N**, there is no state.

(26) Otherwise, the rerouting state code may be:

a) **P** for Produced. There is a valid rerouting going on, waiting to be realised by either an FPL or a CHG.

b) **E** for Executed. Rerouting has been done.

c) **T** for Timed Out. No FPL/CHG received on time.

d) **R** for Rejected. Rerouting proposal has been rejected.

e) **V** for Revoked. Rerouting proposal has been revoked.

f) **N** for No match. Message received did not match the proposal. Rerouting has been invalidated.

(27) The "AO What If Reroute" indicator code may be:

a) **N** for TRY and APPLY are NOT possible.

b) **T** for TRY is allowed.

c) **A** for TRY and APPLY are allowed.

(28) The reroutable indicator grants only access to the initial display of the **Rerouting Function**.

(29) Further detailed analysis of the flight in the **Rerouting Function** might disable some rerouting possibilities.

(30) The types of NM ATFCM Messages that may appear in the column **MSG** can be consulted in the ATFCM Handbook part ATFCM Message Examples.

(31) The initial display sort sequence for the ARCID List is **Ascending** order on the **IOBT** (first column).

(32) The ARCID List allows single or multiple flight selection:

a) Single selection to display additional details or to invoke contextual actions for one flight.

b) Multiple selections to trigger an action on a group of flights (ex: Plot Selected).

(33) As soon as more than one flight is selected, additional details and contextual actions for a single flight are no more available.

(34) When a single flight is selected in the list, the note box at the bottom is filled in with more details:

a) The Initial Estimated Off Block Date (**IOBD**).

b) The last received Estimated Off Block Date and Time (**Last EOBT**).

c) The **Type** of flight, only present if it is a **Proposal**.

d) The title of the **Last Message** sent or received for the flight.

e) The full text corresponding to the state in the list (**Late Filer**, **Late Updater**, **Late Filer and Updater**). Blank if not applicable.

f) The type of flight data (**Flight Type**) (**ATC Activated**, **IFPL**, **PFD**, **MFD**, **RPL**, **TACT Activated**, **Terminated**).

g) The ready state displaying either:

i) **Ready to Depart** when the *readyToDepart* state is true

ii) Or **RFI** when not *readyToDepart* but *readyForImprovement*

iii) Or **SWM** when not *readyToDepart* and not *readyForImprovement*

h) The originally **Filed ADES**: only visible in case the selected flight diverted and the flight list contains the diversion ADES.

i) The name of the **Most Penalising Regulation,** if any,

j) and if **Other Regulations Apply** to the same flight.

k) The Proposed Calculated Take Off Time (**Proposed CTOT**) if present for the flight,

l) with the related limit in time for the response (**Response by**).

m) The **CTOT Limit** reason, if non blank.

See **CTOT Limit** reason in note box of Flight List for list of possible reasons.

n) The suspension status if the flight is currently suspended. Blank if not currently suspended.

See suspension status in note box of Flight List for list of possible reasons.

o) The time limit for the Confirmation message (FCM) after the issuance of a Flight Suspension (**FLS Response by**), if present for the flight.

p) If the flight is subject to a rerouting, the reason and the current state in full text.

i) If the flight has been rerouted, the displayed rerouting reason may be:

- **ATFCM rerouting,**
- **NM rerouting (CWIR) (i.e. former CFMU or CEU rerouting),**
- **Auto rerouting (AWIR),**
- **AO rerouting (AOWIR).**

ii) When there is a rerouting reason, it is followed by one of these rerouting states:

- **Produced** for a valid rerouting going on, waiting to be realised by either an FPL or a CHG.
- **Executed** when the rerouting has been processed.
- **Timed-out** when no FPL/CHG was received on time.
- **Rejected** for a rerouting proposal that has been rejected.
- **Revoked** for a rerouting proposal has been revoked.
- **No-match** when the message received did not match the proposal. Rerouting has been invalidated.

q) The "AO What If Reroute" possibility in full text.

i) **Reroute Try and Apply NOT allowed.**

ii) **Reroute Try possible.**

iii) **Reroute Try and Apply possible.**

r) The CDM Discrepancy details are displayed if any. (see later)

s) The complete **Route (**ICAO Field 15) information comprising of initial requested speed, flight level and route details.

(35) CDM Discrepancy details might include the following:

a) **"IFPS-Airport Discrepancy with Target Off-Block Time"**

b) **"IFPS-Airport Discrepancy with Aircraft Type=**xxxx" where xxxx is the Aircraft Type received in the CDM info.

c) **"IFPS-Airport Discrepancy with Registration Mark=**xxxxxxx" where xxxxxxx is the Registration Mark received in the CDM info.

(36) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

11.5 User interactions

(1) Check or uncheck the **Select on IOBD/T** check box in order to activate or not the IOBD / IOBT selection. This action needs to issue a new query to get up to date data.

(2) Click on a line to **select the flight** and have more detailed information displayed for the flight. If the **Route** and the Tolerance Window (**TW**) information are already available locally on the CHMI workstation, it is displayed immediately.

(3) If not yet available locally and if the user is authorised by security profile and has activated the user preference **Auto Display Extra Info**, the **Display Extra Infos** contextual action is automatically triggered after a small “selection elapse time” (i.e. order of 500 milliseconds) to ensure the user is not just scrolling quickly through the list and the focus remains on the list.

(4) A double click will **select the flight** as described above and it will immediately trigger the default **Display Extra Infos** contextual action if the **Route** and the Tolerance Window (**TW**) information are not yet available locally.

(5) The standard “Ctrl” or “Shift” key combination shall allow the selection of multiple flights.

a) The bottom details pane shall be blanked out when multiple flights are selected in the list.

b) When multiple flights are selected, the contextual popup menu shall only display the actions where the identification of a single flight is not mandatory, if any.

(6) The **ARCID List** shall implement the global **Plot All flights function**.

(7) The **ARCID List** shall implement the global **Plot Selected flights function**.

(8) The **Normal / Proposal** button may be adjusted before issuing a new query.

(9) Following user preferences may be adjusted to associate colours with one of the pre-defined delay categories.

(10) Delay values are coloured using the selected preferences.

a) **CHMI / ATFCM application / Flights / General / Delay less than 1' colour:** Colour used for the display of delays less than 1 minute.

b) **CHMI / ATFCM application / Flights / General / Delay less than 15' colour:** delays between 1 and 15 minutes.

c) **CHMI / ATFCM application / Flights / General / Delay less than 30' colour:** delays between 15 and 30 minutes.

d) **CHMI / ATFCM application / Flights / General / Delay less than 45' colour:** delays between 30 and 45 minutes.

e) **CHMI / ATFCM application / Flights / General / Delay greater or equal 45' colour:** delays greater or equal to 45 minutes.

(11) When the line is selected, colour is used in reverse video (background/foreground).

(12) This window allows to modify a previous query selection and to issue a new request.

(13) On query modification, the **Reset action** is enabled.

(14) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(15) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(16) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(17) To print the ARCID list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(18) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(19) To modify the **Sort sequence** of the flights, click on column headers corresponding to your desired sort criteria.

(20) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(21) Increasing / reducing the window height will impact the size of the main ARCID list control.

(22) Increasing / reducing the window width will impact the size of the main ARCID list and the bottom note box controls.

11.6 Contextual actions

(1) When a line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Flight and may contain the following options:

a) **Flight Data**: displays the details for the selected flight (profiles ...).

b) **Point Profile**: same as Flight Data but activates immediately the Point Profile display when flight details are retrieved.

c) **Airspace Profile**: same as Flight Data but activates immediately the Airspace Profile display when flight details are retrieved.

d) **Flight History**: displays Flight Event History list for one flight.

e) **Operational Log**: lists TACT Operational Log entries for one flight.

f) **Departure Aerodrome Delays**: displays the delays for the departure aerodrome of the flight.

g) **Rerouting Function**: allows to evaluate and request the rerouting for the selected flight.

h) If the flight is not reroutable, the option **Alternate Routes**: allows to list the different route alternative for the selected flight.

i) **Display Extra Infos**: (default action which would be triggered by a double click)

i) Retrieves the ICAO Field 15 format and the Tolerance Window time information for the selected flight, if not yet available locally.

ii) Insert the Tolerance Window time information in the list (**TW**) for the selected flight, if authorised by the user profile.

iii) And append the flight route details as described in the ICAO Field 15 format to the current details displayed in the box at the bottom of the list.

j) **Regulation Delays**: displays the delays due to the regulation, if the selected flight is regulated.

k) **Ready To Depart**: sends an REA message for the selected flight (see conditions in related dialog section).

l) **Request For direct Improvement**: sends an RFI message for the selected flight (see conditions in related dialog section).

m) **SIP Wanted Message**: sends an SWM message for the selected flight (see conditions in related dialog section).

n) **Slot Proposal Accept**: sends an SPA message for the selected flight (see conditions in related dialog section).

o) **Slot Proposal Reject**: sends an SRJ message for the selected flight (see conditions in related dialog section).

p) **Slot Missed Message**: sends an SMM message for the selected flight (see conditions in related dialog section).

q) **Flight Confirmation Message**: sends an FCM message for the flight (see conditions in related dialog section).

12 Suspended Flight List

12.1 Function presentation

(1) This function lists all currently suspended flights with a type different from PFD and RPL based on the Traffic Demand for a combination of several query criteria.

(2) The suspension of Predicted Flight Data (PFD) is justified by internal flight prediction process which shall not interfere with the list of flights suspended for Operational reasons.

(3) The main query criteria include a period and a main selection criterion with related sub-criteria.

(4) In addition, there are two optional criteria: an Aircraft Operator list and a most penalizing FMP.

(5) The flights are selected on:

a) The time over the reference location for a Traffic Volume, a Regulation and a Point.

b) The take off time for the Aircraft Operator, the Departure Aerodrome and Aerodrome Set.

c) The arrival time for the Arrival Aerodrome and Aerodrome Set.

d) The entry time for the Airspace.

12.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Flights / Suspended Flight List**.

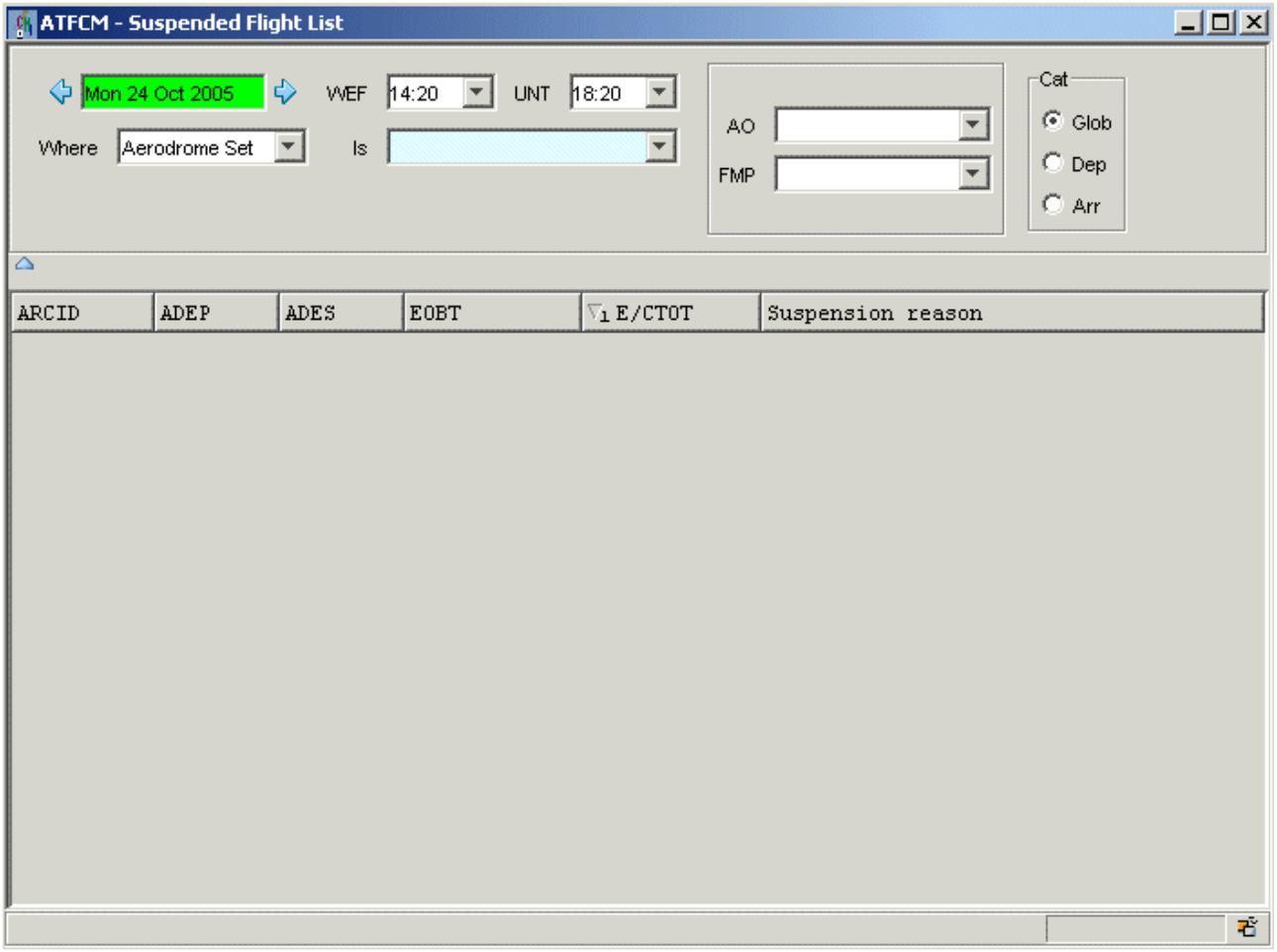


Figure Chmi - 41: Query Suspended Flight List

(2) A new empty **Suspended Flight List** window will be opened with the following defaults:

a) Observed period: current or last queried traffic date is proposed (relative to current UTC date) dependant on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** time: proposed based on corrected current UTC time and substracted by the user adjustable preference **CHMI / ATFCM application / Flights / Suspended flight list / Query lead time**.

c) **UNT** time: proposed based on calculated WEF and user adjustable preference **CHMI / ATFCM application / Flights / Suspended flight list / Query default elapse**.

d) **Where**: last element used with this query This selection is based on the system property **CHMI / ATFCM application / Flights / Suspended flight list / Last Used Identification Type**.

e) **Is**: based on the "Where" selection, the drop-down box will propose the most recent query data elements you have queried before with success.

(3) Current UTC time used is corrected with the **Time adjustment** user preference (CHMI / ATFCM application / General / Dates and times / Time adjustment).

(4) Following non-modifiable query data elements are not displayed:

a) The requested traffic type is always **Traffic Demand** for the selection of the suspended flights.

b) The **Entry** count type is used to allow period selection.

(5) The system property **Query over midnight** which authorises to query over two days, may possibly limit this time calculation process to the start of the day (00:00) (CHMI / ATFCM application / General / Dates and times / Query over midnight).

(6) When Traffic Volume is selected, the **Flow** control is proposed empty. The drop-down list is filled in with the most recent IDs you have queried before with success.

(7) When Aerodrome or Aerodrome Set is selected, the related category controls (**Cat**) are proposed. Global button comes preselected.

(8) When Point is selected, the Maximum (**MAX**) and Minimum (**MNM**) Flight Level (**FL**) controls are proposed. They are initialised with the extreme values (990 for MAX and 000 for MNM).

(9) The **FMP** input box is proposed empty. The drop-down list is filled in with the most recent IDs you have queried before with success.

(10) When **Where** is not set to Aircraft Operator(s), the secondary **AO(s)** input box is proposed empty. The drop-down list is filled in with the most recent IDs you have queried before with success.

(11) All these **Where** query elements may be associated to the ID of one or more Aircraft Operators, except when the main query criterion is already one or more Aircraft Operators.

(12) On initial display, the focus is set on the first mandatory query control with no default value: the **Is** input control.

(13) A contextual Suspended Flight List may be directly requested from a Flight List display if authorised via the FLS button at the top left of the list of flights.In such a case, the query criteria come pre-filled based on the window where the request was initiated.

(14) The observed period is never forced by the calling function. The default values are always proposed as described here above (Traffic date, WEF, UNT).

(15) When requested from a Flight List, the Suspended Flight List will trigger on immediately the **Period Shifting** state.

12.3 Preparing to query

(1) The traffic **Observed period** may be adjusted as follows:

a) The traffic **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

b) The proposed default **WEF** and **UNT** times may be overwritten.

c) Any valid time value from 00:00 to 23:59 included may be entered in the **WEF** and/or the **UNT** entry boxes. This input must respect the HH:MM format.

d) Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(2) The period resulting from the input will be validated taking into account the system property **CHMI / ATFCM application / General / Dates and times / Query over midnight**.

(3) If query over two days is not authorised, the **WEF** time must be smaller than the **UNT** time, except if both times are set to 00:00, which corresponds to a full day query.

(4) The **Where** drop-down list provides the user with following selection:

a) Traffic Volume

b) Airspace

c) Aerodrome

d) Aerodrome Set

e) Point

f) Aircraft Operator(s)

g) Regulation

(5) A tooltip on the **Is** entry box shall describe:

a) The expected input type

b) With a sample format when the user can enter a list of identifiers.

(6) Example: “Enter one or more Aircraft Operator identifiers (ex: ‘AFR,FIN’)”

(7) The identifier corresponding to the selected **Where** shall be keyed in the associated **Is** entry box or selected from the drop-down list (when a previous query was successful).

(8) When Traffic Volume is selected, a **Flow** may be entered or selected from the list of previously queried IDs proposed in the drop-down list.

(9) When Aerodrome or Aerodrome Set is selected, the related category (**Cat**) must be checked (Global, Departure or Arrival).

(10) When Point is selected, the Maximum (**MAX**) and Minimum (**MNM**) Flight Levels (**FL**) may be overwritten to reduce or enlarge selected altitude range.

(11) A valid integer value (maximum of 3 digits) entered in the Maximum (**MAX**) box, will be rounded as follow:

a) 0 remains 0.

b) Input between 1 and 30 is rounded up to the nearest multiple of 10.

c) Input between 31 and 290 is rounded up to the nearest multiple of 5.

d) Input between 291 and 990 is rounded up to the nearest multiple of 10.

e) Above 990 is forced to 990.

(12) A valid integer value (maximum of 3 digits) entered in the Minimum (**MNM**) box, will be rounded as follow:

a) Input between 0 and 34 is rounded down to the nearest multiple of 10.

b) Input between 35 and 294 is rounded down to the nearest multiple of 5.

c) Input between 295 and 999 is rounded down to the nearest multiple of 10.

(13) All these **Where** query elements may be associated to the ID of a most penalising FMP, where the selected flights must have the most penalising Regulation from a given FMP.

(14) All these **Where** query elements may be associated to the ID of one or more Aircraft Operators (**AO(s)**), except when the main query criterion is already one or more Aircraft Operators.

(15) When **Aircraft Operator(s)** is selected from the **Where** drop-down list, the associated **Is** entry box shall accept a comma separated list of 1 to 10 Aircraft Operator Identifiers.

(16) When accessible, the secondary **AO(s)** entry box shall accept a comma separated list of 1 to 10 Aircraft Operator Identifiers.

(17) The comma is the official standard separator character. Other characters which are invalid in AO identifier (ex: semicolon, blank …) could be accepted by the HMI.

(18) After a successful query, valid AO comma separated list shall be proposed in the AO related drop down list when preparing a new query.

(19) Changing the **Where** value could change the settings of the other query criteria except the period.

(20) The **Send** button will send the query to the Network Operations system.

(21) If an Aircraft Operator code is part of the selection criteria, the query period will be limited to the value of the system property **CHMI / ATFCM application / Flights / Flight list / Maximum elapse for AO list**.

(22) Otherwise, the period will be limited to the value of the system property **CHMI / ATFCM application / Flights / Flight list / Maximum elapse**.

(23) This window may be closed before any query is issued.

(24) The **Reset** action common functionality is implemented for this function. All query criteria are considered for activating the Reset action.

12.4 Displaying the query results

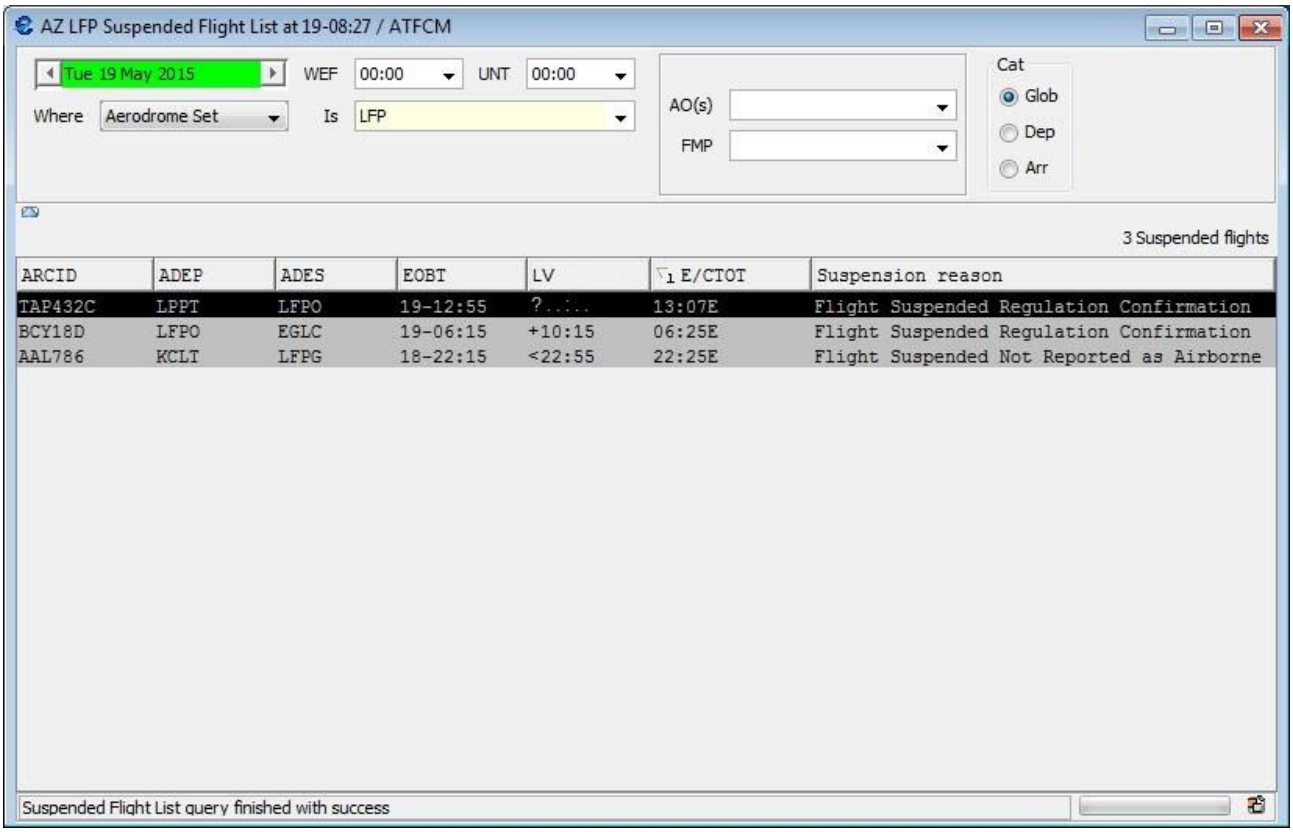


Figure Chmi - 42: Suspended Flight List

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The optional **>>** sign indicates that the **Period Shifting** mechanism will be applied on refresh.

(3) The optional keyword **(Poll)** between parenthesis indicates that the **Polling function** is active for this window.

(4) The header box of the window contains the different **query data elements** used to retrieve the displayed flight data:

a) The observed period (Traffic date, **WEF** and **UNT** times)

b) The main **Where** selection criteria (Traffic Volume, Airspace, Aerodrome or Set with the associated Category, Point with associated Flight Level Range (**FL**), Regulation or Aircraft Operator(s)),

c) The optional selection criteria for the list (Most Penalising **FMP**, Aircraft Operator (**AO(s)**) or **Flow** for a Traffic Volume),

(5) At the top of the Flight List, a counter indicates the **number of suspended flights** retrieved.

(6) This counter may be affected by **filtering** (fff / rrr Flights where fff is the filtered number and rrr is the retrieved number).

(7) On query results display, the focus is set on the list of Suspended Flights after selecting the first flight in the list.

(8) For each flight the **Flight List** display contains following columns:

a) **ARCID**: The aircraft identification.

b) **ADEP**: The ICAO identifier of the Aerodrome of Departure (Only displayed if the list is run for an Aerodrome Set Identifier).

c) **ADES**: The Aerodrome of Destination.

d) **EOBT**: The last received Estimated Off Block Time in the format "DD-HH:MM".

e) **LV**: The Last Valid EOBT acceptable for the flight before triggering IFPS errors

See **LV** column in Flight List for detailed display format, tooltip and ICD source data.

f) **E/CTOT**: Either the Calculated Take Off Time or, if not available, the Estimated Take Off Time. The appended letter indicates if it is the Calculated (**C**) or the Estimated (**E**).

g) The **Suspension reason** (see later).

(9) The flight might be currently suspended for the following reason:

a) **SIT Time Out**

b) **Slot Missed**

c) **FCM Required**

d) **Delay Threshold exceeded, FCM Required**

e) **RVR**

f) **Not Reported as Airborne**

g) **Manual Suspension**

h) **Airport Suspension**

(10) The initial display sort sequence for the Suspended Flight List is **Descending** order on the **E/CTOT** column.

(11) The Suspended Flight List allows single or multiple flight selection:

a) Single selection to invoke contextual actions for one flight.

b) Multiple selections to trigger an action on a group of flights (ex: Plot Selected).

(12) As soon as more than one flight is selected, contextual actions for a single flight are no more available.

(13) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

12.5 User interactions

(1) The standard “Ctrl” or “Shift” key combination shall allow the selection of multiple flights.

(2) When multiple flights are selected, the contextual popup menu shall only display the actions where the identification of a single flight is not mandatory, if any.

(3) The **Suspended Flight List** shall implement the global **Plot All flights function**.

(4) The **Suspended Flight List** shall implement the global **Plot Selected flights function**.

(5) This window allows to modify a previous query selection and to issue a new request.

(6) On query modification, the **Reset action** is enabled.

(7) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(8) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(9) The **Previous / Next period** function may be used to issue the same query but for the previous or the next period in time.

(10) The **Polling function** may be activated to start automatic refresh processing on this window.

(11) The **Period Shifting function** may be activated to start automatic period shifting on subsequent refresh of this window.

(12) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(13) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(14) To modify the **Sort sequence** of the flights, click on column headers corresponding to your desired sort criteria.

(15) To print the Suspended Flight list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(16) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(17) Increasing / reducing the window height or width will impact the size of the main Flight list control.

12.6 Contextual actions

(1) When a line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Flight and may contain the following options:

a) **Flight Data**: displays the details for the selected flight (profiles ...).

b) **Point Profile**: same as Flight Data but activates immediately the Point Profile display when flight details are retrieved.

c) **Airspace Profile**: same as Flight Data but activates immediately the Airspace Profile display when flight details are retrieved.

d) **Flight History**: displays Flight Event History list for one flight.

e) **Operational Log**: lists TACT Operational Log entries for one flight.

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13 Flight Data

13.1 Function presentation

(1) The Flight Data gives the detailed information for one flight.

(2) Four tabs are included on the Flight Data display:

a) **Details**: displays the detailed summary information for the selected flight.

b) **Point Profile**: displays the route steps, the level and the time over the points for the selected flight.

c) **Airspace Profile**: displays the different airspaces intersected by the selected flight.

d) **Restriction Profile**: displays profile information for the different restrictions affecting the selected flight.

13.2 Opening the window

(1) From the Application / ATFCM menu, select either option:

a) **Flights / Flight Data** to open the Flight Data window with the Details tab pre-selected.

b) **Flights / Point Profile** to open the Flight Data window with the Point Profile tab pre-selected.

c) **Flights / Airspace Profile** to open the Flight Data window with the Airspace Profile tab pre-selected.

(2) Immediate shortcut to bypass **Flights / Flight Data** menu selection: **<Ctrl>** + **<Shift>** + **<F11>**.

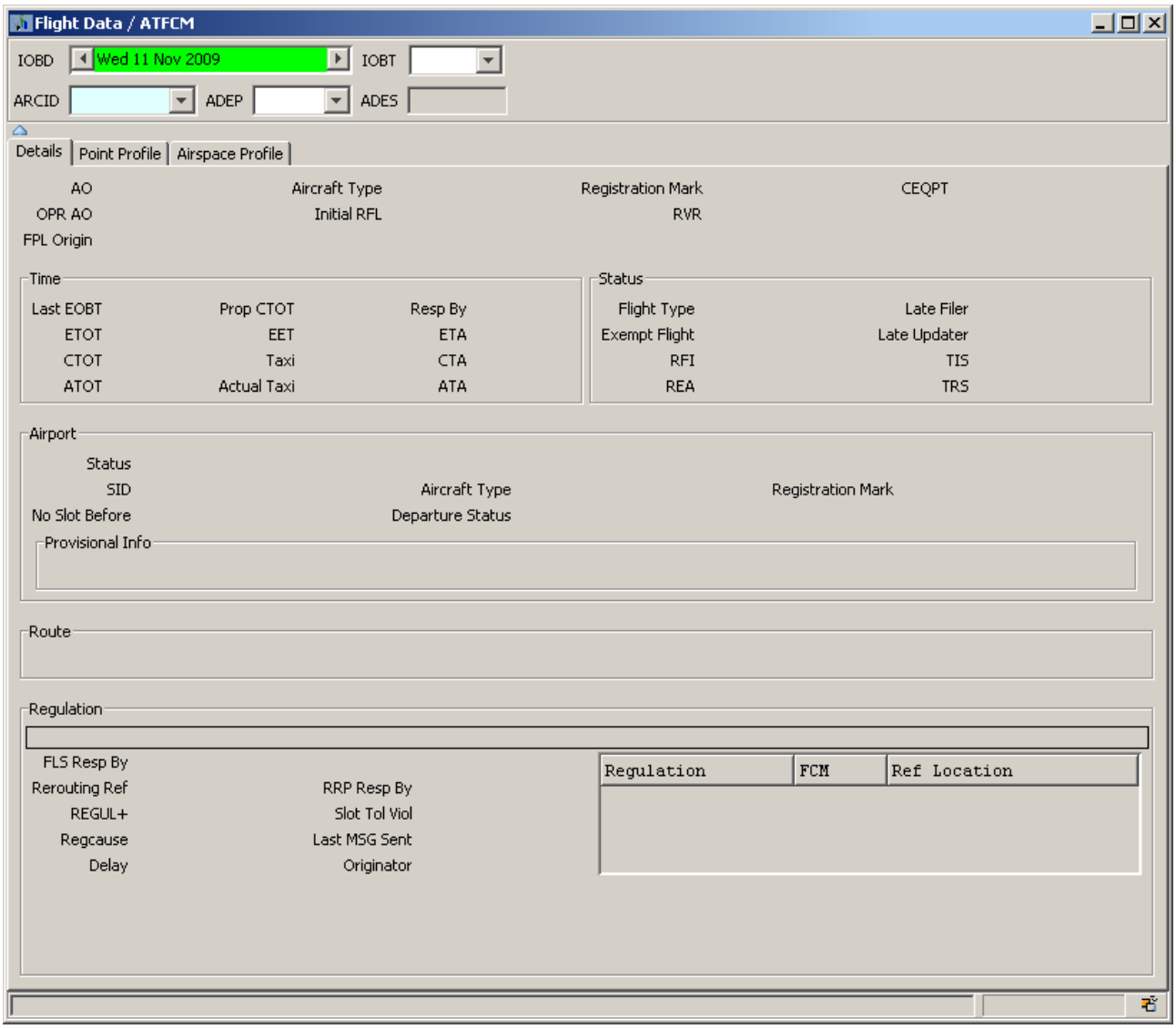


Figure Chmi - 43: Query flight data

(3) A new empty **Flight Data** window will be opened with the following defaults:

a) IOBD: current or last queried traffic date is proposed (relative to current UTC date) dependant on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) IOBT: blank. Drop-down list will propose pre-formatted time on a 20 minutes step base.

c) ARCID: the drop-down box will propose the most recent query data elements you have queried before with success.

d) ADEP: the drop-down box will propose the most recent query data elements you have queried before with success.

(4) On initial display, the focus is set on the first mandatory query control with no default value: the **ARCID** input control

13.3 Preparing to query

(1) The Initial Estimated Off Block Date (**IOBD**) may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The Initial Estimated Off Block Time (**IOBT**) input box may be adjusted to a valid time value from 00:00 to 23:59 included. This time does not have to be exact. The system will allow some flexibility to identify the precise flight you want to query. This is an optional query data element.

(3) The Aircraft Identifier (**ARCID**) shall be filled in or selected from the drop-down list (when a previous query was successful). It is the only mandatory query data element.

(4) The Aerodrome of Departure (**ADEP**) may be filled in with an Aerodrome ICAO code or selected from the drop-down list (when a previous query was successful). This is an optional query data element.

(5) In some circumstances the ARCID identifier is not sufficient to identify a unique flight. The Initial Off Block Time and/or the Aerodrome of Departure are then necessary.

(6) Select the **Proposal** radio button on the button bar to check for a proposed flight before the normal one. Otherwise only the normal flight will be returned.

(7) The **Send** button will send the query to the Network Operations system.

(8) This window may be closed before any query is issued.

(9) The **Reset** action common functionality is implemented for this function. All query criteria except Normal/Proposal selection are considered for activating the Reset action.

13.4 Displaying the query results

13.4.1 Header section



Figure Chmi - 44: Flight data - Header section display

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) Presence of the keyword **(Prop)** in the title indicates that the displayed data might be a "Proposal" flight.

(3) The header section of the window contains the different **query data elements** used to retrieve the displayed flight data:

a) IOBD: Initial Estimated Off Block Date.

b) IOBT: Initial Estimated Off Block Time. Possibly adjusted to its correct value based on the query results data.

c) ARCID: Aircraft Identifier.

d) ADEP: Aerodrome of Departure.

(4) On successful reply, the value received for the Aerodrome of Destination of the highest available model (**ADES**), the filed Aerodrome of Destination **(Filed ADES)** and other optional fields not provided for the query, will be inserted (but protected) next to the input controls.

(5) The controls (static text and value) related to **Filed ADES** are only visible if the flight is diverted and there is a different aerodrome identifier to be displayed.

(6) If there exists more than one flight result for the specified query criteria, then data can be displayed for only one flight (i.e. the first flight when a list of results would be returned).

(7) If that is the case, then a warning alert message will pop up, inviting the user to make a choice of following options to identify the correct flight:

a) Provide all query elements.

b) Use the ARCID List first and select the Flight Data from the list.

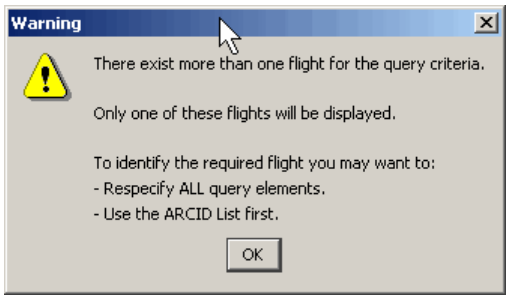


Figure Chmi - 45: Multiple flight warning

13.4.2 Details

13.4.2.1 Introduction

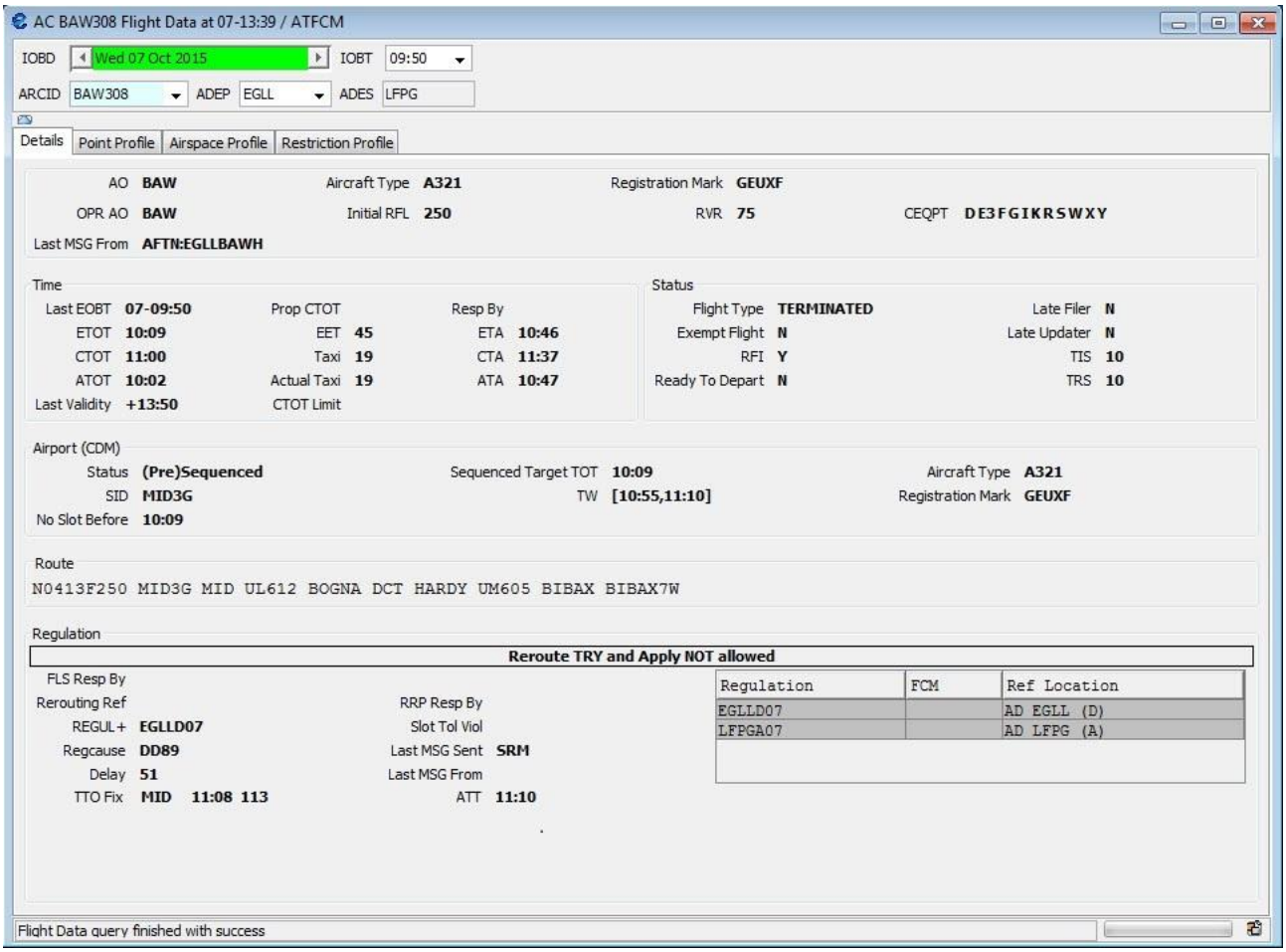


Figure Chmi - 46: Flight data - Details display

(1) This display provides an overview of the major details for the selected flight.

(2) The window title and the query header section are the same as the one described earlier in the **Header** section.

13.4.2.2 General Flight Information

(1) The first section, below the query data section, contains **General Flight Information**:

a) The Aircraft Operator (**AO**)

b) The Operating Aircraft Operator (**OPR AO**)

c) The ICAO identifier of the **Aircraft Type**

d) The Initial Requested Flight Level (Initial **RFL**)

e) The **Registration Mark**

f) The Runway Visual Range (**RVR**)

g) The list of Radio communication, navigation and approach aid equipment types as provided in the flight plan **(CEQPT)**

h) The Originator of the last FPL related message (**Last MSG From**) with the following associated tooltip text: **Originator of the Last FPL Related Message**

i) **CCAMS Code**: SSR code assigned by CCAMS to a flight.
Only showed while the code is active for that flight (status selected).

(2) In case of Proposal flight, the box around the **General Flight Information** section displays the keyword **Proposal** as the title in its upper left corner.

(3) The possible equipment types are:

a) **A** for GBAS

b) **B** for LPV

c) **C** for LORAN C

d) **D** for DME

e) **E1** for FMC WPR ACARS

f) **E2** for D-FIS ACARS

g) **E3** for PDC ACARS

h) **F** for ADF

i) **G** for GNSS

j) **H** for HF RTF

k) **I** for Inertial Navigation

l) **J1** CPDLC ATN VDL Mode 2

m) **J2** CPDLC FANS 1/A HFDL

n) **J3** CPDLC FANS 1/A VDL Mode A

o) **J4** CPDLC FANS 1/A VDL Mode 2

p) **J5** CPDLC FANS 1/A SATCOM (INMARSAT)

q) **J6** CPDLC FANS 1/A SATCOM (MTSAT)

r) **J7** CPDLC FANS 1/A SATCOM (IRIDIUM)

s) **K** for MLS

t) **L** for ILS

u) **M1** for ATC RTF SATCOM (INMARSAT)

v) **M2** for ATC RTF SATCOM (MTSAT)

w) **M3** for ATC RTF SATCOM (IRIDIUM)

x) **N** for No COM/NAV/Approach aid equipment is carried or equipment is U/S.

y) **O** for VOR

z) **P1** Reserved for RCP

aa) **P2** Reserved for RCP

bb) **P3** Reserved for RCP

cc) **P4** Reserved for RCP

dd) **P5** Reserved for RCP

ee) **P6** Reserved for RCP

ff) **P7** Reserved for RCP

gg) **P8** Reserved for RCP

hh) **P9** Reserved for RCP

ii) **R** for RNP type certification; PBN Approved

jj) **S** for Standard COM/NAV/Approach equipment

kk) **T** for TACAN

ll) **U** for UHF RTF

mm) **V** for VHF RTF

nn) **W** for RVSM

oo) **X** for NAT MNPS Airspace

pp) **Y** for 8.33 kHz

qq) **Z** for Other

(4) **S** comprises **V**, **F****, **O** and **L** unless otherwise prescribed by ATS authority.

13.4.2.3 Time

(1) The **Time** related information are grouped in the second section:

a) The Last Estimated Off Block Date and Time (**Last EOBT**).

b) The possible Proposed Calculated Take Off Time (**Prop CTOT**) with the latest time to reply (**Resp By**).

c) The Estimated Take Off Time (**ETOT**).

d) The Calculated Take Off Time (**CTOT**).

e) The Actual Take Off Time (**ATOT**).

f) The Estimated Elapse Time of the flight (**EET**).

g) The **Taxi** time. This value is used by the Network Operations system to calculate the possible slot timing.

h) The **Actual Taxi** time. This value is the latest value identified by the Network Operations system to calculate the Actual profile.

i) The Estimated Time of Arrival (**ETA**).

j) The Calculated Time of Arrival (**CTA**).

k) The Actual Time of Arrival (**ATA**).

l) The **Last Validity**. This is the Last Valid EOBT acceptable for the flight before triggering IFPS errors.

See LV column in Flight List for detailed display format, tooltip and ICD source data.

m) The **CTOT Limit**. This is the possible exceptional reason that may affect the CTOT allocation of the flight, if any. Possible values are:

i) Forced by Tower

ii) Forced by NMOC

iii) Was Forced by NMOC

iv) Slot extension

v) Forced by CHAMAN

vi) Forced by STAM

vii) Limited by Violation

viii) Limited by Violation / Zero Rate / RVR
(i.e. Limited by Violation then Zero Rate or RVR)

ix) " "
(i.e. Blank) when no special CTOT Limit to report

13.4.2.4 Status

(1) The third section contains some flight **Status** information:

a) The **Flight Type** (see later).

b) If the flight is a **Late Filer**.

c) If the flight is a Late **Updater**.

d) If the flight is exempted **(Exempt Flight)**.

e) If the flight is in Request For Improvement mode (**RFI Y**). or SIP Wanted Message mode (**RFI N**).

f) If the flight is reported as **Ready to Depart Y** or **N** (REA message received or set by DPI pre-departure processing (T-DPI_s)).

g) The necessary Time to Insert in Sequence (**TIS**).

h) The necessary Time to Remove from Sequence (**TRS**).

(2) The possible **Flight Types** are :

a) ATC Activated

b) IFPL

c) PFD

d) MFD

e) RPL

f) TACT Activated

g) Terminated

13.4.2.5 Airport

(1) The fourth section contains **Airport** CDM information:

a) The **Airport** section title is appended with the Departure Airport Type: "Airport (CDM)", "Airport (Advanced ATC/TWR)" or "Airport (Standard)".

b) The CDM **Status** when it has one of the following values: "Predicted", "Estimated", "Targeted", "(Pre)Sequenced" or "Actual Off-Block".

c) The **Target Take-Off Times** when they have a defined value: "Turnaround TTOT hh:mm", "Earliest TTOT hh:mm", "Consolidated TTOT hh:mm" and/or "ATC TTOT hh:mm"

d) The ICAO identifier of the **Aircraft Type**, if any. It is appended with the warning symbol **(!)** and both are displayed in **CDM Discrepancy colour** if different from the filed aircraft type.

e) The ICAO designator of the departure terminal procedure (**SID**), if any.

f) The Slot Tolerance Window for a regulated flight or the Departure Tolerance Window for a non regulated flight (**TW**).

i) The **TW** label and value shall only be displayed when authorized by the user profile. A single common use case shall grant the display of the Tolerance Window data in all relevant functionalities (Flight List, Flight Data &).

ii) The Lower and Upper time limit of the Tolerance Window are displayed using the following format: [*low-hh:mm,upper-hh:mm*] (ex: "*TW [10:20,10:40]*"). The complete string is displayed in *italic* when the values are defined as an extended tolerance window.

iii) Following tooltip shall be attached to the TW label: **Slot Tolerance Window for Regulated Flights or Departure Tolerance Window for Non-Regulated Flights.**

g) The **Registration Mark**, if any. It is appended with the warning symbol **(!)** and both are displayed in **CDM Discrepancy colour** if different from the filed registration mark.

h) The Target Off-Block Time (**TOBT** hh:mm), if any. When *Discrepancy with Airport Off Block Time* is detected, the *hh:mm* time is appended with the warning symbol **(!)** and both are displayed in **CDM Discrepancy colour**.

i) The Target Start-up Approval Time (**TSAT** hh:mm), if any.

j) The **Departure Status**, if not Ok.

k) The reason associated to the last received Cancel-DPI message (**C-DPI Reason**), if any**.**

(2) The colour used for discrepancy related information may be adjusted in the Preferences dialog: **CHMI / ATFCM application / Flights / General / CDM Discrepancy colour**.

13.4.2.6 Route

(1) The fifth section contains **Route** related information. Route description is in the format of ICAO Field 15 as output by IFPS to NM external addresses concerned with the flight.

13.4.2.7 Regulation

(1) The sixth section contains Rerouting and **Regulation** information:

a) The Rerouting indicator and AOWIR possibilities (see later).

b) The time limit for the Confirmation message (FCM) after the issuance of a Flight Suspension (**FLS Resp By**).

c) The reference of the Rerouting measure affecting the flight (**Rerouting Ref**).

d) The time limit to reply to the proposed Rerouting (**RRP Resp By**) (visually grouped/aligned with **Rerouting Ref**).

e) The reference of the most penalising Regulation affecting the flight (**REGUL+**).

f) The Slot Tolerance Violation in minutes (ATOT vs CTOT) (**Slot Tol Viol**) (visually grouped/aligned with **REGUL+**).

g) The code of the cause for the Regulation (**Regcause**) (visually grouped/aligned with **REGUL+**).

h) The resulting **Delay** for the flight (visually grouped/aligned with **REGUL+**).

   i) A case sensitive letter may possibly be appended to the delay value to indicate following specific delay characteristics:
      - "D" when the delay value calculated for the flight is exceeding the delay confirmation threshold of a regulation affecting the flight
      - "a" when the delay value of the flight has been adjusted to the clock

   ii) Provisional delays (slot not yet published) are indicated by an asterisk on either side of the value including the possible delay characteristic letter (e.g. *12D*).

i) The Target Time Over Fix Point information (**TTO Fix**).

   i) To preserve readability and intuitive interpretation of this combined TTO Fix information in lists, the following composing three data elements are concatenated using a fixed length format and separated by a blank character:
      - TTO Fix location identification (aerodrome or point id possibly padded with blank characters to the right in order to obtain a fixed length of 5 characters)
      - A blank separator
      - Target time over the TTO Fix location in hh:mm format
      - A blank separator
      - Flight level over the TTO Fix location (three digits numeric value with leading zeros in order to obtain a fixed length of 3 characters ex: 000, 090, 380)

j) The Actual Time at Target (**ATT**) in hh:mm format (visually grouped/aligned with **TTO Fix**) (remark: TTO compliance shall NOT be displayed here).

k) The last message sent (**Last MSG Sent**) or received (**Last MSG Received**) for this flight with the **Originator**, if known (see later).

l) The list of all Regulations affecting the flight (see later).

m) A hidden box at the bottom is made visible when the flight is either currently suspended or had been previously suspended. It will display the **Suspension Reason** (see later).

(2) The types of NM ATFCM Messages that may appear next to **Last MSG Sent** or **Last MSG Received** can be consulted in the ATFCM Handbook part ATFCM Message Examples.

(3) The list of all Regulations affecting the flight contains the following information:

a) The identifier of the **Regulation**.

b) The indication that a Flight Confirmation Message (**FCM**) is still required. In this case, the keyword **Req** is displayed.

c) The related reference location (**Ref Location**).

(4) The Rerouting indicator is composed of a reason and optionally a state.

(5) If the flight has been rerouted, the displayed rerouting reason may be:

a) **ATFCM rerouting**

b) **NM rerouting (CWIR) (i.e. former "CFMU" or "CEU" rerouting)**

c) **Auto rerouting** (AWIR)

d) **AO rerouting** (AOWIR)

(6) When there is a rerouting reason, it is followed by one of these rerouting states:

a) **Produced** for a valid rerouting going on, waiting to be realised by either an FPL or a CHG.

b) **Executed** when the rerouting has been processed.

c) **Timed-out** when no FPL/CHG was received on time.

d) **Rejected** for a rerouting proposal that has been rejected.

e) **Revoked** for a rerouting proposal has been revoked.

f) **No-match** when the message received did not match the proposal. Rerouting has been invalidated.

(7) The "AO What If Reroute" indicator may be one of the following:

a) **Reroute TRY and Apply NOT possible.**

b) **Reroute Try possible.**

c) **Reroute TRY and Apply possible.**

(8) The reroutable indicator grants only access to the initial display of the **Rerouting Function**.

(9) Further detailed analysis of the flight in the **Rerouting Function** might disable some rerouting possibilities.

(10) Possible Suspension Reasons for a flight currently suspended are:

a) **Flight suspended - SIT Time Out**

b) **Flight suspended - Slot Missed**

c) **Flight suspended - FCM Required**

d) **Flight suspended - Delay Threshold exceeded, FCM Required**

e) **Flight suspended - RVR**

f) **Flight suspended - Not Reported as Airborne**

g) **Flight suspended - FP Revalidation**

h) **Flight suspended Manual Suspension**

i) **Flight suspended Airport Suspension**

(11) Possible last Suspension Reasons for a flight that had been previously suspended but in no longer suspended are:

a) **Flight was suspended - SIT Time Out**

b) **Flight was suspended - Slot Missed**

c) **Flight was suspended - FCM Required**

d) **Flight was suspended - Delay Threshold exceeded, FCM Required**

e) **Flight was suspended - RVR**

f) **Flight was suspended - Not Reported as Airborne**

g) **Flight was suspended FP Revalidation**

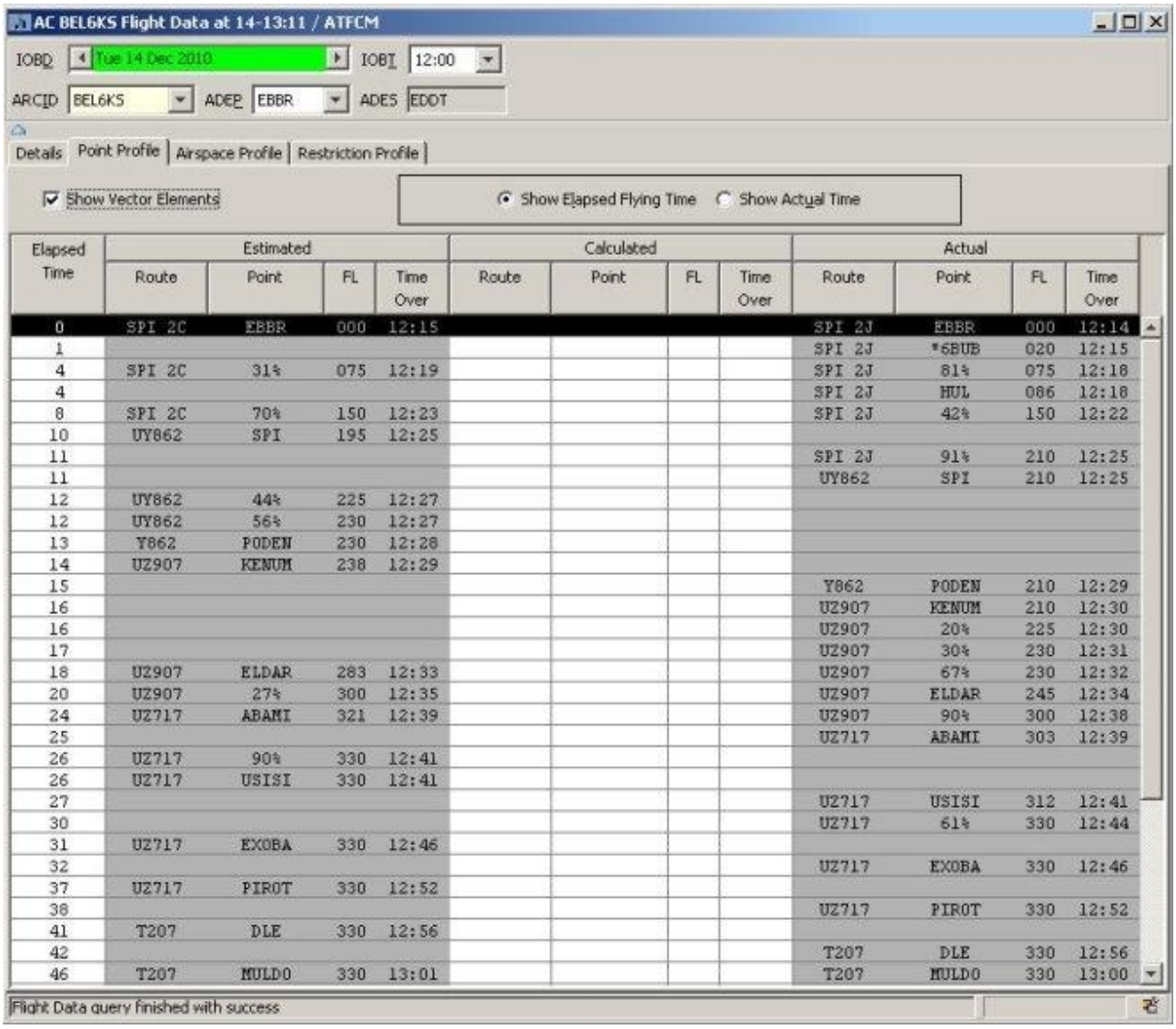
h) **Flight was suspended Manual Suspension**

i) **Flight was suspended Airport Suspension**

(12) On query results display, the focus is set on the regulation list if any present, after selecting the first entry in the list. Otherwise it remains on the query controls.

(13) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

13.4.3 Point Profile



Elapsed Time	Estimated				Calculated				Actual			
	Route	Point	FL	Time Over	Route	Point	FL	Time Over	Route	Point	FL	Time Over
0	SPI 2C	EBBR	000	12:15					SPI 2J	EBBR	000	12:14
1									SPI 2J	*6BUB	020	12:15
4	SPI 2C	31%	075	12:19					SPI 2J	81%	075	12:18
4									SPI 2J	HUL	086	12:18
8	SPI 2C	70%	150	12:23					SPI 2J	42%	150	12:22
10	UY862	SPI	195	12:25								
11									SPI 2J	91%	210	12:25
11									UY862	SPI	210	12:25
12	UY862	44%	225	12:27								
12	UY862	56%	230	12:27								
13	Y862	PODEN	230	12:28								
14	UZ907	KENUM	238	12:29								
15									Y862	PODEN	210	12:29
16									UZ907	KENUM	210	12:30
16									UZ907	20%	225	12:30
17									UZ907	30%	230	12:31
18	UZ907	ELDAR	283	12:33					UZ907	67%	230	12:32
20	UZ907	27%	300	12:35					UZ907	ELDAR	245	12:34
24	UZ717	ABAMI	321	12:39					UZ907	90%	300	12:38
25									UZ717	ABAMI	303	12:39
26	UZ717	90%	330	12:41								
26	UZ717	USISI	330	12:41								
27									UZ717	USISI	312	12:41
30									UZ717	61%	330	12:44
31	UZ717	EXOBA	330	12:46								
32									UZ717	EXOBA	330	12:46
37	UZ717	PIROT	330	12:52								
38									UZ717	PIROT	330	12:52
41	T207	DLE	330	12:56								
42									T207	DLE	330	12:56
46	T207	MULDO	330	13:01					T207	MULDO	330	13:00

Figure Chmi - 48: Flight data - Point Profile display

(1) This display shows the different points that are passed over by the flight.

(2) The window title and the query header section are the same as the one described earlier in the **Header** section.

(3) The Point Profile list includes the following details:

a) Timing information (**Time Over** or **Elapsed Time**).

b) The **Route** identifier (Air Route, Departure / Arrival procedure, ...), sometimes replaced by **DCT** type when flying direct.

c) The **Point** or geographical reference with the **invisibility codification**, if any (see later). This **Point** reference may be a Point or an Aerodrome.

i) In the Calculated profile, the line is highlighted in Italic/Bold when the **Point** reference matches the related reference location (**Ref Location**) of one of the Regulations affecting the Flight.

ii) In the Actual point profile, all Points with distance from the first profile bound on the actual (CTFM) route which is smaller or equal to the distance that has been confirmed by the CPRs and their corresponding lines will be highlighted with a different colour. This colour may be adjusted in the Preferences dialog: **CHMI / ATFCM application / Flights / Flight data / CPR confirmed colour**.

d) The estimated Flight Level (**FL**).

(4) The **invisibility codification** uses the basic rule that a "." (dot) indicates invisibility.

(5) Invisibility means that a change to VFR or OAT is found in the route.

(6) Visibility means that a change to IFR or GAT is found in the route.

(7) Following coding is applied:

a) when the flight becomes invisible after point PPP, the profiles shows "PPP.".

b) when the flight is invisible before and after point PPP, the profiles shows ".PPP.".

c) when the flight becomes visible after point PPP, the profiles shows ".PPP".

d) when the flight is visible before, at and after point PPP, the profiles shows "PPP".

(8) The Vector Elements calculated by the ETFMS system may be displayed in the profile. These Vectors are used to improve system accuracy (plotting, counts...).

(9) When available, the 3 route profiles for the flight are displayed:

a) **ESTIMATED**: profile is based on the flight plan

b) **CALCULATED**: profile is calculated after the ATFCM measures are applied (regulations will modify the times, but also rerouting that may modify the route)

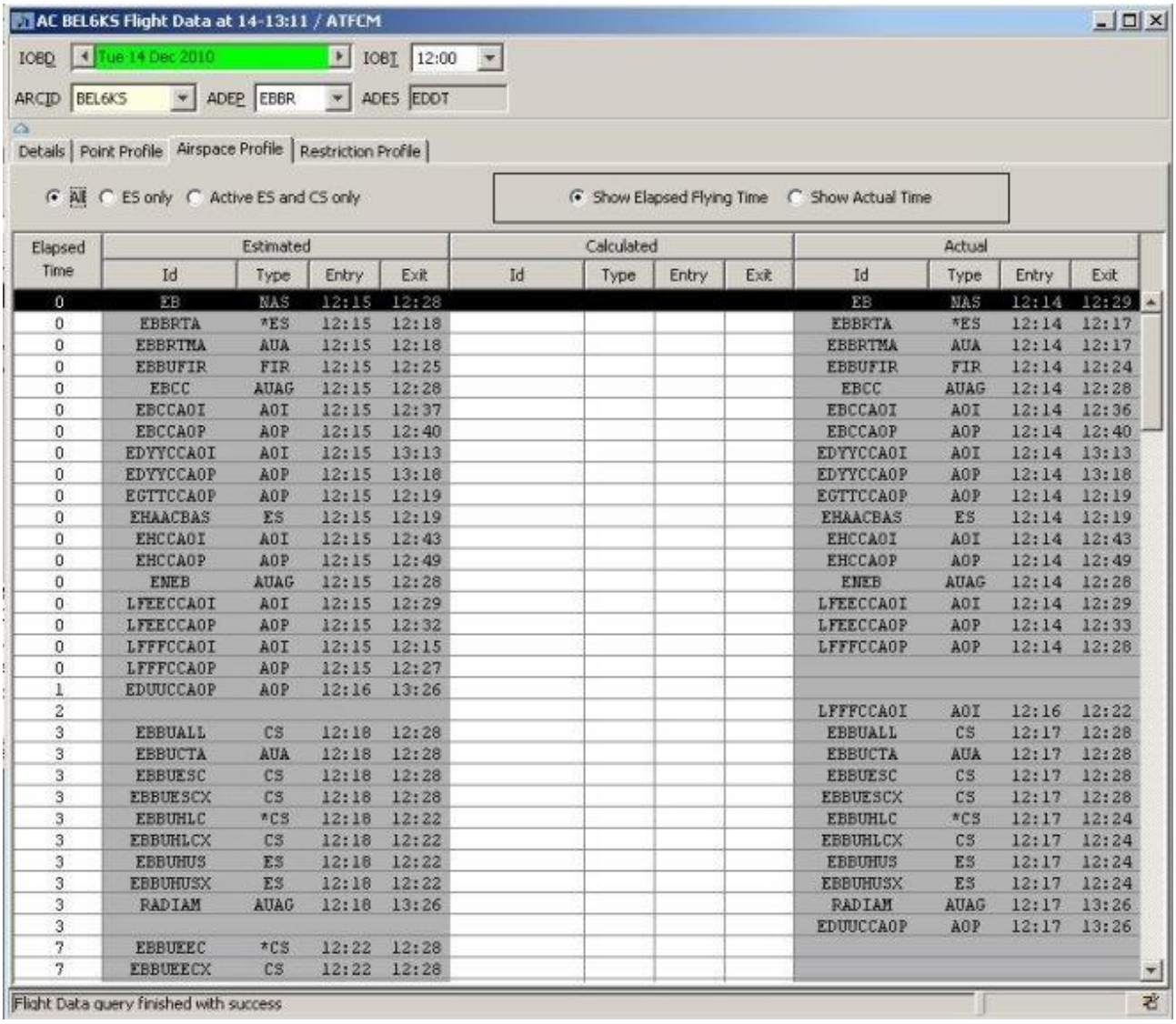
c) **ACTUAL**: profile is calculated according to the effective departure time.

(10) The profiles may be displayed in 2 ways:

a) **Show actual time**: Based on the time over the points which is moved to the 1st column.

b) **Show elapsed flying time**: Based on the elapsed flying time since take off which is moved to the 1st column.

13.4.4 Airspace Profile



Elapsed Time	Estimated				Calculated				Actual			
	Id	Type	Entry	Exit	Id	Type	Entry	Exit	Id	Type	Entry	Exit
0	EB	NAS	12:15	12:28					EB	NAS	12:14	12:29
0	EBBRTA	*ES	12:15	12:18					EBBRTA	*ES	12:14	12:17
0	EBBRTMA	AUA	12:15	12:18					EBBRTMA	AUA	12:14	12:17
0	EBBUFIR	FIR	12:15	12:25					EBBUFIR	FIR	12:14	12:24
0	EBCC	AUAG	12:15	12:28					EBCC	AUAG	12:14	12:28
0	EBCCAOI	AOI	12:15	12:37					EBCCAOI	AOI	12:14	12:36
0	EBCCAOP	AOP	12:15	12:40					EBCCAOP	AOP	12:14	12:40
0	EDYYCCAOI	AOI	12:15	13:13					EDYYCCAOI	AOI	12:14	13:13
0	EDYYCCAOP	AOP	12:15	13:18					EDYYCCAOP	AOP	12:14	13:18
0	EGTTCCAOP	AOP	12:15	12:19					EGTTCCAOP	AOP	12:14	12:19
0	EHAACBAS	ES	12:15	12:19					EHAACBAS	ES	12:14	12:19
0	EHCCAOI	AOI	12:15	12:43					EHCCAOI	AOI	12:14	12:43
0	EHCCAOP	AOP	12:15	12:49					EHCCAOP	AOP	12:14	12:49
0	ENEB	AUAG	12:15	12:28					ENEB	AUAG	12:14	12:28
0	LFEECCAOI	AOI	12:15	12:29					LFEECCAOI	AOI	12:14	12:29
0	LFEECCAOP	AOP	12:15	12:32					LFEECCAOP	AOP	12:14	12:33
0	LFFFCCAOI	AOI	12:15	12:15					LFFFCCAOP	AOP	12:14	12:28
0	LFFFCCAOP	AOP	12:15	12:27								
1	EDUUCCAOP	AOP	12:16	13:26								
2									LFFFCCAOI	AOI	12:16	12:22
3	EBBUALL	CS	12:18	12:28					EBBUALL	CS	12:17	12:28
3	EBBUCTA	AUA	12:18	12:28					EBBUCTA	AUA	12:17	12:28
3	EBBUESC	CS	12:18	12:28					EBBUESC	CS	12:17	12:28
3	EBBUESCX	CS	12:18	12:28					EBBUESCX	CS	12:17	12:28
3	EBBUHLC	*CS	12:18	12:22					EBBUHLC	*CS	12:17	12:24
3	EBBUHLCX	CS	12:18	12:22					EBBUHLCX	CS	12:17	12:24
3	EBBUHUS	ES	12:18	12:22					EBBUHUS	ES	12:17	12:24
3	EBBUHUSX	ES	12:18	12:22					EBBUHUSX	ES	12:17	12:24
3	RADIAM	AUAG	12:18	13:26					RADIAM	AUAG	12:17	13:26
3									EDUUCCAOP	AOP	12:17	13:26
7	EBBUEEC	*CS	12:22	12:28								
7	EBBUEECX	CS	12:22	12:28								

Figure Chmi - 49: Flight data - Airspace profile display

(1) This display shows the different airspaces that are entered by the flight.

(2) The window title and the query header section are the same as the one described earlier in the **Header** section.

(3) The Airspace Profile list includes the following details:

a) Timing information (**Entry Time** or **Elapsed Time**).

b) The Airspace **ID** entered by the flight.

i) In the Calculated profile, the line is highlighted in Italic/Bold when the Airspace **ID** matches the related reference location (**Ref Location**) of one of the Regulations affecting the Flight.

ii) In the Actual airspace profile, all Airspaces with an exit point distance from the first profile bound on the actual (CTFM) route which is smaller or equal to the distance that has been confirmed by the CPRs

and their corresponding lines will be highlighted with a different background colour. If the Airspace entry point is within the CPRs confirmed distance but not yet the Airspace exit point distance, then the corresponding line shall be partially coloured to highlight this fact. This colour may be adjusted in the Preferences dialog : **CHMI / ATFCM application / Flights / Flight data / CPR confirmed colour**).

c) The **Type** of the Airspace (see later).

d) The Time of Entry in the Airspace.

e) The Time of Exit of the Airspace.

(4) The listed airspaces may be of the following different **Type**:

a) **REG**: Region

b) **FIR**: Flight Information Region

c) **AUA**: ATC Unit Airspace

d) **ES**: Elementary Sector

e) **CS**: Collapsed Sector

f) **RSA**: Restricted Airspace

g) **CDA**: Client Defined Airspace

h) **AUAG**: ATC Unit Airspace Group

i) **NAS**: National Airspace

j) **AREA**: NM or EUROCONTROL defined area

k) **CLUS** : Cluster

(5) **Activated ES/CS** airspaces at the moment of the flight are indicated by an asterisk in front of the Airspace type.

(6) A group of pre-defined Airspace local filtering offers the following mutually exclusive options:

a) **Display All** allows to display all line entries of the airspace profile (any Airspace Type)

b) **Display ES only** allows to limit display to the Elementary Sector Airspaces only (activated or not)

c) **Active ES/CS only** allows to limit display to the activated Airspaces only

(7) The user adjustable preference **CHMI / ATFCM application / Flights / Flight data / Airspace filter** allows to automatically check the desired filter checkbox on opening a new Airspace Profile window.

a) Enumerated possible values are:

i) **Display All**

ii) **Display ES only**

iii) **Active ES/CS only**

b) Default value is **Display All**

c) Description: “Initial setting of the Airspace filter when opening a new Flight Data window.”

(8) When available, the 3 airspace profiles for the flight are displayed:

a) **ESTIMATED**: profile is based on the flight plan

b) **CALCULATED**: profile is calculated after the ATFCM measures are applied (regulations will modify the times, but also rerouting that may modify the route)

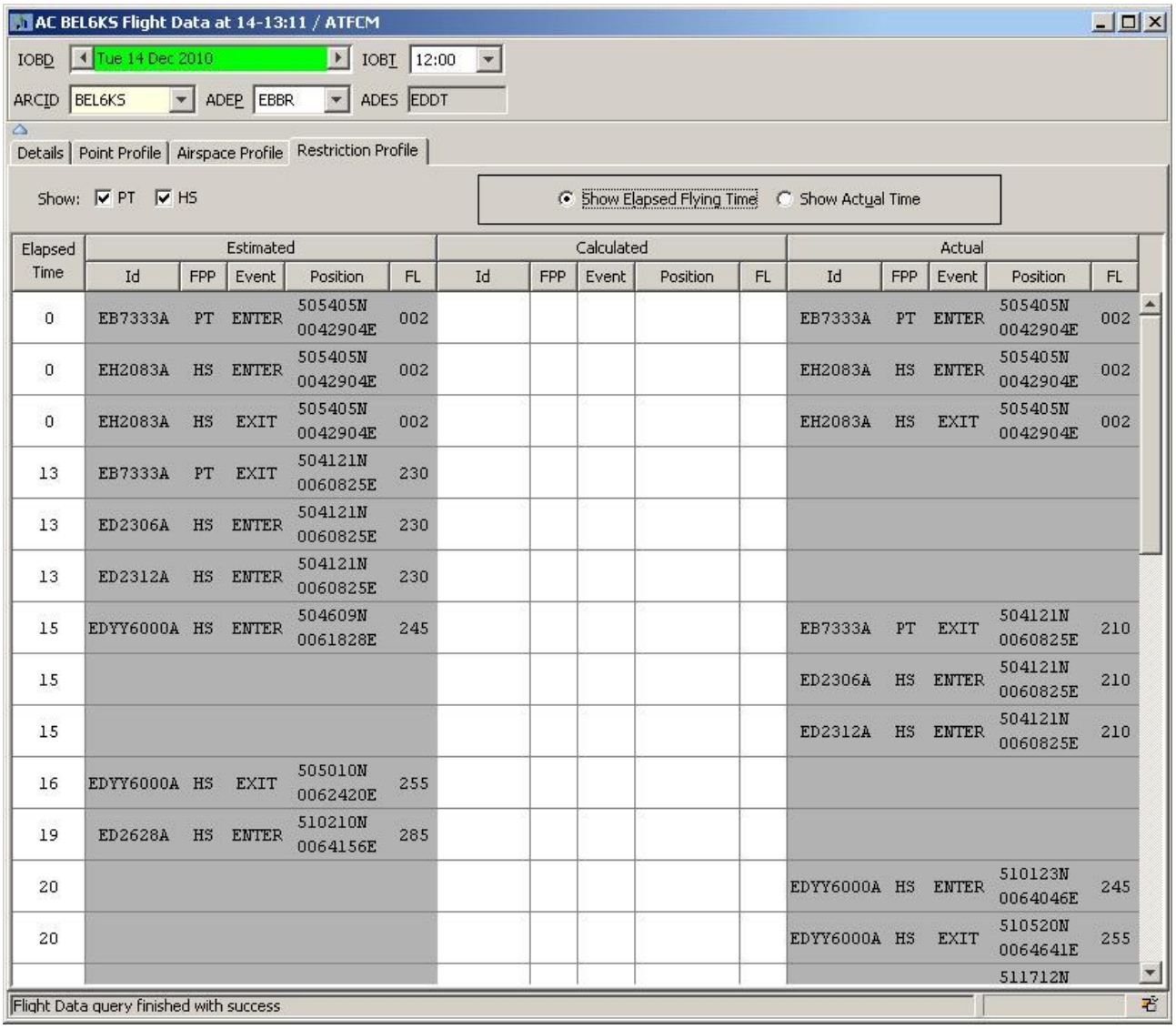
c) **ACTUAL**: profile is calculated according to the effective departure time.

(9) The profiles may be displayed in 2 ways:

a) **Show actual time**: Based on the entry time in the Airspace which is displayed in the 1st column.

b) **Show elapsed flying time**: Based on the elapsed flying time (since take off) when entering the Airspace which is displayed in the 1st column.

13.4.5 Restriction profile



Elapsed Time	Estimated					Calculated					Actual				
	Id	FPP	Event	Position	FL	Id	FPP	Event	Position	FL	Id	FPP	Event	Position	FL
0	EB7333A	PT	ENTER	505405N 0042904E	002						EB7333A	PT	ENTER	505405N 0042904E	002
0	EH2083A	HS	ENTER	505405N 0042904E	002						EH2083A	HS	ENTER	505405N 0042904E	002
0	EH2083A	HS	EXIT	505405N 0042904E	002						EH2083A	HS	EXIT	505405N 0042904E	002
13	EB7333A	PT	EXIT	504121N 0060825E	230										
13	ED2306A	HS	ENTER	504121N 0060825E	230										
13	ED2312A	HS	ENTER	504121N 0060825E	230										
15	EDYY6000A	HS	ENTER	504609N 0061828E	245						EB7333A	PT	EXIT	504121N 0060825E	210
15											ED2306A	HS	ENTER	504121N 0060825E	210
15											ED2312A	HS	ENTER	504121N 0060825E	210
16	EDYY6000A	HS	EXIT	505010N 0062420E	255										
19	ED2628A	HS	ENTER	510210N 0064156E	285										
20											EDYY6000A	HS	ENTER	510123N 0064046E	245
20											EDYY6000A	HS	EXIT	510520N 0064641E	255
														511712N	

Figure Chmi - 50: Flight data - Restriction profile display

(1) The **Restriction Profile** tab of the Flight Data display shows entry and exit information for the different restrictions that affects the flight profile, if any.

(2) The display of the **Restriction Profile** tab shall be subordinated to authorisation via security profile.

(3) The window title and the query header section are the same as the one described earlier in the **Header** section.

(4) The main list of the **Restriction Profile** tab is composed of 4 main columns:

a) Timing information (**Time Over** or **Elapsed Time**)

b) **Estimated**: profile is based on the flight plan

c) **Calculated**: profile is calculated after the ATFCM measures are applied (regulations will modify the times, but also rerouting that may modify the route)

d) **Actual**: profile is calculated according to the effective departure time.

(5) The timing information shall contain the list of times when entering or exiting a restriction:

a) Either displayed in hours and minutes for **Time Over** (hh:mm)

b) Or displayed as a number of minutes for **Elapsed Time**

(6) Each of the other main columns of the **Restriction** profile list (**Estimated**, **Calculated** and **Actual**) shall be composed of the following sub-column details, displaying restriction information for the time identified in the first column:

a) The identifier (**Id**) of the restriction

b) The Flight Plan Processing action type (**FPP +** tooltip on header: **Flight Plan Processing action type**) associated to the Restriction. (see later)

c) The type of **Event**:

i) **Entry** for the time when the flight is entering the Restriction area

ii) **Exit** for the time when the flight is exiting the Restriction area

d) The coordinate **Position** for the corresponding **Event**

e) The flight **Level** at the corresponding **Position**

f) The complementary time value of the first global column: **Elapsed Time** or **Time Over** (i.e. when the first column displays **Time Over,** this column displays **Elapsed Time** and vice versa).

(7) The listed restrictions may be of the following Flight Plan Processing action type (**FPP**):

a) **HS** for Hard/Soft Traffic Flow Restriction

b) **AF** for Aerodrome Flight Rule

c) **PT** for Profile Tuning Restriction

d) **DCT** for DCT Limitation Restriction

e) **AT** for Aircraft Type Restriction on Terminal Procedure

f) **SSR** for SSR Code Allocation

g) **FRA** for FRA DCT Restriction (Free Route Airspace)

(8) A **Flight Plan Processing** filter group box shall be displayed at the top of the **Restriction** list:

a) It shall contain a checkbox for each different **Flight Plan Processing** action type present in the list.

b) Each checkbox shall display the full text of the corresponding **Flight Plan Processing** action type appended with the abbreviated code between parentheses.

c) Those checkboxes shall allow filtering the displayed list of restrictions.

i) As initial default, only Profile Tuning Restriction checkbox shall be checked.

ii) After initial display, the latest filter setup shall be saved as a system property **CHMI / ATFCM application / Flights / Flight Data / Last Used FPP Filter**.

d) The **Restriction** list shall only display restrictions whose Flight Plan Processing action type is the same as a checked Flight Plan Processing action type checkbox, i.e. if a checkbox is checked, restrictions with the corresponding Flight Plan Processing action type are displayed in the list; if a checkbox is unchecked, restrictions with the corresponding Flight Plan Processing action type are removed from the list.

(9) Two exclusive options allow displaying the restriction profiles in 2 different ways:

a) **Show actual time**: Based on the **Time Over** which is then displayed in the 1st column.

b) **Show elapsed flying time**: Based on the elapsed flying time since take off which is then displayed in the 1st column (**Elapsed Time**).

13.5 User interactions

(1) The **Normal / Proposal** button may be adjusted before issuing a new query.

(2) This window allows to modify a previous query selection and to issue a new request.

(3) On query modification, the **Reset action** is enabled.

(4) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(5) To display one of the four detail Flight Data display, click on the desired tab :

a) **Details**: displays the detailed summary information for the selected flight.

b) **Point Profile**: displays the route steps, the level and the time over the points for the selected flight.

c) **Airspace Profile**: displays the different airspaces crossed by the selected flight.

d) **Restriction Profile**: displays the different restrictions affecting the profile of the selected flight.

(6) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(7) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(8) On the Point Profile tab, it is possible to show or hide the Vector Elements in the displayed profile. This is done by checking or unchecking the **Show Vector Elements** checkbox.

(9) To force the initial display value of this **Show Vector Elements** checkbox, just check or uncheck the user preference **CHMI / ATFCM application / Flights / Flight data / Show Vector Elements**.

(10) On the Airspace Profile tab, it is possible to:

a) **Display All** crossed airspaces (any type) identified in the Airspace profile

b) **Display ES only**, to limit display to the Elementary Sector Airspaces only

c) **Active ES/CS only**, to limit display to the activated Airspaces only

(11) This is done by selecting the corresponding Airspace filter radio button.

(12) To force the initial selected value of this Airspace filter checkbox, adjust the corresponding user preference **CHMI / ATFCM application / Flights / Flight data / Airspace filter** to the desired value.

(13) To print the Flight Data details following scenario must be followed:

a) First retrieve the data in the window

b) When ready, click on the **Print** button or select the **File / Print** menu option or select the **File / Print Preview** menu option.

c) A dialog box is displayed to facilitate selective choice of flight data to be printed.

d) The following info can be marked as required to be printed:

i) **Details**: the detailed summary information for the flight can be requested for printing

ii) **Point Profile**: the 3 point profile models (**ESTIMATED, CALCULATED and ACTUAL**) can be requested for printing

iii) **Airspace Profile**: the 3 airspace profile models (**ESTIMATED, CALCULATED and ACTUAL**) can be requested for printing

iv) **Restriction Profile**: the 3 restriction profile models (**ESTIMATED, CALCULATED and ACTUAL**) can be requested for printing

e) A checkbox "ALL" on the dialog box allows to request to print all flight data. When nothing is selected the print button is disabled.

(14) These preferences indicating the kind of flight data desired on print functionality are stored locally in the preference system on the PC.

(15) Those stored values will be used for any user on the same PC to display a default choice on the dialog box if the print or print preview is requested for a flight from within the flight data screen.

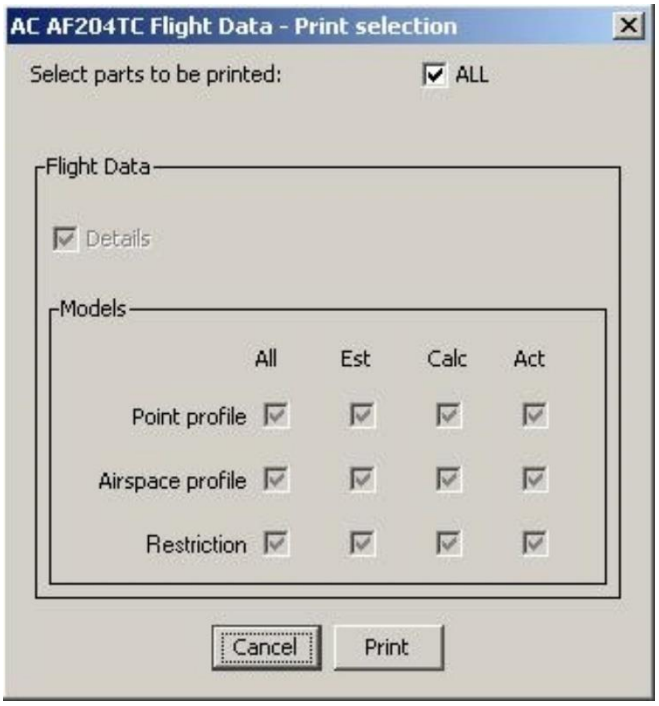


Figure Chmi - 51: Flight data – Print selection dialog

(16) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(17) To plot (Horizontal or Vertical) the displayed details of the flight, click on the **Plot** button(s) or select the **Action / Plot** menu.

(18) In order to allow the Map to highlight geographically restriction profile information of a flight, the data structure passed to the Map on a plot action must also contain the related restriction profile list, if user is authorised to display it.

(19) The Horizontal View will display the Restriction Entry / Exit locations.

(20) The Vertical View will display the Restriction Entry / Exit locations and highlight restricted areas.

(21) In order to allow the Map to highlight geographically on the Horizontal and Vertical view the regulated segments of a flight, the data structure passed to the

Map on a plot action must also contain the list of all Regulations affecting the Flight. For each regulation following info is passed:

a) The identifier of the **Regulation**

b) The related reference location (**Ref Location**)

c) The category attached to Aerodrome and Aerodrome Set (Global, Departure, Arrival)

d) The indication that a Flight Confirmation Message (**FCM**) is still required. In this case, the keyword **Req** is displayed

(22) In order to allow the Map to highlight geographically on the Vertical view the requested Flight Levels, the data structure passed to the Map on a plot Vertical View action must contain the array of requested Flight Levels.

(23) On the Vertical view the requested flight levels are indicated as a black thick horizontal line.

13.6 Contextual actions

(1) When Flight Data details are displayed on any tab, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the displayed Flight and may contain the following options:

a) **Flight History**: displays Flight Event History list for one flight.

b) **Operational Log**: lists TACT Operational Log entries for one flight.

c) **Departure Aerodrome Delays**: displays the delays for the departure aerodrome of the flight.

d) **Rerouting Function**: allows to evaluate and request the rerouting for the selected flight.

e) If the flight is not reroutable, the option **Alternate Routes**: allows to list the different route alternative for the selected flight.

f) ***xxx* Regulation Delays**: displays the delays due to the regulation ***xxx***, if the selected flight is regulated. Where ***xxx*** is the identifier of the regulation currently selected in the list of all Regulations affecting the flight of the Details tab.

g) ***xxx* Regulation Description**: displays detailed information for the regulation ***xxx***, if the selected flight is regulated. Where ***xxx*** is the identifier of the regulation currently selected in the list of all Regulations affecting the flight of the Details tab.

h) **Ready To Depart**: sends an REA message for the selected flight (see conditions in related dialog section).

i) **Request For direct Improvement**: sends an RFI message for the selected flight (see conditions in related dialog section).

j) **SIP Wanted Message**: sends an SWM message for the selected flight (see conditions in related dialog section).

k) **Slot Proposal Accept**: sends an SPA message for the selected flight (see conditions in related dialog section).

l) **Slot Proposal Reject**: sends an SRJ message for the selected flight (see conditions in related dialog section).

m) **Rerouting Reject**: sends an RJT message for the selected flight (see conditions in related dialog section).

n) **Slot Missed Message**: sends an SMM message for the selected flight (see conditions in related dialog section).

o) **Flight Confirmation Message**: sends an FCM message for the flight (see conditions in related dialog section).

p) When the right mouse click is performed on a **Restriction ID** of the Restriction Profile tab, the popup menu may contain the option **Show Detail** to display the CHMI ENV detailed description screen for the selected **Restriction**.

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14 Flight History

14.1 Function presentation

(1) The Flight History window displays the Flight Event History list for one flight.

14.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Flights / Flight History**.

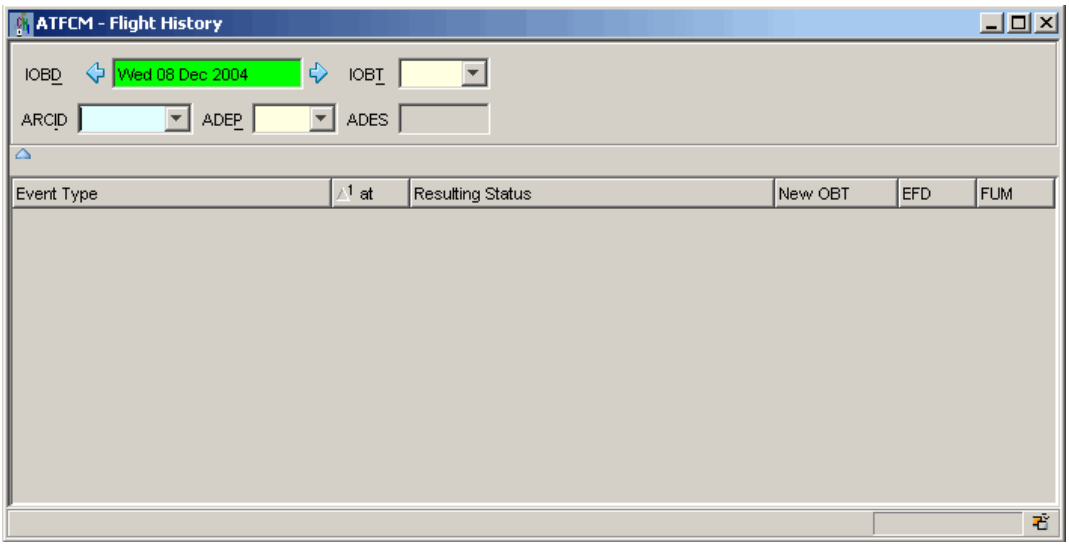


Figure Chmi - 52: Query Flight History

(2) A new empty **Flight History** window will be opened with the following defaults:

a) IOBD: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) IOBT: blank. Drop-down list will propose pre-formatted time on a 20 minutes step base.

c) ARCID: the drop-down box will propose the most recent query data elements you have queried before with success.

d) ADEP: the drop-down box will propose the most recent query data elements you have queried before with success.

(3) On initial display, the focus is set on the first mandatory query control with no default value: the **ARCID** input control.

14.3 Preparing to query

(1) The Initial Estimated Off Block Date (**IOBD**) may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The Initial Estimated Off Block Time (**IOBT**) input box may be adjusted to a valid time value from 00:00 to 23:59 included. This time does not have to be exact. The system will allow some flexibility to identify the precise flight you want to query.

(3) The Aircraft Identifier (**ARCID**) shall be filled in or selected from the drop-down list (when a previous query was successful).

(4) The Aerodrome of Departure (**(ADEP)**) may be filled in with an Aerodrome ICAO code or selected from the drop-down list (when a previous query was successful).

(5) All these query data elements are mandatory in order to identify a unique flight.

(6) Select the **Proposal** radio button on the button bar to check for a proposed flight before the normal one. Otherwise only the normal flight will be returned.

(7) The **Send** button will send the query to the Network Operations system.

(8) This window may be closed before any query is issued.

(9) The **Reset** action common functionality is implemented for this function. All query criteria except Normal/Proposal selection are considered for activating the Reset action.

14.4 Displaying the query results

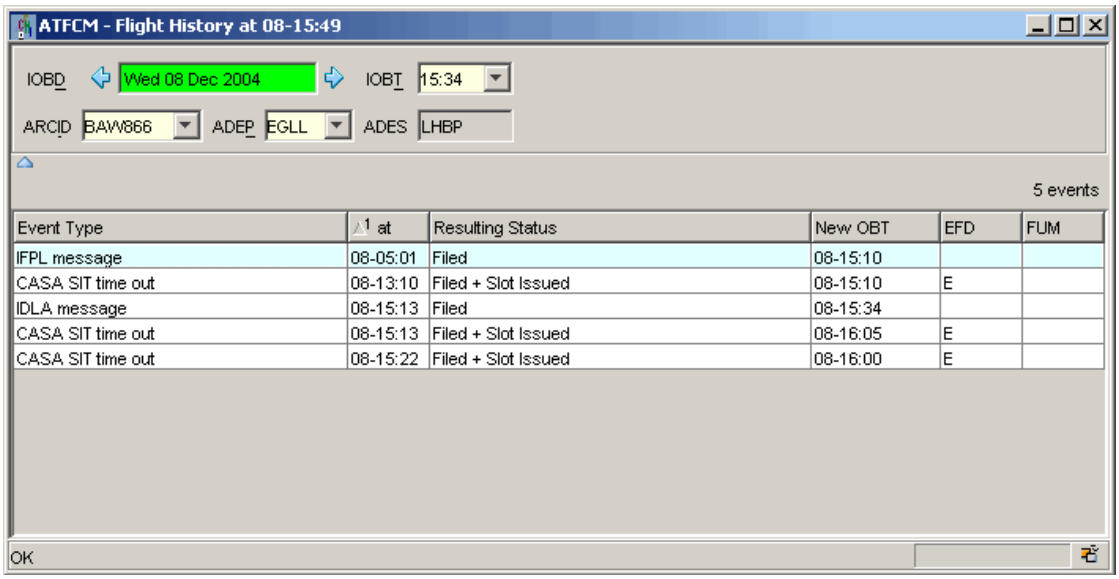


Figure Chmi - 53: Flight History

(1) This display shows the list of TACT Flight Event entries for the flight that corresponds to the selected query crieria.

(2) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(3) Presence of the keyword **(Prop)** in the title indicates that the displayed data might be a "Proposal" flight.

(4) The header section of the window contains the different **query data elements** used to retrieve the displayed flight History:

a) IOBD: Initial Estimated Off Block Date.

b) IOBT: Initial Estimated Off Block Time. Possibly adjusted to its correct value based on the reply data.

c) ARCID: Aircraft Identifier.

d) ADEP: Aerodrome of Departure.

(5) The Aerodrome of Destination (**ADES**) is displayed in the header section when the reply is received from the Network Operations system.

(6) On query results display, the focus is set on the event list after selecting the first entry in the list.

(7) At the top of the event history list, a counter indicates the **number of events** retrieved.

(8) This counter may be affected by **filtering** (fff / rrr events where fff is the filtered number and rrr is the retrieved number).

(9) The list shows for each entry:

a) The **Event Type**.

b) The time when that event was recorded (**at**).

c) The **Resulting Status** after the event.

d) The new Off Block Time after the event (**New OBT**).

e) "**E**" is displayed if an ETFMS Flight Distribution Message (**EFD**) has been sent, otherwise left blank.

f) "**F**" is displayed if a Flight Update Message (**FUM**) has been sent, otherwise left blank.

(10) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

14.5 User interactions

(1) The **Normal / Proposal** button may be adjusted before issuing a new query.

(2) This window allows to modify a previous query selection and to issue a new request.

(3) On query modification, the **Reset action** is enabled.

(4) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(5) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(6) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(7) To print the Event History list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(8) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(9) To modify the **Sort sequence** of the events, click on column headers corresponding to your desired sort criteria.

(10) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(11) Increasing / reducing the window height and width will impact the size of the main Event list control.

15 Flight Operational Log

15.1 Function presentation

(1) The Flight Operational Log displays Operational Log entries for one flight.

15.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Flights / Operational Log**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl>** + **<Shift>** + **<F12>**.

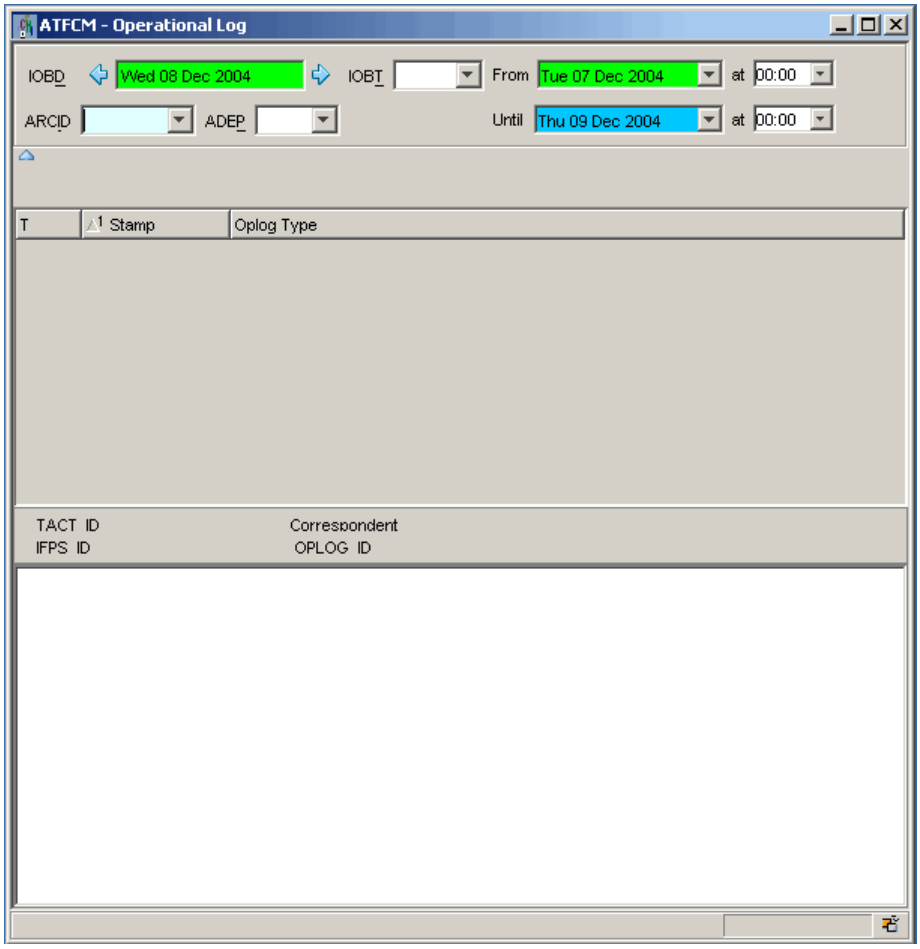


Figure Chmi - 54: Query Flight Operational Log

(3) A new empty flight **Operational Log** window will be opened with the following defaults:

a) IOBD: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) IOBT: blank. Drop-down list will propose pre-formatted time on a 20 minutes step base.

c) ARCID: the drop-down box will propose the most recent query data elements you have queried before with success.

d) ADEP: the drop-down box will propose the most recent query data elements you have queried before with success.

e) From / at: **From** date control will be initialised with the date before current UTC date. Related **at** time control will be intialised with 00:00. The date drop-down list will propose the same dates as in the IOBD list plus one day before. This is needed to query early history of the flight in the Operational Log.

f) Until / at: **Until** date control will be initialised with the date after current UTC date. Related **at** time control will be intialised with 00:00. The date drop-down list will propose the same dates as in the IOBD list plus one day before.

(4) On initial display, the focus is set on the first mandatory query control with no default value: the **ARCID** input control.

15.3 Preparing to query

(1) The Initial Estimated Off Block Date (**IOBD**) may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) Adjusting the IOBD control has no impact on the From and Until date controls.

(3) The Initial Estimated Off Block Time (**IOBT**) input box may be adjusted to a valid time value from 00:00 to 23:59 included.

(4) The Aircraft Identifier (**ARCID**) shall be filled in or selected from the drop-down list (when a previous query was successful).

(5) The Aerodrome of Departure (**(ADEP)**) may be filled in with an Aerodrome ICAO code or selected from the drop-down list (when a previous query was successful).

(6) The **From** date and related **at** time may be adjusted after IOBD is changed to reduce the Operational Log period to be analysed.

(7) The **Until** date and related **at** time may be adjusted after IOBD is changed to reduce the Operational Log period to be analysed.

(8) From and Until dates may be separated by maximum 1 calendar day. In practice From and Until date boxes must display two consecutive date (day / day+1). Except for **Until** midnight on the second day which is expressed as 00:00 on the third day.

(9) There must be at least one minute elapse between the From and the Until date and time.

(10) The **ARCID** and the **From / Until** query data elements are mandatory.

(11) The **IOBT** and the **ADEP** query data elements are optional as a group. Either all of them or none are used to query the Network Operations system.

(12) When trying to issue a query with only one of these elements (ADEP or IOBT), a Warning alert box will invite the user to enter either both or none.

(13) When **IOBT** is not provided, the **IOBD** is not used for flight selection.

(14) The **Send** button will send the query to the Network Operations system.

(15) This window may be closed before any query is issued.

(16) The **Reset** action common functionality is implemented for this function. All query criteria except Normal/Proposal selection are considered for activating the Reset action.

15.4 Displaying the query results

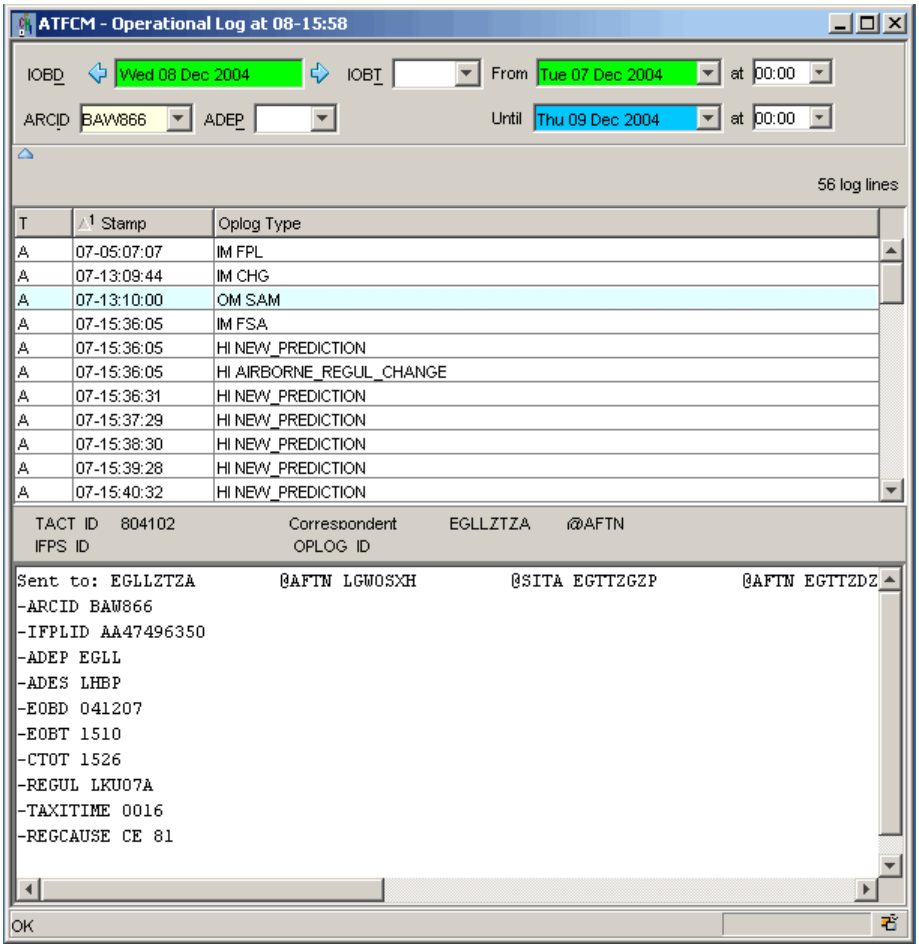


Figure Chmi - 55: Flight Operational Log

(1) This display shows the Operational Log entries for the flight(s) that corresponds to the selected ARCID possibly attached with other query criteria (IOBT/D and ADEP).

(2) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(3) The header section of the window contains the different **query data elements** used to retrieve the displayed flight Operational Log entries:

a) ARCID: The aircraft identifier.

b) IOBD: The Initial Off Block Date.

c) IOBT: The Initial Off Block Time, if initially provided.

d) ADEP: The Aerodrome of Departure, if initially provided,

e) From - Until: The period of the Operational Log scanned for the associated query criteria.

(4) A counter identifies the number of log entries displayed for this ARCID.

(5) This counter may be affected by **filtering** (fff / rrr log lines where fff is the filtered number and rrr is the retrieved number).

(6) The list shows all log entries identified for the selected query criteria.This list displays for each log entry:

a) The one letter code corresponds to a unique TACT flight identifier **(T)** (i.e. under the same callsign, there might be more than one flight and they are here defined by A, B, C, etc). In practice, each single letter corresponds to a different TACT ID. This is done to improve readability of the list.

b) The **Stamp** when that log entry was recorded.

c) The **Oplog Type** composed of a classification code and the event kind description.

(7) the Oplog Type Classification code may be:

a) **UD** for Undefined.

b) **IM** for Incoming Message.

c) **ER** for Erroneous Incoming Message.

d) **OM** for Outgoing Message.

e) **VI** for Violation.

f) **HI** for History.

g) **WA** for Warning.

h) **PE** for Process Error.

i) **EM** for Error Message.

j) **EN** for Environment Message.

k) **UC** for User Command.

l) **TM** for Text Message.

(8) When a log entry is selected in the list, following details are displayed:

a) The TACT unique identification key for the flight **(TACT ID)**.

b) The IFPS unique identification key, if any, for the flight **(IFPS ID)**.

c) The **Correspondent**: Defines the sender of a message (for incoming messages) or first receiver of a message (for outgoing messages), only filled in for an Incoming Message, Erroneous Incoming Message, Outgoing Message or an Error Message.

d) The OPLOG unique identification key **(OPLOG ID)**, if any.

e) The detailed text of the exchanged message in the details pane of the Operational Log window.

(9) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

(10) On initial display, the list is sorted in ascending sequence on the timestamp when the log entry was recorded.

(11) On query results display, the focus is set on the log entry list after selecting the first log entry in the list.

15.5 User interactions

(1) Click on a line in the list to **select the log entry** and have more detailed information displayed below the list.

(2) This window allows to modify a previous query selection and to issue a new request.

(3) On query modification, the **Reset action** is enabled.

(4) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(5) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(6) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(7) To print the Operational Log list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(8) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(9) To modify the **Sort sequence** of the flights, click on column headers corresponding to your desired sort criteria.

(10) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(11) Increasing / reducing the window height and width will impact the size of the main Log list control.

16 Alternate Routes

16.1 Function presentation

(1) This functionality displays a list of alternate routes for a selected flight. The alternate routes may comes from the standard route catalogue, be generated by a route generator or manually added by the user.

(2) On selection of one of the proposed alternate routes, the user will be able to:

a) Request the evaluation of a possible rerouting.

b) Immediately request to reroute the flight without a first evaluation.

c) Request to reroute based on the returned evaluation.

(3) Access to alternate route listing, reroute evaluation, execution, generate and add functions will be restricted by security profile.

16.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Flights / Alternate Routes**.

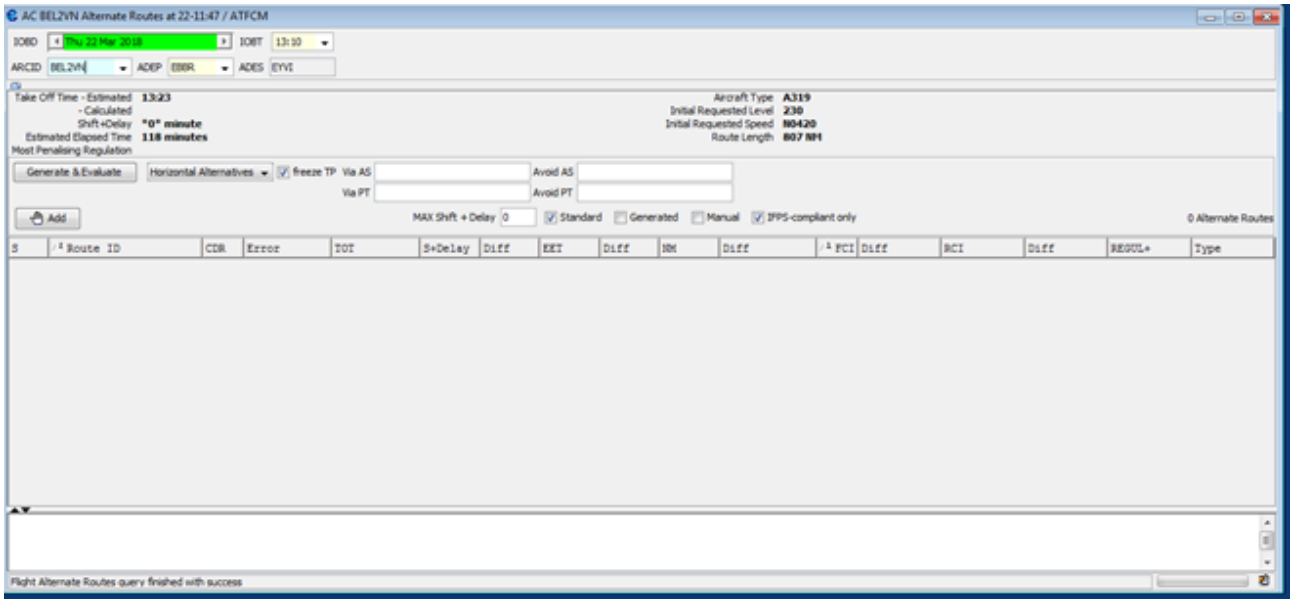


Figure Chmi - 56: Query Alternate Routes

(2) A new empty **Alternate Routes** window will be opened with the following defaults:

a) IOBD: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) IOBT: blank. Drop-down list will propose pre-formatted time on a 20 minutes step base.

c) ARCID: the drop-down box will propose the most recent query data elements you have queried before with success.

d) ADEP: the drop-down box will propose the most recent query data elements you have queried before with success.

(3) On initial display, the focus is set on the first mandatory query control with no default value: the **ARCID** input control.

16.3 Preparing to query

(1) The Initial Estimated Off Block Date (**IOBD**) may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The Initial Estimated Off Block Time (**IOBT**) input box shall be adjusted to a valid time value from 00:00 to 23:59 included. This time does not have to be exact. The system allows some flexibility to identify the precise flight you want to query.

(3) The Aircraft Identifier (**ARCID**) shall be filled in or selected from the drop-down list (when a previous query was successful).

(4) The Aerodrome of Departure (**ADEP**) shall be filled in with an Aerodrome ICAO code or selected from the drop-down list (when a previous query was successful).

(5) All these query data elements are mandatory in order to identify a unique flight.

(6) The **Send** button will send the query to the Network Operations system.

(7) This window may be closed before any query is issued.

(8) The **Reset** action common functionality is implemented for this function. All query criteria are considered for activating the Reset action.

16.4 Displaying the query results

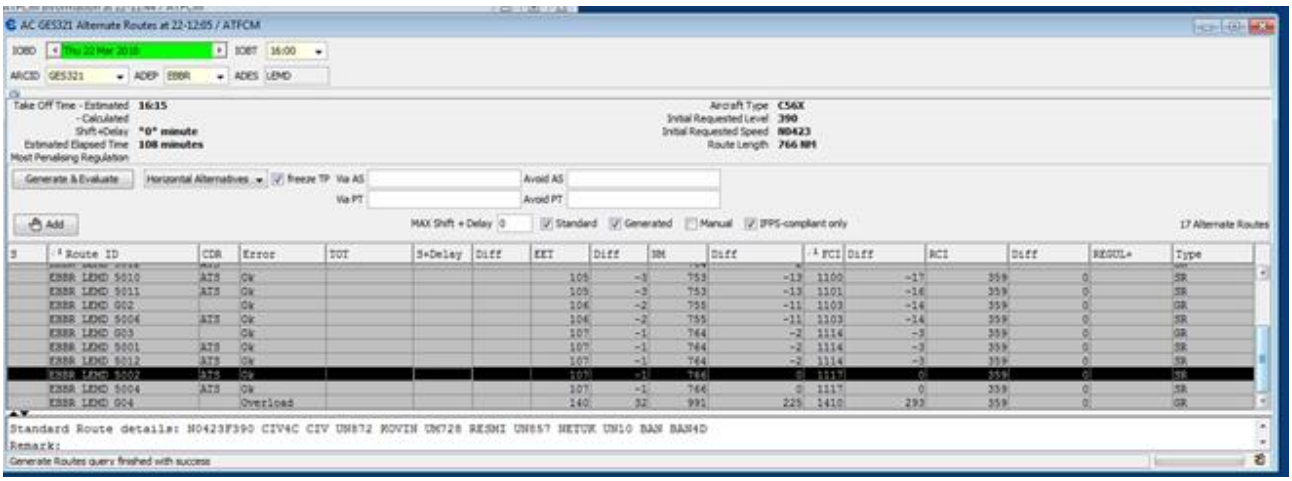


Figure Chmi - 57: Alternate Routes

(1) This display shows the list of alternate routes for the Flight that corresponds to the selected query criteria.

(2) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(3) The header section of the window contains the different **query data elements** used to retrieve the displayed alternate routes:

a) IOBD: Initial Estimated Off Block Date.

b) IOBT: Initial Estimated Off Block Time possibly adjusted to its correct value based on the query results data.

c) ARCID: Aircraft Identifier.

d) ADEP: Aerodrome of Departure.

(4) The Aerodrome of Destination (**ADES**) is displayed in the header section when the reply is received from the Network Operations system.

(5) On query results display, the focus is set on the alternate route list after selecting the first entry in the list.

(6) The frame below the header section displays the following details for the selected flight:

a) The **Estimated Take Off Time**.

b) The **Calculated** Take Off Time, if appropriate.

c) The Calculated **Shift+Delay**, if appropriate.

i) A case sensitive letter may possibly be appended to the delay value to indicate following specific delay characteristics:

- "D" when the delay value calculated for the flight is exceeding the delay confirmation threshold of a regulation affecting the flight
- "a" when the delay value of the flight has been adjusted to the clock

ii) Provisional delays (slot not yet published) are indicated by an asterisk on either side of the value including the possible delay characteristic letter (e.g. "*12D*").

d) The **Estimated Elapsed Time** of the flight.

e) The **Most Penalising Regulation**, if any and possibly if **Other regulations apply**.

f) The ICAO identifier of the **Aircraft Type**.

g) The **Requested Flight Level.**

h) The true airspeed in Kilometers per hour (K), Nautical Miles per hour (N) or Mach number (M) (**Speed**).

i) The current **Route Length** in miles.

(7) Finally, the next part of the window proposes the possible alternate routes.

(8) At the top of the alternate route list, a counter indicates the **number of alternate routes** retrieved.

(9) This counter may be affected by **filtering** (fff / rrr Alternate Routes where fff is the filtered number and rrr is the retrieved number).

(10) The main list box contains the list of proposed Alternate Routes.

(11) The Network Operations system will return all potential alternative routes for the selected flight.

(12) For each alternate route, the following information is displayed:

a) the **Route ID** (see later),

b) the worst **CDR** applied to this route (see later),

c) the length of the proposed alternate route in Nautical Miles (**NM**) and the difference with the current route (**Diff**),

d) the fuel cost indicator (in KG) and route charge indicator (in euro) and their difference with the current route (**Diff**),

e) the **Type** of this alternate route

i) **SR** for Standard Route

ii) **GR** for Generated Route

iii) **UR** for manually added User Route

f) the other columns are also filled in. The presence of the different columns will depend on the evaluation results.

(13) Other columns are used to display the Alternate Route evaluation results. See later **Reroute Evaluation** section for a detailed description.

(14) The **Route ID** is composed of:

a) the route **origin** (Aerodrome),

b) the route **destination** (Aerodrome)

c) and the Route **sequence number**: For Generated route the sequence number shall be preceded by 'G' (e.g.: "LFPG EBBR G01"). For Manually entered route the sequence number shall be preceded by 'U' (E.g.: "LFPG EBBR U01").

(15) Possible **CDR** types are:

a) **ATS** when the ATS route is always plannable (corresponds to CDR 0).

b) **1** for permanently plannable conditional route. Closure is published in NOTAM and EAUP/EUUP.

c) **2** for non permanently plannable conditional route. Opening can be published in EAUP/EUUP.

d) **N** for not available - non plannable.

(16) On initial display, the list is sorted with the following priority from top to bottom

a) Generated Route by ascending order on Route Id

b) Manually entered routes by ascending order on Route Id

c) Standard routes by ascending order on Route Id

(17) Selecting one of the proposed alternate routes will display:

a) the route description using the ICAO field 15 presentation including the associated Cruising Speed / Level. This field is preceded by one of the following static texts:

i) **Standard Route Details** when a standard route entry is selected

ii) **Generated Route Details** when a generated route entry is selected

iii) **User Route Details** when a route entry added by the user is selected

b) the associated **Remark** when a standard route entry is selected.

(18) The **MAX Shift+Delay** initially displays the greatest value accepted by the Network Operations system.

a) This value may be overwritten with the greatest number of minutes representing the Maximum Shift+Delay accepted by the user for the rerouted flight.

b) The application will check that the overwritten value is between 0 and the maximum identified by the Network Operations system.

c) An error message with the corresponding reason will be returned if this limit cannot be respected.

(19) The **Standard and Generated** ilter box is initially on.

(20) The **Manual** filter box are initially off.

(21) The **Manual** and **Generated** filter box are automatically set to on, when the corresponding action is used to add routes to the list.

a) **Add** for **Manual** filter box.

b) **Generate and evaluate** for **Generated** filter box.

(22) When a filter box is unchecked, the corresponding items are filtered out from the list, even if currently selected.

16.5 User interactions

16.5.1 Alternate Routes

(1) Click on a line to **select an Alternate Route** to have more detailed information displayed in the Route details pane (bottom pane).

(2) When a line is selected in the list of Alternate Routes, the right mouse button pops up a **contextual menu**.

(3) This popup menu lists the possible actions for the selected Alternate Route.

(4) Following Plot options are only available when the selected Alternate Route is either extracted from the Route Catalogue (RCAT) or generated from City Pair Stats (CPS):

a) **Plot Route**

b) **Plot All Routes**

(5) Selecting **Plot Route** will display the selected RCAT or CPS route on the map.

(6) Selecting **Plot All Routes** will display all the listed RCAT and CPS routes on the map.

(7) Following user preference may be adjusted to associate a colour with the selected route(s) when displayed on the map: **CHMI / ATFCM application / Flights / General / Alternate Route colour**. The default is white.

(8) The following additional options are only included and active in the menu if the selected flight satisfies the conditions implemented in the Network Operations system to accept a rerouting (Shift + Delay, suspended, IFPS zone coverage...).

a) **Reroute Preparation**: The display of the Reroute Preparation screen.

b) **Reroute Apply**: The immediate Rerouting using that route.

c) **Reroute Apply File**: The immediate Rerouting using that route and filing of the necessary IFPL.

(9) Access to the different evaluation and reroute functions will be restricted by security profile.

(10) When issuing an immediate "**Apply**", the Reroute Preparation dialog is not proposed. The default values will be used for the flight. (Cruising Speed / Flight Level)

(11) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date alternate route list. Be careful, all previous evaluation data will be lost in order to provide a fresh display.

(12) To print the alternate route list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(13) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(14) To modify the **Sort sequence** of the alternate routes, click on column headers corresponding to your desired sort criteria.

(15) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

16.5.2 Add user routes

(1) The **Add** button allows the user to enter its own route. This function is only available for flight which has its “AO What if Reroute” indicator code (**W**) set to **T** (Try allowed) or **A** (Apply and Try is allowed). It opens the Reroute Preparation screen (C.f.: chapter Reroute Preparation).

(2) The **Add** function shall be protected by specific security feature when further AOWIR possibilities are authorised for the displayed flight (Try or Apply are allowed).

(3) The **Manual** filter box is automatically ticked after a route was manually added.

(4) The created route is added to the list of alternate route with a route Id. These routes are presented into the list in ascending order on Route Id, before the Standard routes but after the Generated route.

16.5.3 Ask for generated routes

(1) The **Generate and evaluate** button allows the Operator to request to the Network Operations system dynamically generated routes.

(2) The **Via Point, Via Airspace, Avoid Point** and **Avoid Airspace** fields allow the Operator to specify constraints on the routes to be generated:

a) Via Point: PublishedPoints the proposed routes shall over-fly.

b) Via Airspace: Airspaces (2D projection) the proposed routes shall traverse.

c) Avoid Point: PublishedPoints the proposed routes shall avoid.

d) Avoid Airspace: Airspaces (2D projection) the proposed routes shall avoid.

For performance and overload protection reasons, the NM server will implement a limit on the maximum number of constraints it can accept per request (i.e. per query).

This limiting parameter will be initially set to a maximum of 10 constraints.

An error message will report when this limit is exceeded and invite the user to remove some constraints before re-submitting the generate request.

(3) The freeze TP selection box allows to indicate if the generation needs to respect the original Terminal Procedure or not. By default it is ticked.

(4) The horizontal/vertical selection box allows the user to choose if the system should modify the routes of the flight horizontally or vertically to comply with the constraints.

(5) The generated routes already present from a previous request are removed from the list of alternate routes. The newly generated routes returned by the Network Operations system will be added to the list of alternate routes with a route Id. The number of generated route might be limited by the server to preserve performances (initially limited to 5). These routes are presented into the list in ascending order on Route Id, at the top of the list of alternate routes.

(6) The **Generate** function shall be protected by specific security features for the following two cases:

a) When further AOWIR possibilities are authorised for the displayed flight (Try or Apply are allowed).

b) For all other cases (no further AOWIR possibilities).

16.5.4 Reroute Preparation

(1) The Reroute Preparation screen is an optional intermediate screen in the evaluation process.

(2) Once the list of possible alternate routes is retrieved from the Network Operations system for the selected flight, if it is reroutable and the user is authorised to the reroute functions, the Reroute Preparation dialog may be invoked for one of the proposed alternate routes or via the Add button.

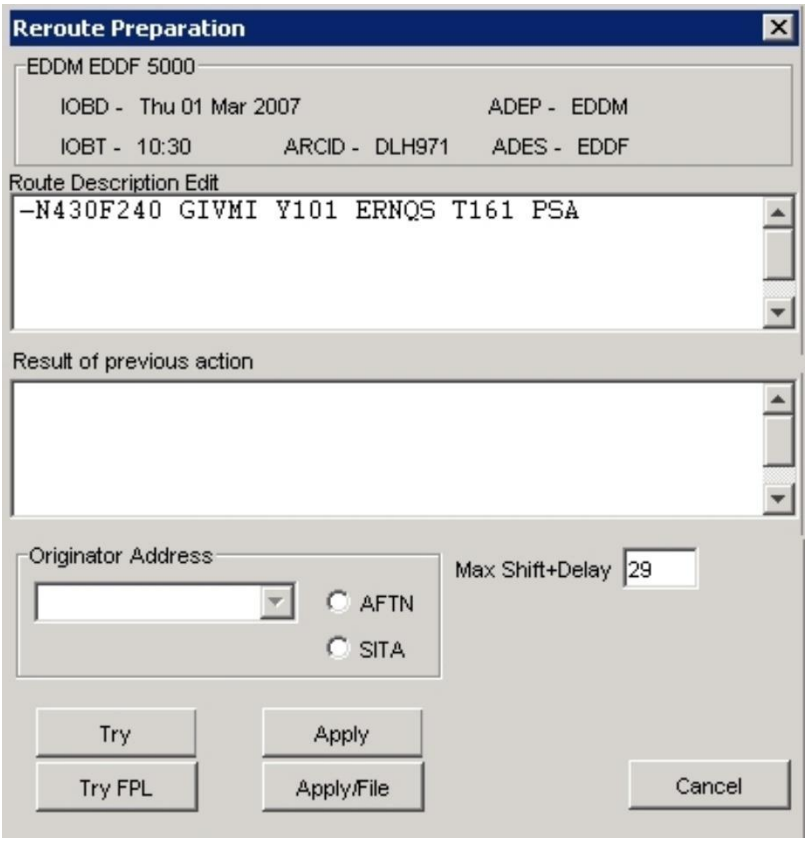


Figure Chmi - 58: Reroute Preparation

(3) This is a modal dialog box! It means that when it is opened, no other function can be invoked.

(4) This dialog allows to update:

a) The route description: cruising speed and level, but also the route itself.

b) The Maximum Shift+Delay accepted by the user for the new route.

c) The associated network address.

(5) The header section contains the selected query criteria used to identify a unique flight:

a) the Initial Estimated Off Block Date **(IOBD)**,

b) the Initial Estimated Off Block Time **(IOBT)**.

c) the Aircraft identifier **(ARCID)**,

d) the ICAO code for the Aerodrome of Departure **(ADEP)**.

(6) The Aerodrome of Destination is displayed for information **(ADES)**.

(7) The **Route ID** is inserted as the title of this header section.

(8) Next to this ID, the stamp of the possible previous evaluation processing time.

(9) When opened from the **Add** button, the **Route Description Edit** is empty.

(10) The Route Description Edit shall allow Copy/Paste functionality.

(11) The "**Result of previous action"** box gives more details on the selected alternate route with possibly the results of a previous evaluation.

a) The evaluation time stamp, status and any relevant error reason message described in the Reroute Evaluation section,

b) the route description using the ICAO field 15 presentation including the associated Cruising Speed / Level. This field is preceded by one of the following static texts:

i) **Standard Route Details** when a standard route entry is selected

ii) **Generated Route Details** when a generated route entry is selected

iii) **User Route Details** when a route entry added by the user is selected

c) the associated **Remark** when a standard route entry is selected.

(12) Before requesting a "**Try**" or an "**Apply**" on this alternate route, the Operator has the possibility to update the following data elements:

a) The **MAX Shift+Delay** may be overwritten with the greatest number of minutes accepted by the user for the new Shift+Delay value. A maximum value is imposed by a Network Operations parameter. The proposed default is this maximum or the last queried Max Shift+Delay value. An error message will be returned if this limit cannot be respected.

b) The network address to be associated with this terminal. This information is mandatory for all "Try" and "Apply" actions. If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preferences (CHMI / ATFCM application / General / Origin / Default Address ID).

c) The **Route description** may be adjusted by the user.

(13) Following possibilities are given to the user for **Route description Edit:**

a) **No Update** in which case the route will be used as proposed by the Network Operations system or as previously created.

b) The Operator changes the **Route Description**:

i) In the case the Reroute Preparation was opened from a Generated or Standard alternate route or from the Add button, a new Manual route

is created. A new line will be added to the alternate route list after having performed a Try or Apply operation.

c) When Reroute Preparation is opened from a Manual route already in the alternate route list, it is replaced by the modified one.

(14) The local ATFCM CHMI will only process syntax checks to validate the entered points. The Network Operations server will perform further context editing.

(15) Click on the **AFTN** or **SITA** radio button to select the address type.

(16) You may introduce an address in the corresponding box or select another one in the list by clicking on the button to drop-down the list. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(17) To request a full reroute evaluation including FPL validation in IFPS, click on the **Try** button.

(18) To request an immediate booking if the selected reroute pass validation with the provided parameters (MAX Shift+Delay and Route Description) click on the **Apply** button. The Operator will then have to file the flight plan by himself.

(19) Click on the **Apply / File** button to request an immediate booking and filing. If the selected reroute is successfully validated with the provided parameters (MAX Shift+Delay and Route Description), the slot will not only be booked in the Network Operations system, but the corresponding flight plan will be filed automatically in IFPS.

(20) To close this Reroute Preparation dialog without any "Try" or "Apply", click on the **Cancel** button. This action will ignore the updates made on this preparation window, if any.

(21) Some buttons may be disabled for the following reasons:

a) The flight, checked by ETFMS, does not meet all requirements.

b) The user has a limited security profile.

16.5.5 Reroute Evaluation

(1) This section describes the functionality implemented to **request the evaluation** of possible reroutings.

(2) This functionality is available only if following conditions are met:

a) The flight satisfies the conditions implemented in the Network Operations system to accept a rerouting evaluation.

i) The **Try** functionality is only accessible for a flight which has its AO What If Reroute indicator code (**W**) set to either **T** (Try allowed) or **A** (Apply and Try allowed).

b) The users' security profile allows him to apply this functionality.

(3) Based on the previous conditions the user is allowed to invoke the **"Try"** functionality to request the evaluation of possible reroutings through following windows:

a) **Reroute Preparation** window command buttons.

(4) When the list of possible alternate routes is retrieved from the Network Operations system and/or added by user for a selected flight, the evaluation of routes is automatically done as well.

(5) On selecting an alternate route, the **MAX Shift+Delay** input box is initialised:

a) With the greatest Shift+Delay time accepted by the Network Operations system to perform a rerouting action, if this route was not yet evaluated.

b) With the last updated **MAX Shift+Delay** value used, if this route was already evaluated.

(6) Before requesting a rerouting evaluation, the user may overwrite the content of the MAX **Shift+Delay** input box with the maximum number of minutes he is ready to accept as the **new Shift+Delay** for the flight.

(7) The **"Generate and evaluate"** and **"Apply"** functions will reject a **Maximum Shift+Delay** if it is greater than the one authorised by the Network Operations system.

(8) The network address to be associated with this terminal is mandatory for the **"Apply"** action.

(9) To have a default address defined, you need to update the **Message Originator** preferences (CHMI / ATFCM application / General / Origin / Default Address ID).

(10) To request an explicit reroute evaluation for the flight, select the desired Alternate Route in the list and press the right button of the mouse. Select one of the following contextual popup or action menu options:

a) **Reroute Preparation**: to request the display of the Reroute Preparation screen. It is used to input route description updates, Maximum Shift+Delay and/or associated network address.

(11) The Reroute Preparation screen is an optional intermediate screen in the evaluation process.

(12) This preparation screen will mainly allow the user to edit and create routes before submitting the "Try" or the "Apply" request.

(13) The Reroute Preparation screen is described in an earlier section of this document.

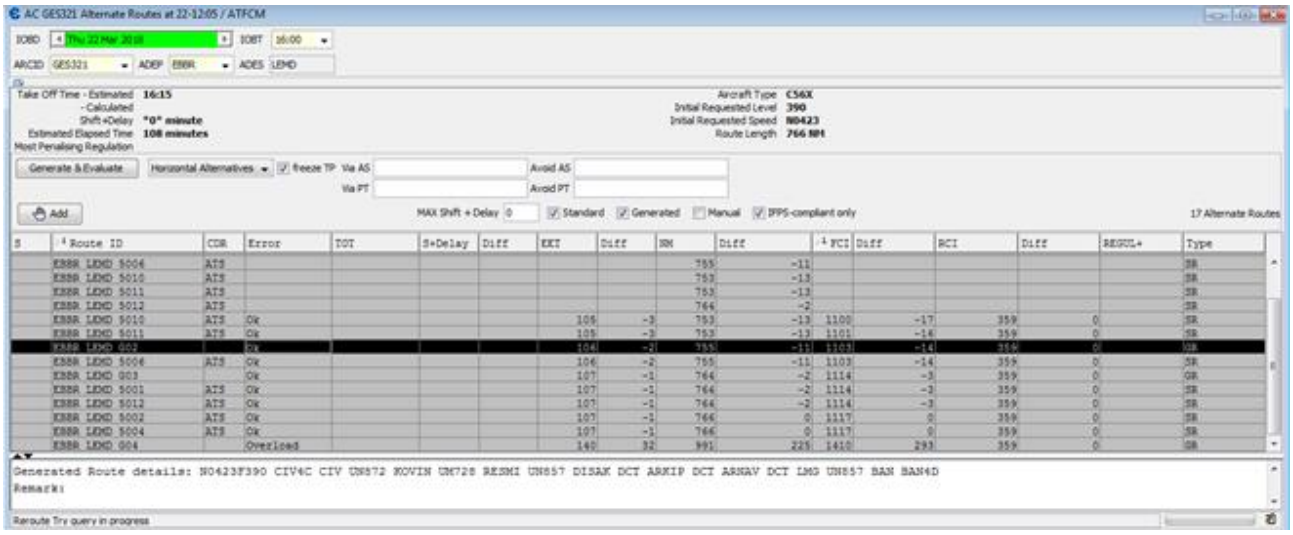


Figure Chmi - 59: Successful Reroute Evaluation

(14) Once a route has been successfully evaluated, the "Reroute Evaluation" results are added in the alternate route list Details pane:

a) The Status column (**S**) indicates the result of Reroute Evaluation process (see list below).

b) The Error column indicates the type of error that an alternate route has (see list below)

c) The new Take Off Time (**TOT**). After the time, a letter **"C"** for Calculated indicates that using this new route, the flight is subject to regulation. A letter **"E"** for Estimated indicates that the flight is out of any regulation.

d) The resulting new calculated Shift+Delay value, if any (**S+Delay**).

e) The difference in minutes compared to the current Shift+Delay **(Diff)**.

f) The Estimated Elapsed Time **(EET)** of the flight based on the new route.

g) The difference in minutes compared to the current EET **(Diff)**.

h) The length of the new Route in Nautical Miles **(NM)**.

i) The difference in Nautical Miles compared to the current length **(Diff)**.

j) The fuel consumption indicator (FCI) of the new route (KG)

k) The difference in KG compared to the current fuel consumption **(Diff)**.

l) The route charge indicator (euro)

m) The difference in euros compared to the current route charges **(Diff)**.

n) The **Last Validity (LV).** This is the Last Valid EOBT acceptable for the flight before triggering IFPS errors.

See LV column in Flight List for detailed display format, tooltip and ICD source data.

o) The identifier of the Most Penalising Regulation based on the new profile, if any (**REGUL+**).

p) The type of route (SR for standard route, GR for generated route, U for user route).

(15) Here follows the list of possible Reroute Evaluation Status:

a) The letter **"T"** indicates that a successful **"Try"** was processed by the Network Operations system.

b) The letter **"E"** indicates that the Reroute Evaluation ended with an error message.

(16) Selecting the evaluated alternate route will display additional details in the Details pane at the bottom.

a) The reroute request error reason, if any (see list below).

b) An error message returned by the TACT system, if applicable.

c) The route description using the ICAO field 15 presentation including the associated Cruising Speed / Level. This field is preceded by one of the following static texts:

i) **Standard Route Details** when a standard route entry is selected

ii) **Generated Route Details** when a generated route entry is selected

iii) **User Route Details** when a route entry added by the user is selected

d) The associated **Remark** when a standard route entry is selected.

e) **Exceptional condition** with the Traffic Volume ID and the Runway Visual Range (RVR) value, if applicable.

f) The **Overloaded Traffic Volume ID**, if applicable.

g) The list of **IFPS error** messages (i.e. "Point for Speed / Level update not found in route details"...), if applicable.

(17) In case the Reroute Evaluation process did not succeed, one of the following error reasons will be displayed inside the Error column:

a) **Wrong input** with the justification message in the details area.

b) **Suspended** if the flight is suspended.

c) **Route not for flight** if the requested alternate route is not valid for the flight.

d) **Not authorised** if you are not authorised to perform the requested action.

e) **Overload** with the overloaded Traffic Volume in the details area

f) **Not IFPS compliant** and the corresponding IFPS messages for rejection in the details area

g) **Bad Shift/Delay** for an excessive new Shift+Delay value.

h) **Not available** if rerouting function has been deactivated by the TACT system.

i) **Already a proposal** when there is still a rerouting proposal outstanding for the selected flight.

j) **Flight Activated is the flight is already TACT or ATC ACTIVATED**

k) **Constraint violated if the route is violating the avoid/via constraints or if a manual route is violating certain constraints. The exact reason is descirbed in the details area**

l) **OBT_Validity reached : if the alternative route gets a new CTOT such that the flight becomes IFPS invalid**

m) Other **TACT Error** with the justification message in the details area.

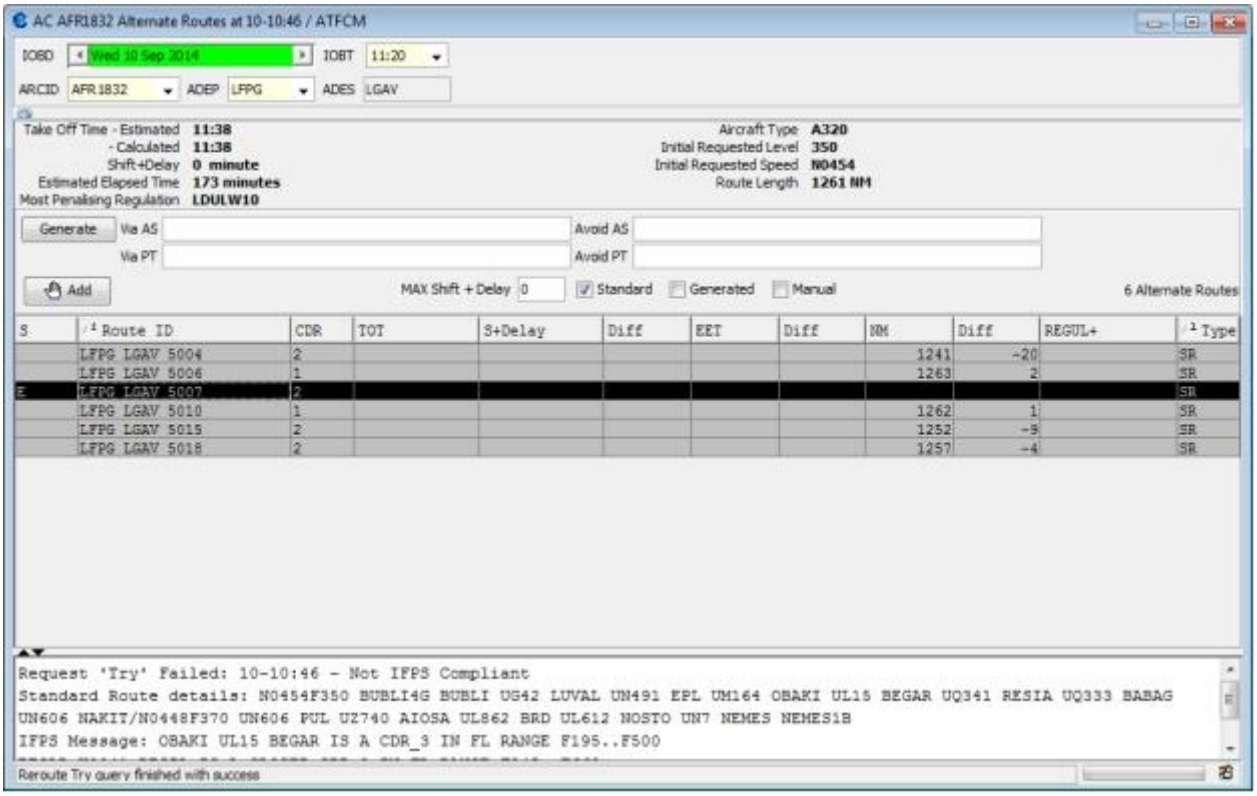


Figure Chmi - 60: Reroute Evaluation Failure

16.5.6 Rerouting request

(1) This section describes the functionality implemented to **request a rerouting**.

(2) This functionality is available only if following conditions are met:

a) The flight satisfies the conditions implemented in the Network Operations system to accept a rerouting.

i) The **Apply** functionality is only accessible for a flight which has its “AO What If Reroute” indicator code (**W**) set to **A** (Apply and Try allowed) and the “Filing Rule” (field 214 of the Flight General ICD message) is either **O** (Operator is obliged to refile) or **F** (filing can be done by AO or NM).

ii) The **Apply / File** functionality is only accessible for a flight which has its “AO What If Reroute” indicator code (**W**) set to **A** (Apply and Try allowed) and the “Filing Rule” (field 214 of the Flight General ICD message) is **F** (filing can be done by AO or NM).

b) The users' security profile allows him to apply this functionality.

(3) Based on the previous conditions the user is allowed to invoke the "**Apply**" functionality to request the evaluation of possible reroutings through following windows:

a) **Alternate Routes** Action menu or contextual menu.

b) **Reroute Preparation** window command buttons.

(4) Once the list of possible alternate routes is retrieved from the Network Operations system for a selected flight, a rerouting may be requested immediately or after one or more initial evaluations.

(5) The Rerouting request is limited to one successful process per flight.

(6) On selecting an alternate route, the **MAX Shift+Delay** input box is initialised:

a) With the greatest Shift+Delay time accepted by the Network Operations system to perform a rerouting action, if this route was not yet evaluated.

b) With the last updated **MAX Shift+Delay** value used, if this route was already evaluated. Before issuing a rerouting request, the **MAX Shift+Delay** input box may be overwritten with the maximum number of minutes accepted by the user as the new Shift+Delay for the flight.

(7) The "**Apply**" functions will reject a maximum Shift+Delay if it is greater than the one authorised by the Network Operations system.

(8) The network address to be associated with this terminal is mandatory for the "**Try**" and "**Apply**" actions.

(9) To have a default address defined, you need to update the **Message Originator** preferences (CHMI / ATFCM application / General / Origin / Default Address ID).

(10) To issue a reroute request for the flight, select the desired Alternate Route in the list and press the right button of the mouse. Select one of the following contextual popup or action menu options:

a) **Reroute Preparation**: to request the display of the Reroute Preparation screen. It is used to input Route description updates, Maximum Shift+Delay and/or associated network address.

b) **Reroute Apply**: for an immediate booking if the selected reroute pass validation with the provided parameters (Maximum Shift+Delay, Route description and associated network address). This "Apply" will only reserve a slot in the Network Operations system but the Operator has still to file the new flight plan via AFTN / SITA.

c) **Reroute Apply File**: for an immediate booking including filing of the new flight plan in IFPS. The selected reroute must pass validation with the provided parameters. (Maximum Shift+Delay, Route description and associated network address).

(11) This route may have been evaluated or not before its booking.

(12) The Reroute Preparation screen is an optional intermediate screen in the evaluation process.

(13) This preparation screen will mainly allow the user to modify the Route description before submitting the "Try" or the "Apply" request.

(14) The Reroute Preparation screen is described in an earlier section of this document.

(15) This rerouting request will issue a query to the Network Operations system.

(16) If the rerouting fails to pass validation, the request is then considered as another evaluation process. Error messages will be displayed as described above in the **Reroute Evaluation** section.

(17) Because a rerouting request may result in important modifications for the flight, a Rerouting Confirmation dialog will be presented to the Operator before the request is sent to the Network Operations system.

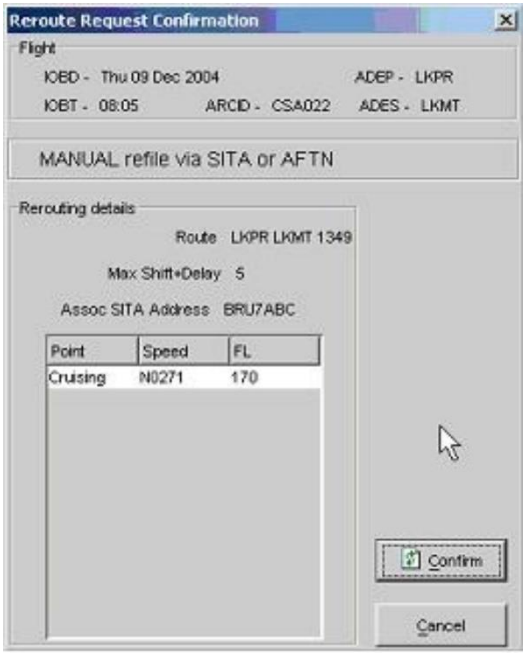


Figure Chmi-61: Reroute Request Confirmation dialog

(18) This is a modal dialog box ! It means that when it is opened, no other function can be invoked.

(19) This confirmation dialog box contains a summary of the rerouting request:

a) Identification of the unique flight to be rerouted (IOBD, IOBT, ARCID, ADEP, ADES).

b) Type of "Apply" (including automatic filing of flight plan or not).

c) Rerouting details (alternate Route, Maximum Shift+Delay, associated network address).

(20) If the **AUTOMATIC** filing was selected by using the **Reroute Apply File** option, no action is required from the user if the "Apply File" process is successful.

(21) With the **MANUAL** filing was selected by using the **Reroute Apply** option, if the "Apply" process is successful, a new slot will be booked and the user will have to REFILE a new FPL via SITA or AFTN networks.

(22) The user has two options to continue:

a) Choose the **Confirm** button to confirm the rerouting request.

b) Choose the **Cancel** button to cancel the rerouting request and come back to the window where the "Apply" was requested. The Operator may then possibly change some details and issue a new request.

(23) Once the **Confirm** button is chosen, then the Rerouting Request will be either:

a) **ACCEPTED**. In this case the following "Booked Reroute" confirmation dialog is displayed (see below).

b) **REJECTED**. This means that the "Apply" request is considered as a Reroute Evaluation request (i.e. "Try"). In this case the Rerouting Request window closes and the Alternate Routes display is activated again. The evaluation results are displayed as described in the **Reroute Evaluation** section.

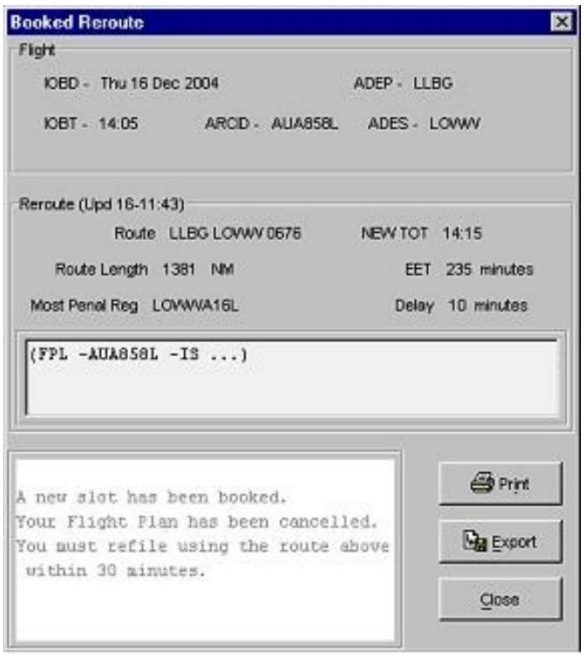


Figure Chmi-62: Booked Reroute dialog

(24) After a successful confirmation the "Booked Reroute" modal dialog box is displayed. In that case no other function can be invoked.

(25) Contrary to the succesful Rerouting Evaluation results that are of an informative nature, the results of the Rerouting Request are the ones applied to the Network Operations system.

(26) The displayed details correspond to the newly calculated values based on the booking of the new flight plan:

a) The selected Route identifier for the rerouting (**Route**).

b) The length of the new route in Nautical Miles (**Route Length**).

c) The **Most Penalising Regulation**, if any.

d) The newly obtained Shift+Delay (might be absent if flight no more regulated) (**Delay**).

e) The new Take Off Time (Calculated TOT or Estimated TOT).

f) The new Estimated Elapsed Time (EET).

g) The complete flight plan details used in IFPS.

h) If applicable, any relevant IFPS message.

i) If applicable, any relevant, Exceptional condition warning message...).

j) One of the following slot warning messages based on new **Delay** value:

i) "**No slot required for this rerouting**", if **Delay** is blank (contains no numeric value).

ii) "**A new slot has been booked**", if **Delay** >= 0 (contains a numeric value including 0 (zero)).

k) One of the following warning messages will be displayed based on the reroute request type:

i) If the page is invoked through an **Apply** action, the message "**Your Flight Plan has been cancelled, you must refile with the above route within 30 minutes**".

ii) If the page is invoked through an **Apply/File** action, the message "**Your Flight Plan has been refiled using the route above".**

(27) Following actions are possible on the **Booked Reroute** dialog box:

a) Clicking the **Print** button will produce a paper copy of the new flight details as they appear on the window.

b) Clicking the **Export** button will save the new flight plan details displayed on the screen into a file. A special dialog box will pop up to ask the file name to the user. The Booked Reroute dialog box is not closed immediately in order for the user to issue an additional print request.

c) Clicking the **Close** button without an initial print or export request will cause a confirmation dialog box to pop up: **You have not printed/exported a copy of the new flight details. Do you want to PRINT/EXPORT it ?**

d) After a print/export, the **Close** button will close automatically both the "Booked Reroute" dialog box and the "Alternate Routes" window.

(28) The box containing the new flight plan details allows to **Copy/Paste** its content.

17 Tower Departure List

17.1 Function presentation

(1) The Tower Departure List gives the list of flights departing from a pre-selected Aerodrome or Aerodrome Set.

(2) In addition to customisable automatic monitoring of the departure phase, this function will allow to:

a) Display list of currently Suspended flights.

b) Display detailed information on selected flights.

c) Issue some specific messages to keep sharp control on effective departing information.

(3) The interface design was done taking into account the requested implementation of a sensitive display allowing operations without traditional peripherals like keyboard and mouse.

(4) To avoid confusions when the Tower Departure List is referring to the current UTC, based on the PC time, the user adjustable preference **CHMI / ATFCM application / General / Dates and times / Time adjustment** will not be taken into account. The UTC time referred to in the following paragraphs is displayed in the top right corner of the Tower Departure List window. This preference is only used when a function has to propose a default but adjustable time value for query preparation.

17.2 Opening the window

(1) The Tower Departure List function may be started manually from the Application / ATFCM menu. Select the option **Flights / Tower Departure List**.

(2) On opening, the Tower Departure List window will check the following user adjustable preference to customise the display and start flight data retrieval:

a) **CHMI / ATFCM application / Flights / Tower departure list / Departure List Identifier**

(3) If already specified, this preference is used to retrieve the initial list of flights.

(4) If not yet specified, a modal dialog box will invite the user to input the desired identifier type and value (Aerodrome or Aerodrome Set).



Figure Chmi-63: Identifier input dialog

(5) Upon choosing the button OK, the Tower Departure List window is being displayed and related User Preference is updated accordingly.

(6) Cancel closes the dialog box and the Tower Departure List is not started.

(7) The user adjustable preference **CHMI / ATFCM application / Flights / Tower departure list / Open at startup** allows, when checked, to request immediate start-up of the Tower Departure List on ATFCM Collaboration Human Machine Interface application start. In this case, the ATFCM Information window and the Release Notifier will not be opened automatically at the same time.

(8) On opening, the Tower Departure List window will restore its previous state based on system properties:

a) **CHMI / ATFCM application / Flights / Tower departure list / Last window state**

b) **CHMI / ATFCM application / Flights / Tower departure list / Last window position**

c) **CHMI / ATFCM application / Flights / Tower departure list / Last window view type**

(9) It will be maximised if it was maximised when it was last closed.

(10) If it was in Normal state, it will be restored with the last displayed size and position.

(11) If it was minimised, it will use preferred size and position.

(12) It will be externalised if it was externalised when it was last closed.



Figure Chmi-64: Tower Departure List

17.3 Preparing to query

17.3.1 Introduction

(1) The Tower Departure List is designed to run unattended with pre-defined default values.

(2) Consequently, the window will immediately issue a first set of queries as soon as it is opened.

17.3.2 Departure list

(1) For the first query, the following values are used to retrieve the Departure List:

a) The query **WEF date and time** is calculated based on the current UTC stamp minus the **Departure lead** time specified in the toolbox at the bottom of the window.

b) The query **UNT date and time** is calculated based on the WEF stamp obtained above plus the **Departure elapsed** time specified in the toolbox at the bottom of the window.

c) Only **departing** flights are requested for the Aerodrome or Aerodrome Set identified here above in the user adjustable preferences.

d) The displayed flights are based on the **Traffic Load** type. **Proposal** flights are never requested for the Tower Departure List.

(2) For the Traffic Load on a Departure Flight List, the flights are selected on the ATOT, then on the CTOT if ATOT is not yet available, and finally on the ETOT if none before are available.

17.3.3 Suspended list

(1) For the second query, the following values are used to retrieve the Suspended Flight List:

a) The query **WEF date and time** is calculated based on the current UTC stamp minus the **FLS lead** time specified in the toolbox at the bottom of the window.

b) The query **UNT date and time** is calculated based on the WEF stamp obtained above plus the **FLS elapsed** time specified in the toolbox at the bottom of the window.

c) Only **departing** flights are requested for the Aerodrome or Aerodrome Set identified here above in the user adjustable preferences.

d) The displayed flights are based on the **Traffic Demand** type.

e) The list is only filled in with the currently **suspended flights**.

17.4 Displaying the query results

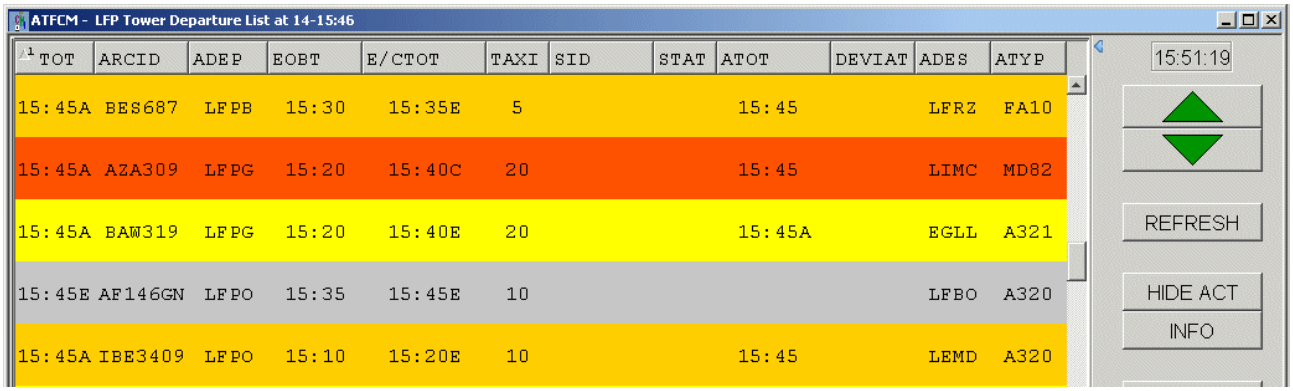


Figure Chmi-65: Tower Departure List - List details

17.4.1 Header data

(1) The window title clearly identifies:

a) It is a Tower Departure List.

b) The Aerodrome ICAO identifier or the Aerodrome Set Identifier for which the list is displayed.

c) The stamp when the list was last updated with fresh data.

(2) The current PC time (UTC) is displayed in the up-right corner of the window.

17.4.2 Departure list

(1) The main part of the Tower Departure List is dedicated to the **Departure list** of flights.

(2) For each flight, the **main list** contains:

a) **TOT**: The Take Off Time for the Departure Aerodrome or the Departure Set of Aerodromes (see later).

b) **ARCID**: The aircraft identification.

c) **ADEP**: The ICAO identifier of the Aerodrome of Departure (Only displayed if the list is run for an Aerodrome Set Identifier).

d) **EOBT**: The last received Estimated Off Block Time.

e) **E/CTOT**: Either the Calculated Take Off Time or, if not available, the Estimated Take Off Time. The appended letter indicates if it is the Calculated (**C**) or the Estimated (**E**).

f) **TAXI**: The Actual Taxi time is the latest value identified by the Network Operations system to calculate the Actual profile.

g) **SID**: The Fixed connecting point of the Departure Procedure.

h) **STAT**: The Status of the flight (see later).

i) **A/TTOT**: The Take Off Time information from the Aerodrome of Departure (see later).

j) **DEVIAT**: The difference between the Actual Take Off Time and the planned Calculated Take Off Time. While ATOT is not yet available, the Anticipated Actual Take Off Time is used instead. The Deviation is only displayed when the flight is regulated and the value is outside the -5/+10 minutes boundaries.

k) The Slot Tolerance Window for a regulated flight or the Departure Tolerance Window for a non regulated flight (**TW**).

i) The **TW** column shall only be displayed when authorized by the user profile. A single common use case shall grant the display of the Tolerance Window data in all relevant functionalities (Flight List, Flight Data …).

ii) The Lower and Upper time limit of the Tolerance Window are displayed using the following format: “[*low-hh:mm*,*upper-hh:mm*]” (ex: "*[10:20,10:40]*").

iii) The complete string is displayed in *italic* when the values are defined as an extended tolerance window.

iv) Following tooltip shall be implemented on the column header: “Slot Tolerance Window for Regulated Flights or Departure Tolerance Window for Non-Regulated Flights”.

v) Each time newly retrieved Tolerance Window values are displayed, the flight list shall be resorted if the **TW** column is part of the selected sort criteria for ordering the list.

l) Note: initial sort direction for **TW** column is: empty first (i.e. TW not yet retrieved) followed by increasing time order taking date extrapolated from IOBD into account (start time of the Tolerance Window first, than on the end time, if start times are equals).

m) **ADES**: The Aerodrome of Destination.

n) **ATYP**: The ICAO identifier of the Aircraft Type.

(3) The Take Off Time (**TOT**) is the time used for the selection of the flight in the currently displayed flight list. It may be of three sources:

a) Estimated (**E**) when based on the Estimated flight profile.

b) Calculated (**C**) when based on the Calculated flight profile (allocated slot).

c) Actual (**A**) when based on the Actual flight profile (after departure).

(4) Following different statuses may appear in the Tower Departure List before the flight is TACT Activated, ATC Activated or Terminated:

a) **TSH**: TOT Shifted status is displayed when a shifting of the CTOT has already been applied for this flight.

b) **REA**: If not TSH, REA status is displayed when when the flight is REAdy to depart (REA message received or set by DPI pre-departure processing (T-DPI_s)).

c) **FRZ**: If not TSH or REA, Slot Frozen status will be displayed when Current UTC time > (Calculated TOT - Taxi time - Time to Remove from sequence). Frozen status is only considered before the flight is TACT Activated.

d) **RSI**: If not TSH, REA or FRZ, Regulated Slot Issued status will be displayed if a slot has been published for this flight.

e) **blank**: If not TSH, REA, FRZ or RSI, status column will remain blank. This concerns the remaining cases after above checks are performed in sequence.

(5) If the take off time displayed in the **A/TTOT** column is the effective ATOT and its value is in the future of the current UTC time and the flight type is TACT activated, the entire entry in the **A/TTOT** column is displayed in italic.

(6) If there are CDM Discrepancies, the entire entry in the **A/TTOT** column is displayed in **CDM Discrepancy colour**.

(7) The colour used may be adjusted in the Preferences dialog: **CHMI / ATFCM application / Flights / Tower departure list / CDM Discrepancy colour**.

(8) The list is initially sorted in ascending sequence on the Take Off Time (**TOT**).

(9) This flight List allows only single selection.

(10) On initial display after any refresh (automatic or manual), the list is positioned on the first flight with an **TOT** greater or equal to the Current UTC time minus the **Shift Display** Time specified in the toolbox at the bottom of the window.

(11) Automatic refresh of the list is processed based on the **Refresh interval** specified in the toolbox at the bottom of the window.

17.4.3 Suspended list

(1) The **FLS** button at the bottom right of the window, just above the Toolbox button, allows to show or hide the Suspended list.

(2) This **FLS** button indicates the current number of currently suspended flights to be displayed in the list (i.e. "<< 5 FLS").

(3) When the Suspended list counter becomes a non zero value (from zero to non zero), the following user adjustable preference allows to show up the suspended list automatically:

a) **CHMI / ATFCM application / Flights / Tower departure list / Suspended list auto show**

(4) When it is requested or automatically, the list of currently **suspended flights** is displayed in the lower left part of the display (**FLS**).

(5) For each flight, the **list** contains the following columns:

a) **ARCID**: The aircraft identification.

b) **ADEP**: The ICAO identifier of the Aerodrome of Departure (Only displayed if the list is run for an Aerodrome Set Identifier).

c) **ADES**: The Aerodrome of Destination.

d) **EOBT**: The last received Estimated Off Block Time in the format "DD-HH:MM".

e) **E/CTOT**: Either the Calculated Take Off Time or, if not available, the Estimated Take Off Time. The appended letter indicates if it is the Calculated (**C**) or the Estimated (**E**).

f) **U**: The Suspension reason code (see later).

(6) The flight might be currently suspended for the following reason:

a) **ST** for SIT Time Out

b) **SM** for Slot Missed

c) **RC** for FCM required

d) **DC** for Delay Threshold exceeded, Flight Confirmation Message (FCM) required

e) **TV** for RVR

f) **NR** for Not Reported as Airborne

g) **RV** for FP Revalidation

h) **MS** for Manual Suspension

i) **AS** for Airport Suspension

(7) The initial display sort sequence for the list of Suspended flights is **Descending** order on the **E/CTOT** column.

(8) This flight List allows only single selection.

(9) On initial display after any refresh (automatic or manual), the list is positioned at the top with the first flight selected.

(10) The vertical scroll bar at the right side of the Suspended list allows to:

a) Move up and down selection in the Suspended list.

b) Request detailed flight info for the currently selected flight.

(11) Automatic refresh of the list is processed based on the **Refresh interval** specified in the toolbox at the bottom of the window.

17.5 User interactions

17.5.1 General tools

(1) If the key Aerodrome or Aerodrome Set Identifier of the Tower Departure List needs to be changed, just open the Preferences dialog, update related preferences with the newly desired Identifier and refresh the Tower Departure List after closing the preference dialog. These preferences are:

a) **CHMI / ATFCM application / Flights / Tower departure list / Departure List Identifier**

(2) Clicking on the **Refresh** button will force immediately the retrieval of up to date data for the currently displayed window (Flight lists or currently displayed flight Info).

(3) To modify the **Sort sequence** of the flights, click on column headers corresponding to your desired sort criteria.

(4) A **TOOLBOX** button at the bottom right of the window allows to show or hide the toolbox.

(5) The different parameters in the toolbox at the bottom of the window may be adjusted by the user.



Figure Chmi-66: Tower Departure List - Toolbox parameters

(6) The value assigned to these parameters is stored in the following user preferences.

a) **CHMI / ATFCM application / Flights / Tower departure list / Refresh interval**: The interval in minutes between two automatic refreshes of the Tower Departure List.

i) Proposed pre-defined values in minutes are: 1, 2, 5, 10, 15, 30

ii) Initial default **Refresh interval** value is 5 minutes.

b) **CHMI / ATFCM application / Flights / Tower departure list / Departure lead time**: The lead time in minutes applied to current UTC time when calculating the WEF time of the Departure list query.

i) Proposed pre-defined values in minutes are: 0, 5, 10, 15, 30, 60

ii) Initial default **Departure lead time** value is 30 minutes.

c) **CHMI / ATFCM application / Flights / Tower departure list / Departure elapsed time**: The elapsed time in minutes used when calculating the UNT time for the Departure list query period based on above calculated WEF time.

i) Proposed pre-defined values in minutes are: 15, 30, 60, 90, 120

ii) Initial default **Departure elapsed time** value is 90 minutes.

d) **CHMI / ATFCM application / Flights / Tower departure list / Departure shift display**: For new Departure list initial display positioning, lead time in minutes before current UTC time.

i) Proposed pre-defined values in minutes are: 0, 5, 10, 15, 30

ii) Initial default **Departure shift display** value is 10 minutes.

e) **CHMI / ATFCM application / Flights / Tower departure list / FLS lead time**: The lead time in minutes applied to current UTC time when calculating the WEF time of the Suspended list query.

i) Proposed pre-defined values in minutes are: 0, 5, 10, 15, 30, 60

ii) Initial default **FLS lead time** value is 30 minutes.

f) **CHMI / ATFCM application / Flights / Tower departure list / FLS elapsed time**: The elapsed time in minutes used when calculating the UNT time for the Suspended list query period based on above calculated WEF time.

i) Proposed pre-defined values in minutes are: 15, 30, 60, 90, 120

ii) Initial default **FLS elapsed time** value is 90 minutes.

(7) The use of these parameters is detailed in the previous sections of this document.

(8) The font used for the buttons on the right side may be adjusted with the following user preference:

a) **CHMI / ATFCM application / Flights / Tower departure list / Button font.**

17.5.2 Departure list

(1) Clicking on the large **UP** / **DOWN** buttons on the right side of the Departure list, moves backward / forward by 1 line in the list (hold click repeats).

(2) Clicking the **SHOW ACT** or **HIDE ACT** button (2 states button) will turn on and off the display of the activated flights in the list.

(3) Clicking on the **INFO** button on the right side of the Departure list, displays Flight Data details for the selected flight (See below for Flight info section).

(4) If the user needs to display earlier or later Departing flights, he has to adjust the Departure **Departure lead** time and / or the **Departure elapsed** time with the Toolbox and request an immediate refresh.

(5) Increasing / reducing the window height and width will impact the size of the Departure list control.

(6) Following user preferences may be adjusted to associate colours with the current state of the departing flight:

a) **CHMI / ATFCM application / Flights / Tower departure list / Airborne colour**

b) **CHMI / ATFCM application / Flights / Tower departure list / Past CTOT colour:** used for the flights with a CTOT in the past, not yet departed. This rule is applied with a tolerance of 10 minutes (CTOT + 10 minutes).

c) **CHMI / ATFCM application / Flights / Tower departure list / Shifted by FAM:** selection criteria is FAM status = "S".

(7) All other flights in the Departure list will use the normal Tower Departure List foreground and background colours:

a) **CHMI / ATFCM application / Flights / Tower departure list / Departure list foreground colour**

b) **CHMI / ATFCM application / Flights / Tower departure list / Departure list background colour**

(8) The font used to list the Departing flights may be adjusted with the following user preference:

a) **CHMI / ATFCM application / Flights / Tower departure list / Departure list font.**

(9) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

17.5.3 Suspended list

(1) A button at the bottom right of the window, above the toolbox button, allows to show or hide the Suspended flight list (**FLS**).

(2) The automatic show up of the suspended flight list when the list counter becomes a non zero value may be enabled or disabled using the preference:

a) **CHMI / ATFCM application / Flights / Tower departure list / Suspended list auto show**

(3) If the user needs to display earlier or later currently Suspended flights, he has to adjust the **FLS lead** time and / or the **FLS elapsed** time with the Toolbox and request an immediate refresh.

(4) Clicking on the **UP** / **DOWN** buttons on the right side of the Suspended list, moves selection backward / forward by 1 line in the list (hold click repeats).

(5) Clicking on the **INFO** button on the right side of the Suspended list, displays Flight Data details for the selected flight (See below for Flight info section).

(6) Colour background of the suspended flight list may be adjusted using the preferences:

a) **CHMI / ATFCM application / Flights / Tower departure list / Suspended list foreground colour**

b) **CHMI / ATFCM application / Flights / Tower departure list / Suspended list background colour**

(7) The font used to list the Suspended flights may be adjusted with the following user preference:

a) **CHMI / ATFCM application / Flights / Tower departure list / Suspended list font.**

(8) Increasing / reducing the window width will impact the size of the Suspended list control.

17.5.4 Flight info

(1) On selecting a flight in the Departure list or the Suspended list, the corresponding **INFO** button is enabled.

(2) On clicking the **INFO** button on the right side of the Departure list or the **INFO** button on the vertical scrollbar of the Suspended list, detailed **Flight Data:** is retrieved and displayed for the selected flight:

a) The Departure list display is replaced by the tabbed **Flight info** presentation.

b) The **INFO** button is switched to **LIST** to allow returning to the Departure list USER.

c) The **UP**, **DOWN** and **SHOW ACT** / **HIDE ACT** buttons are disabled on detailed Info display since not of any use in this context.

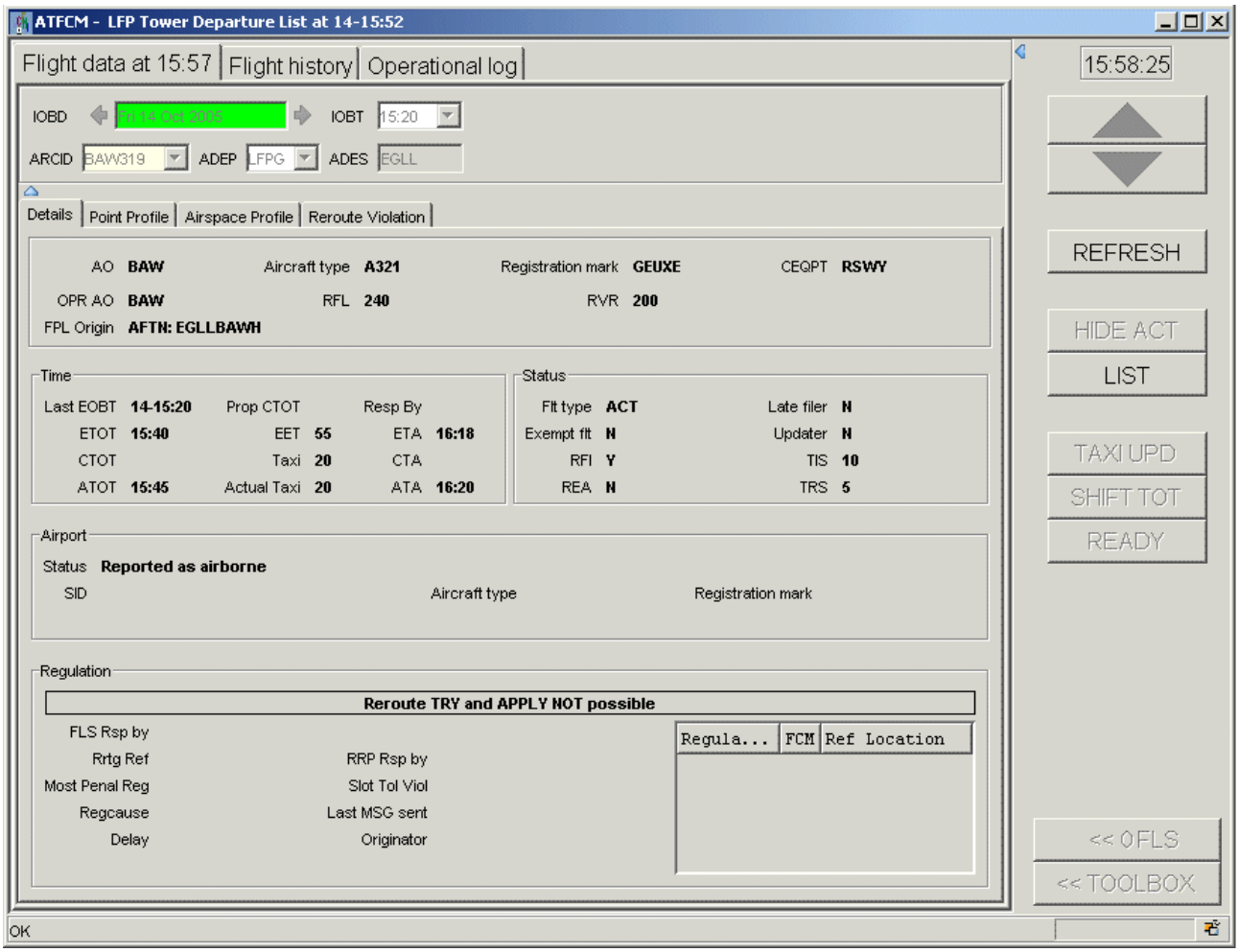


Figure Chmi-67: Tower Departure List - Flight Info display

(3) Following tabs are presented on the tabbed **Flight info** presentation:

a) The tabbed **Flight data** display.

b) The **Flight history** display.

c) The **Operational log** display.

(4) **Flight history** and **Operational log** data are only retrieved when the corresponding tab is selected for display.

(5) After a first retrieval, the **Flight history** and **Operational log** data are only refreshed on request.

(6) Switching between the different Flight info tabs triggers initial data retrieval but will not trigger subsequent refreshes.

(7) Subsequent refresh for the currently displayed tab is requested via a manual click on the **REFRESH** button.

(8) Clicking the **REFRESH** button will trigger an immediate query of the currently displayed detailed flight information.

17.6 Contextual actions

17.6.1 General

(1) On selecting a flight in the Departure list or when the tabbed info window is displayed for a flight, the different Action buttons are activated or de-activated based on the flight data context.

(2) On pushing an Action button while a flight is selected in the Departure List or displayed on the tabbed info window, an action bar is made visible at the bottom of the current display (LIST or INFO).

(3) If the Toolbox is displayed when an Action is triggered for a selected flight, the Toolbox is automatically closed.

(4) Following Action buttons may be enabled or disabled for the selected flight:

a) **TAXI UPD**: to issue an update request of the Taxi Time for the flight.

b) **SHIFT TOT**: to issue a shift request of the Take Off Time (TOT) for the flight.

c) **READY**: to issue a Ready to Depart message for the flight.

(5) The action bar includes:

a) Identifier of the selected Action and related flight.

b) Input controls for the data element related to the selected Action.

c) Send and Close buttons.

(6) When the action bar is opened, the ARCID column background of the selected flight is highlighted in a specific colour associated to the user preference:

a) **CHMI / ATFCM application / Flights / Tower departure list / Action colour.**

(7) This colour may be adjusted in the **Preferences** dialog.

(8) It will remain highlighted until the corresponding action bar is removed (Close or next Action).

(9) After the desired Action request is issued, the **Send** action button is disabled and its caption is set to **Processing ...** while the message is issued.

a) If successful, the Send action button text on the opened action bar is replaced by the word **Accepted**.

b) If failed, a dialog box is displayed with the error message returned by the server. The Send action button text on the opened action bar is replaced by the word **Rejected**.

(10) On successful Action:

a) The Tower Departure List is reactivated if it was not the case (back from Info display).

b) An immediate refresh is triggered automatically.

c) The ARCID identifier of the flight remains highlighted in BOLD for the next 5 minutes in the Tower Departure List.

(11) The action bar opened for any Action is removed:

a) If the user clicks the Close button on the right of the bar before or after Action request.

b) If the user clicks an Action button for the same or another flight in the list.

(12) Every time a refresh of the list (manual or automatic) or a refresh of the flight info display is successfully completed, any action bar opened based on the previous data is validated against the newly retrieved data.

a) If the flight is no more satisfying the conditions for the Action, the opened action bar is disabled except the Close button. The text on the Send button is replaced by the word **Obsolete**.

b) If the opened action bar remains valid, the ARCID column background of the corresponding flight is highlighted again in the list with the **Action colour**.

17.6.2 TAXI UPD

(1) The **TAXI UPD** button is activated when the Current UTC time is smaller than (Calculated TOT - Taxi time - Time to Remove from Sequence). When Calculated TOT is not available, the Estimated TOT is used instead. This button is disabled when the time limit to issue the **TAXI UPD** is reached.

(2) Clicking the TAXI UPD button, will display the **Change Taxi Time** action bar at the bottom of the screen.

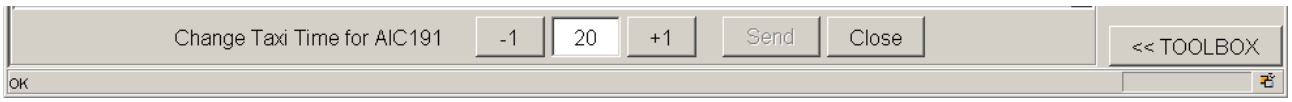


Figure Chmi-68: Tower Departure List - TAXI UPD action bar

(3) The selected ARCID is displayed on the action bar.

(4) The Taxi Time box is presented with the current Taxi Time displayed in the TXT column.

(5) This Change Taxi Time action bar will allow to adjust the Taxi Time value by:

a) Clicking one or more times on the -1 / +1 buttons (hold click repeats).

b) Each time the button is clicked, the Taxi Time is adjusted with the clicked value.

c) Accepted values are between 00 and the maximum identified by the system property **CHMI / ATFCM application / Flights / General / Maximum MINLINEUP**. When reaching the limits, the corresponding -1 / +1 button is disabled and re-enabled when appropriate.

d) The **Send** button remains disabled when the Taxi Time value on the action bar corresponds to the Taxi time in the TXT column.

(6) Once updated, you can issue the **Change Taxi Time** message for the indicated flight by clicking on the **Send** button.

(7) This button is then disabled and its caption is set to **Processing ...** while the message is issued.

(8) On reply, the button caption is changed either to **Accepted** or **Rejected** (See above).

(9) You can close the action bar without issuing any message by clicking on the Close button.

17.6.3 SHIFT TOT

(1) The **SHIFT TOT** button is activated when:

a) The selected flight in the Departure list is Actual Off-Block (ATC DPI received)

b) Or Not reported as Airborne

c) Or not departing from a CDM airport and after pushback (Current UTC time < CTOT and Current UTC time >= (CTOT - Taxi Time)).

(2) Clicking the SHIFT TOT button, will display the **ATC Target TOT** action bar at the bottom of the screen.



Figure Chmi-69: Tower Departure List - SHIFT TOT action bar

(3) The selected ARCID is displayed on the action bar.

(4) The new TOT box is presented with the current TOT displayed in the E/CTOT column.

(5) This SHIFT TOT action bar will allow to adjust the new TOT value by:

a) Clicking one or more times on the -5 / +5 buttons (hold click repeats).

b) Each time the button is clicked, the new TOT is adjusted with the clicked value in minutes.

c) Accepted values are between current CTOT-10 minutes and current CTOT+20 minutes. When reaching the limits, the corresponding -5 / +5 button is disabled and re-enabled when appropriate.

d) If the CTOT was already adjusted by a previous SHIFT TOT on the same ARCID, the authorised boundaries are still based on the initial CTOT value.

e) The Send button remains disabled when the New TOT value on the action bar corresponds to the CTOT value of the E/CTOT column. Once updated, you can issue the **SHIFT TOT** message for the indicated flight by clicking on the Send button.

(6) This button is then disabled and its caption is set to **Processing ...** while the message is issued.

(7) On reply, the button caption is changed either to **Accepted** or **Rejected** (See above).

(8) You can close the action bar without issuing any message by clicking on the Close button.

17.6.4 READY

(1) The **READY** button is activated for a regulated flight not yet airborne in the Departure list (presence of a CTOT with no ATOT yet).

(2) Clicking the READY button, will display the **Ready message** action bar at the bottom of the screen.



Figure Chmi-70: Tower Departure List - READY action bar

(3) The selected ARCID is displayed on the action bar.

(4) The Minlineup box is presented empty which implies Network Operations system will take the Taxi time as default value for MinLineUp.

(5) **Default** button is consequently disabled.

(6) This READY action bar will allow to adjust the Minlineup by:

a) Clicking one or more times on the -1 / +1 buttons (hold click repeats).

b) Each time the button is clicked, the Minlineup is adjusted with the clicked value.

c) Accepted values are between 00 and the maximum identified by the system property **CHMI / ATFCM application / Flights / General / Maximum MINLINEUP**. When reaching the limits, the corresponding -1 / +1 button is disabled and re-enabled when appropriate.

d) Clicking on the **Default** button will suppress any value in the MinLineUp box. This is used to tell the Network Operations system to use the Taxi time by default.

e) When there is no value in the MinLineUp box, the Default and the -1 buttons are disabled.

f) After this, on the first click on the +1 button, the first value displayed is 1 and buttons are re-enabled as appropriate.

(7) Once updated, you can issue the **READY** message for the indicated flight by clicking on the Send button.

(8) This button is then disabled and its caption is set to **Processing ...** while the message is issued.

(9) On reply, the button caption is changed either to **Accepted** or **Rejected** (See above).

(10) You can close the action bar without issuing any message by clicking on the Close button.

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18 CASA Regulations

18.1 Function presentation

(1) This display gives the **list of regulations** linked to all or selected FMPs for the specified date.

(2) The query dialog allows to select **All** Regulations or **Active only**.

18.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Measures / CASA Regulations**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl> + <Shift> + <F1>**.

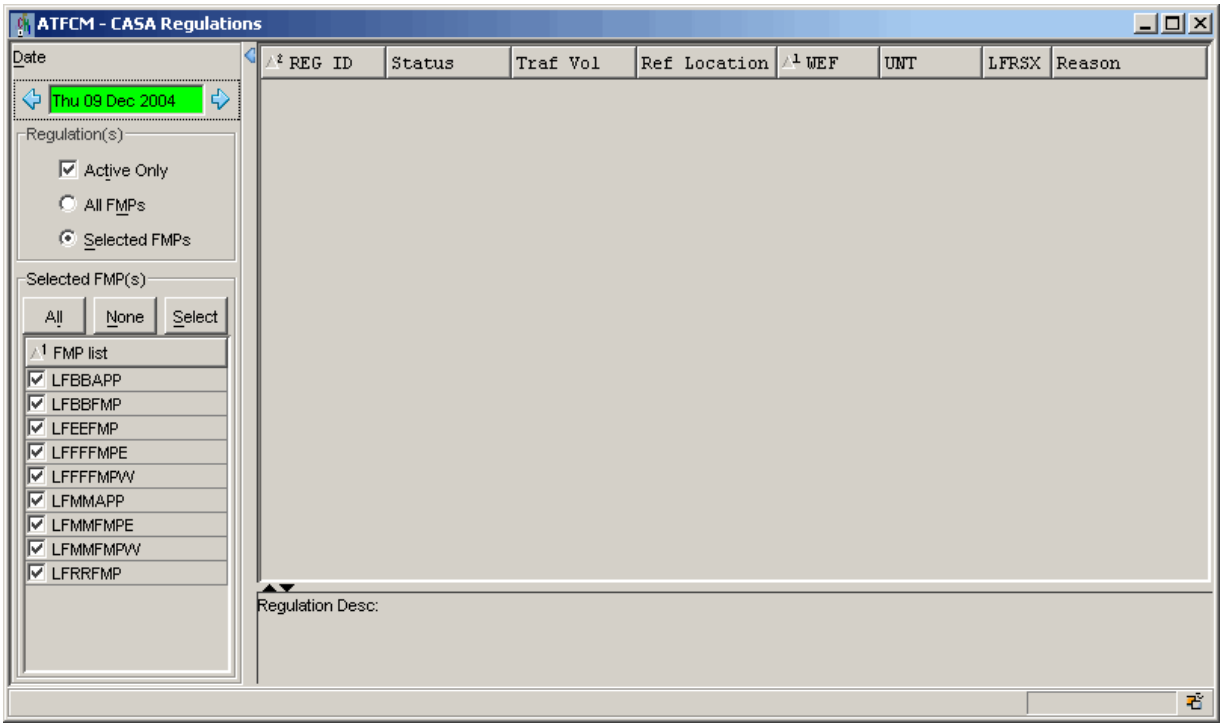


Figure Chmi-71: Query CASA Regulations

(3) A new empty **CASA Regulations** window will be opened with the following defaults:

a) **Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) "**Active only**" comes selected or unselected, same as last query.

c) "**Selected FMPs**" radio button is always initially checked.

d) "**All FMPs**" radio button is never switched on initially.

e) "**Selected FMP(s)**" list is filled in with the last updated list maintained using the **FMP Selection** dialog.

f) FMPs are checked in the "**Selected FMP(s)**" list based on the last query issued that uses FMP selection.

(4) When no FMPs have been selected yet using the **FMP Selection** dialog (this is the case after application installation),

a) A message box will warn you and invite you to select first one or more FMPs.

b) On clicking OK, the **FMP Selection** dialog is opened automatically.

c) After adding one or more FMPs to the "**Selected list**", click on the OK button.

d) This will close the dialog and update the "**Selected list**" on the calling window.

(5) On initial **CASA Regulations** query preparation, the focus is set on the date control.

(6) After this, it is moved logically between the query controls in position sequence (left to right and then top to bottom).

18.3 Preparing to query

(1) The traffic **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The "**Active only**" checkbox may be checked if only active regulations shall be retrieved. Otherwise this control shall remain unchecked.

(3) The "**All FMPs**" radio button shall be checked if no selection on FMP shall be performed (retrieve regulations for all FMPs). This hides the Selected FMP(s) data group. Otherwise "**Selected FMPs**" radio button shall be checked.

(4) When the "**Selected FMPs**" radio button is checked, the Selected FMP(s) data group is displayed.

(5) FMPs in the "**Selected list**" may then be checked or unchecked as desired.

(6) Click on the "**Selected list**" title to adjust sort sequence (alternatively ascending and descending sequence). This shall not impact the list of selected FMPs and the fact that a given FMP is checked or not.

(7) Click the "**All**" button to check all the FMPs present in the Selected list.

(8) Click the "**None**" button to uncheck all the FMPs present in the Selected list.

(9) To add or remove one or more FMPs in the Selected list, the **FMP selection** dialog shall be used. Click on the "**Select**" button to open this dialog.

(10) When the Selected list is up to date on the **FMP selection** dialog, click on the OK button to save the list.

(11) Rules for updating the **Selected list** on all opened windows using an FMP **Selected list** for the query is described in **FMP selection** dialog.

(12) Clicking the Cancel button of the **FMP selection** dialog stops editing the "**Selected list**" and come back to the previous screen without saving any modification to the list.

(13) **CASA Regulations** window may be closed before issuing any query.

(14) A maximum of 12 FMPs may be checked when issuing a query. Depending on your access rights, this limit may be extended.

(15) The **Send** button will send the query to the Network Operations system.

(16) The **Reset** action common functionality is implemented for this function.

(17) Following query data elements are considered for activating the Reset action:

a) Date.

b) Active only.

c) All or Checked FMPs in the Selected list.

18.4 Displaying the query results

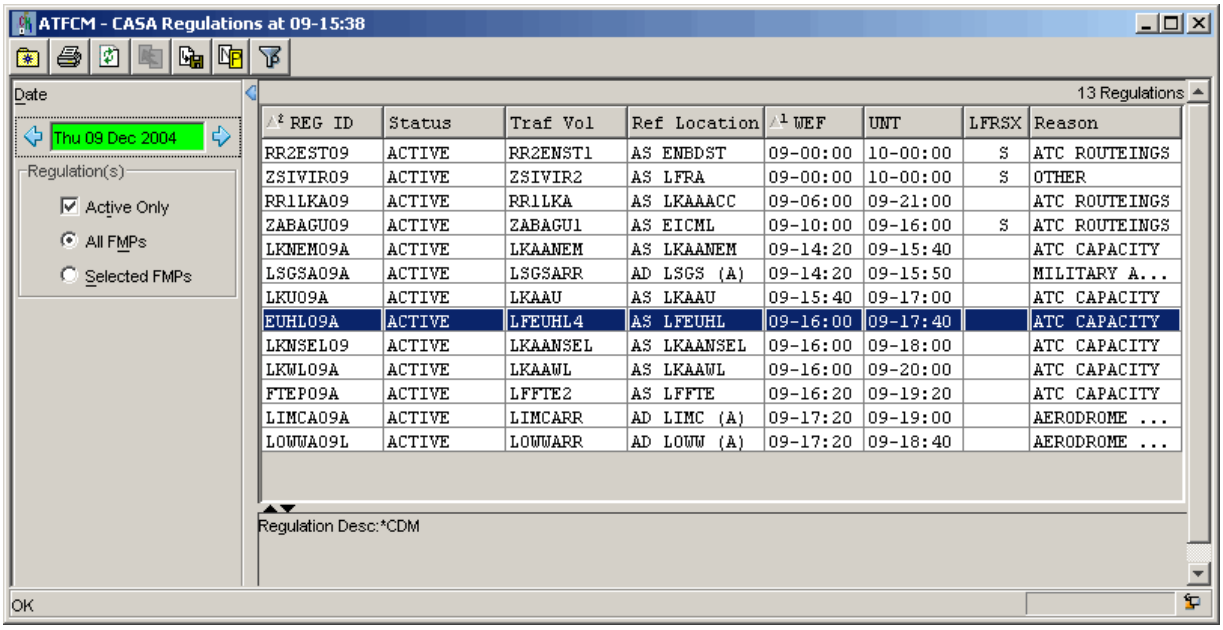


Figure Chmi-72: CASA Regulations

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The left part of the window contains the different **query data elements** used to retrieve the displayed regulation list:

a) The observed **Date**.

b) If the retrieved list contains **Active only** Regulations.

c) If Regulations were retrieved for **All FMPs**.

d) Otherwise if FMP selection was applied, the list of **Selected FMPs** is displayed where FMPs included in the query are checked.

(3) At the top of the regulation list, a counter indicates the **number of regulations** retrieved.

(4) This counter may be affected by **filtering** (fff / rrr Regulations where fff is the filtered number and rrr is the retrieved number).

(5) On query results display, the focus is set on the regulation list after selecting the first regulation in the list.

(6) For each regulation, the **main list** contains:

a) The identifier of the regulation (**REG ID**),

b) The **Status** of the regulation (see later),

c) The traffic volume on which it applies (**Traf Vol**),

d) The related reference location (**Ref Location**),

e) The Start and End time of the regulation (**WEF** and **UNT**),

f) The link state indicates that the regulation is linked to other regulations to calculate the delays (**L**) (see later).

g) The indication that a Confirmation message is required when a flight is affected by this regulation (**F**). Set to "F" if applicable, otherwise remains blank.

h) The indication that a Runway Visual Range (RVR) is associated to one or more sub-periods (**R**). Set to "R" if applicable, otherwise remains blank.

i) The indication that flights affected by this regulation have to be shifted for one or more sub-periods (**S**). Set to "S" if applicable, otherwise remains blank.

j) The Regulation **Reason** (see later).

k) The related **FMP** identifier, if the query was performed on selected FMPs. This information is not available when query is performed on **All FMPs**.

(7) The initial display sort sequence for the regulation list is:

a) First ascending on WEF.

b) Second ascending on REG ID.

(8) The list allows only single selection.

(9) When a regulation is selected in the list, the details pane of the Regulation List is filled in with:

a) the description and comments, if any, made by the NM Operator.

(10) The possible activity values for the **Status** are:

a) Not Active

b) Activating

c) Active

d) Modifying

e) Terminated

f) Cancelling

g) Cancelled

(11) The possible Link states are:

a) **A** for Automatic Link,

b) **M** for Manual Link,

c) **L** when Both Automatic and Manual Links are present.

18.5 User interactions

(1) The **Normal / Proposal** two states button may be adjusted before selecting an option of the popup menu described in the "Contextual actions" section.

(2) The selected **Normal / Proposal** value will be transferred to the invoked function.

(3) This window allows to modify a previous query selection and to issue a new request.

(4) On query modification, the **Reset action** is enabled.

(5) To display / hide the query part, click respectively the Right / Left arrow at the top next to the split bar.

(6) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(7) To print the Regulation list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(8) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(9) To modify the **Sort sequence** of the regulations, click on column headers corresponding to your desired sort criteria.

(10) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(11) Increasing / reducing the window height will impact the size of the FMP Selected list and the Regulation list controls.

(12) Increasing / reducing the window width will impact the size of the Regulation list and the details pane of the Regulation List controls.

18.6 Contextual actions

(1) When a line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Regulation and may contain the following options:

a) **Regulation Description**: displays detailed information for the selected regulation.

b) **Regulation Delays**: displays the delays for the selected regulation and its activation period.

c) **Regulation Flight List**: displays the Provisional Slot Allocation List (Traffic Load) for the selected regulation and its activation period starting either with the current time or the start time of the regulation.

d) **Traffic counts**: displays the count graph for the regulated traffic volume for the selected day.

(3) For the Regulation Delays, Regulation Flight List and Traffic Counts options, the transferred period may be affected by user preferences related to **Default query period identification**.

19 Regulation Description

19.1 Function presentation

(1) This display gives the different data elements for **one Regulation** associated to a selected date.

(2) The same functionality is used for the display of the **ADP Current** in the context of the ATM Messages menu.

19.2 Opening the window

(1) From the Application / ATFCM menu, select the option:

a) **Measures / Regulation Description** for the Regulation Description display.

b) **ATM Messages / ADP Current** for the display of the detailed ADP Current.

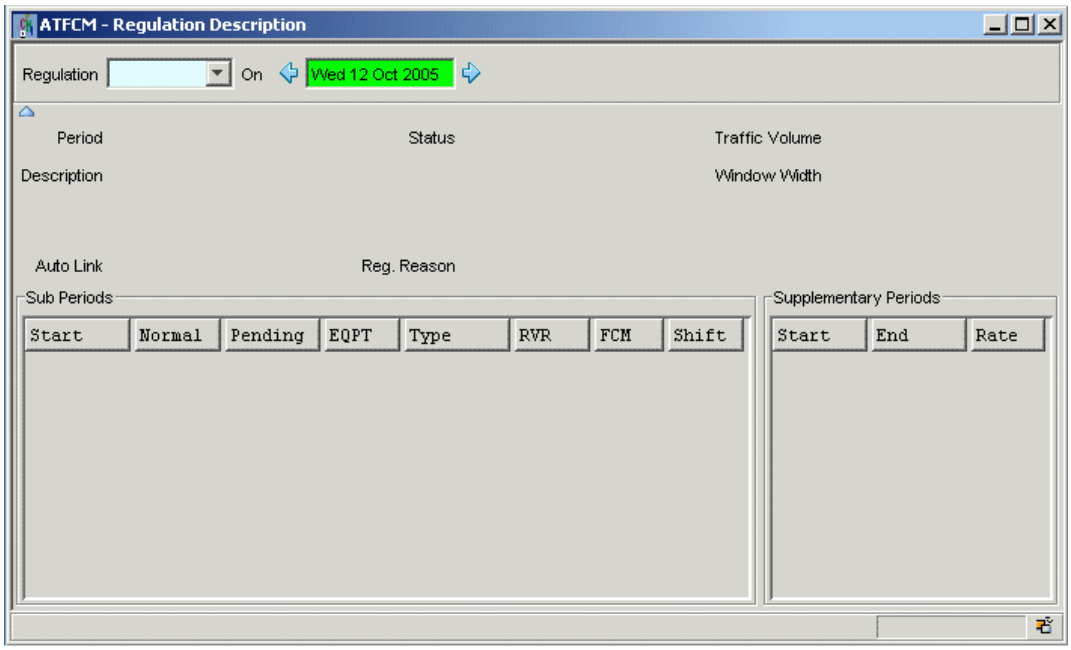


Figure Chmi-73: Query Regulation Description

(2) A new empty **Regulation Description** or **ADP Current** window will be opened with the following defaults:

a) The **Regulation** drop-down box will propose the most recent Regulation identifiers you have queried before with success.

b) **On**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

(3) On initial display of the **Regulation Description** or **ADP Current** window, the focus is set on the **Regulation** input control.

(4) After this, it is moved logically between the active and visible controls in position sequence. (left to right and then top to bottom)

19.3 Preparing to query

(1) The **Regulation** control may be adjusted by selecting one of the IDs proposed in the drop-down list or by keying in a new Regulation identifier in the dedicated input box.

(2) The Regulation identifier has a maximum of 8 characters.

(3) The traffic date (**On**) may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(4) The **Send** button will send the query to the Network Operations system.

(5) **Regulation Description** and **ADP Current** windows may be closed before issuing any query.

(6) The **Reset** action common functionality is implemented for this function.

(7) Following query data elements are considered for activating the Reset action:

a) **Regulation**

b) Date (**On**)

19.4 Displaying the query results

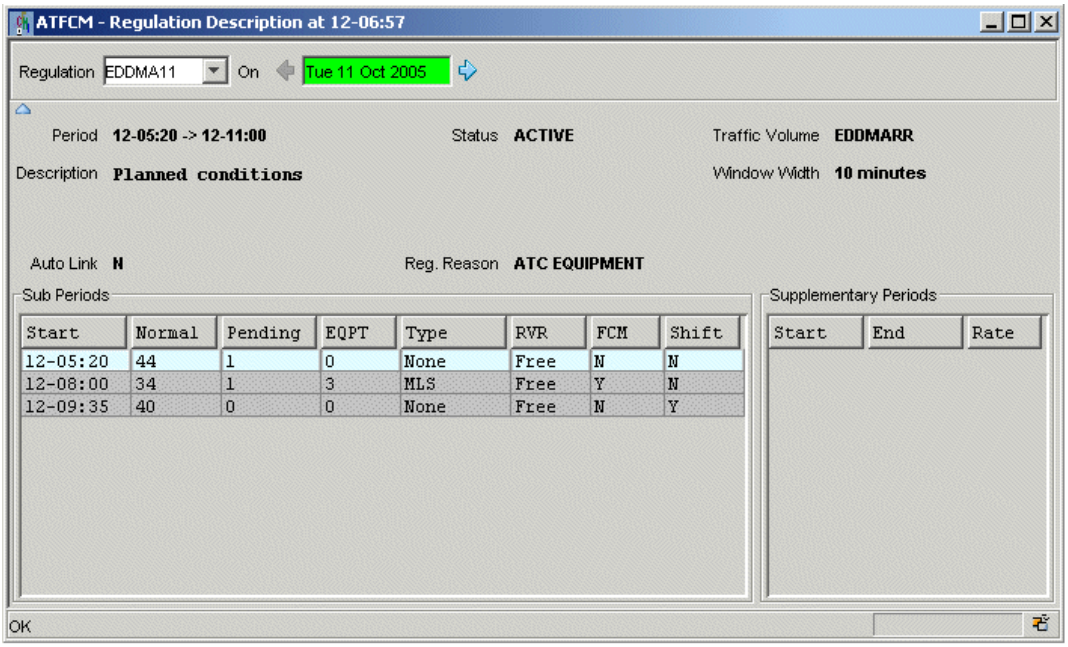


Figure Chmi-74: Regulation Description

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The header box of the window contains the different **query data elements** used to retrieve the displayed regulation data:

a) The Regulation identifier (**Regulation**).

b) The observed date (**On**).

(3) Below the query header box, the display provides the different **Regulation Description reply data elements**:

a) The activation **Period** of the selected Regulation using the format DD-HH:MM for the From and the Until times.

b) The **Status** of the Regulation.

c) The **Traffic Volume** to which the Regulation applies,

d) The full **Description** text of the selected Regulation, if any.

e) The **Window Width** related to the Regulation.

f) The **Auto Link** yes/no state indicates if the local delay given by this regulation will be taken into account to compute the delay in other regulations.

g) The Regulation reason (**Reg. Reason**) (see below).

(4) The possible **Regulation Reasons** are:

a) ACCIDENT/INCIDENT

b) AERODROME CAPACITY

c) AERODROME SERVICES

d) AIRSPACE MANAGEMENT

e) ATC CAPACITY

f) ATC EQUIPMENT

g) ATC IND ACTION

h) ATC ROUTEINGS

i) ATC STAFFING

j) DE-ICING

k) ENVIRONMENTAL ISSUES

l) IND ACTION NON-ATC

m) OTHER

n) SPECIAL EVENT

o) WEATHER

(5) The main part of the display shows two listboxes, which provides the periods.

(6) The listbox on the left contains the **Sub Periods** definition.

(7) Each line contains:

a) The **Start** date and time (DD-HH:MM) within the Regulation activity period. This sub-period ends when the next one starts or at the end of the Regulation activity period if no other sub-period follows.

b) The associated **Normal** rate for this part of the Regulation period.

c) The associated **Pending** rate.

d) The equipment rate (**EQPT**), if any.

e) With the associated Radio communication, navigation and appraoch aid equipment **Type**.

f) The Runway Visual Range associated to the sub-period (**RVR**).

g) If Flight Confirmation is mandatory Yes or No (**FCM**).

h) If flights affected by this regulation must be shifted (**Shift**).

(8) The listbox on the right contains the **Supplementary Periods** definition.

(9) Each line contains:

a) the **Start** date and time (DD-HH:MM),

b) the **End** date and time (DD-HH:MM),

c) the associated **Rate** for the supplementary sub-period.

(10) On query results display, the focus is set on the Sub Periods list with the first line selected.

(11) After this it moves to the Supplementary Periods list where the first line is also selected, if not empty.

(12) After this, back to the query data elements.

19.5 User interactions

(1) The **Normal / Proposal** two states button may be adjusted before selecting an option of the popup menu described in the Contextual actions section.

(2) The selected **Normal / Proposal** value will be transferred to the invoked function.

(3) This window allows to modify a previous query selection and to issue a new request.

(4) On query modification, the **Reset action** is enabled.

(5) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(6) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(7) To print the Regulation Description details, first retrieve the desired details. When ready, click on the **Print** button or select the **File / Print** menu option.

(8) To modify the **Sort sequence** of the Regulation sub-periods, click the column headers corresponding to your desired sort criteria.

(9) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(10) Increasing / reducing the window height will impact the size of both Regulation Sub Periods and Supplementary Periods list controls.

(11) Increasing / reducing the window width will impact the size of the Regulation Sub Periods list controls.

19.6 Contextual actions

(1) Once Regulation Description details are displayed for one Regulation, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the displayed regulation and may contain the following options:

a) **Regulation Delays**: displays the delays for the selected regulation and its activation period.

b) **Regulation Flight List**: displays the Flight List (Traffic Load) for the selected regulation and its activation period starting either with the current time or the start time of the regulation.

c) **Traffic Counts**: displays the count graph for the regulated traffic volume for the selected day.

d) **ADP History**: lists the history log for the selected ADP Regulation and date (only available in the ADP Current display).

(3) The transferred period may be affected by user preferences related to **Default query period identification**.

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20 Rerouting Measures

20.1 Function presentation

(1) This display gives the **list of rerouting measures** linked to all or selected FMPs for the specified date.

(2) The terminated rerouting measures are not displayed in the list.

20.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Measures / Reroutings**.

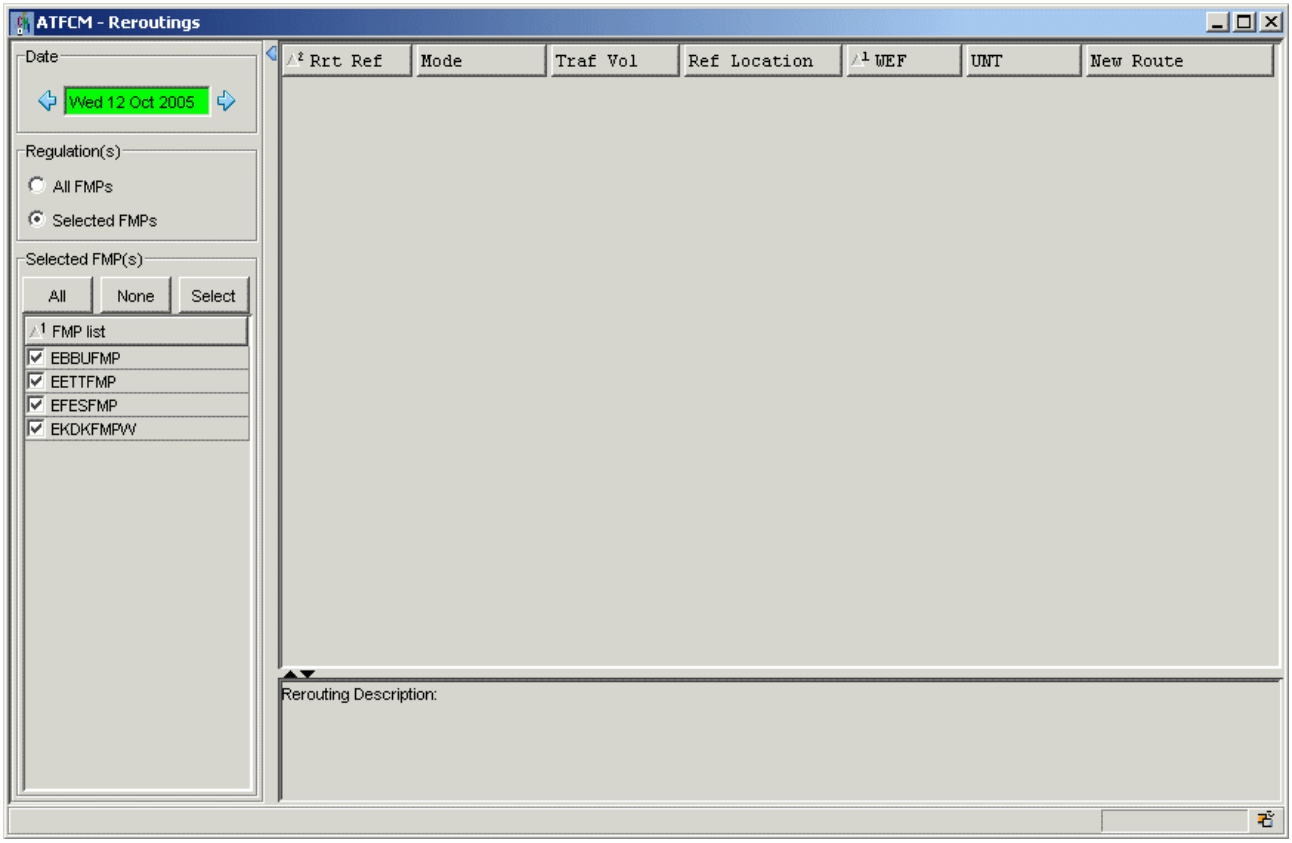


Figure Chmi-75: Query Reroutings

(2) A new empty **Reroutings** window will be opened with the following defaults:

a) **Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) "**Selected FMPs**" radio button is always initially checked.

c) "**All FMPs**" radio button is never switched on initially.

d) "**Selected FMP(s)**" list is filled in with the last updated list maintained using the **FMP Selection** dialog.

e) FMPs are checked in the "**Selected FMP(s)**" list based on the last query issued that uses FMP selection.

(3) When no FMPs have been selected yet using the **FMP Selection** dialog (this is the case after application installation),

a) A message box will warn you and invite you to select first one or more FMPs.

b) On clicking OK, the **FMP Selection** dialog is opened automatically.

c) After adding one or more FMPs to the "**Selected list**", click on the OK button.

d) This will close the dialog and update the "**Selected list**" on the calling window.

(4) On initial **Reroutings** query preparation, the focus is set on the date control.

(5) After this, it is moved logically between the query controls in position sequence (left to right and then top to bottom).

20.3 Preparing to query

(1) The traffic **Date** may be adjusted to one of the proposed values.

(2) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(3) The "**All FMPs**" radio button shall be checked if no selection on FMP shall be performed (retrieve reroutings for all FMPs). This hides the Selected FMP(s) data group. Otherwise "**Selected FMPs**" radio button shall be checked.

(4) When the "**Selected FMPs**" radio button is checked, the Selected FMP(s) data group is displayed.

(5) FMPs in the "**Selected list**" may then be checked or unchecked as desired.

(6) Click on the "**Selected list**" title to adjust sort sequence (alternatively ascending and descending sequence). This shall not impact the list of selected FMPs and the fact that a given FMP is checked or not.

(7) Click the "**All**" button to check all the FMPs present in the Selected list.

(8) Click the "**None**" button to uncheck all the FMPs present in the Selected list.

(9) To add or remove one or more FMPs in the Selected list, the **FMP selection** dialog shall be used. Click on the "**Select**" button to open this dialog.

(10) When the Selected list is up to date on the **FMP selection** dialog, click on the OK button to save the list.

(11) Rules for updating the **Selected list** on all opened windows using an FMP **Selected list** for the query is described in **FMP selection** dialog.

(12) The FMPs checked in the "**Selected list**" before opening the **FMP selection** dialog shall remain checked.

(13) Clicking the Cancel button of the **FMP selection** dialog stops editing the "**Selected list**" and come back to the previous screen without saving any modification to the list.

(14) **Reroutings** window may be closed before issuing any query.

(15) A maximum of 12 FMPs may be checked when issuing a query. Depending on your access rights, this limit may be extended.

(16) The **Send** button will send the query to the Network Operations system.

(17) The **Reset** action common functionality is implemented for this function.

(18) Following query data elements are considered for activating the Reset action:

a) Date

b) All or Checked FMPs in the Selected list

20.4 Displaying the query results

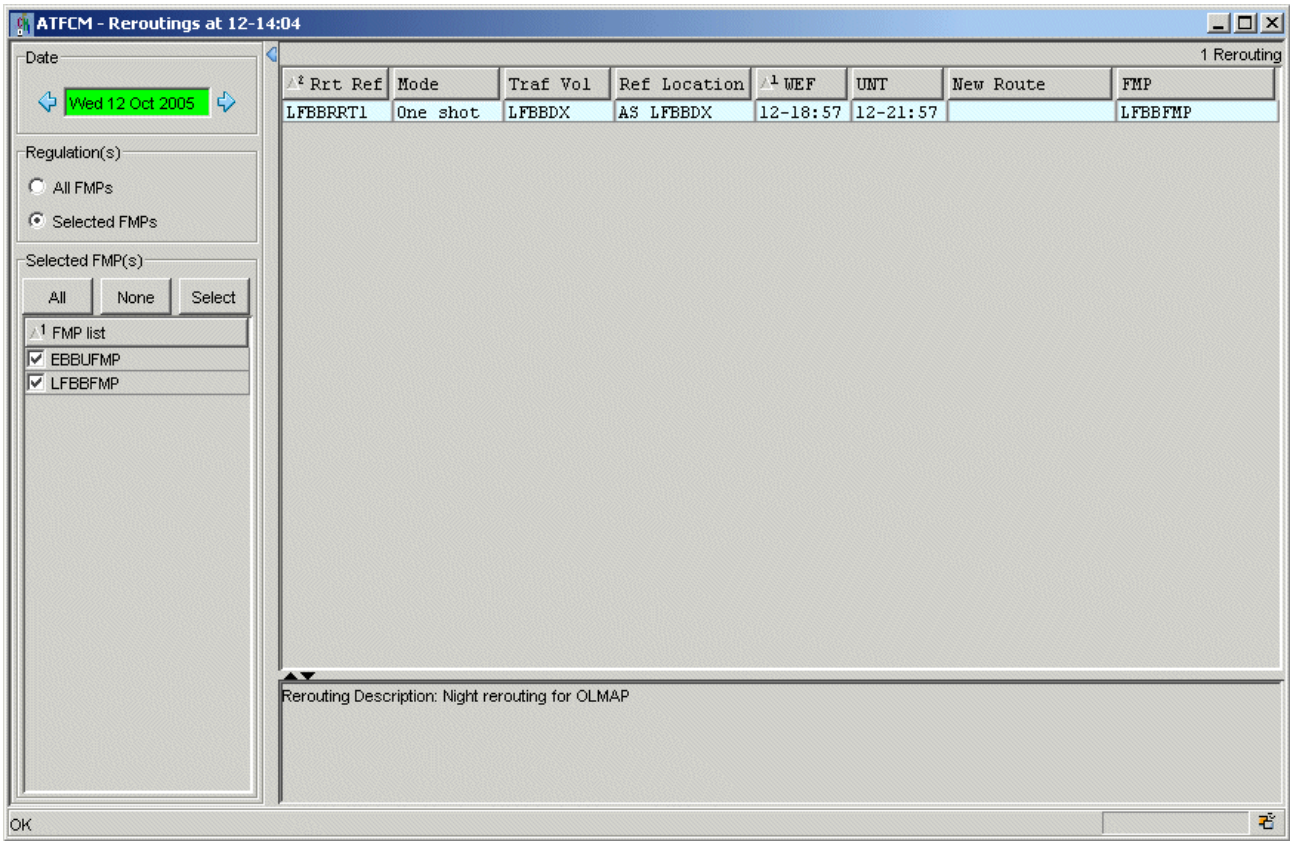


Figure Chmi-76: Reroutings

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The left part of the window contains the different **query data elements** used to retrieve the displayed rerouting:

a) The observed **Date**,

b) If rerouting measures were retrieved for **All FMPs**,

c) Otherwise if FMP selection was applied, the list of **Selected FMPs** is displayed where FMPs included in the query are checked.

(3) At the top of the rerouting list, a counter indicates the **number of reroutings** retrieved.

(4) This counter may be affected by **filtering** (fff / rrr Reroutings where fff is the filtered number and rrr is the retrieved number).

(5) On query results display, the focus is set on the rerouting list after selecting the first rerouting in the list.

(6) For each rerouting, the **main list** may contain the following data elements, if available:

a) The identifier of the rerouting measure (**Rrt Ref**).

b) The rerouting measure **Mode**.

c) The Traffic Volume on which it applies (**Traf Vol**), if any.

d) The Reference Location (**Ref Location**), if any.

e) The start and end time of the rerouting measure (**WEF / UNT**), if any.

f) The new Route identifier (**New Route**), if any.

g) The related **FMP** identifier, if the query was performed on selected FMPs. This information is not available when query is performed on **All FMPs**.

(7) The possible values for the rerouting measure **Mode** are:

a) **One shot** when the rerouting measure will be applied only once on the selected flights.

b) **Recurrent** when the rerouting measure will be applied on selected flights and will be applicable to all future received flights that are affected by the measure. A recurrent mode rerouting measure will be effective until the end of the reference period or until it is cancelled.

(8) The list allows only single selection.

(9) When a rerouting is selected in the list, the details pane at the bottom is filled in with:

a) the description and comments, if any, made by the NM Operator.

(10) The initial display sort sequence for the rerouting list is:

a) First ascending on WEF.

b) Second ascending on Rrt Ref.

20.5 User interactions

(1) This window allows to modify a previous query selection and to issue a new request.

(2) On query modification, the **Reset action** is enabled.

(3) To display / hide the query part, click respectively the Right / Left arrow at the top next to the split bar.

(4) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(5) To print the Rerouting list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(6) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(7) To modify the **Sort sequence** of the reroutings, click the column headers corresponding to your desired sort criteria.

(8) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(9) Increasing / reducing the window height will impact the size of the FMP Selected list and the Rerouting list controls.

(10) Increasing / reducing the window width will impact the size of the Rerouting list and the bottom note box controls.

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21 ATM Messages Release Notifier

21.1 Function presentation

(1) This display notifies the user when some new ATM Messages have been released.

(2) When started, this window will be refreshed on a regular time interval identified by the system property:

a) **CHMI / ATFCM application / ATM Messages / Release Notifier / Refresh interval**

(3) To preserve server performance, the automatic refresh might be interrupted after a given elapse time of inactivity with no other manual Send query action in-between. This elapse time is identified by the system property:

a) **CHMI / ATFCM application / ATM Messages / Release Notifier / Limited auto refresh period**

(4) Once interrupted, the Release Notifier window is **forced on top** of all other windows, but it will not remain on top when another window gets the focus ! A large button covering all reply data will be displayed with the text "**Click here to restart auto-refresh**".

(5) The user may request to retrieve at any time, latest release information from the Network Operations server by clicking the Send button.

(6) This notifier window may be opened automatically when the ATFCM Collaboration Human Machine Interface application is started after signon.

(7) This window may only be started once.

21.2 Opening the window

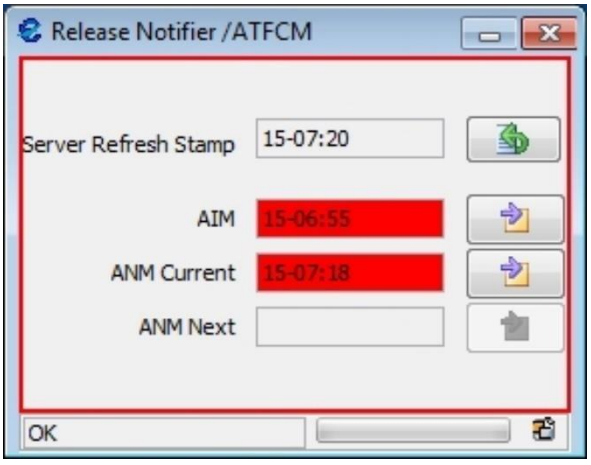


Figure Chmi-77: ATM Messages Release Notifier

(1) From the Application / ATFCM menu, select the option **ATM Messages / Release Notifier**.

(2) On opening, the latest retrieved stamps are automatically restored from your system properties.

(3) On first opening after application installation, no stamps are available locally.

(4) Stamps are displayed using the DD-HH:MM format.

(5) A stamp may remain blank if no related ATM Messages have been published yet for the corresponding day.

(6) The Server Refresh Stamp indicates when the server retrieved the latest release information (DD-HH:MM format).

(7) Every time this window is re-opened, latest ATM Messages release stamps are retrieved. These stamps are based on the latest server refresh which happens at regular intervals (currently every 15 minutes).

(8) The user adjustable preference **CHMI / ATFCM application / ATM Messages / Release Notifier / Open at startup** allows, when set to TRUE, to request immediate start-up of the **ATM Messages Release Notifier** on ATFCM Collaboration Human Machine Interface application start.

21.3 Displaying the query results

(1) The Server Refresh Stamp contains the time stamp when the server retrieved the latest release information.

(2) If any of the locally stored stamps is later than the newly retrieved stamp:

a) The newly retrieved stamp, if available, replaces the previously displayed value.

b) The background colour of the line is switched to the **Newly released colour** user preference.

c) The **ATFCM - Release Notifier** window is forced on top of all other windows but it will not remain on top when another window gets the focus!

(3) When no ATM message has been released yet for the given day, the corresponding stamp box remains empty.

(4) The different **retrieve buttons** next to each displayed stamp are only enabled when a valid stamp is displayed.

21.4 User interactions

(1) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will refresh the displayed stamp based on the latest data available on the server.

(2) Click on one of the following **retrieve buttons** to automatically open the corresponding window with an immediate query for the latest released ATM Messages for the given date. This action will also reset the stamp background to normal colour for the corresponding line.

a) **AIM** to show the latest list of released AIM.

b) **ANM Current** to show the latest updates to the ANM for the current day.

c) **ANM Next** to show the latest updates to the ANM for the next day.

(3) Click on the **Reset** button at the top, next to the Server Refresh Stamp, to restore the background colour of all lines to the normal colour.

(4) To print the currently displayed list of stamps, click on the **Print** button or select the **File / Print** menu option.

(5) To export the currently displayed list of stamps, click on the **Export** button or select the **Action / Export** menu option.

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22 AIM Display

22.1 Function presentation

(1) This display gives the list of ATFCM Information Messages (AIM) valid on a selected date or later.

(2) Based on this list, the user is able to retrieve all details for a selected AIM entry.

(3) Data retrieved by this query are extracted from the ATM Message Exchange (AME) database.

22.2 Opening the window

(1) From the Application / ATFCM menu, select the option **ATM Messages / AIM List**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl>** + **<Shift>** + **<F2>**.

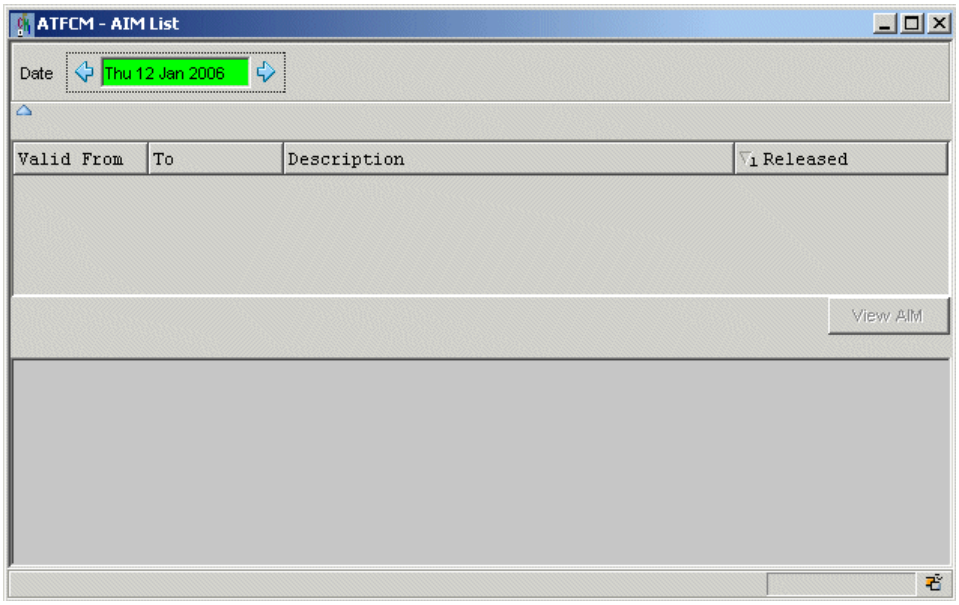


Figure Chmi-78: Query AIM Display

(3) A new empty **AIM Display** window will be opened with the following defaults:

a) **Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

(4) On initial display, the focus is set on the only query control: the **Date** input control.

22.3 Preparing to query

(1) The **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The **Send** button will send the query to the Network Operations system.

(3) This window may be closed before any query is issued.

(4) The **Reset** action common functionality is implemented for this function.

22.4 Displaying the query results

22.4.1 AIM List

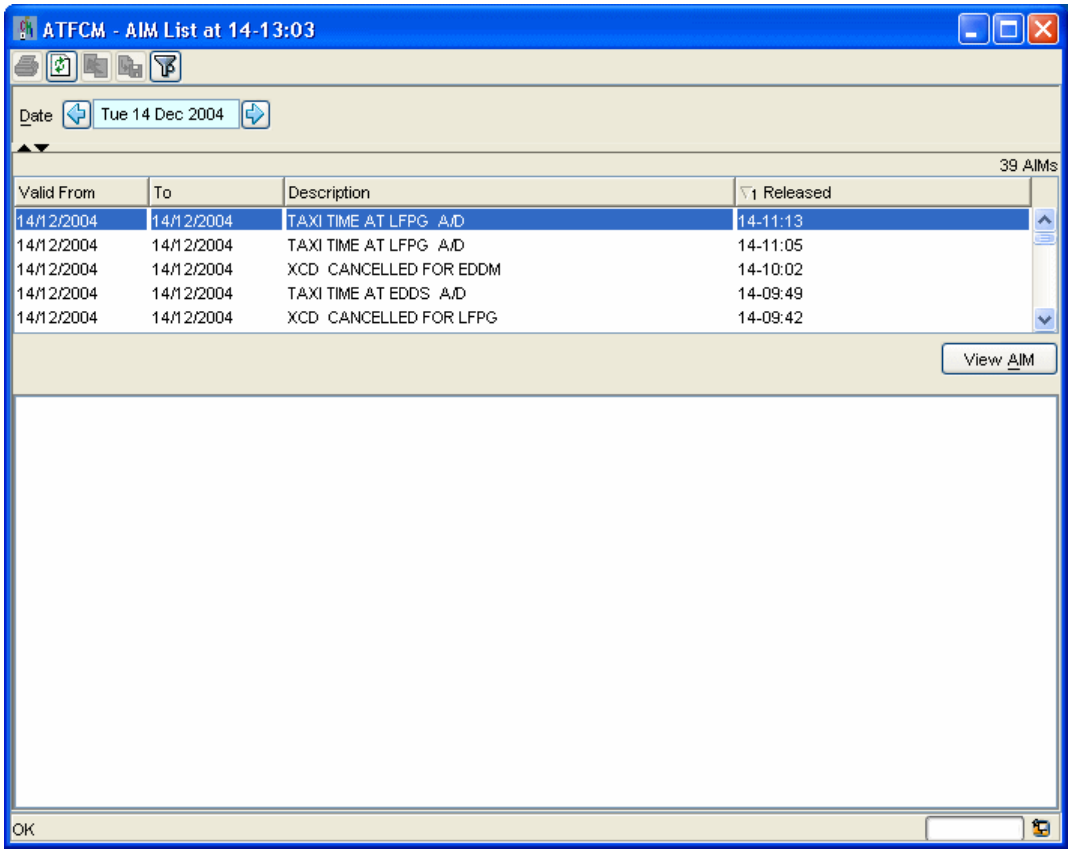


Figure Chmi-79: AIM List Display

(1) The title contains the time stamp when the AIM List was retrieved from the Network Operations system.

(2) The header box contains the selected query criteria: the observed Date.

(3) At the top of the AIM list, a counter indicates the **number of AIMs** retrieved.

(4) This counter may be affected by **filtering** (fff / rrr AIMs where fff is the filtered number and rrr is the retrieved number).

(5) The list shows all AIMs valid on the selected date or later.

(6) The list shows following details for each AIM entry:

a) The **Valid From** date (DD/MM/YYYY).

b) The valid **To** date (DD/MM/YYYY or **UFN** for Until Further Notice).

c) The **Description** associated to the AIM entry.

d) The stamp when the AIM was **Released** (DD-HH:MM).

(7) On AIM list query results display, the focus is set on the AIM list after selecting the first line in the list.

(8) After this, the focus moves to the "View AIM" button.

22.4.2 AIM Details

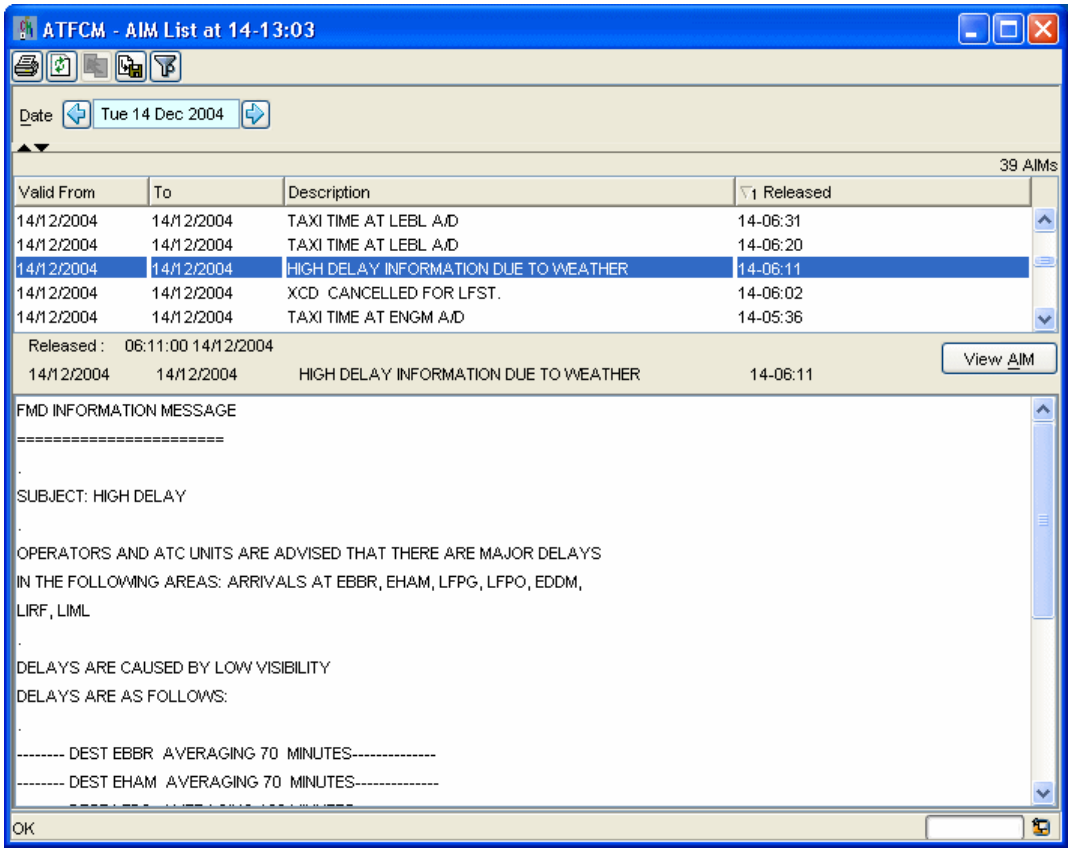


Figure Chmi-80: Display AIM details

(1) Once the AIM list has been retrieved for a selected date and later, the details for one AIM entry may be retrieved as follow:

a) Select an AIM entry in the list by clicking on it or moving the selection with up and down arrows while the focus is on the list.

b) When an entry is selected in the list, the **View AIM** button is enabled.

c) Clicking on this button or simply double-clicking the AIM line in the list, will request to retrieve and display the full AIM details.

(2) When all details are retrieved for the selected AIM in the list:

a) A first line indicates the date and time when the AIM was released (HH:MM:SS DD/MM/YYYY).

b) A second line repeat the information selected from the AIM list. This will allow to identify the retrieved AIM details when another AIM line is selected in the list.

c) The listbox at the bottom displays the description text linked to the retrieved AIM.

(3) On AIM detail query results display, the focus is set on the AIM List with the previous line selection kept.

(4) After this, the focus moves to the "View AIM" button and then on the AIM detail listbox control at the bottom.

22.5 User interactions

(1) This window allows to modify a previous query selection and to issue a new request.

(2) On query modification, the **Reset action** is enabled.

(3) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(4) To update the AIM list, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(5) To view the details for one AIM, select the desired line in the list and choose the **View AIM** button or double-click the line.

(6) To print AIM details, first retrieve the desired AIM detailed data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(7) To export the retrieved AIM details, first retrieve the desired AIM detailed data in the window. When ready, click on the **Export** button or select the **Action / Export** menu option.

(8) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(9) To modify the **Sort sequence** of the AIMs, click on column headers corresponding to your desired sort criteria.

(10) Increasing / reducing the window height will impact the size of the AIM details control.

(11) The window size proportion dedicated to the AIM list part and the AIM details part shall be adjustable by the user.

(12) Increasing / reducing the window width will impact the size of the AIM list and details controls.

23 ANM Display

23.1 Function presentation

(1) This display gives the ATFCM Notification Message (ANM) details for a selected date.

(2) Data retrieved by this query are extracted from the ATM Message Exchange (AME) database.

23.2 Opening the window

(1) From the Application / ATFCM menu, select the option **ATM Messages / ANM**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl> + <Shift> + <F3>**.

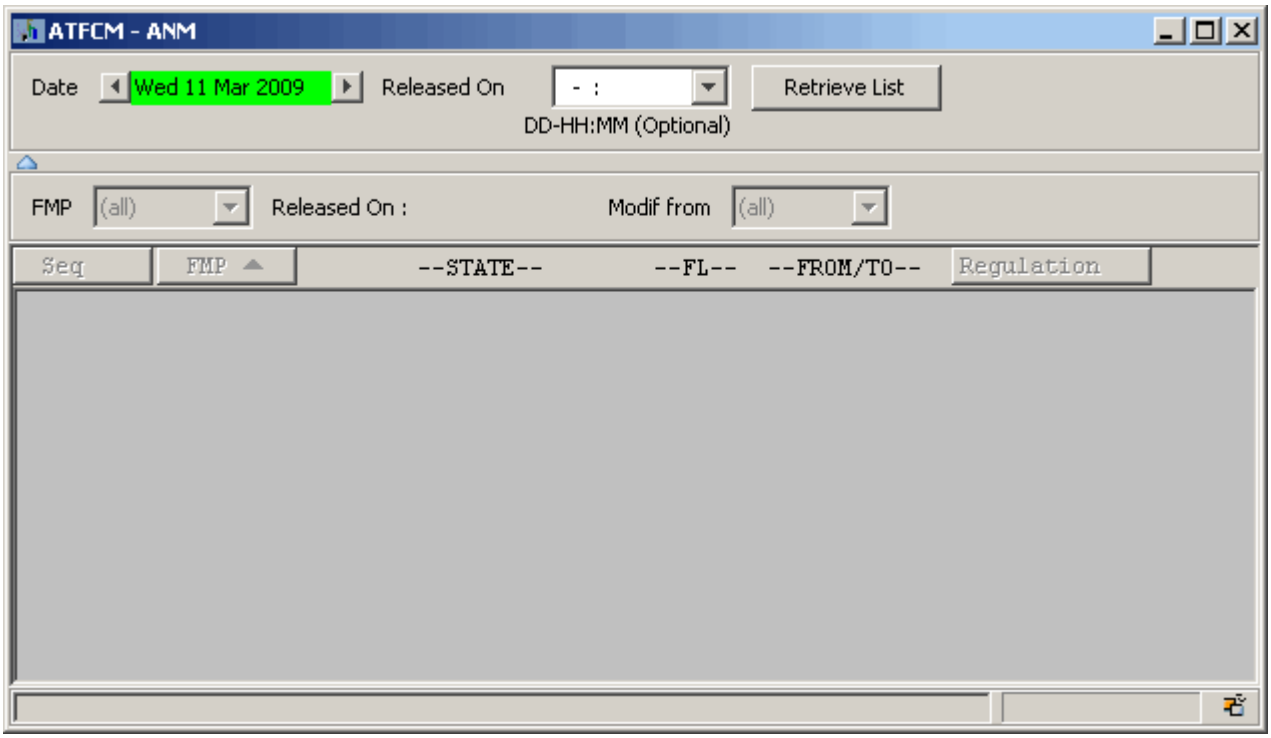


Figure Chmi-81: Query ANM Display

(3) A new empty **ANM Display** window will be opened with the following defaults:

a) **Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **Released On**: proposed empty (optional query criteria).

(4) On initial display, the focus is set on the Date control.

(5) After this it is moved to the Released On edit control of the combo box and finally on the Retrieve List button.

23.3 Preparing to query

(1) The **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The **Released On** query criteria is optional.

(3) It allows the selection of the latest ANM released on a specified date and time. (DD-HH:MM)

(4) By clicking on the **Retrieve List** button, the system will issue a query to propose the list of effective release stamps for the selected date.

(5) On reply of the Retrieve List, the focus is set on the **Released On** edit control of the combo box.

(6) When the Release stamp is not provided, the system will return the latest released ANM data for the selected date.

(7) The **Send** button will send the query to the Network Operations system.

(8) This window may be closed before any query is issued.

(9) The **Reset** action common functionality is implemented for this function.

23.4 Displaying the query results

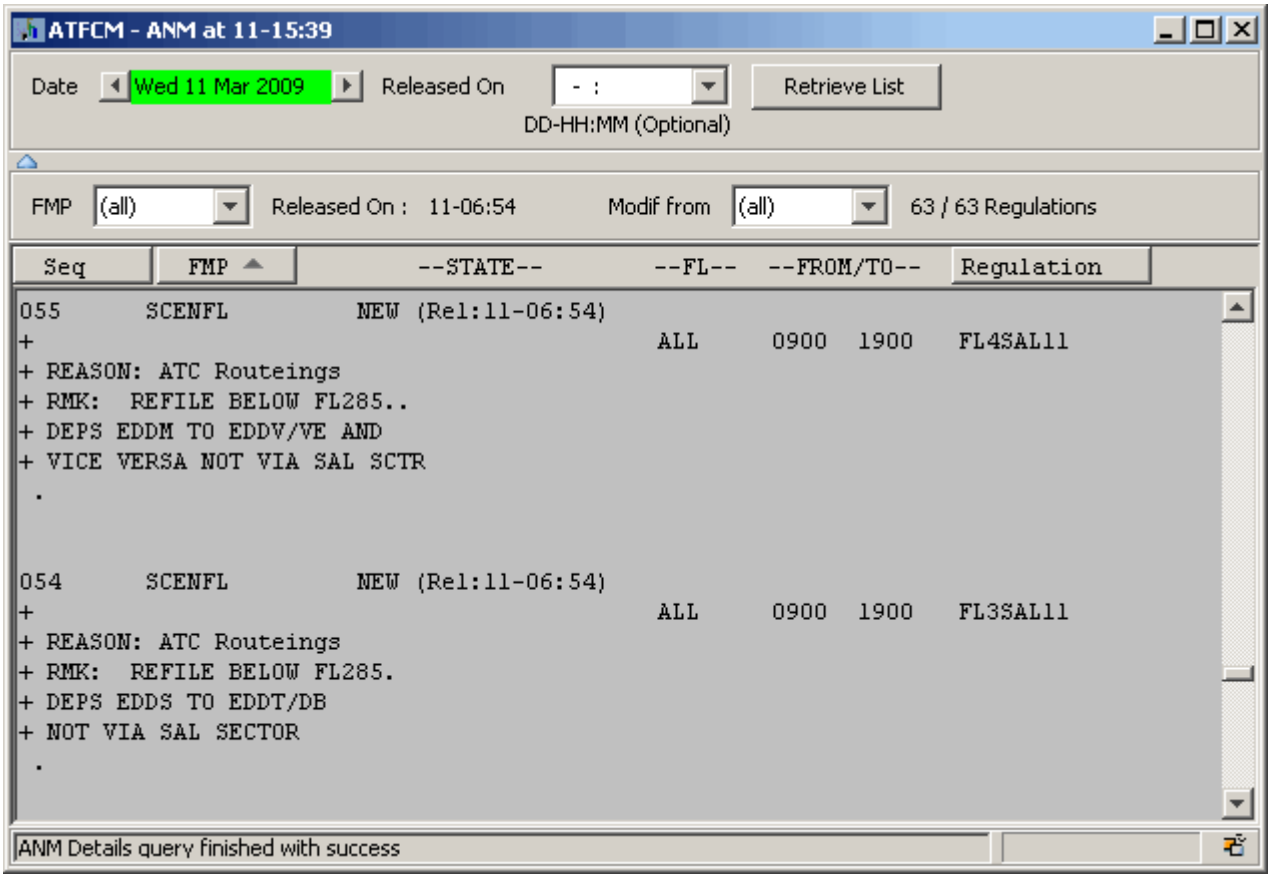


Figure Chmi-82: ANM Display

(1) The title contains the time stamp when the information was retrieved from the Network Operations system.

(2) The header box contains the selected query criteria:

a) The observed **Date**.

b) And possibly the **Released On** stamp.

(3) Just below the header box, a combo box contains all the **FMPs** related to the different regulations listed in the displayed ANM.

(4) This combo box allows sub-selection on all or one specific FMP.

(5) Next to the FMP combo box, the **"Released On"** text gives the day and time when the displayed ANM message has been released (DD-HH:MM).

(6) The **Modif from** combo box is filled in with the different release dates of the displayed list of regulations.

a) By default, the entry **(all)** displays the full list.

b) If you change to one of the specific Release dates in the combo box, the list will be filtered to leave only the regulations created or modified from the selected release date onward.

c) This filter is a sub-selection of the FMP filter described above. It means that the filtering on FMP will happen first and the selection on release dates will be done on the remaining regulations.

(7) Note: When the ANM Display is invoked from the ATM Messages Release Notifier, after the immediate query, the **Modif from** combo box will be set to pre-select the latest Release time stamp.

(8) A counter indicates the number of regulations displayed.

(9) This counter might be affected by the "FMP" and/or the "Modif from" filter sub-selections. (fff / rrr Regulations where fff is the filtered number and rrr is the retrieved number).

(10) Three different sort sequences may be alternatively selected by clicking on the corresponding header button:

a) The displayed regulations are initially sorted on the **FMP identifier** and for the same FMP, by **Regulation** identifier.

b) Alternatively, the displayed list may be sorted directly by **Regulation** identifier.

c) A third possibility is to sort the displayed list on the **Regulation number**.

(11) Subsequent clicks on these header buttons swap between ascending and descending sequences.

(12) Selecting one of these sort sequence will reset the filtering criteria to "(all)" for both the FMP and the Modif from.

(13) The title of the active sort sequence is followed by an arrow indicating the sort direction.

(14) For each Regulation, the detailed group of lines contains:

a) The global sequence number (**SEQ**) of a regulation (001 for the first regulation). If the regulation is cancelled, its number is not re-used.

b) The Flow Management Position (**FMP**) identifier

c) The regulation **STATE**. Possible values are NEW, CANCEL or CHANGE.

d) Followed by the stamp (**Rel:***dd-hh:mm*) when this Regulation was last modified based from the currently displayed ANM version.

e) **FL** is the description of the Flight Levels concerned by the regulation.
This can be any one of the following:

i) "minimum and maximum flight levels" separated by "-" meaning all ICAO FL between and including the indicated values, e.g. 165-335.

ii) "flight level minus" meaning all ICAO FL below and including the indicated flight level value, e.g. 240- means at and below FL240.

iii) "flight level plus" meaning all ICAO FL above and including the indicated flight level value, e.g. 240+ means at and above FL240.

iv) "ALL" to represent all ICAO flight levels.

f) **FROM/TO** gives the start and end time of the regulation (CNL for a cancelled regulation).

g) **Regulation** is the regulation identifier.

h) **REASON** indicates the initial cause of the regulation.

i) **RMK** is an optional remark string.

j) The description of the Traffic Volume where the regulation is applied. The Traffic Volume description format should comply with the formatting rules.

k) In front of all detail line(s) for one regulation, the presence of a "**+**" flag indicates whether this regulation was modified and consequently published in the displayed released ANM. No "**+**" flag indicates that the regulation was published by an earlier ANM and not modified since then.

(15) Following example assists the layout specification of the ANM regulation constituent fields in an ANM message. Note that the description of the Traffic Volume should occupy the last lines from the detailed group of lines for each Regulation.

Seq	FMP	STATE	FL	-FROM/TO-	Regulation
001	EHAAFMP	NEW (Rel:26-16:10)	ALL	0600 2300	ZN85227
+ REASON: Others					
+ RMK: CDR ROUTE CLOSURE SEE AIP NL					
+ EHAM DEPARTURES VIA N852-LUTOM NOT ALLOWED					

(16) On ANM detail query results display, the focus is set on the ANM detail listbox control after selecting the first line in the listbox.

(17) After this the focus moves to the Seq, FMP and Regulation header buttons, then on the query data elements FMP and finally to the FMP and the Release on filter selection.

23.5 User interactions

(1) This window allows to modify a previous query selection and to issue a new request.

(2) The **Normal / Proposal** two states button may be adjusted before selecting an option of the popup menu described in the "Contextual actions" section.

(3) The selected **Normal / Proposal** value will be transferred to the invoked function.

(4) On query modification, the **Reset action** is enabled.

(5) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(6) To update the ANM details, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(7) Clicking on the **Retrieve List** button, will issue a query to propose in the corresponding drop-down combo box, the list of effective release stamps for the selected date.

(8) Clicking on the **Seq** button just above the listbox, will sort the displayed regulations on the regulation sequence number.

(9) Clicking on the **FMP** button just above the listbox, will sort the displayed regulations on the FMP identifier, and for the same FMP, on the Regulation identifier.

(10) Clicking on the **Regulation** button just above the listbox, will sort the displayed regulations on the Regulation identifier.

(11) To view the ANM regulations for only one FMP or (all), **select the FMP** in the combo box.

(12) To print the ANM details, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(13) To export the retrieved ANM details, click on the **Export** button or select the **Action / Export** menu option.

(14) Increasing / reducing the window height and width will impact the size of the ANM details control.

23.6 Contextual actions

(1) When a line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Regulation and may contain the following options:

a) **Regulation Description**: displays detailed information for the selected regulation.

b) **Regulation Delays**: displays the delays for the selected regulation and its activation period.

c) **Regulation Flight List**: displays the Provisional Slot Allocation List (Traffic Load) for the selected regulation and its activation period starting either with the current time or the start time of the regulation.

(3) For the Regulation Delays and the Regulation Flight List, the transferred period may be affected by user preferences related to **Default query period identification**.

24 ADP List

24.1 Function presentation

(1) This display lists the regulations in the ATFCM Daily Plan (ADP) for the selected FMPs and the specified date.

24.2 Opening the window

(1) From the Application / ATFCM menu, select the option **ATM Messages / ADP List**.

(2) Immediate shortcut to bypass menu selection: **<Ctrl>** + **<Shift>** + **<F4>**.

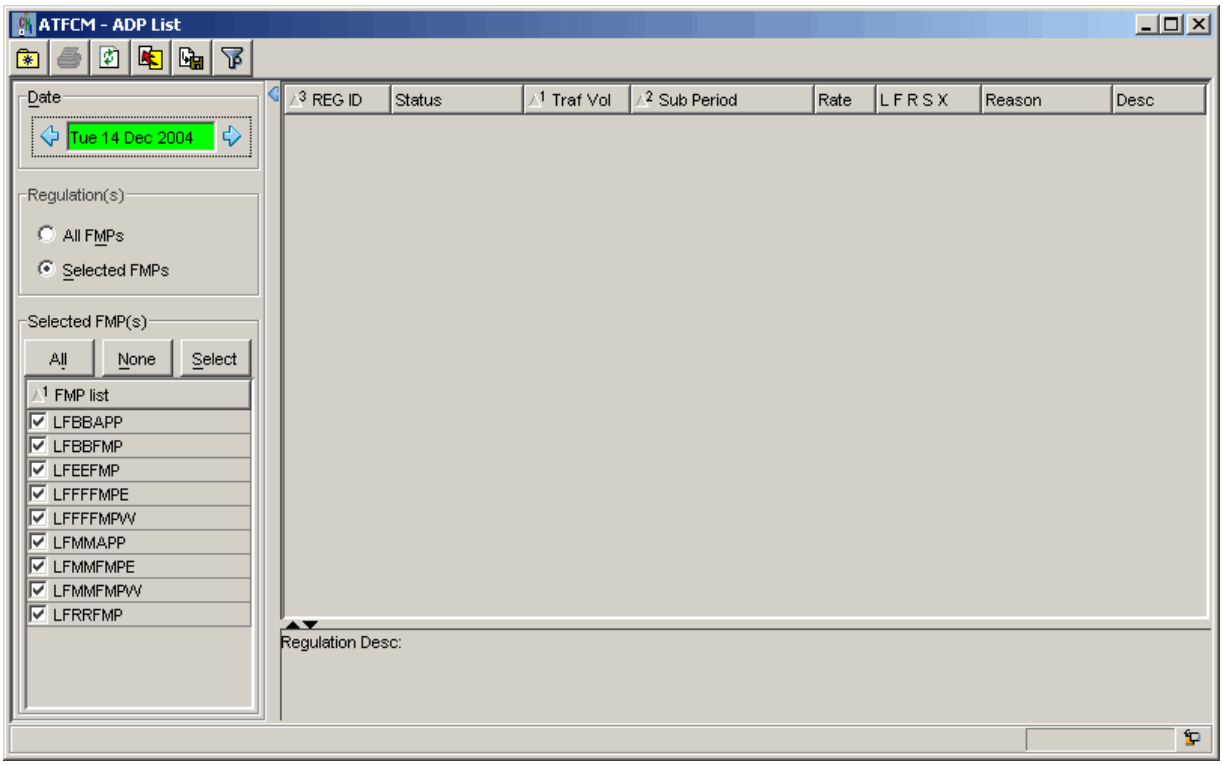


Figure Chmi-83: Query ADP

(3) A new empty **ADP List** window will be opened with the following defaults:

a) **Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **"Selected FMPs"** radio button is always initially checked.

c) **"All FMPs"** radio button is never switched on initially.

d) **"Selected FMP(s)"** list is filled in with the last updated list maintained using the **FMP Selection** dialog.

e) FMPs are checked in the "**Selected FMP(s)**" list based on the last query issued that uses FMP selection.

(4) When no FMPs have been selected yet using the **FMP Selection** dialog (this is the case after application installation):

a) A message box will warn you and invite you to select first one or more FMPs.

b) On clicking OK, the **FMP Selection** dialog is opened automatically.

c) After adding one or more FMPs to the "**Selected list**", click on the OK button.

d) This will close the dialog and update the "**Selected list**" on the calling window.

(5) On initial **ADP List** query preparation, the focus is set on the date control.

(6) After this, it is moved logically between the query controls in position sequence (left to right and then top to bottom).

24.3 Preparing to query

(1) The traffic **Date** may be adjusted to one of the proposed values.

(2) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(3) The "**All FMPs**" radio button shall be checked if no selection on FMP shall be performed (retrieve regulations for all FMPs). This hides the Selected FMP(s) data group. Otherwise "**Selected FMPs**" radio button shall be checked.

(4) When the "**Selected FMPs**" radio button is checked, the Selected FMP(s) data group is displayed.

(5) FMPs in the "**Selected list**" may then be checked or unchecked as desired.

(6) Click on the "**Selected list**" title to adjust sort sequence (alternatively ascending and descending sequence). This shall not impact the list of selected FMPs and the fact that a given FMP is checked or not.

(7) Click the "**All**" button to check all the FMPs present in the Selected list.

(8) Click the "**None**" button to uncheck all the FMPs present in the Selected list.

(9) To add or remove one or more FMPs in the Selected list, the **FMP selection** dialog shall be used. Click on the "**Select**" button to open this dialog.

(10) When the Selected list is up to date on the **FMP selection** dialog, click on the OK button to save the list.

(11) Rules for updating the **Selected list** on all opened windows using an FMP **Selected list** for the query is described in **FMP selection** dialog.

(12) The FMPs checked in the "**Selected list**" before opening the **FMP selection** dialog shall remain checked.

(13) Clicking the Cancel button of the **FMP selection** dialog stops editing the "**Selected list**" and come back to the previous screen without saving any modification to the list.

(14) **ADP List** window may be closed before issuing any query.

(15) A maximum of 12 FMPs may be checked when issuing a query. Depending on your access rights, this limit may be extended.

(16) The **Send** button will send the query to the Network Operations system.

(17) The **Reset** action common functionality is implemented for this function.

(18) Following query data elements are considered for activating the Reset action:

a) Date

b) All or Checked FMPs in the Selected list

24.4 Displaying the query results

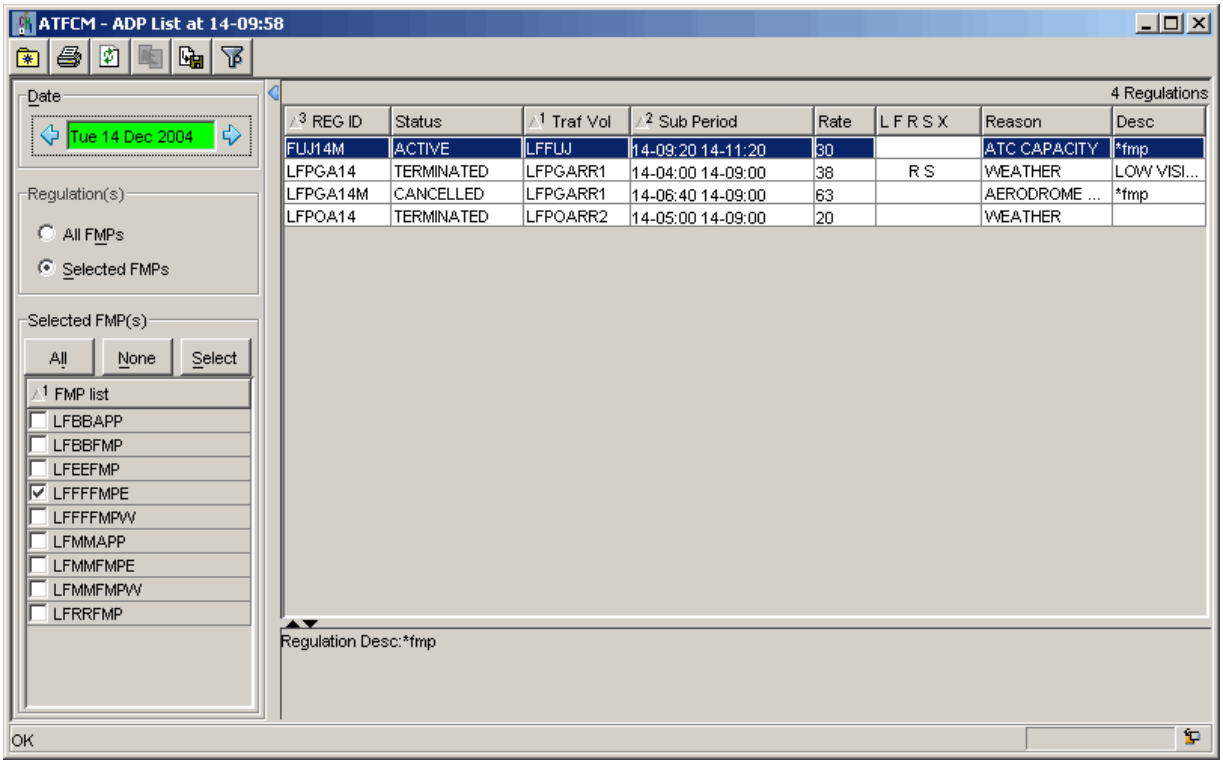


Figure Chmi-84: ADP List

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The left part of the window contains the different **query data elements** used to retrieve the displayed regulation list:

a) The observed **Date**.

b) If Regulations were retrieved for **All FMPs**.

c) Otherwise if FMP selection was applied, the list of **Selected FMPs** is displayed where FMPs included in the query are checked.

(3) At the top of the regulation list, a counter indicates the **number of regulations** retrieved.

(4) This counter may be affected by **filtering** (fff / rrr Regulations where fff is the filtered number and rrr is the retrieved number).

(5) On query results display, the focus is set on the regulation list after selecting the first regulation in the list.

(6) For each regulation, the **main list** contains:

a) The name of the regulation (**REG ID**).

b) The current Activity **Status** of the Regulation (see below).

c) The traffic volume on which it applies (**Traf Vol**).

d) The different **Sub Periods** defined for the regulation.

e) The corresponding total **Rate** for the different sub-periods.

f) The link state indicates that the regulation is linked to other regulations to calculate the delays (**L**) (see below).

g) The indication that a Confirmation message is required when a flight is affected by this regulation (**F**). Set to F if applicable.

h) The indication that a Runway Visual Range (RVR) is associated to one or more sub-periods (**R**). Set to R if applicable.

i) The indication that flights affected by this regulation have to be shifted for one or more sub-periods (**S**). Set to S if applicable.

j) The indication for exemption from other regulations (**X**). Set to X if applicable.

k) The Regulation **Reason** (see below).

l) The Regulation description (**Desc**).

(7) The regulation list is initially sorted using the following ascending sequence:

a) Traffic Volume

b) Start time of the Regulation

c) Regulation identifier

(8) The list allows only single selection.

(9) When a regulation is selected in the list, the note box at the bottom is filled in with:

a) the description and comments, if any, made by the NM Operator.

(10) The possible Regulation Activity **Status** are:

a) Active

b) Cancelled

c) Activating

d) Cancelling

e) Modifying

f) Not Active

g) Terminated

(11) The possible Link states are:

a) **A**: Automatic Link.

b) **M**: Manual Link.

c) **L**: Both Automatic and Manual Links are present.

24.5 User interactions

(1) This window allows to modify a previous query selection and to issue a new request.

(2) On query modification, the **Reset action** is enabled.

(3) To display / hide the query part, click respectively the Right / Left arrow at the top next to the split bar.

(4) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(5) To print the ADP list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(6) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(7) To modify the **Sort sequence** of the regulations, click the column headers corresponding to your desired sort criteria.

(8) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(9) The ADP List export process shall generate one record per sub-period for Regulations defined with several sub periods, duplicating all other unique data elements.

(10) Example:

```
EDDFA31,ACTIVE,EDDFARR,31-04:00 31-05:00,39,    ,WEATHER,EDDF ARRIVALS
EDDFA31,ACTIVE,EDDFARR,31-05:00 31-06:40,39,    ,WEATHER,EDDF ARRIVALS
EDDFA31,ACTIVE,EDDFARR,31-06:40 31-07:40,41,    ,WEATHER,EDDF ARRIVALS
EDDFA31,ACTIVE,EDDFARR,31-07:40 31-16:00,43,    ,WEATHER,EDDF ARRIVALS
```

(11) Increasing / reducing the window height will impact the size of the FMP Selected list and the Regulation list controls.

(12) Increasing / reducing the window width will impact the size of the Regulation list and the bottom note box controls.

24.6 Contextual actions

(1) When a line is selected in the list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Regulation and may contain the following options:

a) **ADP Current**: displays current detailed information for one ADP Regulation identifier.

b) **ADP History**: lists the history log for the selected ADP Regulation and date.

25 ADP Current

25.1 Function presentation

(1) This display gives the different data elements for **one Regulation** associated to a selected date.

(2) The same functionality is used for the display of the **Regulation Description** in the context of the Measures menu.

(3) Therefore I refer you to the **Regulation Description** pages for the detailed function description.

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26 ADP History

26.1 Function presentation

(1) This display lists the different updates that have been processed on one selected regulation in the ATFCM Daily Plan (ADP) for the specified date.

26.2 Opening the window

(1) From the Application / ATFCM menu, select the option **ATM Messages / ADP History**.

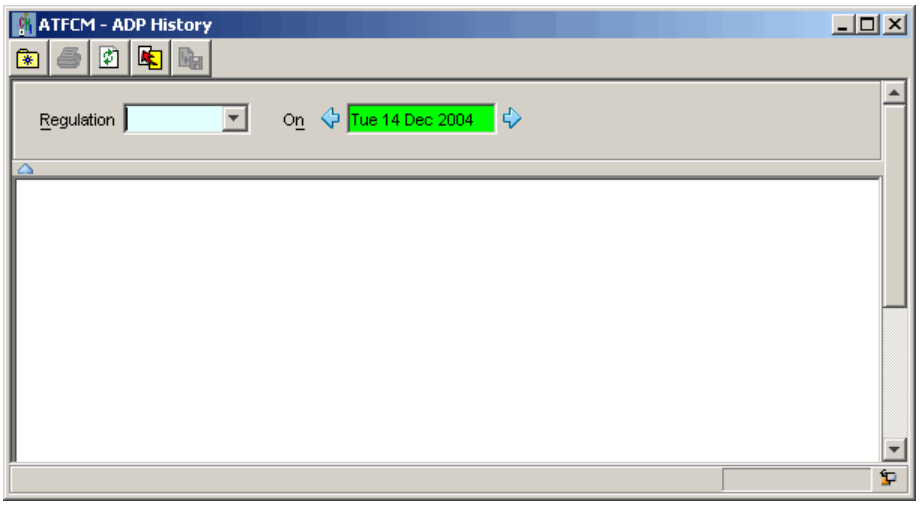


Figure Chmi-85: Query ADP History

(2) A new empty **ADP History** window will be opened with the following defaults:

a) The **Regulation** drop-down box will propose the most recent Regulation identifiers you have queried before with success.

b) **On**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

(3) On **ADP History** query preparation, the focus is set on the **Regulation** input control.

(4) After this, it is moved logically between the active and visible controls in position sequence. (left to right and then top to bottom)

26.3 Preparing to query

(1) The **Regulation** control may be adjusted by selecting one of the identifiers proposed in the drop-down list or by keying in a new Regulation identifier in the dedicated input box.

(2) The Regulation identifier has a maximum of 8 characters.

(3) The traffic date (**On**) may be adjusted to one of the proposed values.

(4) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(5) The **Send** button will send the query to the Network Operations system.

(6) **ADP History** window may be closed before issuing any query.

(7) The **Reset** action common functionality is implemented for this function.

(8) Following query data elements are considered for activating the Reset action:

a) **Regulation**

b) Date (**On**)

26.4 Displaying the query results

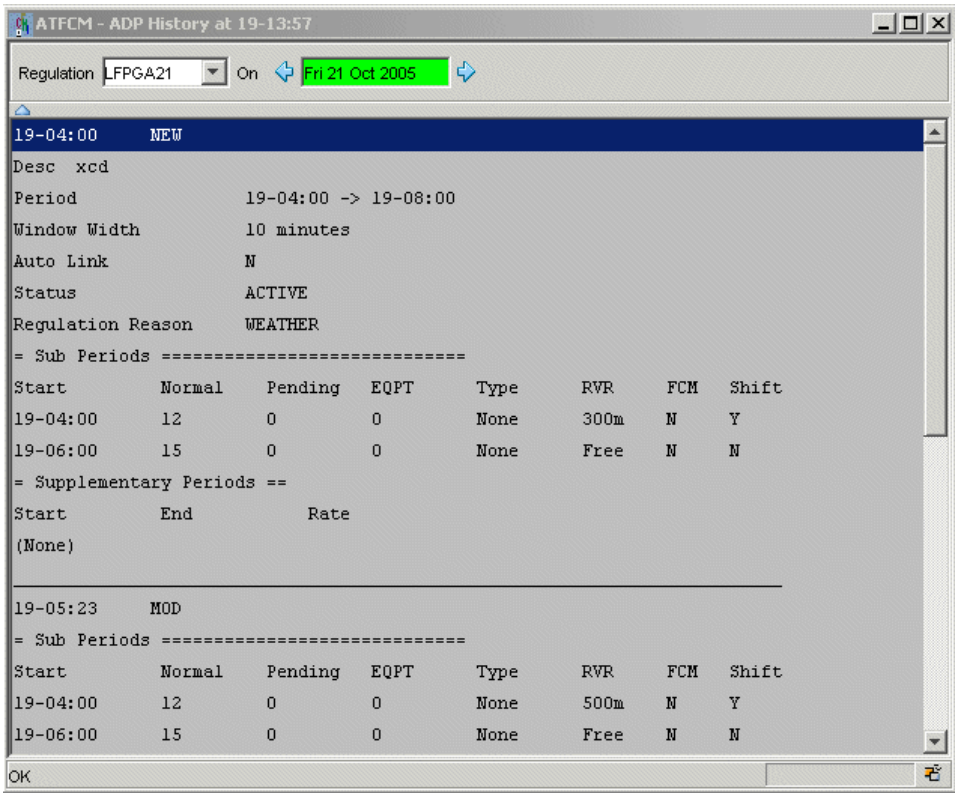


Figure Chmi-86: ADP History

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The header box of the window contains the different **query data elements** used to retrieve the displayed regulation data:

a) The Regulation identifier (**Regulation**).

b) The observed date (**On**).

(3) Below the query header box, the main part of the display provides the different **ADP History reply data elements**:

a) The update stamp in the format DD-HH:MM.

b) The update state (NEW, MOD, CNL).

c) The full text description of the selected Regulation if present (**Desc**).

d) The activation **Period** of the selected Regulation using the format DD-HH:MM for the From and the Until stamps.

e) The related **Window Width** for the Regulation.

f) The **Auto Link** state indicates that the local delay given by this regulation will be taken into account to compute the delay in other regulations.

g) The new regulation **Status** after the update is processed.

h) The **Regulation Reason.**

i) The regulation **Sub-Periods** definition (see below).

j) The regulation **Supplementary-Periods** definition including the Start time, End time and the Rate.

(4) The regulation **Sub-Periods** definition includes:

a) The **Start** time.

b) The **Normal** Rates.

c) The **Pending** Rates.

d) The Equipment Rates **(EQPT)**.

e) The Equipment **Type**.

f) The Runway Visual Range (**RVR**), if any.

g) If **FCM** is mandatory.

h) If Regulation has to shift flights (**Shift**).

(5) The first update block of data contains all the data elements described above.

(6) For the subsequent update blocks, only the modified information will be displayed.

(7) These blocks are displayed from the older to the latest.

(8) On query results display, the focus is set on the History list with the first line selected.

26.5 User interactions

(1) This window allows to modify a previous query selection and to issue a new request.

(2) On query modification, the **Reset action** is enabled.

(3) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(4) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(5) To print the ADP History details, first retrieve the desired details. When ready, click on the **Print** button or select the **File / Print** menu option.

(6) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(7) Increasing / reducing the window height and width will impact the size of the history lists list controls.

27 ATFCM Information

27.1 Function presentation

(1) The **ATFCM Information** function displays a broadcast message prepared by the Network Operations server for the current date.

(2) This displayed **ATFCM Information** is retrieved automatically when the ATFCM Collaboration Human Machine Interface application is started after sign on.

(3) The user may request the **ATFCM Information** function to retrieve up to date broadcast information.

27.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Environment / ATFCM Information**.

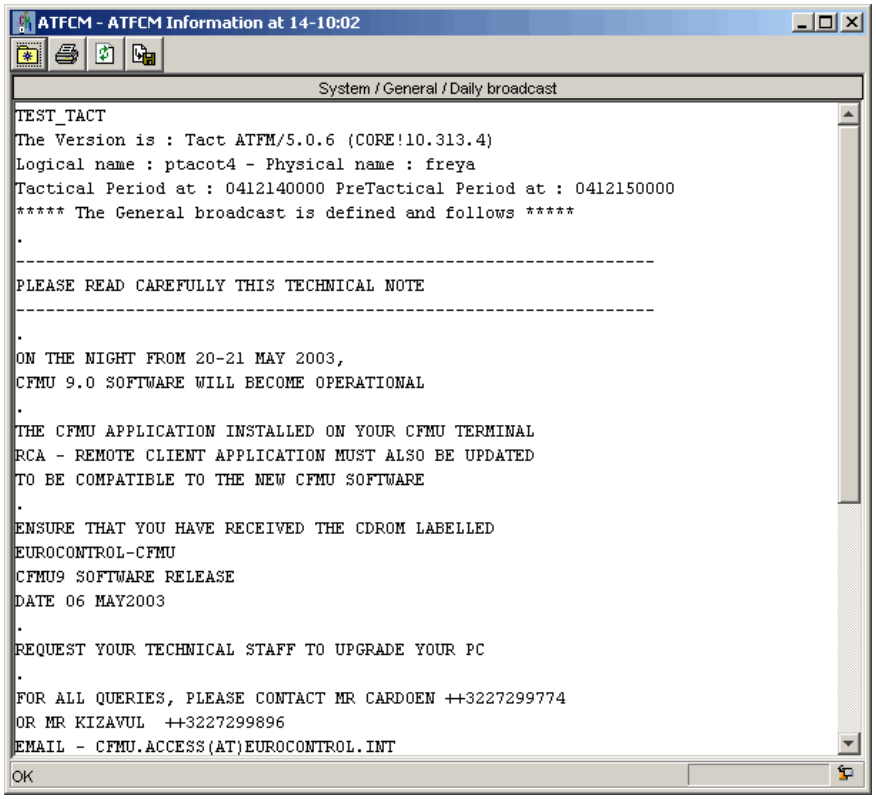


Figure Chmi-87: Query ATFCM Information

(2) Every time this window is re-opened, the query is immediately issued for the current date.

27.3 Displaying the query results

(1) The title contains the time stamp when the information was retrieved from the Network Operations system.

(2) On query results display, the focus is set on the main list after selecting the first line in the list.

(3) The main listbox of the window shows the retrieved broadcast / information message.

(4) This information message is composed of three parts:

a) Technical information on the connected Network Operations server. (TACT system, version, machine, system dates...)

b) An optional general NM ATFCM broadcast message.

c) An optional daily NM ATFCM broadcast message.

27.4 User interactions

(1) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date data.

(2) To print the ATFCM Information, first retrieve the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(3) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(4) Increasing / reducing the window height will impact the size of the list control.

28 Traffic Capacity

28.1 Function presentation

(1) The **Traffic Capacity** function displays the capacity values for the selected date and reference location.

(2) Data retrieved by the **Traffic Capacity** query are extracted from the ETFMS Environment database and Tactical updates.

28.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Environment / Traffic Capacity**.

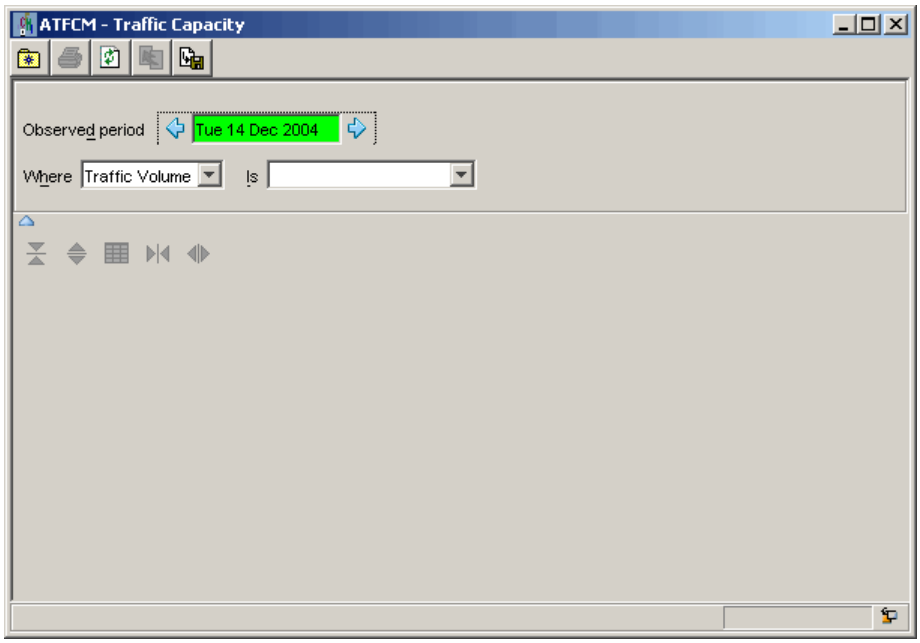


Figure Chmi-88: Query Traffic Capacity

(2) A new empty **Traffic Capacity** window will be opened with the following defaults:

a) **Observed period**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **Where**: last reference location type used with this query. This selection is based on the system property **CHMI / ATFCM application / Environment / Capacity value / Last Used Capacity type**.

c) **Is**: based on the "Where" selection, the drop-down box will propose the most recent query data elements you have queried before with success.

(3) When Aerodrome or Aerodrome Set is selected, the related category controls are proposed. Global button comes preselected.

(4) On initial display, the focus is set on the first mandatory query control with no default value: the **Is** input control.

28.3 Preparing to query

(1) The date of the **Observed Period** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The **Where** parameter may be adjusted to one of the following reference location types:

a) Traffic Volume

b) Airspace

c) Aerodrome

d) Aerodrome Set

e) Point

(3) The identifier corresponding to the selected **Where** shall be keyed in the associated **Is** entry box or selected from the drop-down list when a previous query was successful.

(4) When Aerodrome or Aerodrome Set is selected, the related category (**Cat**) must be checked:

a) Global (**Glob**)

b) Departure (**Dep**)

c) Arrival (**Arr)**

(5) The **Send** button will send the query to the Network Operations system.

(6) This window may be closed before any query is issued.

(7) The **Reset** action common functionality is implemented for this function. All query criteria are considered for activating the Reset action.

28.4 Displaying the query results

28.4.1 Common Display

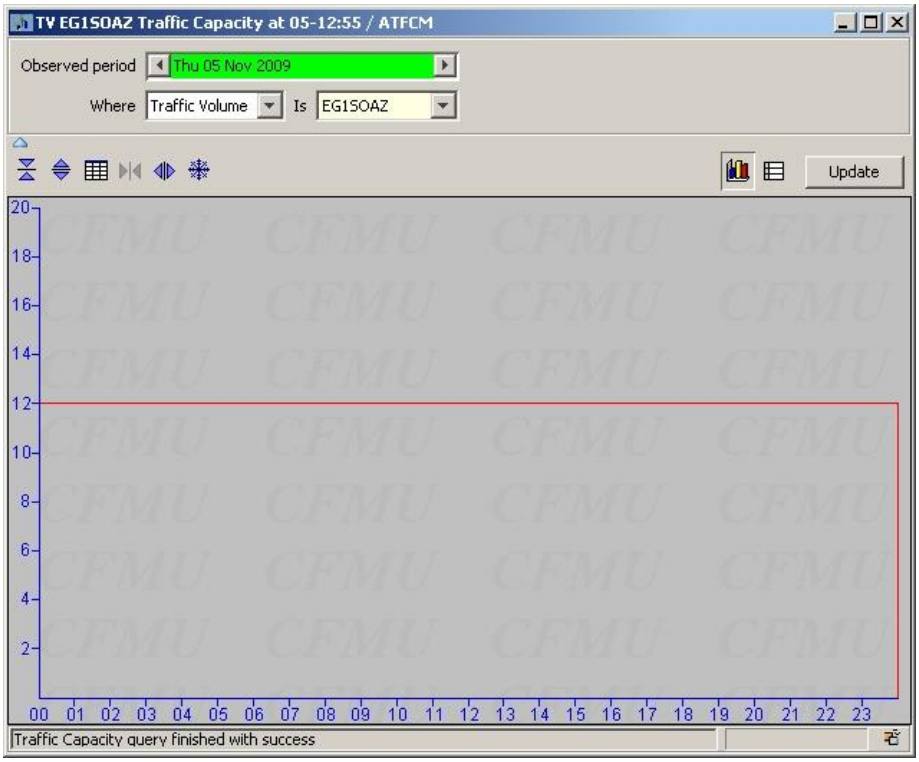


Figure Chmi-89: Traffic Capacity- graph display

(1) The header box contains the selected query criteria:

a) The **Observed period.**

b) The main **Where** selection criteria (Traffic Volume, Aerodrome or Set with the associated **Category**, Airspace, Point).

c) The identifier corresponding to the selected "Where" reference (**Is**) with the additional criteria (Category), if any.

(2) Underneath and to the right of the split bar that separates the query fields from the query results, there are two data presentation buttons; they allow the user to select one of the following display types:

a) Graph: bar chart representation of the capacities.

b) List: list presentation of the different capacity values.

(3) The Graph display is shown initially.

(4) When the user chooses to change the display type then the active window will use the data previously retrieved.

(5) On query results display, the focus is set on the Capacity graph after selecting the first period in the graph or the Capacity List depending on the selected type.

(6) The first bar of the Graph or the first line of the List is then selected.

(7) Next to the display type, access to update the traffic capacity functionality via an **Update** button is restricted by security profile.

28.4.2 Graph Display

(1) The graph contains a series of linked horizontal lines representing the capacity expressed as a number of flights per hour.

(2) The colour of the capacity lines is identified by the user adjustable preference **CHMI / ATFCM application / General / Graph / Capacity colour**.

(3) The **vertical axis** indicates the number of flights.

(4) The **horizontal axis** indicates the time periods.

28.4.3 List Display

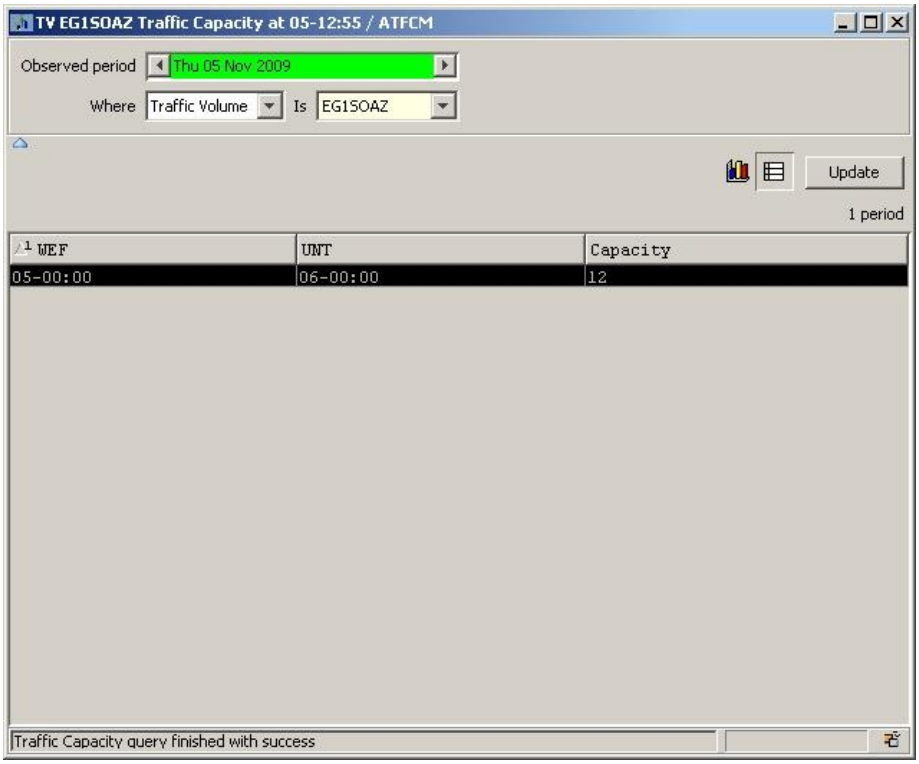


Figure Chmi-90: Traffic Capacity- list display

(1) A counter identifies the number of Capacity entries.

(2) Each detailed Capacity line contains:

a) The With Effect From day and time (**WEF**).

b) The Until day and time (**UNT**).

c) The corresponding capacity value applied (**Capacity**).

(3) A tactical update performed by an authorised user is displayed in:

a) user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Constraint by Regulation Colour** if the tactical update was constrained by a regulation, otherwise

b) user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour**.

28.5 User interactions

28.5.1 Common

(1) This window allows to modify a previous query selection and to issue a new request.

(2) On query modification, the **Reset action** is enabled.

(3) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(4) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(5) To print the Traffic Capacity, first retrieve the data in the window and select the desired display type (Graph, list). Adjust possibly the displayed period, scale and other details. When ready, click on the **Print** button or select the **File / Print** menu option.

(6) To export the retrieved Capacity details, click on the **Export** button or select the **Action / Export** menu option.

(7) Increasing / reducing the window height and width will impact the size of the main display control (graph or list).

(8) Click the **Update** button to open the editing area of the **Traffic Capacity** window (see below).

(9) The **Update** button is disabled.

28.5.2 Graph

(1) Clicking on a bar puts the focus on the graph and sets the bar as selected item.

(2) When the focus is on the graph, the left and right arrows on the keyboard moves selection from one bar to the other.

(3) The right mouse button pops up a **contextual menu** for the selected bar.

(4) The **bottom scroll bar** within the graph pane is available only when the scope of the shown period is reduced or enlarged.

(5) Another scroll bar may appear at the bottom of the Traffic Capacity component when the window is too small to see all contents; it will disappear again when all of the windows contents are visible.

(6) A split bar separates the Query fields from the query results. It includes on the left, a small arrow button (Up/Down) to Hide/Show the query part.

(7) Underneath the split bar that separates the Query fields from the query results several buttons may be displayed. When the graph presentation button on the right side of the pane is selected, then graph related buttons appear on the left of the pane to allow the user to adjust the following graph characteristics:

a) **To enlarge the scope of the vertical axis**: click on the two vertical arrows pointing to each other with an horizontal line in the middle.

b) **To reduce the scope of the vertical axis**: click on the two vertical arrows pointing away from each other with an horizontal line in the middle.

c) **To display / hide a grid on the graph**: click on the toggle button with a table representation.

d) **To enlarge the scope of the shown period**: click on the two horizontal arrows pointing to each other with an vertical line in the middle.

e) **To reduce the scope of the shown period**: click on the two horizontal arrows pointing away from each other with an vertical line in the middle.

f) **To freeze / unfreeze the current level of vertical and horizontal zooming and the current time scrolling**: click on the **Zoom Freeze ON / OFF** toggle button.

28.5.3 List

(1) To modify the **Sort sequence** of the Capacity list, click on column headers corresponding to your desired sort criteria.

28.5.4 Editing Area of the Traffic Capacity

28.5.4.1 Displaying the Window

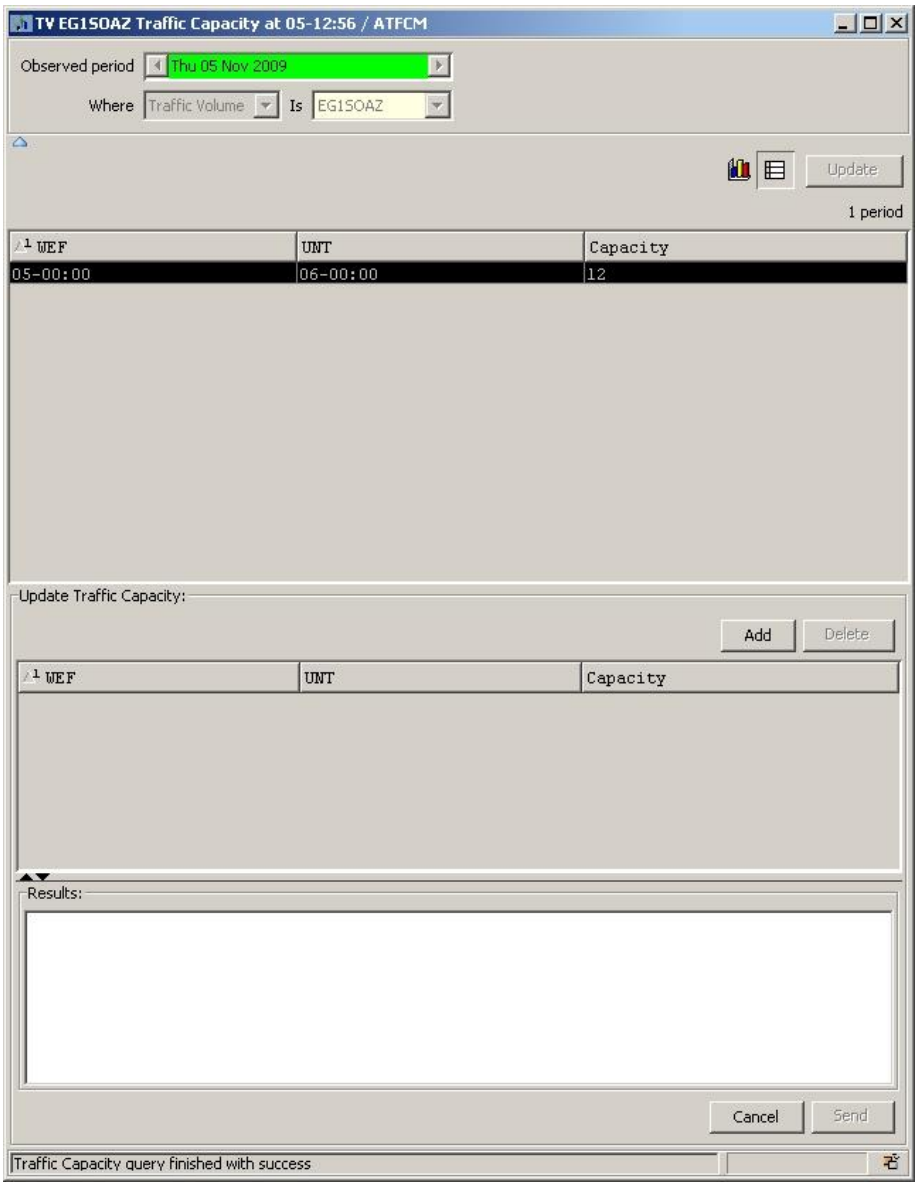


Figure Chmi-91: Update Traffic Capacity

(1) The **Update Traffic Capacity** window will be opened with the following defaults:

a) The main listbox of the editing area of the window only lists the periods whose capacity has been updated tactically (i.e. periods containing data shown in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Constraint by Regulation Update Colour** or user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour**). If no tactical updates have been performed for the **Observed Period**, the listbox will be empty.

b) The listbox is sorted by **WEF** (earliest at the top of the list).

c) By default, the first period in the list that has not started (i.e. whose **WEF** is after the current UTC) is selected.

d) The **Cancel** button is enabled.

e) The **Send** button is disabled.

(2) Each detailed Capacity line contains:

a) The With Effect From day and time (**WEF**).

b) The Until day and time (**UNT**).

c) The corresponding capacity value applied (**Capacity**).

(3) If a period has expired (i.e. the **UNT** of the period is before the current UTC time), all the fields of the period are shown as read-only.

(4) If a period has started but not yet expired, (i.e. current UTC time is between the **WEF** and **UNT**), except for the **UNT** field, all the other fields of the period are shown as read-only.

(5) If a period has not yet started (i.e. the **WEF** is after the current UTC time), all the fields of the period are editable.

28.5.4.2 User Interactions

28.5.4.2.1 To Select a Period

(1) To select a period, click on any field of the period. The period is shown selected and the previously selected period, if any, is shown deselected.

28.5.4.2.2 To Add a Period

(1) To add a new period, click the **Add Period** button.

(2) A new period is inserted in the list – below the previously selected period, if present.

(3) The new period is automatically set to be the selected period.

(4) The **Send** button is enabled.

(5) The default values for the new period are:

a) **WEF**: To value of:

i) the previously selected period, if present, otherwise

ii) the current UTC time.

b) **UNT**: From of the new period plus 3 hours limited to the end of the Observed Period.

c) **Capacity**: set to the value defined in the main **Traffic Capacity** window (i.e. merged view) at the start time of the new period.

28.5.4.2.3 To Delete a Period

(1) Select the period you wish to delete, if not already currently selected.

(2) The **Delete Period** button is only enabled when either:

a) the **WEF** of the currently selected period is later than the current UTC time or

b) the selected period has been added in the current window session (i.e. not already submitted into the system).

(3) Click the **Delete Period** button if enabled.

(4) The selected period is deleted from the list.

(5) The **Send** button is enabled.

28.5.4.2.4 To Modify a Period

(1) The **WEF** field, when editable, shall allow the user to change the start time of the period.

(2) The **UNT** field, when editable, shall allow the user to change the end time of the period.

(3) The **Capacity** field, when editable, shall allow the user to change the number of flights per hour for the reference location type.

(4) If any of the editable fields have been modified, the **Send** button is enabled.

28.5.4.2.5 To Send the Updates

(1) Click the **Send** button, if enabled.

(2) The results area is cleared and both the **Cancel** and the **Send** buttons are disabled.

(3) The following validation checks are performed and reported in the results area:

a) If update period was already submitted (i.e. from previous update) and is expired (**WEF** & **UNT** are before current UTC time), no validations error should be reported.

b) If update period was already submitted but only **WEF** is expired (**WEF** before current UTC time), **UNT** must be later than the current UTC time and within the **Observed Period**.

c) If a new update period is added or for an already submitted period that has not started, both **WEF** & **UNT** have to be later than the current UTC time, and within the **Observed Period**. **UNT** has also to be later than the **WEF**.

d) There are no overlaps between the periods.

(4) Upon reporting of any validation errors, the **Cancel** button is re-enabled.

(5) The user shall be able to correct any validation errors.

(6) On any subsequent edit(s), the results area is cleared.

(7) If there are no validation errors:

a) The **Update** button is enabled,

b) the non-editing part of the window (i.e. the initial query/reply part) is automatically refreshed with the latest system known situation,

c) the editing part of the window becomes read-only,

d) the **Send** button is renamed to **Close** and is enabled,

e) the results area displays a message indicating the status of the last submission:

i) if no errors are returned, the message "**Update successful**" is displayed,

ii) if the update was successfully merged with parallel updates from another user, the message "**Update successfully merged with updates from other users**" is displayed,

iii) if the update was performed on the Ops (ETFMS) system for the pre-tactical day and the plan had not yet been transferred from Predict and applied, the message "**Updates for tomorrow could not be applied – the plan from Predict has not been transferred to Ops (ETFMS) and applied**" is displayed

iv) if the update was performed on the Predict system for the pre-tactical day after the "cut-off" time – a duration parameter (TBD) related to a specific UTC time, the message "**Updates could not be applied due to late submission – plan being finalized in preparation for transfer to the Ops (ETFMS) system**" is displayed.

v) if the system is not able to merge the submitted update with an update made by another user, the message "**The updates conflict with updates received from other users. The results of the query is refreshed with the latest information. Please review your updates and re-submit if necessary.**" Is displayed.

28.5.4.2.6 To Cancel the Updates

(1) Click the **Cancel** button, when available.

a) All modifications, if any, are discarded

b) The editing area of the **Traffic Capacity** window is closed and

c) the **Update** button is re-enabled.

28.5.4.2.7 To Close the Editing Area

(1) Click the **Close** button, when available.

(2) The editing area of the **Traffic Capacity** window is closed.

(3) The **Update** button is re-enabled.

28.6 Contextual actions

(1) When a bar is selected on the graph, the right mouse button pops up a **contextual menu** for the corresponding period.

(2) The main popup menu displays:

a) The **time slice** corresponding to the selected bar.

b) The **Hourly Capacity** value for the period of the selected bar.

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29 Occupancy Traffic Monitoring Values

29.1 Function presentation

(1) The **Occupancy Traffic Monitoring Values** function displays the Occupancy Traffic Monitoring Values (OTMV) for the selected date and Traffic Volume.

(2) Data retrieved by the **Occupancy Traffic Monitoring Values** query are extracted from the ENV OPS database for the Permanent OTMV definitions and from the ETFMS Environment database for the Daily OTMV Applicability definitions.

29.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Traffic / Monitor / Occupancy Traffic Monitoring Values**.

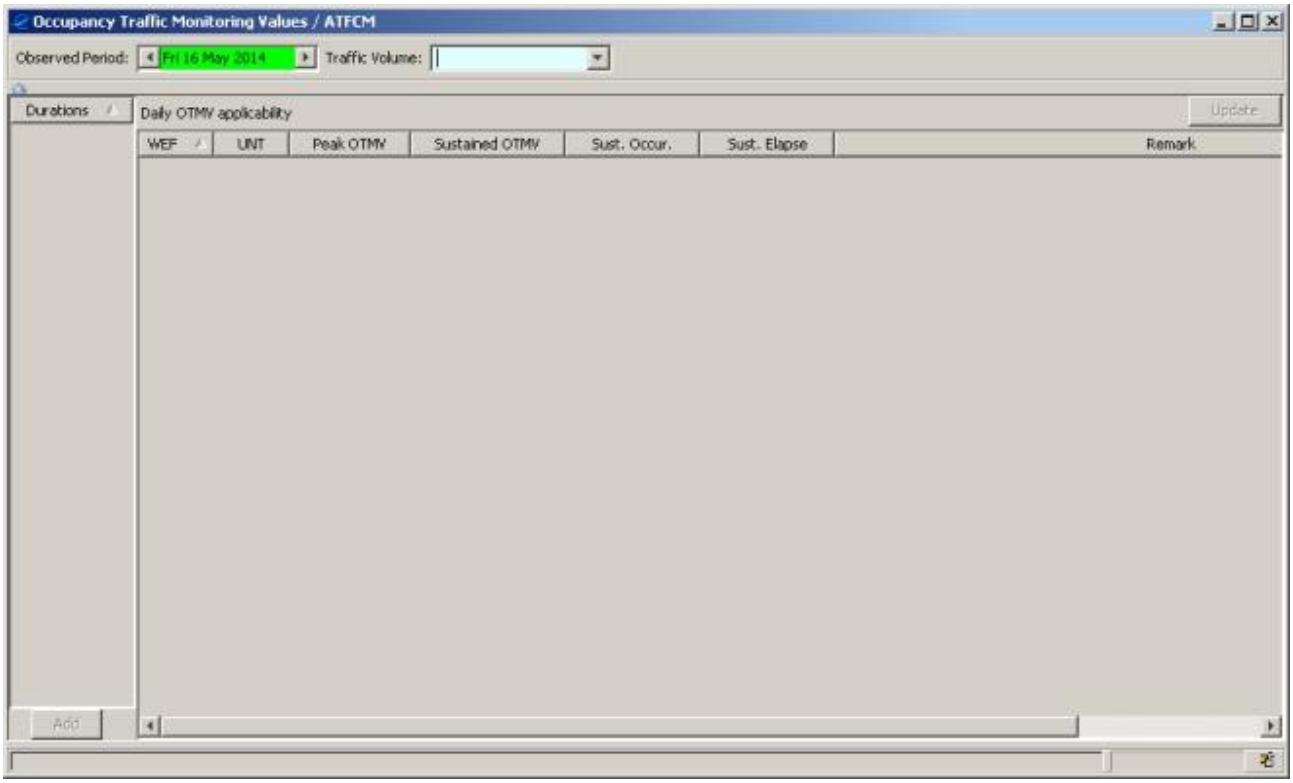


Figure Chmi-92: Query Occupancy Traffic Monitoring Values

(2) A new empty **Occupancy Traffic Monitoring Values** window will be opened with the following defaults:

a) **Observed period**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **for Traffic Volume**: the drop-down box will propose the most recent Traffic Volumes you have queried before with success.

(3) On initial display, the focus is set on the first mandatory query control with no default value: the **Traffic Volume** input control.

29.3 Preparing to query

(1) The date of the **Observed Period** may be adjusted to one of the proposed values.

(2) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(3) The desired **Traffic Volume** identifier shall be keyed in or selected from the drop-down list (when a previous query was successful).

(4) The **Send** button will send the query to the Network Operations system.

(5) The **Occupancy Traffic Monitoring Values** may be closed before any query is issued.

(6) The **Reset** action common functionality is implemented for this function.

(7) All query criteria are considered for activating the Reset action.

29.4 Displaying the query results

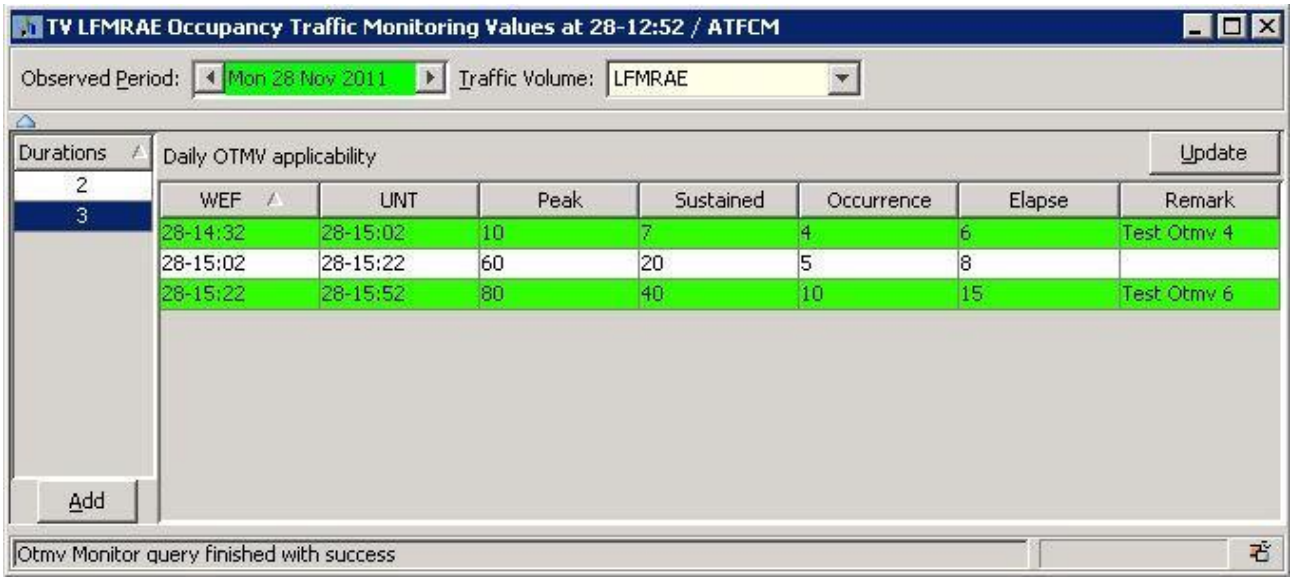


Figure Chmi-93: Occupancy Traffic Monitoring Values display

(1) The title contains the **time stamp** when the displayed OTMV were retrieved from the Network Operations system.

(2) The header box of the window contains the different **query data elements** used to retrieve the displayed OTMV:

a) The **Observed period**.

b) The **Traffic Volume** identifier.

(3) Underneath the split bar that separates the query fields from the query results, the left hand part of the window shows the list of **Durations** for which OTMV are defined. It allows structuring the display of the OTMV definitions on the right side per Occupancy Counts duration (i.e. no durations listed when no OTMV defined yet).

(4) The **Add** button, just below the Durations list, allows starting OTMV definition for a new Duration value.

(5) The detailed OTMV definition is displayed on the right side for the Duration currently selected in the list.

(6) The OTMV definition per duration is composed of the following two main display parts:

a) The **Daily OTMV applicability for *dd/mm/yyyy*** obtained from the merge of the possible Temporary OTMV overwrites, if any, over the Permanent OTMV definition for the date selected in the **Observed period** (i.e. ***dd/mm/yyyy***).

b) The list of **Permanent OTMV definitions**: it shall list all defined Permanent OTMV periods which include any current (i.e. current UTC date) and future date of applicability.

(7) The **Daily OTMV applicability** list is composed of the following data elements:

a) The With Effect From (**WEF**) and the Until (**UNT**) times (dd-hh:mm) when the attached OTMV definition becomes applicable

b) The **Peak** OTMV threshold parameter expressed as a number of flights, when defined.

c) The **Sustained** OTMV threshold parameter expressed as a number of flights, when defined.

d) The Sustained **Occurrence** parameter expressed as a number of occurrences, when defined.

e) The **Sustained Elapse** (column title may be abbreviated to "**Sust. Elapse**") expressed as a number of consecutive minutes for checking the presence of the minimum **Sustained_Occurrence** (column title may be abbreviated to "**Sust. Occur.**") for reporting an alert, when defined.

f) The **Remark**, if any, giving additional information on a temporary overwrite.

(8) The lines in the **Daily OTMV applicability** that correspond to **Temporary OTMV** overwrites are displayed using the user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour** background.

(9) The list of **Daily OTMV applicability** is initially sorted in ascending sequences of With Effect From and Until times.

(10) The **Update** button, just above the **Temporary OTMV definitions** list, allows entering the update mode with the **Temporary OTMV editing** window (see below). This functionality is restricted by security profile.

(11) The **Permanent OTMV definition** line that includes the date selected in the **Observed period** is displayed using the user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour** background.

(12) The list of **Permanent OTMV definitions** is composed of the following data elements:

a) **WEF**: the date (dd/mm/yyyy) and time (hh:mm) when the Permanent OTMV definition becomes applicable

b) **UNT**: the date (dd/mm/yyyy) and time (hh:mm) when the Permanent OTMV definition stops being applicable. This UNT column may contain "UFN" (Until Further Notice) when the applicability is not limited in time.

c) The **Peak OTMV** threshold parameter expressed as a number of flights, when defined.

d) The **Sustained OTMV** threshold parameter expressed as a number of flights, when defined.

e) The **Sustained Occurrence** parameter expressed as a number of occurrences (column title may be abbreviated to "**Sust. Occur.**"), when defined.

f) The **Sustained Elapse** expressed as a number of consecutive minutes (column title may be abbreviated to "**Sust. Elapse**") for checking the presence of the minimum **Sustained_Occurrence** for reporting an alert, when defined.

(13) The list of **Permanent OTMV definitions** is initially sorted in ascending sequence of **WEF** date and time.

(14) The **Update** button, just above the **Permanent OTMV definitions** list, allows entering the update mode with the **Permanent OTMV editing** window (see below). This functionality is restricted by security profile.

29.5 User interactions

29.5.1 OTMV display

(1) This window allows to modify a previous query selection and to issue a new request.

(2) On query modification, the **Reset action** is enabled.

(3) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(4) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(5) To print the **Occupancy Traffic Monitoring Values**, first retrieve and sort the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(6) To export the retrieved Capacity details, click on the **Export** button or select the **Action / Export** menu option.

(7) Click the **Update** button above the **Permanent OTMV definitions** list to open the **Permanent OTMV editing** window (see below).

(8) Click the **Update** button above the **Daily OTMV applicability** list to open the Temporary OTMV editing window (see below).

(9) To modify the **Sort sequence** of the **Permanent OTMV definitions** list, click on column headers corresponding to your desired sort criteria.

(10) To modify the **Sort sequence** of the **Daily OTMV applicability** list, click on column headers corresponding to your desired sort criteria.

29.5.2 Permanent OTMV editing

29.5.2.1 Displaying the Window

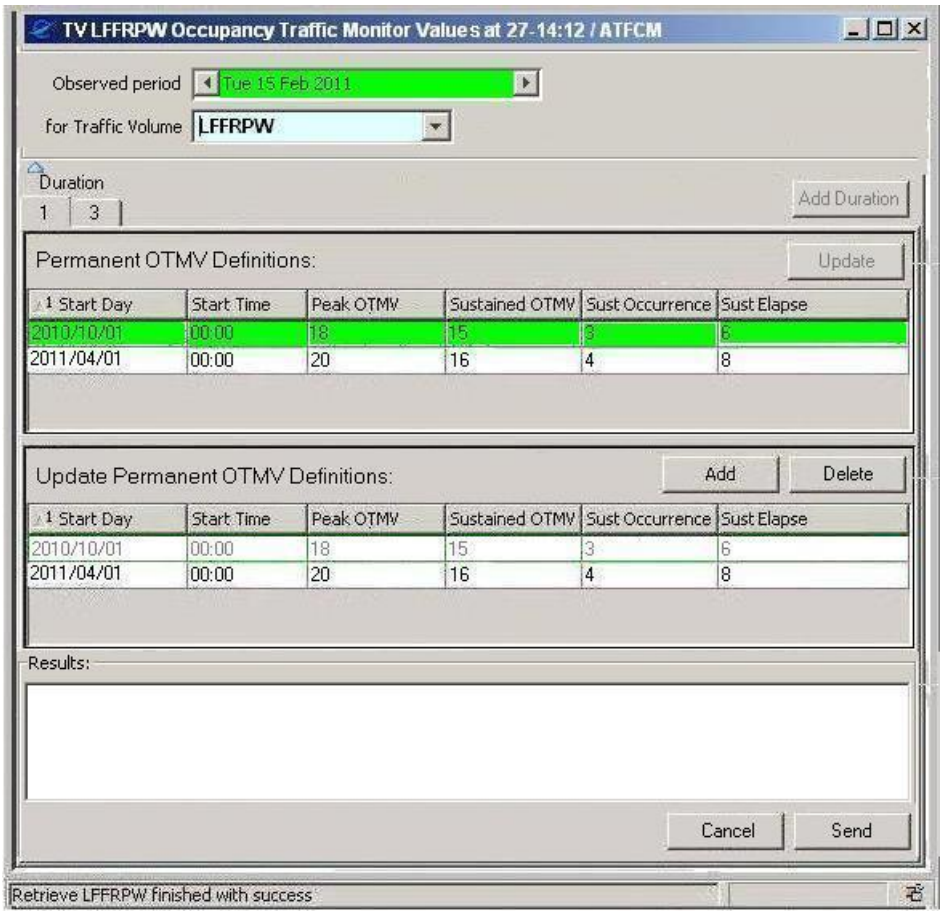


Figure Chmi-94: Permanent OTMV editing

(1) The **Permanent OTMV editing** window will be opened with the following defaults:

a) The window title and the header box keeps displaying the different **query data elements** used to retrieve the displayed OTMV. The query controls are disabled.

b) The OTMV definition for the selected/displayed **Duration** remains the same as before the Update button was clicked. The **Duration** list and the **Add Duration** button are disabled.

c) The **Update** button at the top of the **Permanent OTMV Definitions** list is disabled.

d) The rest of the display is divided into following parts:

i) At the top, the **Permanent OTMV Definitions** list remains as on the OTMV display. It reflects the current **Permanent OTMV Definitions** as they are in the Network Operations ENV server.

ii) A new **Update Permanent OTMV Definitions** list is displayed next. It allows entering the updates (see below). It is associated to an **Add** action buttons which is initially enabled and a **Delete** action buttons which is initially disabled.

iii) A **Results** box is displayed next. It will be used to display resulting states when update requests are issued.

iv) A **Cancel** button to navigate back to the **OTMV display** without applying any update. It is initially enabled.

v) .A **Send** button to issue updates to the Network Operations ENV server. It is initially disabled (i.e. Only enabled when some valid updates have been entered in the **Update Permanent OTMV Definitions** list).

(2) The **Update Permanent OTMV Definitions** list displays the same columns as the **Permanent OTMV Definitions** list (see earlier definition).

(3) Initially, the **Update Permanent OTMV Definitions** list content is identical to the **Permanent OTMV Definitions** list, except that:

a) The line that includes the date selected in the **Observed period** is NOT highlighted with a specific background colour.

b) The line that includes the current UTC time (i.e. UTC time is after **WEF** Date and Time of the line and before the **WEF** Date and Time of the next line in chronological sequence) is disabled for any update processing and its content is greyed (i.e. all the fields are shown as read-only).

(4) The list is initially sorted in ascending sequence on **WEF** Date and Time.

(5) By default, the first line in the list that has not started (i.e. whose **WEF** Date and Time is after the current UTC) is selected. In such case, the **Delete** action button is enabled.

(6) If a period has not yet started (i.e. the **WEF** Date and Time is after the current UTC time), all the fields of the period are editable.

29.5.2.2 User Interactions

29.5.2.2.1 To Select a Period

(1) To select a Permanent Definition period, click on any field of the desired line. This period is then shown selected and the previously selected period, if any, is shown deselected.

29.5.2.2.2 To Add a Period

(1) To add a new period, click the **Add** button above the list.

(2) A new period is inserted in the list – below the previously selected period, if present.

(3) The new period is automatically set to be the selected period.

(4) The **Send** button is enabled.

(5) The default values for the new period are:

a) **WEF** Date

i) Current UTC date + 1 day, if the previous period in chronological sequence has a Start Date and Time before current UTC date and time,

ii) Otherwise Start Date of the previous period in chronological sequence + 1 day.

b) **WEF** Time

i) Forced to 00:00

c) **Peak OTMV, Sustained OTMV, Sustained Occurrence and Sustained Elapse**

i) Set to the respective values defined in the current **Permanent OTMV Definitions** at the Start Date and Time of the newly proposed period.

(6) The proposed default values may be adjusted before the **Send** button is clicked.

29.5.2.2.3 To Delete a Period

(1) Select the period you wish to delete, if not already selected.

(2) The **Delete Period** button is only enabled when either:

a) the **WEF** Date and Time of the currently selected period is later than the current UTC time

b) Or the selected period has been added in the current window session (i.e. not already submitted into the system using the Send action).

(3) Click the **Delete** button above the list, if enabled.

(4) The selected period is deleted from the list.

(5) The **Send** button is enabled.

29.5.2.2.4 To Modify a Period

(1) The **WEF** Date and Time fields, when editable, shall allow the user to change the start time of the period (i.e. only editable when referring to future date and time).

(2) The **Peak OTMV, Sustained OTMV, Sustained Occurrence** and **Sustained Elapse** fields, when editable, shall allow the user to change the OTMV thresholds parameters for the displayed Traffic Volume.

(3) If any of the editable fields have been modified, the **Send** button is enabled.

29.5.2.2.5 To Send the Updates

(1) Click the **Send** button, if enabled.

(2) The results area is cleared and both the **Cancel** and the **Send** buttons are disabled.

(3) The following validation checks are performed and reported in the results area:

a) The stamp derived from **WEF** Date and Time must be unique for the displayed Traffic Volume and Duration (i.e. two lines may never become active at the same time).

b) Updates may only be issued for future **WEF** Date and Time.

c) When defined, **Peak OTMV** and **Sustained OTMV** values shall be greater or equal to zero.

d) When defined, **Sustained Occurrence** and **Sustained Elapse** values shall be greater than zero.

e) **Sustained** parameters are optional as a group. All (**Sustained OTMV, Sustained Occurrence** and **Sustained Elapse**) shall be defined or none.

f) When both are defined, **Peak OTMV** value shall be strictly greater than **Sustained OTMV** value.

g) When defined, **Sustained Elapse** shall be greater or equal to **Sustained Occurrence**.

(4) Upon reporting of any validation errors, the **Cancel** button is re-enabled.

(5) The user shall be able to correct any validation errors.

(6) On any subsequent edit(s), the **Send** button is re-enabled.

(7) If there are no validation errors:

a) The non-editing part of the window (i.e. the initial query/reply part) is automatically refreshed with the latest system known situation,

b) the editing part of the window becomes read-only,

c) the **Send** button is renamed to **Close** and is enabled,

d) the results area displays a message indicating the status of the last submission:

i) if no errors are returned, the message "**Update successful**" is displayed,

ii) if the system is not able to merge the submitted update with an update made by another user, the message "**Your updates conflict with updates received from other users. The results of the query is refreshed with the latest Permanent OTMV information. Please review current definition and re-submit your updates if necessary.**" is displayed,

iii) if the system is currently not able to process submitted update due to locked database state, the following message is displayed:

"**Operational Environment Database is temporarily locked – Permanent OTMV updates cannot be processed.**"

"Temporary circumstances prevent us to accept updates to the Permanent OTMV definitions. Please try again in a few minutes."

"As stated in the operational procedures, during the Tactical and Pre-Tactical phases, urgent short term OTMV updates shall be applied as Temporary OTMV overwrites."

"If you encounter problems updating Permanent OTMV definitions, you can always obtain help from Network Management Operations."

29.5.2.2.6 To Cancel the Updates

(1) Click the **Cancel** button, when available.

a) All modifications pending a Send action, if any, are discarded

b) The editing area of the **Permanent OTMV editing** window is closed and

c) the **Update** button is re-enabled.

29.5.2.2.7 To Close the Editing Area

(1) Click the **Close** button, when available.

a) The editing area of the **Permanent OTMV editing** window is closed,

b) the **Update** button is re-enabled.

29.5.3 Temporary OTMV editing

29.5.3.1 Displaying the Window

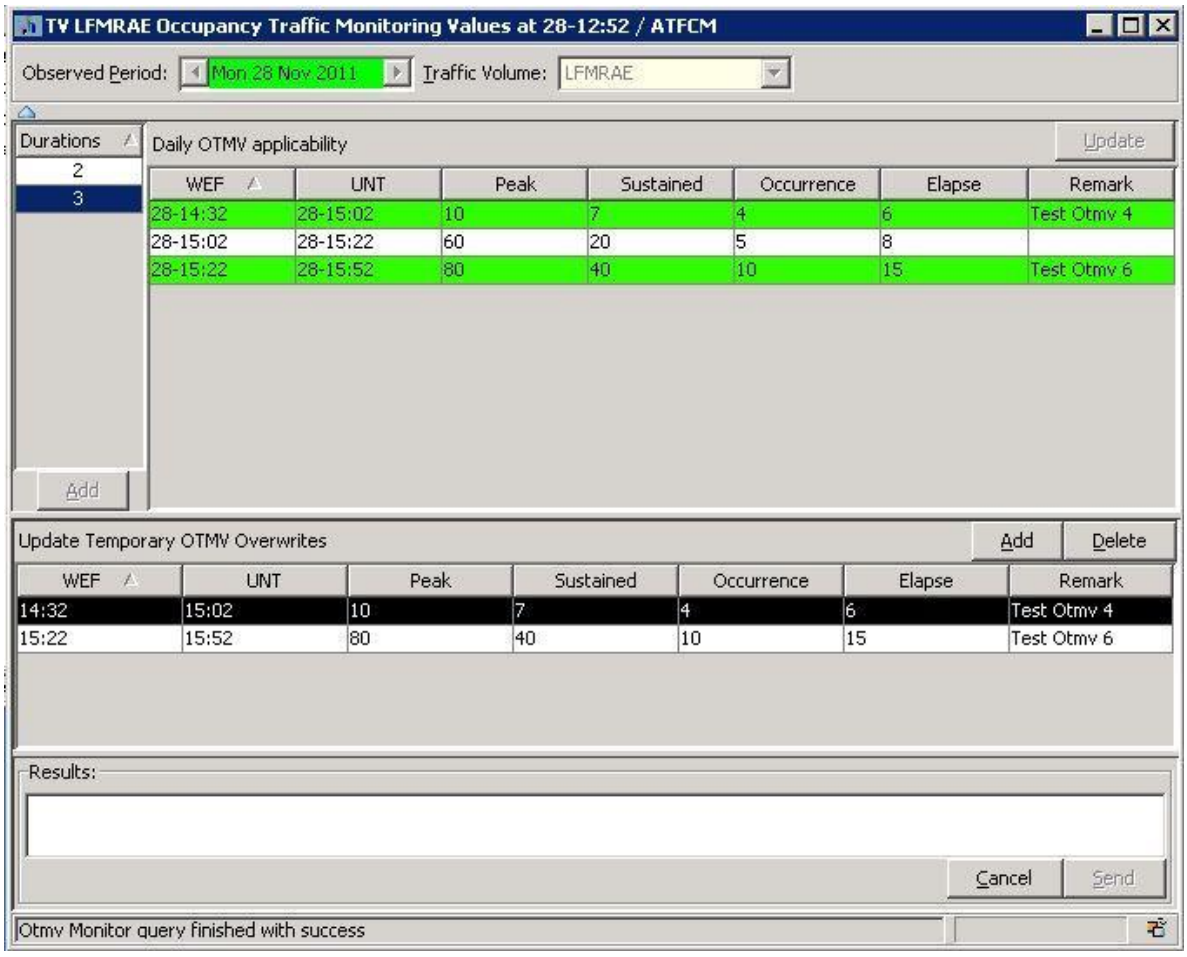


Figure Chmi-95: Temporary OTMV editing

(1) The **Temporary OTMV editing** window will be opened with the following defaults:

a) The window title and the header box keeps displaying the different **query data elements** used to retrieve the displayed OTMV. The query controls are disabled.

b) The OTMV definition for the selected/displayed **Duration** remains the same as before the Update button was clicked. The **Duration** list and the **Add** button are disabled.

c) The **Update** button at the top of the **Temporary OTMV Definitions** list is disabled.

d) The rest of the display is divided into following parts:

i) At the top, the **Daily OTMV applicability for *dd/mm/yyyy*** list remains as on the OTMV display. It reflects the merge of the current possible Temporary OTMV overwrites, if any, with the current Permanent OTMV definition for the date selected in the **Observed period** (i.e. ***dd/mm/yyyy***).

ii) A new **Update Temporary OTMV Overwrites for *dd/mm/yyyy*** list is displayed next. It allows entering the updates (see below). It is associated to an **Add** action button which is initially enabled and a **Delete** action button which is initially disabled.

iii) A **Results** box is displayed next. It will be used to display resulting states when update requests are issued.

iv) A **Cancel** button to navigate back to the **OTMV display** without applying any update. It is initially enabled.

v) A **Send** button to issue updates to the Network Operations ETFMS server. It is initially disabled (i.e. Only enabled when some valid updates have been entered in the **Update Temporary OTMV Overwrites** list).

(2) The **Update Temporary OTMV Overwrites** list displays the same columns as the **Daily OTMV applicability** list (see earlier definition).

(3) WEF and UNT format are limited to hh:mm to facilitate user input. Each time is always relative to the date of the current **Observed period** (note: UNT = 00:00 refers to the end of the day i.e. 00:00 of "D+1").

(4) The **Update Temporary OTMV Overwrites** list contains initially the overwrites as currently known to the ETFMS system for the date selected in the **Observed period**, if any.

(5) If no Temporary OTMV Overwrites have been defined yet for the **Observed Period**, the list is presented empty.

(6) The list is initially sorted by **WEF** (i.e. earliest at the top of the list).

(7) By default, the first period in the list that has not started (i.e. whose **WEF** is after the current UTC) is selected. In such case, the **Delete** action button is enabled.

(8) If a period has expired (i.e. the **UNT** of the period is before the current UTC time), all the fields of this **Temporary OTMV Overwrites** line are disabled for any update processing and its content is greyed (i.e. all the fields are shown as read-only).

(9) If a period has started but not yet expired, (i.e. current UTC time is between the **WEF** and **UNT**), except for the **UNT** field, all the other fields of this **Temporary OTMV Overwrites** line are disabled for any update processing and its content is greyed (i.e. all the fields are shown as read-only).

(10) If a **Temporary OTMV Overwrites** period has not yet started (i.e. the **WEF** is after the current UTC time), all the fields of the period are editable.

29.5.3.2 User Interactions

29.5.3.2.1 To Select a Period

(1) To select a **Temporary OTMV Overwrites** period, click on any field of the desired line. This period is then shown selected and the previously selected period, if any, is shown deselected.

29.5.3.2.2 To Add a Period

(1) To add a new **Temporary OTMV Overwrites** period, click the **Add** button above the list.

(2) A new period is inserted in the list:

a) Either below the previously selected period

The previous selection is the reference period for allocating default values for the new period (see later).

b) Or at the end of the list, if no line was selected or none existed when the **Add** button was clicked.

In such case, there is NO reference period for allocating default values for the new period (see later).

(3) The new period is automatically set to be the selected period.

(4) The **Send** button is enabled.

(5) The default values for the new period are:

a) **WEF**

i) If there is NO reference period OR **UNT** value of the reference period is 00:00 OR is before current UTC,

>> if **Observed period** is the current UTC date

WEF is set to current UTC time, rounded to the next hour if UTC time < 23:00 (ex: 11:34 is rounded to 12:00 but 23:10 is not rounded and stays 23:10).

>> if **Observed period** is for a date in the future

WEF is forced to 00:00

ii) Otherwise, **WEF** is initialised with the **UNT** value of the reference period.

b) **UNT**

i) Initialised to **WEF** + 1 hour, but forced to 00:00 when the result is above 23:59

(note: UNT = 00:00 indicates the end of the day)

c) **Peak OTMV, Sustained OTMV, Sustained Occurrence** and **Sustained Elapse**

i) Set to the respective values defined in the current **Daily OTMV applicability** at the start time of the newly proposed period (i.e. proposed values may be empty).

d) **Remark** is proposed empty.

(6) The proposed default values may be adjusted before the **Send** button is clicked.

Note: Several Add, Delete and/or Modify actions may be prepared before the Send action is triggered.

29.5.3.2.3 To Delete a Period

(1) Select the **Temporary OTMV Overwrites** period you wish to delete, if not already selected.

(2) The **Delete Period** button is only enabled when either:

a) the **WEF** of the currently selected period is later than the current UTC time

b) Or the selected period has been added in the current window session (i.e. not already submitted into the system using the Send action).

(3) Click the **Delete** button above the list, if enabled.

(4) The selected period is deleted from the list.

(5) The **Send** button is enabled.

Note: Several Add, Delete and/or Modify actions may be prepared before the Send action is triggered.

29.5.3.2.4 To Modify a Period

(1) The **WEF** and **UNT** fields, when editable, shall allow the user to change the start and end date and time of the **Temporary OTMV Overwrites** period (i.e. only editable when referring to a period in the future).

(2) The **Peak OTMV, Sustained OTMV, Sustained Occurrence**, **Sustained Elapse** and **Remark** fields, when editable, shall allow the user to change the OTMV thresholds parameters for the displayed Traffic Volume.

(3) If any of the editable fields have been modified, the **Send** button is enabled.

Note: Several Add, Delete and/or Modify actions may be prepared before the Send action is triggered.

29.5.3.2.5 To Send the Updates

(1) Click the **Send** button, if enabled.

(2) The results area is cleared and both the **Cancel** and the **Send** buttons are disabled.

(3) The following validation checks are performed and reported in the results area:

a) If update period was already submitted (i.e. from previous update) and is expired (**WEF** & **UNT** are before current UTC time), no validations error should be reported.

b) If update period was already submitted but only **WEF** is expired (**WEF** before current UTC time), **UNT** must be later than the current UTC time and within the **Observed Period**.

c) If a new update period is added or for an already submitted period that has not started, both **WEF** & **UNT** have to be later than the current UTC time, and within the **Observed Period**. **UNT** has also to be later than the **WEF**.

d) There are no overlaps between the periods. Note that each period is an open-ended interval:[WEF, UNT[

e) When defined, **Peak OTMV** and **Sustained OTMV** values shall be greater or equal to zero.

f) When defined, **Sustained Occurrence** and **Sustained Elapse** values shall be greater than zero.

g) **Sustained** parameters are optional as a group. All (**Sustained OTMV, Sustained Occurrence** and **Sustained Elapse**) shall be defined or none.

h) When both are defined, **Peak OTMV** value shall be strictly greater than **Sustained OTMV** value.

i) When defined, **Sustained Elapse** shall be greater or equal to **Sustained Occurrence.**

(4) Upon reporting of any validation errors, the **Cancel** button is re-enabled.

(5) The user shall be able to correct any validation errors.

(6) On any subsequent edit(s), the **Send** button is re-enabled.

(7) If there are no validation errors:

a) The non-editing part of the window (i.e. the initial query/reply part) is automatically refreshed with the latest system known situation,

b) the editing part of the window becomes read-only,

c) the **Send** button is renamed to **Close** and is enabled,

d) the results area displays a message indicating the status of the last submission:

i) if no errors are returned, the message "**Update successful**" is displayed.

ii) if the update was successfully merged with parallel updates from another user, the message "**Update successfully merged with updates from other users**" is displayed.

iii) if the update was performed on the Ops (ETFMS) system for the pre-tactical day and the plan had not yet been transferred from Predict and applied, the message "**Updates for tomorrow could not be**

applied – the plan from Predict has not been transferred to Ops (ETFMS) and applied" is displayed.

iv) if the update was performed on the Predict system for the pre-tactical day after the "cut-off" time – a duration parameter (TBD) related to a specific UTC time, the message "**Updates could not be applied due to late submission – plan being finalized in preparation for transfer to the Ops (ETFMS) system**" is displayed.

v) if the system is not able to merge the submitted update with an update made by another user, the message "**The updates conflict with updates received from other users. The results of the query is refreshed with the latest information. Please review current definition and re-submit your updates if necessary.**" is displayed.

29.5.3.2.6 To Cancel the Updates

(1) Click the **Cancel** button, when available.

a) All modifications pending a Send action, if any, are discarded

b) The editing area of the **Temporary OTMV editing** window is closed and

c) the **Update** button is re-enabled.

29.5.3.2.7 To Close the Editing Area

(1) Click the **Close** button, when available.

a) The editing area of the **Temporary OTMV editing** window is closed,

b) the **Update** button is re-enabled.

29.6 Contextual actions

(1) None

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30 ATC Airspace OTMV List

30.1 Function presentation

(1) The **ATC Airspace OTMV List** function lists OTMV definition state on a selected date and time for all Traffic Volumes in the area of responsibility of one non clustered ATC Unit Airspace (AUA) or a Cluster.

(2) Data retrieved by the **ATC Airspace OTMV List** query are extracted from the ETFMS Environment database.

30.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Traffic / Monitor / ATC Airspace OTMV List**.

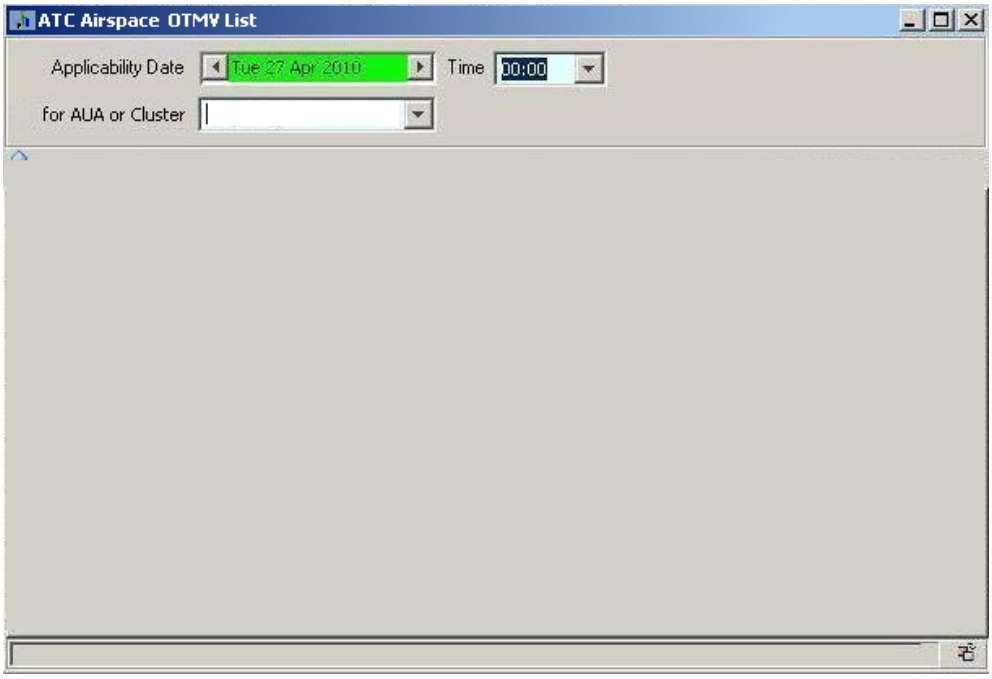


Figure Chmi-96: Query ATC Airspace OTMV List

(2) A new empty **ATC Airspace OTMV List** window will be opened with the following defaults:

a) **Applicability Date**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **Time**: initialised to 00:00

c) The **for AUA or Cluster** drop-down box will propose the most recent identifiers you have queried before with success.

(3) On query preparation, the focus is set on the first input control with no default value: **for AUA or Cluster** input control.

30.3 Preparing to query

(1) The **Applicability Date** may be adjusted to one of the proposed values.

(2) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(3) The proposed default **Time** may be forced to a specific time value. Any valid time value from 00:00 to 23:59 included may be entered. This input must respect the HH:MM format. Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(4) The **for AUA or Cluster** control may be adjusted by selecting one of the ID proposed in the drop-down list or by keying in a new ATC Unit Airspace (AUA) or a Cluster identifier in the dedicated input box.

(5) The **Send** button will send the query to the Network Operations system.

(6) **ATC Airspace OTMV List** window may be closed before issuing any query.

(7) The **Reset** action common functionality is implemented for this function.

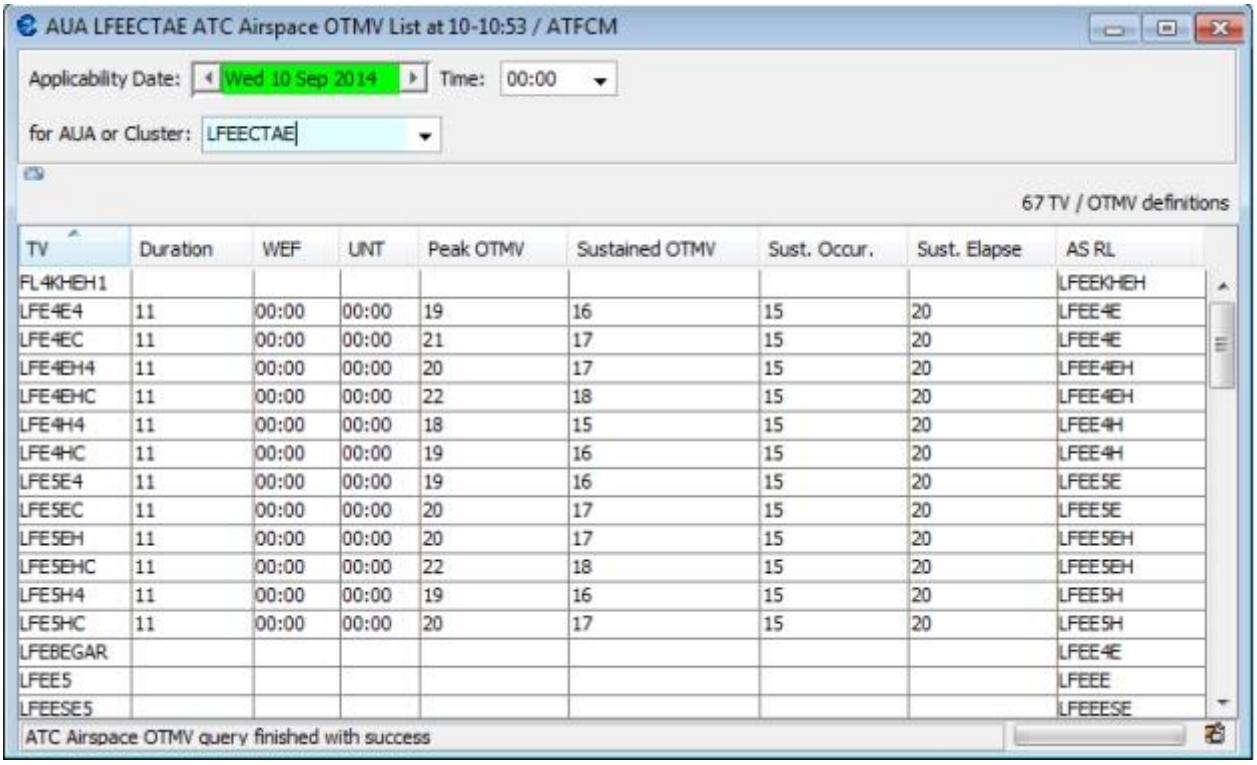
(8) Following query data elements are considered for activating the Reset action:

a) **Applicability Date**

b) **Time**

c) **for AUA or Cluster**

30.4 Displaying the query results



TV	Duration	WEF	UNT	Peak OTMV	Sustained OTMV	Sust. Occur.	Sust. Elapse	AS RL
FL4KHEH1								LFEEKHEH
LFE4E4	11	00:00	00:00	19	16	15	20	LFEE4E
LFE4EC	11	00:00	00:00	21	17	15	20	LFEE4E
LFE4EH4	11	00:00	00:00	20	17	15	20	LFEE4EH
LFE4EHC	11	00:00	00:00	22	18	15	20	LFEE4EH
LFE4H4	11	00:00	00:00	18	15	15	20	LFEE4H
LFE4HC	11	00:00	00:00	19	16	15	20	LFEE4H
LFE5E4	11	00:00	00:00	19	16	15	20	LFEE5E
LFE5EC	11	00:00	00:00	20	17	15	20	LFEE5E
LFE5EH	11	00:00	00:00	20	17	15	20	LFEE5EH
LFE5EHC	11	00:00	00:00	22	18	15	20	LFEE5EH
LFE5H4	11	00:00	00:00	19	16	15	20	LFEE5H
LFE5HC	11	00:00	00:00	20	17	15	20	LFEE5H
LFEBEGAR								LFEE4E
LFEE5								LFEEE
LFEESE5								LFEEESE

Figure Chmi-97: ATC Airspace OTMV List

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The header box of the window contains the different **query data elements** used to retrieve the displayed configuration:

a) The **Applicability Date** and **Time**

b) The **for AUA or Cluster** identifier.

(3) The OTMV State list is displayed below the query header box.

(4) Each row may contain the following data elements:

a) The Traffic Volume identifier (**TV**)

b) The **Duration** for which OTMV are defined, if any.

c) The **WEF** and **UNT** times when the corresponding OTMV definition is applicable during the queried **Applicability Date**.

d) The **Peak OTMV** threshold parameter expressed as a number of flights, when defined.

e) The **Sustained OTMV** threshold parameter expressed as a number of flights, when defined.

f) The **Sustained Occurrence** parameter expressed as a number of occurrences (column title may be abbreviated to "**Sust. Occur.**"), when defined.

g) The **Sustained Elapse** expressed as a number of consecutive minutes (column title may be abbreviated to "**Sust. Elapse**") for checking the presence of the minimum **Sustained_Occurrence** for reporting an alert, when defined.

h) The identifier of the Airspace Reference Location on which the Traffic Volume is defined (**AS Ref Loc**).

(5) The OTMV State list displays at least one row per Traffic Volume identified in the area of responsibility of the selected non clustered ATC Unit Airspace (AUA) or the Cluster.

(6) When no OTMV definitions exist for the Traffic Volume, a single row is displayed with empty Duration and OTMV related columns.

(7) When OTMV definitions exist for the Traffic Volume, one OTMV state row is displayed per Duration for which OTMV are defined.

(8) AS Ref Loc columns is filled in independently of the presence of OTMV definitions.

(9) At the top of the OTMV State list, a counter indicates the number of TV / OTMV definitions rows retrieved.

(10) This counter may be affected by **filtering** (fff / rrr TV / OTMV definitions where fff is the filtered number and rrr is the retrieved number).

(11) The OTMV State list is initially sorted in ascending sequence on:

a) The Traffic Volume identifier (TV)

b) Then on the **Duration**

(12) On query results display, the focus is set on the OTMV State list after selecting the first row in the list.

30.5 User interactions

(1) This window allows to modify a previous query selection and to issue a new request.

(2) On query modification, the **Reset action** is enabled.

(3) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(4) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(5) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(6) To modify the **Sort sequence** of the OTMV State list, click on the column headers corresponding to your desired sort criteria.

(7) To print the OTMV State list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(8) To export the OTMV State list, click on the **Export** button or select the **Action / Export** menu option.

(9) Increasing / reducing the window height or width will impact the size of the OTMV State list control.

30.6 Contextual actions

(1) When a line is selected in the OTMV State list, the right mouse button pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the selected Traffic Volume and may contain the following options:

a) **Occupancy Traffic Monitoring Values**: lists detailed OTMV definitions for the selected Traffic Volume.

b) **Occupancy Counts**: displays the Occupancy Counts for the selected Traffic Volume.

i) The Counts observed date is forced to the selected applicability Date.

ii) The Counts observed time period is using default period assignment like when a new Occupancy Counts window is opened.

c) **ATC Configuration** requests ATC configuration details and the activation table for the queried applicability Date and the non clustered ATC Unit Airspace (AUA) or the Cluster.

d) **Show Traffic Volume Description** to display the CHMI ENV detailed description screen for the selected Traffic Volume.

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31 ATC Configuration

31.1 Function presentation

(1) The **ATC Configuration** function displays the ATC configurations and the activation table for one non clustered ATC Unit Airspace (AUA) or a Cluster associated to a selected date.

(2) Data retrieved by the **ATC Configuration** query are extracted from the ETFMS Environment database and Tactical updates.

31.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Traffic / Monitor / ATC Configuration**.

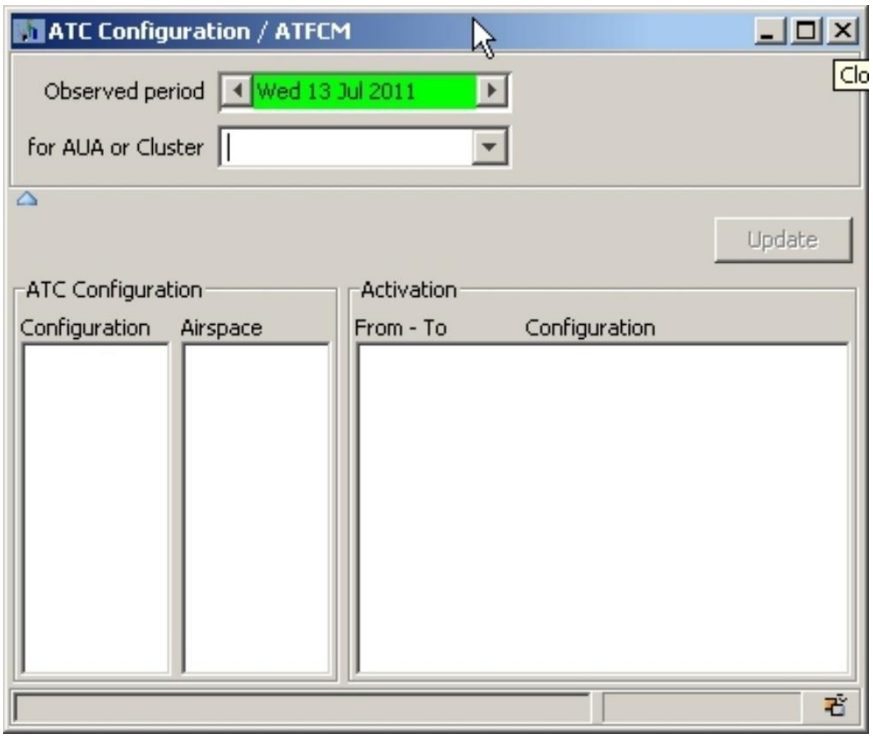


Figure Chmi-98: Query ATC Configuration

(2) A new empty **ATC Configuration** window will be opened with the following defaults:

a) **Observed period**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) The **for AUA or Cluster** drop-down box will propose the most recent identifiers you have queried before with success.

(3) On query preparation, the focus is set on the first input control with no default value: **for AUA or Cluster** input control.

31.3 Preparing to query

(1) The **Observed period** may be adjusted to one of the proposed values.

(2) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(3) The **for AUA or Cluster** control may be adjusted by selecting one of the ID proposed in the drop-down list or by keying in a new ATC Unit Airspace (AUA) or a Cluster identifier in the dedicated input box.

(4) The **Send** button will send the query to the Network Operations system.

(5) **ATC Configuration** window may be closed before issuing any query.

(6) The **Reset** action common functionality is implemented for this function.

(7) Following query data elements are considered for activating the Reset action:

a) **Observed period**

b) **for AUA or Cluster**

31.4 Displaying the query results

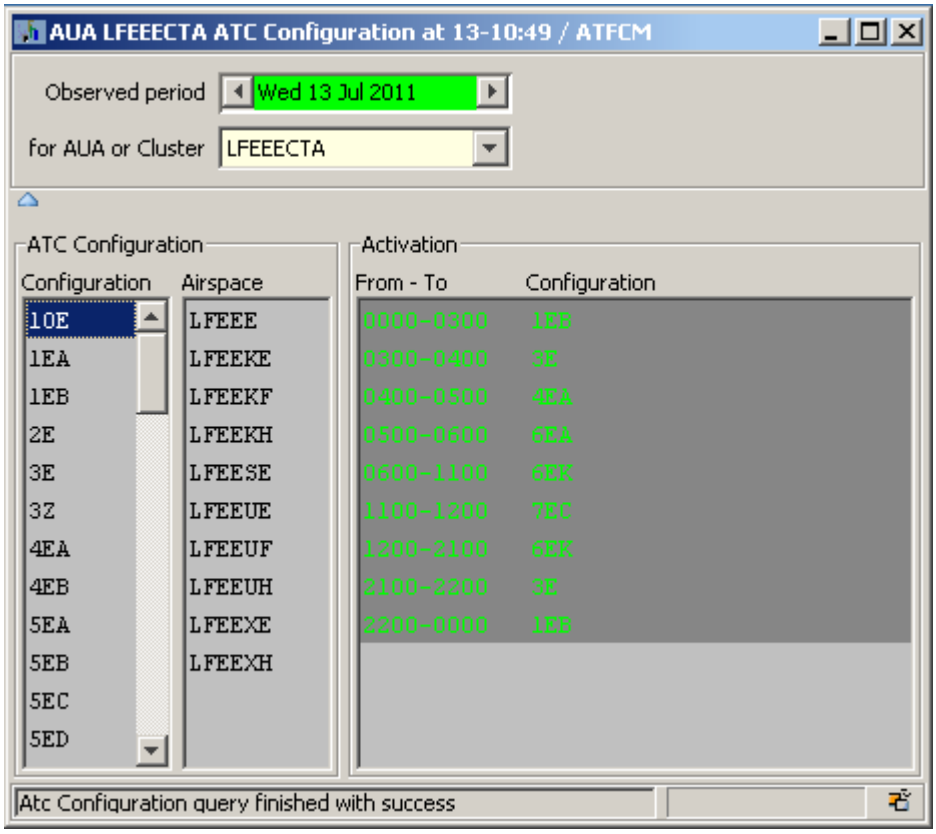


Figure Chmi-99: ATC Configuration

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The header box of the window contains the different **query data elements** used to retrieve the displayed configuration:

a) The **Observed period.**

b) The for **AUA or Cluster** identifier.

(3) Below the query header box, the main part of the display provides three lists.

(4) The left hand part of the window shows the list of existing **Configurations** for the ATC Unit Airspace (AUA) or a Cluster.

(5) On query results display, the focus is set on the Configuration list after selecting the first **Configuration** in the list.

(6) When you select one configuration in the list, you get in the adjacent box, the list of **Airspaces** that are included in the selected configuration.

(7) The right hand part of the window shows the **Activation** table.

(8) This **Activation** list displays:

a) The **From** and the **To** times of the activation period.

b) The identifier of the activated **Configuration**.

(9) The data source for the activation period can be either from the ETFMS Environment database or from a tactical update performed by an authorised user.

(10) Tactical updates of **Configuration** are shown in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour**.

(11) Access to update the configuration activation periods functionality via an **Update** button is restricted by security profile.

31.5 User interactions

31.5.1 General

(1) Click the **Update** button to open the editing area of the **ATC Configuration Activations** window (see below).

(2) The **Update** button is disabled.

(3) Select one activation in the activation table, to get the corresponding configuration in the left hand part of the window.

(4) When either one of the items in the three panes is selected, the right button of the mouse gives access to a sub menu.

(5) The **Normal / Proposal** two states button may be adjusted before selecting an option of the popup menu described in the "Contextual actions" section.

(6) The selected **Normal / Proposal** value will be transferred to the invoked function.

(7) This window allows to modify a previous query selection and to issue a new request.

(8) On query modification, the **Reset action** is enabled.

(9) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(10) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(11) To print the Regulation Description details, first retrieve the desired details. When ready, click on the **Print** button or select the **File / Print** menu option.

(12) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(13) Increasing / reducing the window height will impact the size of the three list controls.

(14) Increasing / reducing the window width will impact the size of the Activation lists control.

31.5.2 Editing Area of the ATC Configuration Activation window

31.5.2.1 Displaying the Window

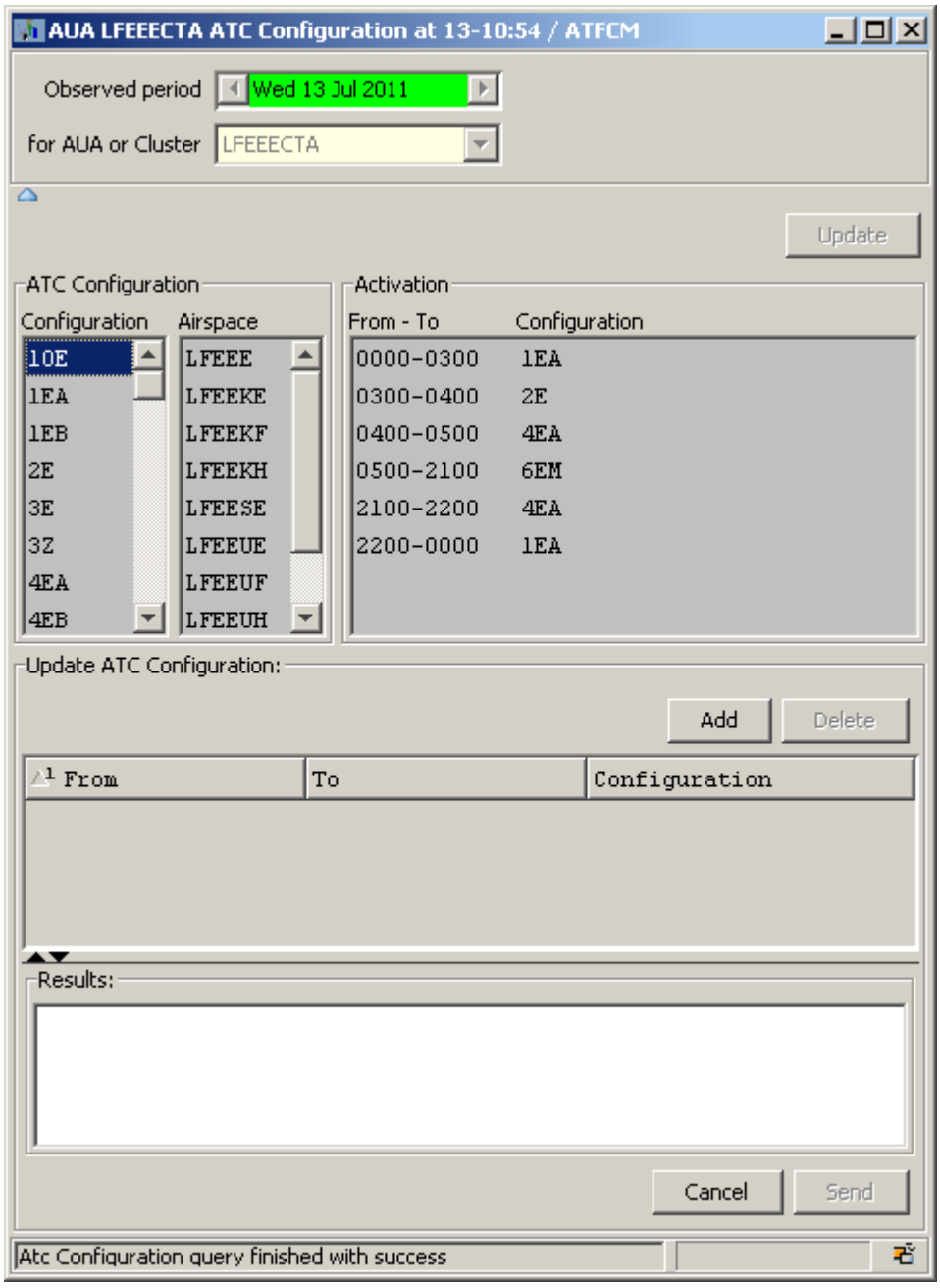


Figure Chmi-89: Update ATC Configuration Activation

(1) The editing area of the **ATC Configuration Activation** window will be opened with the following defaults:

a) The main listbox in the editing area of the window only lists the periods whose configuration has been updated tactically (i.e. periods containing data shown in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour**). If no tactical updates have been performed for the **Observed Period**, the listbox will be empty.

b) The listbox is sorted by **From** (earliest at the top of the list).

c) By default, the first period in the list that has not started (i.e. whose **From** is after the current UTC) is selected.

d) The **Cancel** button is enabled.

e) The **Send** button is disabled.

(2) This main listbox displays:

a) The **From** and the **To** times of the activation period.

b) The identifier of the activated **Configuration**.

(3) If an activation period has expired (i.e. the **To** of the period is before the current UTC time), all the fields of the activation period are shown as read-only.

(4) If an activation period has started but not yet expired, (i.e. current UTC time is between the **From** and **To**), except for the **To** field, all the other fields of the activation period are shown as read-only.

(5) If an activation period has not yet started (i.e. the **From** is after the current UTC time), all the fields of the activation period are editable.

31.5.2.2 User Interactions

31.5.2.2.1 To Select a Period

(1) To select an activation period, click on any field of the period. The period is shown selected and the previously selected period, if any, is shown deselected.

31.5.2.2.2 To Add a Period

(1) To add a new period, click the **Add Period** button.

(2) A new period is inserted in the list - below the previously selected period, if present.

(3) The new period is automatically set to be the selected period.

(4) The **Send** button is enabled.

(5) The default values for the new period are:

a) **From**: To value of:

i) the previously selected period, if present, otherwise

ii) the current UTC time plus 10 minutes.

b) **To**: From of the new period plus 3 hours limited to the end of the Observed Period.

c) **Configuration**: set to the value defined in the main **ATC Configuration** window (i.e merged view) at the start time of the new period.

31.5.2.2.3 To Delete a Period

(1) Select the period you wish to delete, if not already currently selected.

(2) The **Delete Period** button is only enabled when either:

a) the **From** of the currently selected period is later than the current UTC time or

b) the selected period has been added in the current window session (i.e. not already submitted into the system).

(3) Click the **Delete** button if enabled.

(4) The selected period is deleted from the list.

(5) The **Send** button is enabled.

31.5.2.2.4 To Modify a Period

(1) The **From** field, when editable, shall allow the user to change the start time of the configuration activation period.

(2) The **To** field, when editable, shall allow the user to change the end time of the configuration activation period.

(3) The **configuration** field, when editable, shall allow the user to set it to any one of the system known configurations available for the ATC Unit Airspace for the observed period.

(4) If any of the editable fields have been modified, the **Send** button is enabled.

31.5.2.2.5 To Send the Updates

(1) Click the **Send** button, if enabled.

(2) The results area is cleared and both the **Cancel** and the **Send** buttons are disabled.

(3) The following validation checks are performed and reported in the results area:

a) If update period was already submitted (i.e. from previous update) and is expired (**From** & **To** are before current UTC time), no validations error should be reported.

b) If update period was already submitted but only **From** is expired (**From** before current UTC time), **To** must be later than the current UTC time and within the **Observed Period**.

c) If a new update period is added or for an already submitted period that has not started, both **From** & **To** have to be later than the current UTC time and within the **Observed Period**. **To** has also to be later than the **From**.

d) There are no overlaps between the periods.

(4) Upon reporting of any validation errors, the **Cancel** button is re-enabled.

(5) The user shall be able to correct any validation errors.

(6) On any subsequent edit(s), the results area is cleared.

(7) If there are no validation errors:

a) The **Update** button is enabled,

b) the non-editing part of the window (i.e. the initial query/reply part) is automatically refreshed with the latest system known situation,

c) the editing part of the window becomes read-only,

d) the **Send** button is renamed to Close and is enabled,

e) the results area displays a message indicating the status of the last submission:

i) if no errors are returned, the message "**Update successful**" is displayed,

ii) if the update was successfully merged with parallel updates from another user, the message "**Update successfully merged with updates from other users**" is displayed,

iii) if the update was performed on the Ops (ETFMS) system for the pre-tactical day and the plan had not yet been transferred from Predict and applied, the message "**Updates for tomorrow could not be applied – the plan from Predict has not been transferred to Ops (ETFMS) and applied**" is displayed

iv) if the update was performed on the Predict system for the pre-tactical day after the "cut-off" time – a duration parameter (TBD) related to a specific UTC time, the message "**Updates could not be applied due to late submission – plan being finalized in preparation for transfer to the Ops (ETFMS) system**" is displayed.

v) if the system is not able to merge the submitted update with an update made by another user, the message "**The updates conflict with updates received from other users. The results of the query is refreshed with the latest information. Please review your updates and re-submit if necessary.**" Is displayed.

31.5.2.2.6 To Cancel the Updates

(1) Click the **Cancel** button, when available.

a) All modifications, if any, are discarded

b) The editing area of the **ATC Configuration Activation** window is closed and

c) The **Update** button is re-enabled.

31.5.2.2.7 To Close the Editing Area

(1) Click the **Close** button, when available.

(2) The editing area of the **ATC Configuration Activation** window is closed.

(3) The **Update** button is re-enabled.

31.6 Contextual actions

(1) When an identifier is selected in the **Configuration** list, the right mouse button pops up a contextual menu.

(2) This popup menu lists the following possible actions:

a) **ATC Airspace Monitor** for the selected ATC Unit Airspace (AUA) or a Cluster, configuration and default period

b) **ATC Configuration Optimizer** requests optimal ATC Configuration list for the ATC Unit Airspace (AUA) or a Cluster and for default interval and period.

(3) By dragging the left mouse button on the **Activation** list, it is possible to select one or multiple adjacent periods.

(4) When one or more periods are selected in the **Activation** list, the right mouse button pops up a contextual menu.

(5) This popup menu lists the following possible actions:

a) **ATC Airspace Monitor**

i) Only available if a single activation line is selected.

ii) To query the ATC Airspace Monitor for the selected ATC Unit Airspace (AUA) or a Cluster, configuration and period

b) **ATC Configuration Optimizer** requests:

i) To open the ATC Configuration Optimizer without an initial query

ii) Following query criteria are forced: **ATC Unit Airspace** (AUA) or a Cluster and **Observed Period, WEF and UNT** from the combined period selected in the **Activation** table.

iii) The **Configuration Order Criteria** dialog is opened for the selected period and, if only one period was selected in the **Activation** table, the selected Configuration identifier is automatically inserted in the **Forbidden Configurations** input box.

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32 ATC Configuration Optimizer

32.1 Function presentation

(1) The **ATC Configuration Optimizer** function displays the optimal list of configurations for selected:

a) Time interval (step and duration)

b) Period

c) AUA or Cluster

d) Traffic Type

(2) The **ATC Configuration Optimizer** application customisation and access rights will be controlled by dedicated security profile for each of the following possible Traffic Types (**TFC Type**):

a) Traffic load

b) Traffic demand

c) Regulated demand

32.1.1 Pitfalls to avoid when reading the ATC Configuration Optimizer display

(1) Following important remarks are copied since we believe it is important to recall these principles. ETFMS requirement labels are left for reference to the original text.

a) The global results output by Opticon looks like a solution matrix, the rows being the time intervals and the columns the configurations sorted by ascending cost. However, it is important to remember that the optimisation process is done line by line, i.e. time interval by time interval. It is even more correct to say that it is done block of intervals by block of intervals, the default display interval by interval corresponding to a block size of 1.

b) When looking at Opticon results, one should always remember the “overload shift” mechanism explained in section 32.1.5.2. Basically, this mechanism consists in adding the traffic overload (i.e. the worst excess traffic if positive) at one interval (if any) to the traffic counts of all traffic volumes at the next interval. Hence the list of best configurations at a given interval always depends on the best configuration choice from the previous interval. However, if the user defines blocks of intervals by means of the “Minimum number of intervals” parameter, it should be borne in mind that within each block, the overload shift is done “column by column”, i.e. from and to the same configuration. It is only the worst excess traffic from the best configuration at the last interval of the block that will be shifted to all configurations at the beginning of the next block of intervals.

32.1.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Traffic / Monitor / ATC Configuration Optimizer**.

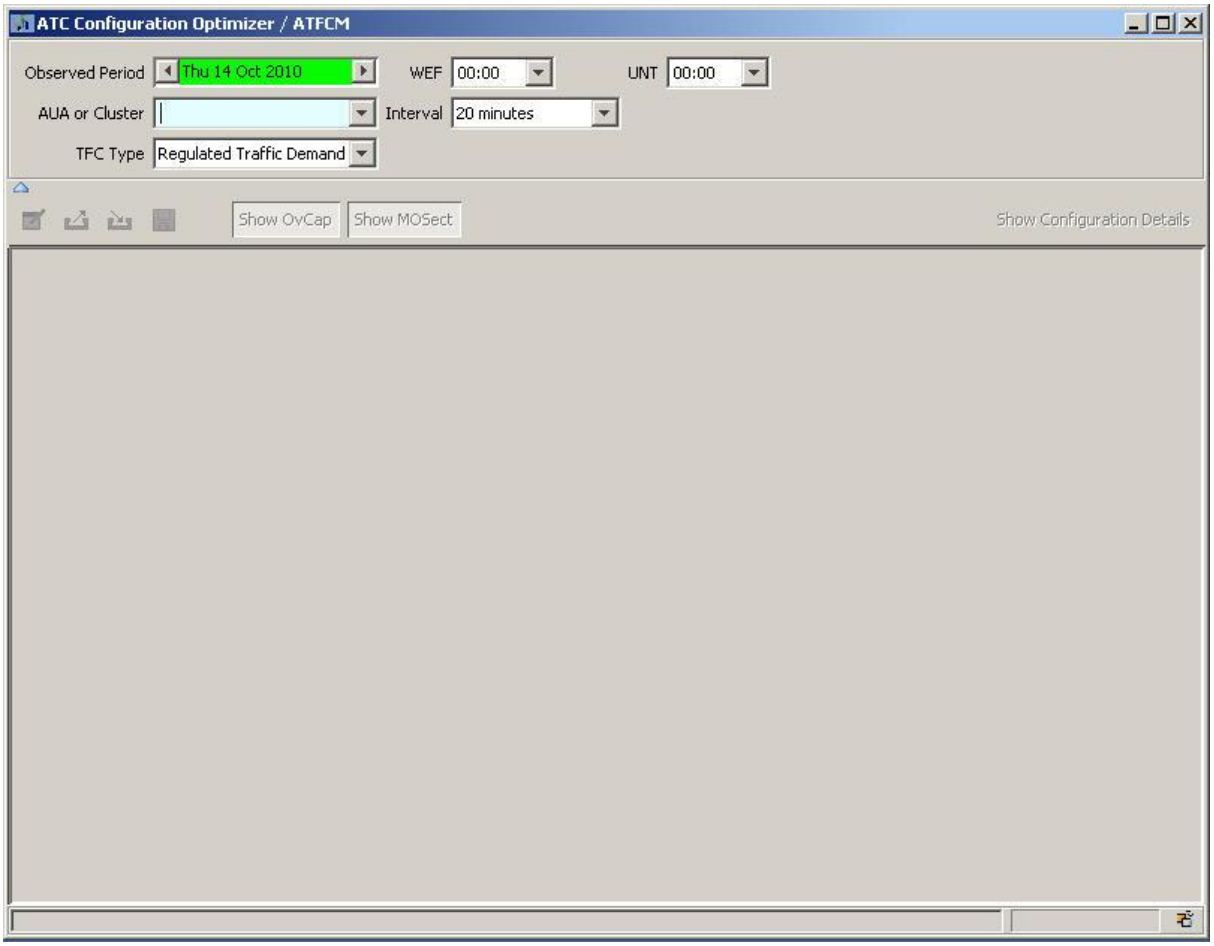


Figure Chmi-100: Query ATC Configuration Optimizer

(2) A new empty **ATC Configuration Optimizer** window will be opened with the following defaults:

a) **Observed period**: Current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **WEF** and **UNT** time: Proposed for full day query (00:00 to 00:00).

c) **AUA or Cluster**: Drop-down box will propose the most recent identifiers you have queried before with success.

d) Traffic Type (**TFC Type**): Last traffic type used with this query. This selection is based on the system property **CHMI / ATFCM application / Traffic / ATC Configuration Optimizer / Last Used Traffic Type**. Available types may be limited by security profile.

e) **Interval**: Last selection used with your last similar query. This selection is based on the system property **CHMI / ATFCM application / Traffic / ATC Configuration Optimizer / Last Used Interval.**

(3) On query preparation, the focus is set on the first input control with no default value: **AUA or Cluster** input control.

32.1.3 Preparing to query

(1) The **Observed period** may be adjusted to one of the proposed values.

(2) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(3) The traffic **Observed period** may be adjusted as follow:

a) The traffic **Date** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

b) The proposed default **WEF** and **UNT** times may be overwritten.

c) Any valid time value from 00:00 to 23:59 included may be entered in the **WEF** and/or the **UNT** entry boxes. This input must respect the HH:MM format.

(4) Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(5) The **AUA or Cluster** control may be adjusted by selecting one of the ID proposed in the drop-down list or by keying in a new ATC Unit Airspace identifier in the dedicated input box.

(6) The Traffic Type (**TFC Type**) parameter may be adjusted to one of the following values, when authorised:

a) Traffic load,

b) Traffic demand,

c) Regulated demand.

(7) The **Interval** parameter may be adjusted to one of the following values:

a) Hour every 20 Min

b) Hour every 10 Min

c) Hour

d) 20 Minutes

(8) The **Send** button will send the query to the Network Operations system.

(9) **ATC Configuration Optimizer** window may be closed before issuing any query.

(10) The **Reset** action common functionality is implemented for this function.

(11) Following query data elements are considered for activating the **Reset** action:

a) **Observed period, WEF and UNT**

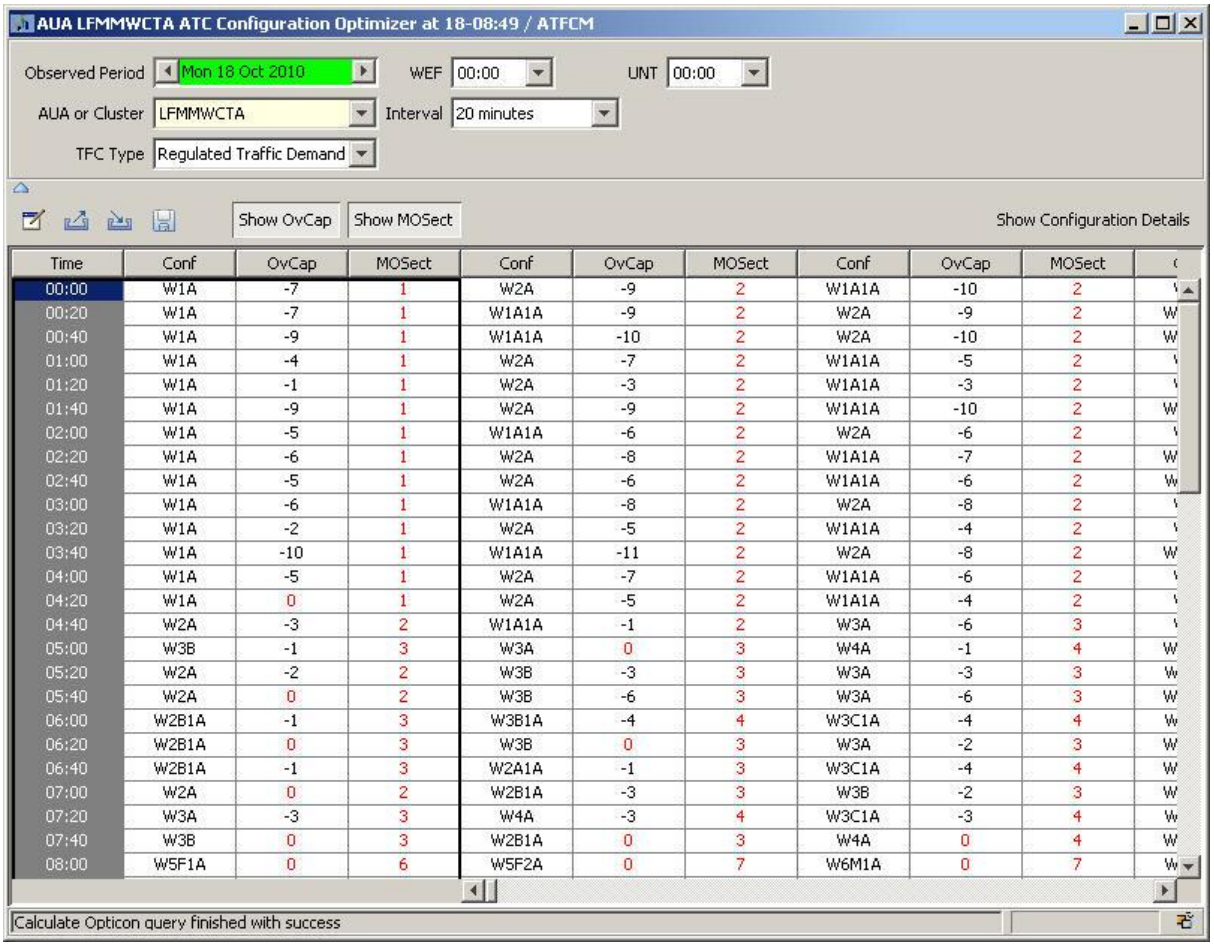
b) **AUA or Cluster**

c) **TFC Type**

d) **Interval**

(12) The service *CALCULATE_OPTICON* is available from ETFMS to evaluate the different ATC Configurations per time interval based on the current optimizer criteria definition.

32.1.4 Displaying the query results



Time	Conf	OvCap	MOSect	Conf	OvCap	MOSect	Conf	OvCap	MOSect
00:00	W1A	-7	1	W2A	-9	2	W1A1A	-10	2
00:20	W1A	-7	1	W1A1A	-9	2	W2A	-9	2
00:40	W1A	-9	1	W1A1A	-10	2	W2A	-10	2
01:00	W1A	-4	1	W2A	-7	2	W1A1A	-5	2
01:20	W1A	-1	1	W2A	-3	2	W1A1A	-3	2
01:40	W1A	-9	1	W2A	-9	2	W1A1A	-10	2
02:00	W1A	-5	1	W1A1A	-6	2	W2A	-6	2
02:20	W1A	-6	1	W2A	-8	2	W1A1A	-7	2
02:40	W1A	-5	1	W2A	-6	2	W1A1A	-6	2
03:00	W1A	-6	1	W1A1A	-8	2	W2A	-8	2
03:20	W1A	-2	1	W2A	-5	2	W1A1A	-4	2
03:40	W1A	-10	1	W1A1A	-11	2	W2A	-8	2
04:00	W1A	-5	1	W2A	-7	2	W1A1A	-6	2
04:20	W1A	0	1	W2A	-5	2	W1A1A	-4	2
04:40	W2A	-3	2	W1A1A	-1	2	W3A	-6	3
05:00	W3B	-1	3	W3A	0	3	W4A	-1	4
05:20	W2A	-2	2	W3B	-3	3	W3A	-3	3
05:40	W2A	0	2	W3B	-6	3	W3A	-6	3
06:00	W2B1A	-1	3	W3B1A	-4	4	W3C1A	-4	4
06:20	W2B1A	0	3	W3B	0	3	W3A	-2	3
06:40	W2B1A	-1	3	W2A1A	-1	3	W3C1A	-4	4
07:00	W2A	0	2	W2B1A	-3	3	W3B	-2	3
07:20	W3A	-3	3	W4A	-3	4	W3C1A	-3	4
07:40	W3B	0	3	W2B1A	0	3	W4A	0	4
08:00	W5F1A	0	6	W5F2A	0	7	W6M1A	0	7

Figure Chmi-101: ATC Configuration Optimizer

(1) The title contains the **time stamp** when the optimizer information was retrieved from the Network Operations system.

(2) The header box of the window contains the different **query data elements** used to retrieve the displayed optimizer configuration list:

a) **Observed period, WEF and UNT**

b) **AUA or Cluster**

c) **TFC Type**

d) **Interval**

(3) On successful query / reply, the WEF and UNT query data elements are aligned on the selected interval.

a) If WEF is not a multiple of the specified interval, it is rounded to the lower multiple.

b) If UNT is not a multiple of the specified interval, it is rounded to the upper multiple.

Example:	*Interval*	*WEF*	*UNT*
	Hour every 20 Min	*00:35*	*12:10*
Becomes after valid query/reply:			
	Hour every 20 Min	*00:20*	*12:20*

(4) Below the query header box, the main part of the display provides the optimal list of configurations for each interval of the selected period. This list is similar to a matrix with rows and columns.

(5) Each row gives the list of configurations, ordered from the “best” (i.e. smallest cost) to the “worst” (i.e. highest cost), for one time interval.

(6) The first row from the top corresponds to the first interval in the period, the second row to the next interval and so on until the last interval at the bottom of the window.

(7) Each column index corresponds to the rank of the configuration in the list for the considered time interval.

(8) In fact, each “matrix column” is made up of three columns used to display:

a) The name of the configuration (**Conf**).

b) The worst excess traffic (**OvCap**) along with the local excess traffic between brackets, if any. The worst excess traffic is highlighted in red when positive in order to draw the user’s attention to the risk of overload.

c) The number of active sectors (**MOSect**) in the configuration.

(9) The first three columns of the Optimizer contain the information related to the best configuration (the cells are highlighted in orange to emphasise that).

(10) The next three columns contain the information related to the second best and so on…

(11) The initial list of configurations to be sorted depends on the airspace for which the optimisation of configurations will be computed:

a) AUA having no clusters: The list is given by all the configurations of the AUA

b) Cluster: The list is given by all the configurations of the AUA

c) Multicluster AUA: The list is given by the combination of the configurations of its Clusters. For an AUA with n clusters, the resulting configurations are named, and displayed when appropriate, as follows:

<Configuration of cluster 1>/<Configuration of cluster2>/…/<Configuration of cluster n>

Where cluster 1 to cluster n are ordered in alphabetical order.

d) Example: AUA XXXX is formed has Clusters XXXWEST and XXXEAST

Cluster XXXXWEST is associated to configurations CONF1, CONF2

Cluster XXXXEAST is associated to configurations CONF1, CONF2, CONF3

e) The resulting list of configurations to be sorted, i.e to be fed to the optimisation process, is formed by:

CONF1/CONF1

CONF1/CONF2

CONF2/CONF1

CONF2/CONF2

CONF3/CONF1

CONF3/CONF2

From now onwards, within this “ATC Configuration Optimizer” chapter, the term configuration refers, unless otherwise stated, indistinctly to “simple” configurations, those associated to one AUA or one cluster, or to these “multicluster” configurations.

(12) The list of **Configurable Airspaces** for the selected **AUA or Cluster** shall be displayed at the top of the optimal list of configurations.

(13) In case of multiple **Configurable Airspaces** for a clustered AUA, the same presentation format applies as for multiple configurations in the **Conf** cells in the optimal list of configurations.

(14) Ex: “**Configurable Airspaces** LECMNCTA/ LECMSCTA”

(15) See also 32.1.1 Pitfalls to avoid when reading the ATC Configuration Optimizer display.

(16) The configurations that do not comply with at least one parameter (e.g. forced sectors, minimum number of active sectors…) will be displayed with a greyed font in order to single them out from the rest of the “valid” configurations.

(17) On query results display, the focus is set on the matrix containing the optimal list of configurations.

(18) The first cell of the matrix is selected.

(19) The **Show OvCap** display option is initially checked.

(20) When unchecked, worst excess traffic (**OvCap**) columns are NOT displayed in the matrix.

(21) The **Show MOSect** display option is initially checked.

(22) When unchecked, number of active sectors (**MOSect**) columns are NOT displayed in the matrix.

(23) If a TV has no capacity during a certain interval, but this TV would be activated if the configuration of the Optimizer cell is applied, then the traffic volume is considered as inactive during this interval

a) The background of the OvCap column for the Optimizer cell shall be coloured in light gray (RGB=192,192,192). This colour shall be linked to the adjustable user preference: **CHMI / ATFCM application / Traffic / ATC Configuration Optimizer / TV without Capacity Background Colour.**

b) When the **Configuration Details** list box is displayed, it shall include the list of TVs without capacity that would be activated if the configuration of the selected cell would be applied (**TV without capacity**).

(24) In other words, a TV without capacity does not play a role in the choice of a configuration and this is made visible in the ATC Configuration Optimizer screen.

(25) If a sector of a configuration has 0 (zero) active traffic volume (a TV without capacity being considered as inactive, see previous requirement), then:

a) The background of the MOSect column for the Optimizer cell shall be coloured in light gray (RGB=192,192,192). This colour shall be linked to the adjustable user preference: **CHMI / ATFCM application / Traffic / ATC Configuration Optimizer / No Active TV Background Colour.**

b) When the **Configuration Details** list box is displayed, it shall include the list of sectors with 0 traffic volumes active for the selected cell (**Sector with no active TV**).

(26) The **Show Configuration Details** display option is initially unchecked.

(27) When checked, a **Configuration Details** list box is displayed over the right part of the matrix. It displays the detailed criteria used to identify the optimal ATC configuration for the currently selected interval in the period.

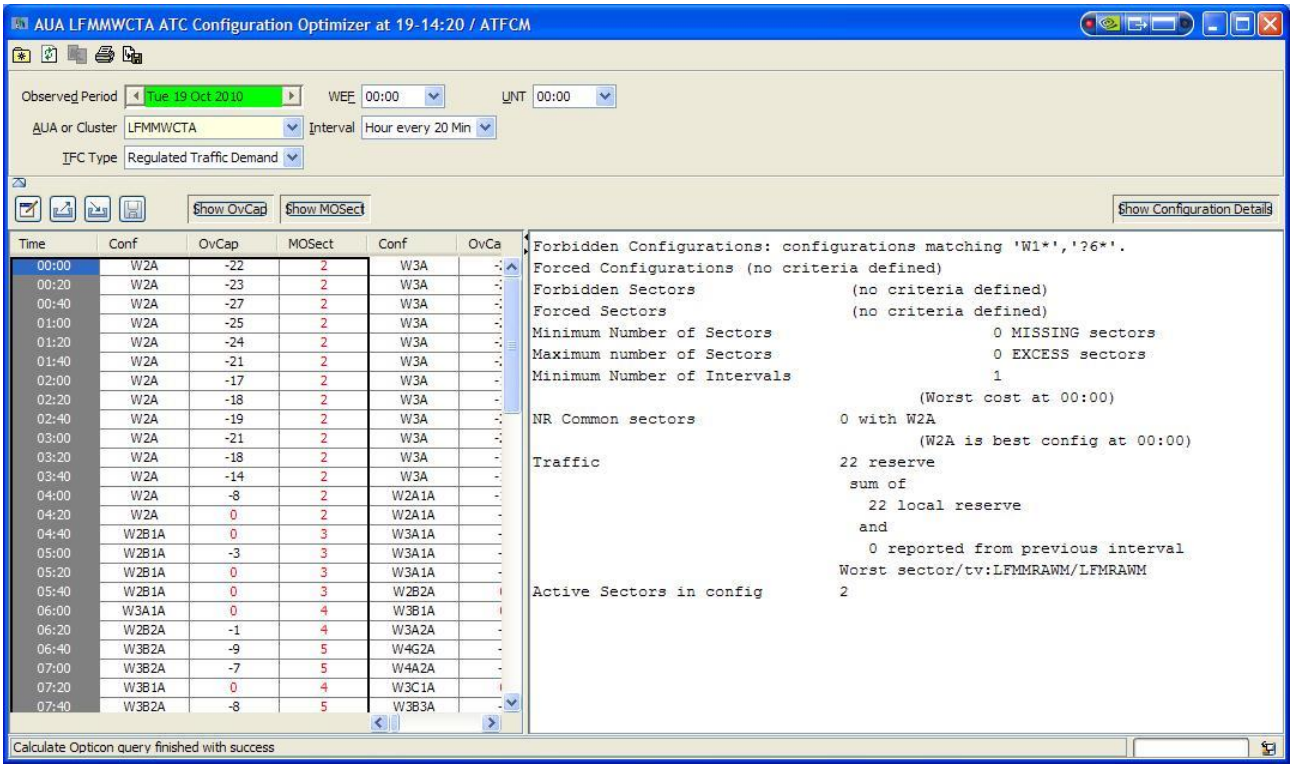


Figure Chmi-102: ATC Configuration Optimizer with Configuration Details

32.1.5 User interactions

32.1.5.1 Main window

(1) Check / Un-Check the **Show OvCap** display option to show / hide the worst excess traffic (**OvCap**) columns in the matrix.

(2) Check / Un-Check the **Show MOSect** display option to show / hide the number of active sectors (**MOSect**) columns in the matrix.

(3) Check / Un-Check the **Show Configuration Details** display option to show / hide the **Configuration Details** list box over the right part of the matrix.

(4) The proportion of the horizontal size dedicated to the matrix versus the list box is adjustable by the user.

(5) When the **Show Configuration Details** display option is turned on, click on the desired Configuration cell to see the **Configuration Details** associated to the selected ATC Configuration and Interval.

(6) This window allows to modify a previous query selection and to issue a new request.

(7) On query modification, the **Reset action** is enabled.

(8) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(9) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(10) To print the ATC Configuration Optimizer details, first retrieve the optimal list of configurations for selected criteria. Update criteria if needed and refresh optimal list. When ready, click on the **Print** button or select the **File / Print** menu option.

(11) In addition to the standard printer options, the print dialog will allow to select between:

a) **Print**: To print only the ATC Configuration Optimizer matrix content as displayed.

b) **Print detail**: To print the ATC Configuration Optimizer matrix content as displayed in addition to the specific configuration order criteria entered for each time interval.

(12) To export the full ATC Configuration Optimizer details, including the specific configuration order criteria entered for each time interval, click on the **Export** button or select the **Action / Export** menu option.

(13) Increasing / reducing the window height or width will impact the size of the matrix control.

32.1.5.2 Maintaining Configuration Order Criteria

(1) By dragging the left mouse button on the leftmost column (time intervals column), it is possible to select one or multiple adjacent intervals.

(2) A click on the *Edit Order Criteria* button located on the left above the Time column header will display the Configuration Order Criteria dialog. It is also accessible via *Edit Order Criteria* option in the popup menu over the selected intervals.

(3) This dialog contains all the order criteria parameters for the selected intervals. Those parameters enable the user to modify and/or customise the configuration optimisation criteria.

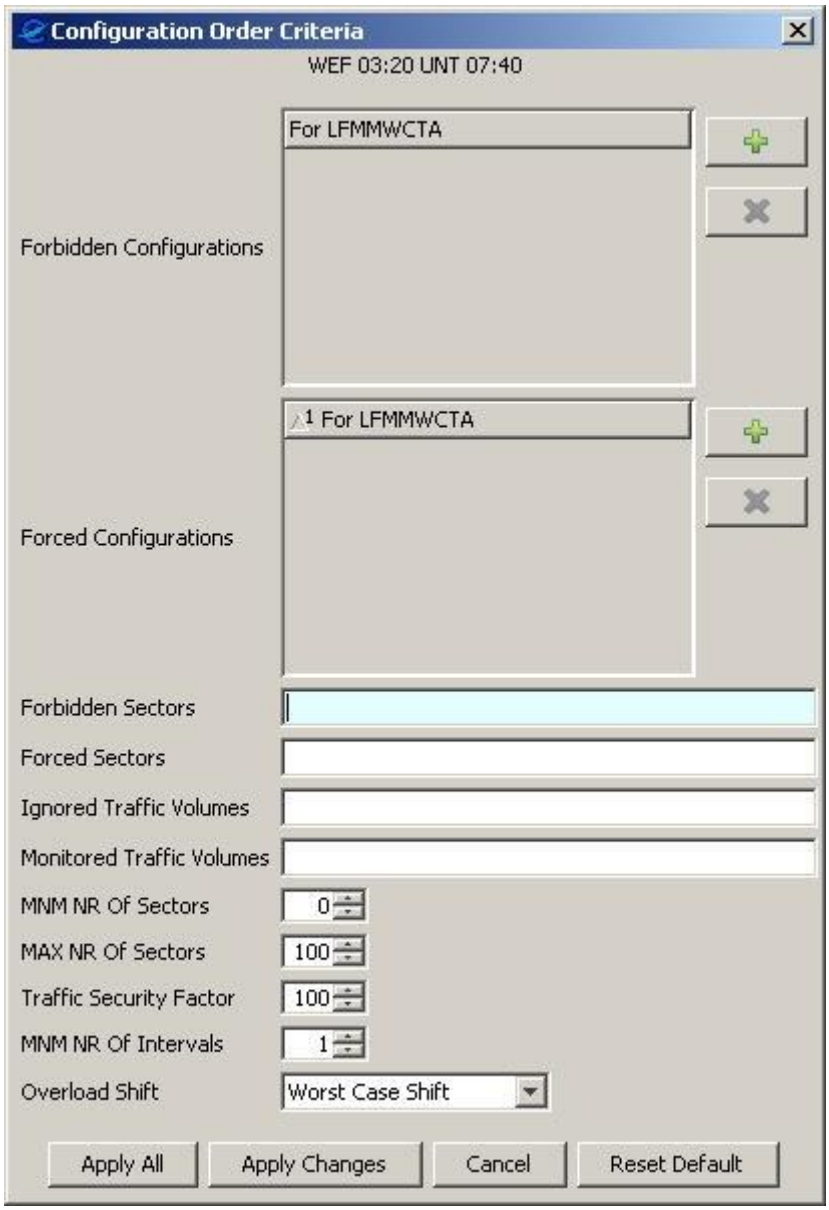


Figure Chmi-103: Configuration Order Criteria dialog

(4) Similarly to the main display, the list of **Configurable Airspaces** for the selected **AUA or Cluster** shall be repeated just above the input list for **Forbidden Configurations** and **Forced Configurations**.

(5) The order criteria window contains the following parameters (input criteria fields marked with a * means wildcard allowed as input):

a) **Forbidden Configurations (*)**: The user can provide the list of forbidden configurations to be used by the order function. The names of the configurations must be in upper case. Following tooltip shall explain the possible input characters: **"Enter the ID of the Configuration to forbid. Use ? and * as single and multiple letter wild cards (Ex: 'CS5A' 'CE4*' 'C?8*')".**

b) **Forced Configurations (*)**: The user can provide the list of forced configurations to be used by the order function. The names of the configurations must be in upper case. Following tooltip shall explain the

possible input characters: **"Enter the ID of the Configuration to force. Use ? and * as single and multiple letter wild cards (Ex: 'CS5A' 'CE4*' 'C?8*')".**

c) The **Forbidden Configurations** and **Forced Configurations** are using list input control:

i) The header row identifies the list of *Configurable Airspaces* for the selected **AUA or Cluster** (one per column).

ii) To add a new line to the **Forbidden** or **Forced Configurations** input list, click on the add button (**+**) and fill in the configuration identification in the corresponding *Configurable Airspaces* column.

iii) Each elementary configuration identifier (i.e. for "one cluster of a clustered AUA" or for "one non clustered AUA") can be specified either explicitly with the full identifier or implicitly using a wildcard (i.e. A wildcard character is either a "*" replacing 0 to n characters, or "?" replacing exactly 1 character).

iv) To remove a line from the **Forbidden Configurations** or **Forced Configurations** input lists, select the desired line and click on the delete button (**x**).

v) A blank input cell shall be forced by the system to a "*" wild card value to emphasize to the user this will select any configuration value for that column.

vi) A warning dialog shall inform the user when he requests to apply changes to the **Configuration Order Criteria** while a line of either the **Forbidden** or **Forced Configurations** input list contains ONLY "*" (i.e. "You submitted a line with full '*' wild card only which will overwrite any other more specific line. Do you want to proceed?").

d) **Forbidden Sectors**: The user can provide the list of forbidden sectors to be used by the order function. The names of the sectors must be in upper case and separated by one space.

e) **Forced Sectors**: The user can provide the list of forced sectors to be used by the order function. The names of the sectors must be in upper case and separated by one space.

f) **Ignored Traffic Volumes**: The user can provide a list of traffic volumes he wants to de-activate regardless the classical "monitorable" property. This will **simulate** (only during the computation of the opticon) that the user had selected the "de-activate tvs" feature from the ETFMS MMI. The names of the traffic volumes must be in upper case and separated by one space.

g) **Monitored Traffic Volumes**: The user can provide a list of traffic volumes he wants to activate regardless the classical "monitorable" property. This will **simulate** (only during the computation of the opticon) that the user had selected the "activate tvs" feature from the ETFMS MMI. The names of the traffic volumes must be in upper case and separated by one space.

h) **Minimum / Maximum number of sectors**: The user can provide with the minimum and maximum number of active sectors to be used by the order function.

i) **Traffic security factor (%)**: The user can provide a traffic security factor, which is a percentage to apply to the counts of flights of the concerned interval when computing the excess traffic. By default, the value is 100, which means no security margin. With a traffic security factor equal to 120, all the counts will be increased by 20%. For instance, a count of 5 flights will become 6. This feature is useful to quickly simulate an increase or decrease of traffic in the original data.

j) **Minimum number of intervals**: The user can provide the minimum number of intervals, that is to say the size of blocks of intervals over which only one best configuration will be computed. The default value is 1, which corresponds to the standard case where the optimizer computes the sorted list of best configurations interval by interval.

k) **Overload shift**: Basically, this feature aims at making the optimizer model more realistic when the worst excess traffic of the best configuration at a given interval I is positive, thus reflecting traffic on overload. The user may choose between three kinds of overload shift mechanism, each of them representing a different way to shift the “excess flights” to the next interval:

i) **Do not shift**: This option is self-explanatory and consists in not shifting the excess flights. This enables the user to de-activate the overload shift mechanism.

ii) **Worst Case Shift**: When this option is selected and the best configuration at a given interval I has a positive worst excess traffic, it shifts this “potential overload” value to the next interval by adding it to the “local excess traffic” of each traffic volume of each configuration, the result of this addition being stored in the Worst Excess Traffic value.

iii) **Overload Spread On ES**: This method has been designed to provide the user with a more realistic approach of the overload shift mechanism in order to offer an alternative for the main shortcoming of the previous method (the “pessimistic” count figures). Here, the basic principle is to consider the sector responsible for the overload of the best configuration at interval I and to compute the proportion of this overload on each elementary sectors making up this sector based on the traffic counts. Then each “partial” overload on each elementary sector is shifted to the interval I+1 by increasing the count on those elementary sectors accordingly, thus increasing also the counts on the sectors containing those elementary sectors at interval I+1.

(6) The default value for the **Overload shift** mechanism is the **Worst Case Shift**.

(7) The user can provide a different set of order criteria parameters for each interval. As a consequence, when selecting a range of adjacent intervals, it may be possible that each of the selected intervals is already associated with a different

set of parameters from the others. This would of course make it impossible to display a unique set of parameters in the window. In that case, the set of parameters of the first interval from the selection is displayed, but a **Multiple Values** warning text appears in front of each parameter to notify the user.

(8) The user can choose between two options to validate his changes in the Configuration Order Criteria dialog:

a) By clicking on **Apply All**, all the values currently displayed in the Configuration Order Criteria dialog will be used to update the associated order criteria on the selected intervals.

b) By clicking on **Apply Changes**, only the values modified by the user in the Configuration Order Criteria dialog will be taken into account to update the associated order criteria on the selected intervals.

32.1.5.3 Saving Configuration Order Criteria

(1) It is very likely that the definition of order criteria on a given day of traffic may also be applicable on other similar days. In fact, there might be recurrent order criteria patterns relevant for certain time and traffic conditions that the user would like to define once for all and apply on any day. To address this issue, a feature has been implemented to enable the user to save and load a set of order criteria via a simple text file.

(2) By dragging the left mouse button on the leftmost column (time intervals column), it is possible to select one or multiple adjacent intervals.

(3) A click on the *Save Order Criteria* button located above the Time column header will display the **Save Order Criteria** dialog. It is also accessible via *Save Order Criteria* option in the popup menu over the selected intervals.

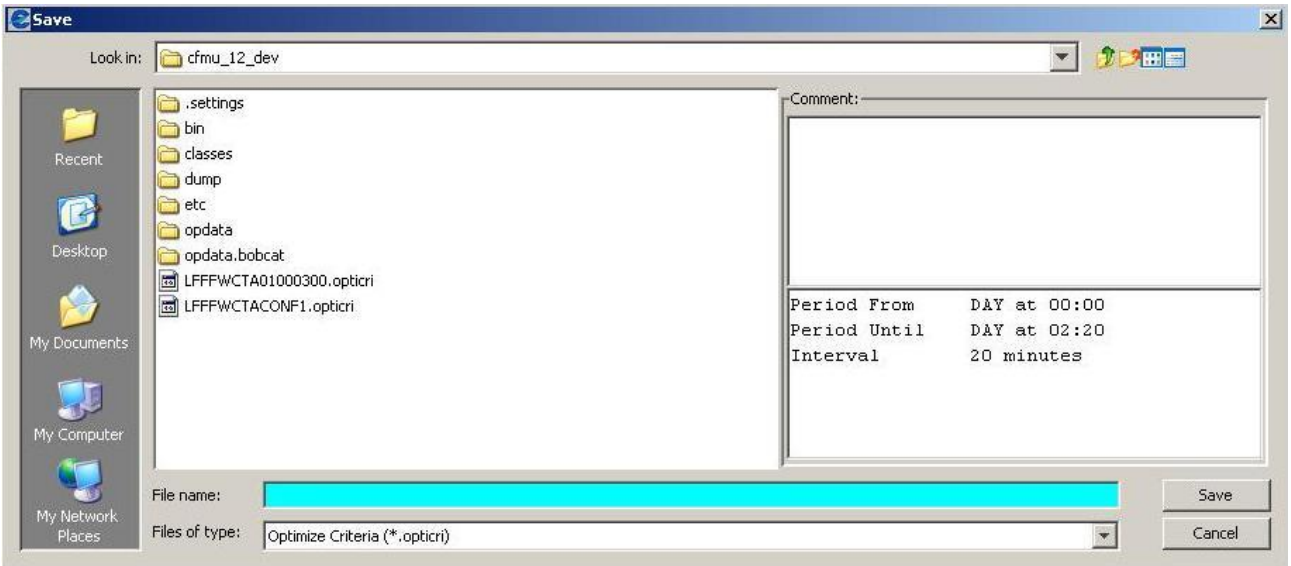


Figure Chmi-104: Save Order Criteria dialog

(4) The **Save Order Criteria** dialog displays in the bottom box information related to the period the user is intending to save:

a) The relative start date and time of the period (**Period From**).

b) The relative end date and time of the period (**Period Until**).

c) The **Interval** used by the optimizer for the period.

(5) The **Save Order Criteria** dialog allows the user to identify:

a) The location on the disk where the file will be saved

b) The name to be given to the file to recognize it for later reference

c) A comment statement that will be associated to the file in order to facilitate later reference

(6) When ready, the user shall click the **Save** button to initiate the saving of the Configuration Criteria with the attached comment into the specified file and location.

(7) The file extension will be forced to "*.***opticri***" for OPTImizer CRIteria.*

(8) The service *CONVERT_TO_CRITERIA_FILE_FORMAT* is available from ETFMS to convert normal optimizer input into standard storable file format.

(9) The file format is perfectly readable and re-uses the lisp-like TACOT script format.

(10) It is made up of a two parts:

a) A header containing:

i) The time at which the order criteria file was created

ii) A comment the user may input

iii) The overall time period for the order criteria definition and the interval frequency (i.e. the length of each time interval)

b) A list of order criteria definition for each interval, starting with the interval start time and followed by the full list of order criteria

(11) It must be stressed that all the time values recorded in the cost criteria file are "relative" values rather than "absolute values".

(12) To avoid duplication of code and format knowledge with the server, to preserve ETFMS encoding / decoding independency and the possibility to exchange criteria files between different platforms, the standard storable file format shall be used as a black box by the client application and stored untouched and standalone in the specified file.

32.1.5.4 Loading Configuration Order Criteria

(1) Once an order criteria file is created, it is of course possible to load it into any ATC Configuration Optimizer window

a) By dragging the left mouse button on the leftmost column (time intervals column), it is possible to select one or multiple adjacent intervals.

b) A click on the *Load Order Criteria* button located above the Time column header will display the **Load Order Criteria** dialog. It is also accessible via *Load Order Criteria* option in the popup menu over the selected intervals.

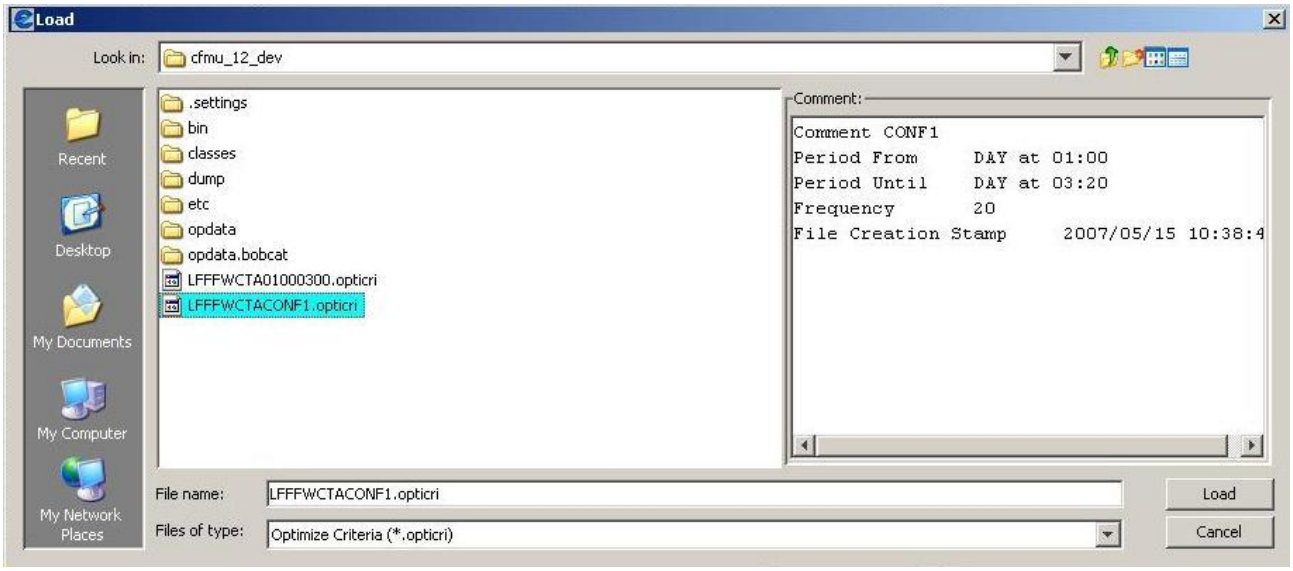


Figure Chmi-105: Load Order Criteria dialog

(2) The **Load Order Criteria** dialog allows the user to identify:

a) The location on the disk where the file can be loaded from

b) The name of the file to be loaded

(3) The **Load Order Criteria** dialog displays in the bottom box information related to the currently selected file:

a) The **Comment** provided by the user when the file was saved

b) The relative start date and time of the period (**Period From**)

c) The relative end date and time of the period (**Period Until**)

d) The **Interval** used by the optimizer for the period.

e) The **File Creation Stamp**

(4) To avoid duplication of code and format knowledge with the server, to preserve ETFMS encoding / decoding independency and the possibility to exchange criteria files between different platforms, the standard storable file format shall be used as a black box by the client application.

(5) To preserve this independency, each time a file is selected in the displayed list, the ETFMS service *CONVERT_FROM_CRITERIA_FILE_FORMAT* shall be used based on the file content to obtain and display above detailed information.

(6) Once the desired Optimized Criteria file has been identified, the user shall click the **Load** button to request the merge of the criteria stored in the file on top of the criteria of the current ATC Configuration Optimizer session.

(7) To be successful, the load of an order criteria file into an ATC Configuration Optimizer window must meet two conditions:

a) The interval frequency from the file must be equal to the interval frequency used to define the current ATC Configuration Optimizer window

b) The load period defined by the set of selected intervals in the current ATC Configuration Optimizer window must be contained within the global period from the cost criteria file header.

(8) The service *MERGE_OPTICON_CRITERIA* is available from ETFMS to merge the current optimizer criteria definition with the criteria extracted from the file to be loaded.

(9) Once the merge of the current and newly loaded criteria is completed, the service *CALCULATE_OPTICON* shall be used to obtain a new matrix in line with the modified criteria.

32.1.5.5 Applying ATC Configuration Choice

(1) Once satisfied with the complete or part of the displayed optimized list of ATC Configurations for the queried ATC Unit Airspace, period and intervals, the user may request to apply the desired period of the optimized list.

a) By dragging the left mouse button on the leftmost column (time intervals column), it is possible to select one or multiple adjacent intervals.

b) A click on the *Apply ATC Configuration Choice* button located above the Time column header will display the **Apply ATC Configuration Choice** dialog. It is also accessible via *Apply ATC Configuration Choice* option in the popup menu over the selected intervals.

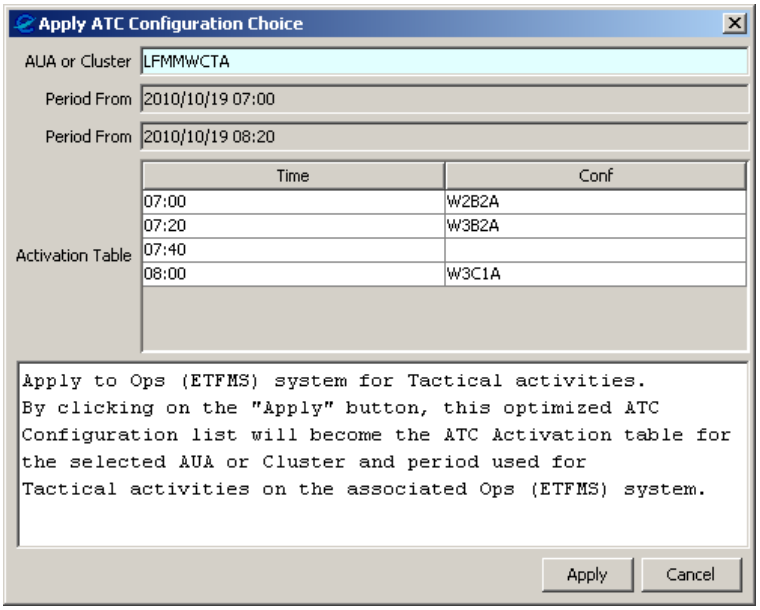


Figure Chmi-106: Apply ATC Configuration Choice dialog

(2) The **Apply ATC Configuration Choice** dialog displays following information related to the ATC Configuration choice:

a) The **AUA or Cluster** identifier related to these configurations

b) The start date and time of the period that will be applied (**Period From**)

c) The end date and time of the period that will be applied (**Period Until**)

d) The list of **Time** intervals and, when changed from previous interval, the corresponding ATC Configuration identifier (**Conf**). These might be single or multi-cluster configuration identifiers.

(3) On the Ops (ETFMS) system, the following statement warns the user of important subsequent impact linked to the confirmation of this dialog: **"Apply to Ops (ETFMS) system for Tactical activities. By clicking on the "Apply" button, this optimized ATC Configuration list will become the ATC Activation table for the selected ATC Unit Airspace (or Cluster) and period used for Tactical activities on the associated Ops (ETFMS) system."**

(4) On the Predict system, the following statement warns the user of important subsequent impact linked to the confirmation of this dialog: **"Apply to PREDICT system for Pre-Tactical activities. By clicking on the "Apply" button, this optimized ATC Configuration list will become the ATC Activation table for the selected ATC Unit Airspace (or Cluster) and period used for Pre-Tactical activities on the associated PREDICT system."**

(5) Clicking on the **Apply** button will update the related operational ATC Configuration activation table.

(6) The service *APPLY_OPTICON_CHOICE* is available from ETFMS:

a) To apply the optimized ATC Configuration Choice for the selected period

b) And log for possible later reference, the full Configuration Criteria used to identify the Optimized Configuration list

(7) Clicking on the **Cancel** button will close the **Apply ATC Configuration Choice** dialog and come back to the ATC Configuration Optimizer display without any update on the server.

(8) The access rights provided to modify the sector configurations in Ops (ETFMS) is granted on:

a) on the day of operations (D Day) and

b) on the pre-tactical day (D -1 day) if the plan has been transferred from the Predict system.

(9) The access rights provided to modify the sector configurations in PREDICT is granted until a certain period before the day of operations (D Day).

(10) After the "cut-off" date is passed, an error message will be returned and the modified ATC Configuration Choice will not be applied.

(11) This "cut-off" date is implemented as a parameter in the server. The initial value is 36 hours.

32.1.6 Contextual actions

(1) When a single Configuration identifier is selected in the matrix, the right mouse button pops up a **contextual menu** with the following options (where "*Configuration Name*" is replaced by the currently selected Configuration identifier):

a) **Show All *Configuration Name***: when the selected *Configuration Name* is NOT yet highlighted, this action will highlight the same configuration name in the whole ATC Configuration Optimizer matrix. This feature is very handy to quickly locate the rank of a given configuration in successive intervals.

b) **Monitor *Configuration Name***: to display the ATC Airspace Monitor for the selected ATC Unit Airspace, Configuration and period. In line with the capacities currently provided by the ATC Airspace Monitor, the **Monitor *Configuration Name*** popup option is not proposed when working on a multicluster AUA.

c) **Cancel Show All *Configuration Name***: when a *Configuration Name* is currently highlighted, this action will undo the highlighting activated by means of Show all *Configuration Name*.

(2) By dragging the left mouse button on the leftmost column (time intervals column) of the matrix, it is possible to select one or multiple adjacent intervals.

(3) When one or more time intervals are selected in the matrix, the right mouse button pops up a contextual menu.

(4) This popup menu lists the following possible actions:

a) **Edit Order Criteria** to display the Configuration Order Criteria dialog for the selected period.

b) **Save Order Criteria** to save the currently defined configuration order criteria for the selected period.

c) **Load Order Criteria** to load for the selected period, configuration order criteria saved during an earlier session.

d) **Apply ATC Configuration Choice** to apply the selected period of the optimized list.

e) **Monitor *Configuration Name*** (where "*Configuration Name*" is replaced by the currently selected Configuration identifier) to query the ATC Airspace Monitor for the selected AUA (without clusters) or Cluster, the best Configuration and the highlighted period. If several configuration identifiers appear in the best configuration column (first **Conf** column) for the highlighted period, the first of the list is taken. In line with the capacities currently provided by the ATC Airspace Monitor, the **Monitor *Configuration Name*** popup option is not proposed when working on a multicluster AUA.

33 Aerodrome Details

33.1 Function presentation

(1) The **Aerodrome Details** display gives environment details for one Aerodrome associated to a selected date.

(2) Data retrieved by the **Aerodrome Details** query are extracted from the ETFMS Environment database and Tactical updates.

33.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Environment / Aerodrome Details**.

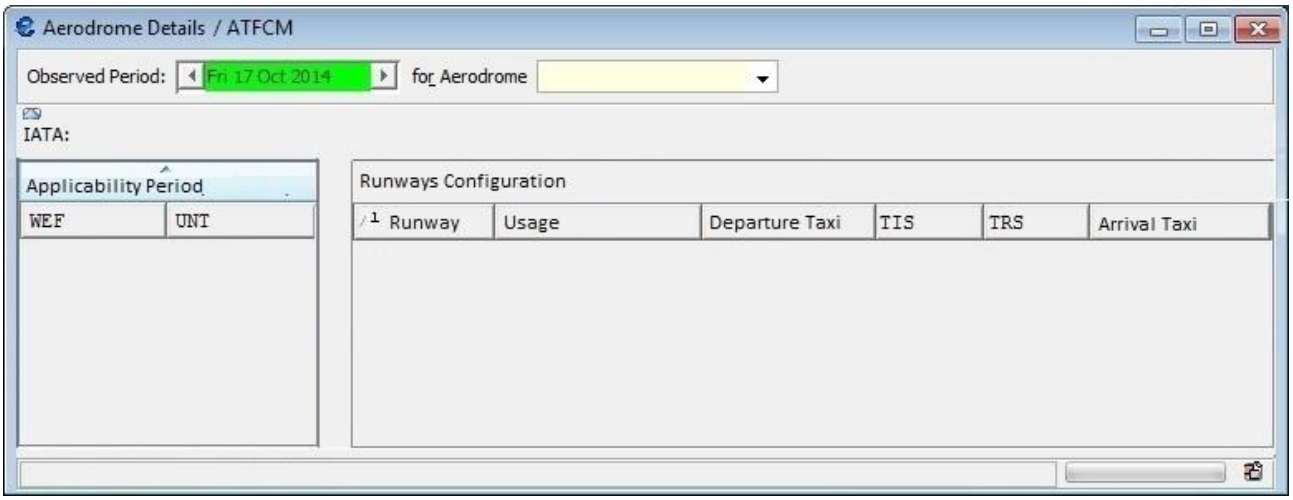


Figure Chmi-107: Query Aerodrome Details

(2) A new empty **Aerodrome Details** window will be opened with the following defaults:

a) **Observed period**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) The **for Aerodrome** drop-down box will propose the most recent identifiers you have queried before with success.

(3) On **Aerodrome Details** query preparation, the focus is set on the first input control with no default value: **Aerodrome** input control.

33.3 Preparing to query

(1) The **Observed period** may be adjusted to one of the proposed values.

(2) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(3) The **Aerodrome** control may be adjusted by selecting one of the ID proposed in the drop-down list or by keying in a new Aerodrome ICAO identifier in the dedicated input box.

(4) The **Send** button will send the query to the Network Operations system.

(5) **Aerodrome Details** window may be closed before issuing any query.

(6) The **Reset** action common functionality is implemented for this function.

(7) Following query data elements are considered for activating the Reset action:

a) **Observed period**

b) **Aerodrome**

33.4 Displaying the query results

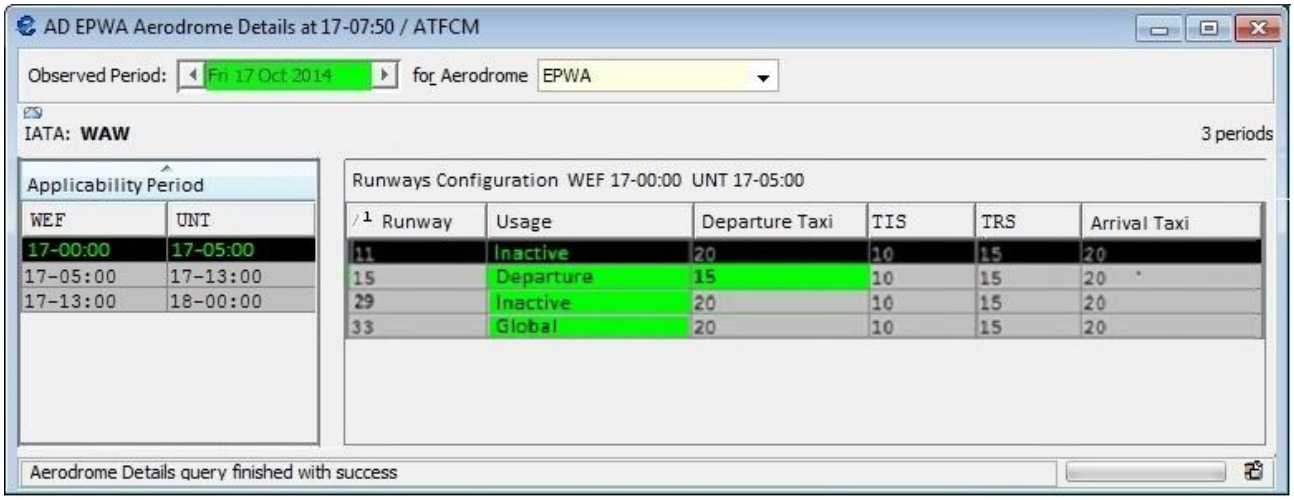


Figure Chmi-108: Aerodrome Details

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The header box of the window contains the different **query data elements** used to retrieve the displayed configuration:

a) The **Observed period.**

b) The **Aerodrome** identifier.

(3) Below the query header box, the equivalent **IATA** codification for the selected aerodrome.

(4) At the top of the list, a counter indicates the **number of periods** retrieved.

(5) On query results display, the focus is set on the **Applicability Period** list after selecting the first period.

(6) Detailed applicable runway usage for the selected date and aerodrome are displayed in two lists using a master-slave mechanism as follow:

a) The master listbox shows the list of defined **Applicability Periods**

b) The slave listbox shows all (i.e. active and non-active runways) **Runways Configuration** details for the currently selected applicability period.

(7) Each line in the **Applicability Period** listbox contains:

a) The start time of the period in *dd-hh:mm* format (**WEF**).

b) The end time of the period in *dd-hh:mm* format (**UNT**).

(8) By default, the listbox is sorted by **WEF** (earliest at the top of the list).

(9) By default, the first period in the list that has not started is selected (i.e. whose **WEF** is after the current UTC).

(10) Each line in the **Runways Configuration** listbox contains:

a) The **Runway** identifier (i.e. runway direction possibly appended with Left / Right / Center indicator).

Column header tooltip: "Runway Direction"

b) The associated **Usage** type

Column header tooltip: "Runway Usage (Inactive, Departure, Arrival, Global)"

c) The associated **Departure Taxi** time.

Column header tooltip: "Departure Taxi Time (MIN)"

d) The associated Time to Insert in Sequence (**TIS**).

Column header tooltip: "Time to Insert in Sequence"

e) The associated Time to Remove from Sequence (**TRS**).

Column header tooltip: "Time to Remove from Sequence"

f) The associated **Arrival Taxi** time.

Column header tooltip: "Arrival Taxi Time (MIN)"

(11) A header line indicates the period for which details are currently listed:

a) **Runways Configuration WEF** dd-hh:mm **UNT** dd-hh:mm.

(12) By default, the listbox is sorted by **Runway** identifier (lower value at the top of the list).

(13) The possible values for the runway **Usage** type are:

a) **Inactive** when the runway is not used by default (except if DPI explicitly states that it is used)

b) **Arrival** when the runway is used for arrival only

c) **Departure** when the runway is used for departure only

d) **Global** when the runway is used both for departure and arrival

(14) The data source for the **Applicability Period** and related **Runways Configuration** details can be either from the ETFMS Environment database or from a tactical update (i.e. merged view).

(15) An **Applicability Period** line background is coloured in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour** if any related **Runways Configuration** details have been tactically updated.

(16) A **Runways Configuration** cell background is coloured in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour** if the contained value has been tactically updated (i.e. if this value was explicitly tactically updated).

(17) Access to update the runway configuration data functionality via the **Update** button is restricted by security profile.

33.5 User interactions

33.5.1 General

(1) To select a period, click on any field of the period. The period is shown selected and the previously selected period, if any, is shown deselected.

(2) The **Runways Configuration** detail is displayed for the newly selected period.

(3) Click the **Update** button to open the editing area of the **Aerodrome Details** window (see below *"Editing Area of the Aerodrome Details window*).

(4) The **Update** button is disabled while the editing area is visible.

(5) The **Applicability Period** *list* and the **Runways Configuration** *list* of the display part remains enabled to allow selecting another period for consultation of current configuration during the input of new tactical updates.

(6) When a period is selected, the right button of the mouse gives access to a sub menu over both lists (**Applicability Period** and **Runways Configuration**).

(7) The **Normal / Proposal** two states button may be adjusted before selecting an option of the popup menu described in the "Contextual actions" section.

(8) The selected **Normal / Proposal** value will be transferred to the invoked function.

(9) This window allows to modify a previous query selection and to issue a new request.

(10) On query modification, the **Reset action** is enabled.

(11) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(12) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(13) To print the **Aerodrome Details**, first retrieve and sort the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(14) The printout will display in sequence:

a) Query details followed by elementary reply data (i.e. IATA)

b) Then followed sequentially by as many tables as there are applicability periods:

i) Each table contains the **Runways Configuration** for one period as on display

ii) And is preceded by its header line.

(15) To modify the **Sort sequence** of the periods, click the column headers corresponding to your desired sort criteria.

(16) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(17) Increasing / reducing the window height and width will impact the size of both list controls.

33.5.2 Editing Area of the Aerodrome Details window

33.5.2.1 Displaying the Window

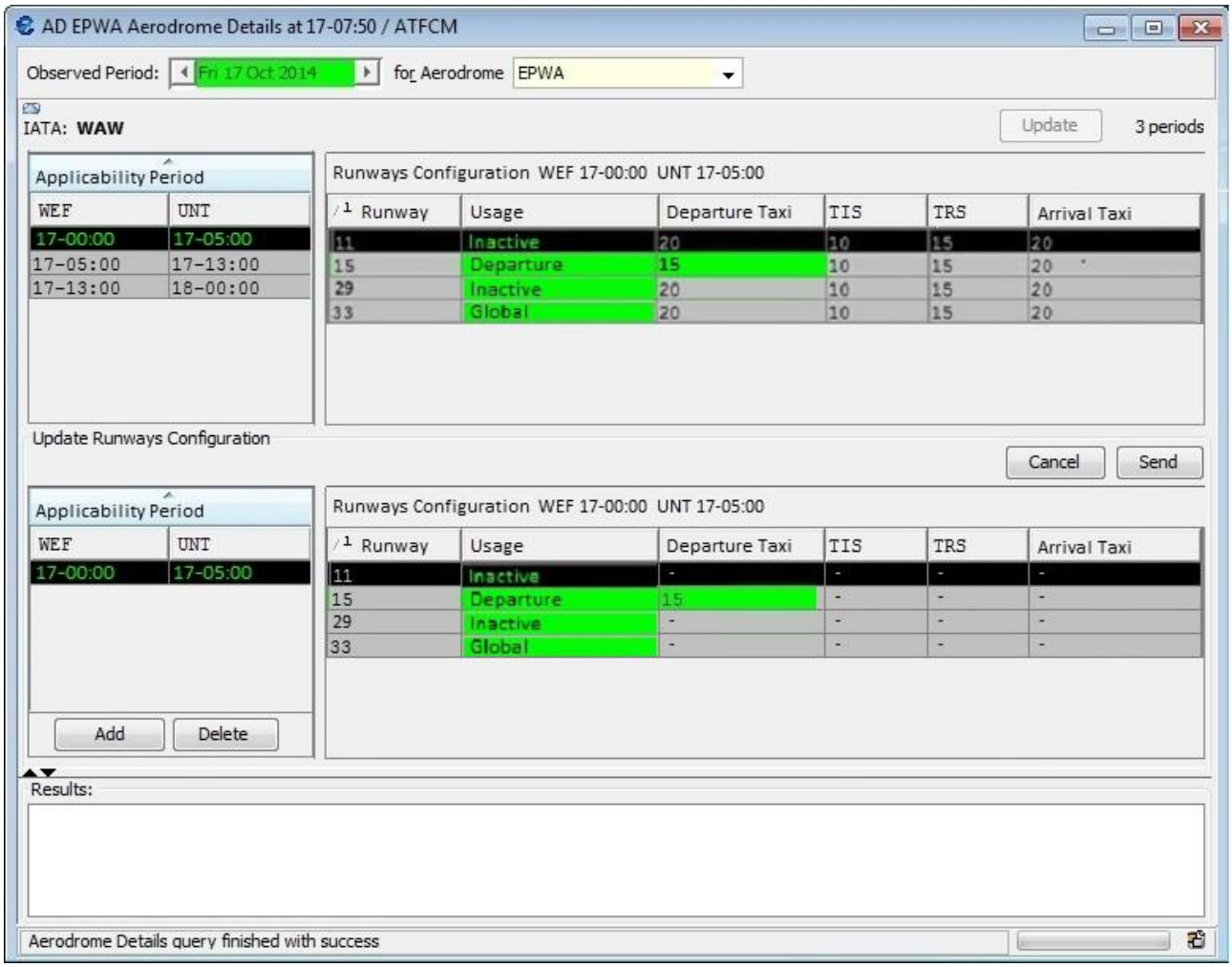


Figure Chmi-98: Editing Area of the Aerodrome Details window

(1) The editing area of the **Aerodrome Details** window will be opened with the following defaults:

a) The **Applicability Period** update listbox in the editing area of the window:

i) Only lists the periods if any related **Runways Configuration** details have been updated tactically (i.e. periods shown in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour**).

ii) If no tactical updates have been performed for the **Observed Period** at the aerodrome, the listbox will be empty.

iii) The listbox is sorted by **WEF** (earliest at the top of the list).

iv) By default, the first period in the list that has not started is selected (i.e. whose **WEF** is after the current UTC).

b) The **Runways Configuration** update listbox in the editing area of the window:

i) The list is composed of the same columns as in the display view (i.e. merged view).

- Note that the hyphen symbol "-" possibly displayed based on the current tactical updates, is also added to the possible *runway* ***Usage*** *type* available to the user to further update the configuration.
- The list is presented empty when no period is selected.

ii) The list always contains all **Runway** identifiers of the aerodrome as defined in ENV.

iii) All fields except the **Runway** identifier may contain either the tactically updated value or a hyphen symbol "-" (i.e. indicates to reuse the original configuration values as defined in the Environment system).

iv) The cell background is coloured in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour** if the contained value has been tactically updated (i.e. if this value was explicitly tactically updated).

v) A header line indicates the period for which details are currently listed: **Runways Configuration WEF** *dd-hh:mm* **UNT** *dd-hh:mm*.

vi) By default, the listbox is sorted by **Runway** identifier (lower value at the top of the list).

c) The **Cancel** button is enabled.

d) The **Send** button is disabled.

(2) On the editing area of the **Aerodrome Details** window, the Runways Configuration update list only displays values that correspond to explicit tactical updates. Unchanged runways configuration values are displayed as "-".

(3) On the editing area of the **Aerodrome Details** window:

a) An **Applicability Period** line background is coloured in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour** if any related **Runways Configuration** details have been tactically updated.

b) A **Runways Configuration** cell background is coloured in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour** if it has been tactically updated.

(4) If an applicability period has expired (i.e. the **UNT** of the period is before the current UTC time), all the fields of the period and of the corresponding **Runways Configuration** are shown as read-only.

(5) If an applicability period has started but not yet expired, (i.e. current UTC time is between the **WEF** and **UNT**), except for the **UNT** field, the **WEF** and all the fields of the corresponding **Runways Configuration** are shown as read-only (i.e. **UNT** remains modifiable).

(6) If an applicability period has not yet started (i.e. the **WEF** is after the current UTC time), all the fields of the period and corresponding **Runways Configuration** are updatable except the **Runway** identifier which is displayed as read-only.

33.5.2.2 User interactions

33.5.2.2.1 To Select a Period

(1) To select an Applicability Period, click on any field of the period. The period is shown selected and the previously selected period, if any, is shown deselected.

(2) The Runways Configuration detail is displayed for the newly selected period, possibly including latest input which is not sent yet.

33.5.2.2.2 To Add a Period

(1) To add a new period, click the **Add** Applicability Period button.

(2) A new period is inserted in the list - in ascending sequence based on the newly proposed WEF (see below for default value).

(3) The new period line background is coloured in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour**.

(4) The new period is automatically set to be the selected period.

(5) The **Send** button is enabled.

(6) The default values when adding a new Applicability Period are:

a) **WEF**:

i) UNT value of the previously selected period, if any and if this time is in the future (i.e. later than current UTC) but still in the current **Observed Period**,

ii) Otherwise the current UTC time plus 10 minutes, if in the current **Observed Period**.

iii) Otherwise the start of the current **Observed Period** (i.e. 00:00).

b) **UNT**: WEF of the new period plus 30 minutes limited to the end of the **Observed Period**.

c) The **Runways Configuration** *update list* is initialised as follow:

i) One line per **Runway** identifier defined for the aerodrome.

ii) The **Usage** for the different runways is initialised to the explicit original configuration value as defined in the Environment system at

the start of this newly defined Applicability Period (i.e. NOT the hyphen symbol "-".

iii) All other configuration fields are initialised to the default hyphen symbol "-" to reuse the original configuration values as defined in the Environment system.

33.5.2.2.3 To Delete a Period

(1) Select the period you wish to delete, if not already currently selected.

(2) The **Delete Period** button is only enabled when either:

a) the **WEF** of the currently selected period is later than the current UTC time or

b) the selected period has been added in the current window session (i.e. not already submitted to the backend system).

(3) Click the **Delete Period** button if enabled.

(4) The selected period is deleted from the list.

(5) The **Send** button is enabled.

33.5.2.2.4 To Modify a Period

(1) Select the period you wish to modify, if not already currently selected.

(2) Directly edit the accessible fields as required.

(3) A **Runways Configuration** cell background is coloured in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour** if it set to a value other than that defined in the Environment database (i.e. other than "-").

(4) Any field blanked out (i.e. null or blank) is forced back to the default "-" value.

(5) The runway **Usage** field, when editable, shall allow the user to select the runway usage from the list of possible usage types, including the special hyphen symbol "-" value.

(6) To ensure coherency in case of tactical update of some runway **Usage**, the system will impose that within an applicability period:

a) Either the **Usage** for ALL runways is set to "-"

b) Or the **Usage** for ALL runways is set to one of the possible explicit ***Usage type*** (i.e. other than "-").

(7) If any of the editable fields have been modified, the **Send** button is enabled.

33.5.2.2.5 To Send the Updates

(1) Click the **Send** button, if enabled.

(2) Several periods and related runways configuration can be added, modified or deleted before updates are sent.

(3) The results area is cleared and both the **Cancel** and the **Send** buttons are disabled.

(4) The following validation checks are performed and reported in the results area:

a) If either **WEF** and/or **UNT** time values were changed or newly entered:

i) Any new time value must be later than the current UTC time and within the **Observed Period**.

ii) **UNT** time must always be later than **WEF** time.

b) Any **Departure** or **Arrival Taxi** time that was modified must be in the range 0 to 90 minutes.

c) The Time to Insert in Sequence (**TIS**) that was modified must be in the range 0 to 30 minutes.

d) The Time to Remove from Sequence (**TRS**) that was modified must be in the range 0 to 20 minutes.

e) **Applicability Periods** may not overlap.

f) No period allowed where **Usage** is tactically updated for some runways but NOT ALL (i.e. in one period, all **Usage** are updated or they are all equal to "-").

g) No period allowed where all updatable fields are equal to "-" (i.e. if all defaults apply then this period should be deleted).

(5) Upon reporting of any validation errors, the **Send** and the **Cancel** button are re-enabled.

(6) The user shall be able to correct any validation errors.

(7) On any subsequent **Send** or **Cancel**, the results area is cleared.

(8) If there are no validation errors:

a) the non-editing part of the window (i.e. the initial query/reply part) is automatically refreshed with the latest system known situation,

b) the editing part of the window becomes read-only,

c) the **Send** button is renamed to **Close** and is enabled,

d) the results area displays a message indicating the status of the last submission:

i) if no errors are returned, the message **Update successful** is displayed,

ii) if the update was successfully merged with parallel updates from another user, the message **Update successfully merged with updates from other users** is displayed,

iii) if the update was performed on the Ops (ETFMS) system for the pre-tactical day and the plan had not yet been transferred from Predict and applied, the message **Updates for tomorrow could not be applied - the plan from Predict has not been transferred to Ops (ETFMS) and applied** is displayed

iv) if the update was performed on the Predict system for the pre-tactical day after the cut-off time a duration parameter (TBD) related to a specific UTC time, the message **Updates could not be applied due to late submission plan being finalized in preparation for transfer to the Ops (ETFMS) system** is displayed.

v) if the system is not able to merge the submitted update with an update made by another user, the message **The updates conflict with updates received from other users. The results of the query is refreshed with the latest information. Please review your updates and re-submit if necessary.** is displayed.

33.5.2.2.6 To Cancel the Updates

(1) Click the **Cancel** button, when available.

a) All modifications, if any, are discarded

b) The editing area of the **Aerodrome Details** window is closed and

c) The **Update** button is re-enabled.

33.5.2.2.7 To Close the Editing Area

a) Click the **Close** button, when available.

b) The editing area of the **Aerodrome Details** window is closed.

c) The **Update** button is re-enabled.

33.6 Contextual actions

(1) When a line is selected in the **Applicability Period** or in the **Runways Configuration** list, the right mouse button on either lists pops up a **contextual menu**.

(2) This popup menu lists the possible actions for the queried Aerodrome with the selected applicability period and may contain the following options:

a) **Traffic Capacity**: displays the capacity graph for the selected Aerodrome.

b) **Traffic Counts**: displays the count graph for the Aerodrome for the selected period.

c) **Departure Delays**: displays the delays for the Departure Aerodrome for the selected period.

d) **Flight List** to query Traffic Load list of flights for the selected Aerodrome and period.

e) **Departure List** to query Traffic Demand list of flights for the selected Departure Aerodrome and period.

f) **Arrival List** to query Traffic Load list of flights for the selected Arrival Aerodrome and period.

(3) For the counts, the delays and the flight Lists, the transferred period may be affected by user preferences related to **Default query period identification**.

34 Route Catalogue

34.1 Function presentation

(1) The **Route Catalogue** function list routes from the Route Catalogue.

(2) The **Route Catalogue** function lists all **Routes** that match the provided city pair for the selected date.

(3) Data retrieved by the **Route Catalogue** query are extracted from the ETFMS Environment database.

34.2 Opening the window

(1) From the Application / ATFCM menu, select the option **Environment / Route Catalogue**.

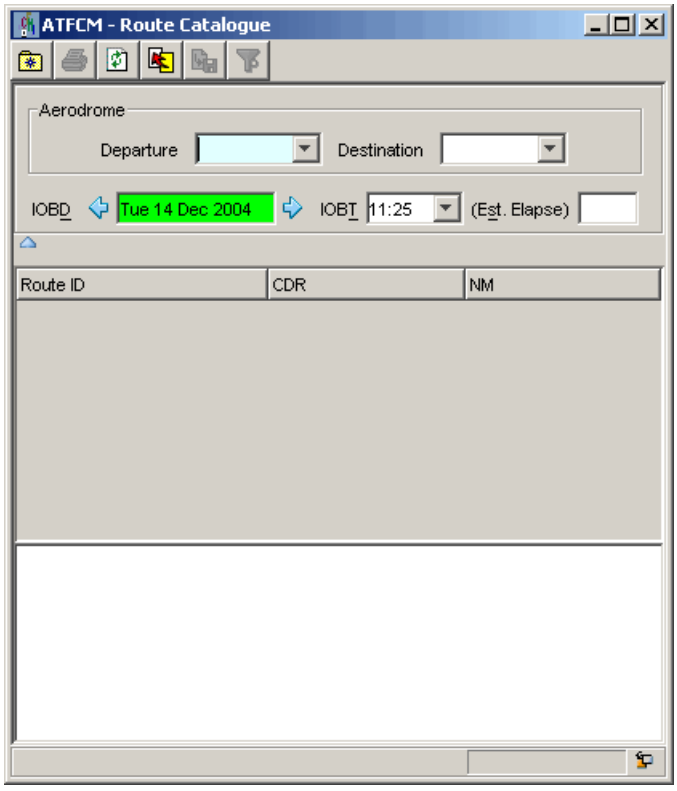


Figure Chmi-109: Query Route Catalogue

(2) A new empty **Route Catalogue** window will be opened with the following defaults:

a) The **Departure** and **Destination** Aerodrome drop-down box will propose the most recent identifiers you have queried before with success.

b) **IOBD**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

c) **IOBT**: current UTC time plus one hour.

d) Estimated Elapsed Time: No default value is proposed (**Est. Elapse**).

(3) On query preparation, the focus is set on the first input control with no default value: **Departure** Aerodrome input control.

34.3 Preparing to query

(1) The **Departure** and **Destination** Aerodrome controls may be adjusted by selecting one of the IDs proposed in the drop-down list or by keying in a new Aerodrome ICAO identifier in the dedicated input box.

(2) The **Departure** and **Destination** Aerodrome identifiers must be different from each other.

(3) The **IOBD** may be adjusted to one of the proposed values.

(4) Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(5) The proposed default **IOBT** may be forced to a specific time value. Any valid time value from 00:00 to 23:59 included may be entered. This input must respect the HH:MM format. Drop-down list may be used to select pre-formatted time on a 20 minutes step base.

(6) The **Est. Elapse** may be forced to any valid number of minutes with a maximum of four digits (9999).

(7) The Estimated Elapsed Time is used to identify the worst CDR category applying to the route.

(8) This is the only optional query criteria on this display. If this criteria is not provided, the Network Operations system will use a default parameter value for it.

(9) The **Send** button will send the query to the Network Operations system.

(10) The **Route Catalogue** window may be closed before issuing any query.

(11) The **Reset** action common functionality is implemented for this function.

(12) All query data elements are considered for activating the Reset action.

34.4 Displaying the query results

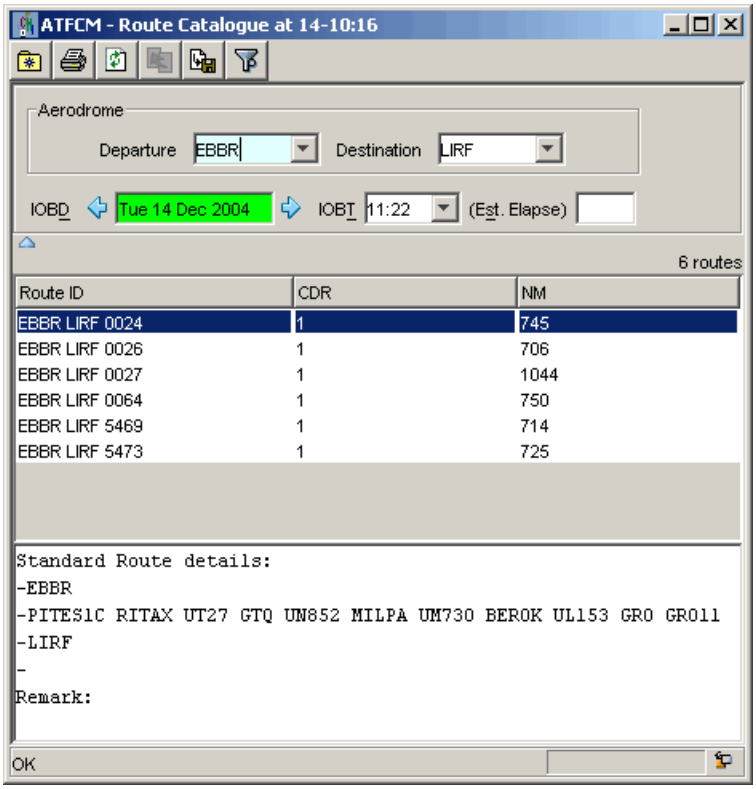


Figure Chmi-110: Route Catalogue Display

(1) The title contains the **time stamp** when the information was retrieved from the Network Operations system.

(2) The header box of the window contains the different **query data elements** used to retrieve the list of standard routes:

a) The Departure and Destination Aerodrome ICAO identifiers.

b) The Initial Estimated Off Block Date and Time used for the route selection in the catalogue.

c) The optional Estimated Elapse time used to identify the worst CDR category.

(3) At the top of the route list, a counter indicates the **number of routes** retrieved from the catalogue.

(4) This counter may be affected by **filtering** (fff / rrr Routes where fff is the filtered number and rrr is the retrieved number).

(5) On query results display, the focus is set on the list after selecting the first route.

(6) For each route, the detail listbox contains:

a) The Identifier of the route (**Route ID**) (see below).

b) The worst **CDR** applied to this route based on the IOBD/T and the Estimated Elapse time specified or the default used by ETFMS (see below).

c) The route length in Nautical Miles (**NM**).

(7) The Identifier of the route (**Route ID**) is composed of:

a) The route origin (Aerodrome)

b) The route destination (Aerodrome)

c) The route sequential number.

(8) Possible **CDR** types are:

a) **ATS** when the ATS route is always plannable (corresponds to CDR 0).

b) **1** for permanently plannable conditional route. Closure is published in NOTAM and EAUP/EUUP.

c) **2** for non-permanently plannable conditional route. Opening can be published in EAUP/EUUP.

d) **N** for not available - not-plannable.

(9) Routes are initially sorted on the Route ID.

(10) Selecting one of the proposed routes will display in the box at the bottom of the window:

a) **Route details** with the description as a point sequence.

b) The **Remark** associated to the Standard Route.

34.5 User interactions

(1) Click on a line to know more information about one route. This will select the route and more detailed information will be shown in the details pane of the Route Catalogue window.

(2) This window allows to modify a previous query selection and to issue a new request.

(3) On query modification, the **Reset action** is enabled.

(4) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(5) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(6) To print the list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(7) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(8) To modify the **Sort sequence** of the routes, click on column headers corresponding to your desired sort criteria.

(9) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(10) Increasing / reducing the window height will impact the size of the list control.

(11) Increasing / reducing the window width will impact the size of the list and the detail box controls.

34.6 Contextual actions

(1) When a line is selected in the list of Route Catalogue display, the right mouse button pops up a contextual menu.

(2) This popup menu lists the possible actions for the selected Route:

a) **Plot Route**

b) **Plot All Routes**

(3) Selecting **Plot Route** will display the selected route on the map.

(4) Selecting **Plot All Routes** will display all the listed routes on the map.

(5) Following user preference may be adjusted to associate a colour with the selected route(s) when displayed on the map: **CHMI / ATFCM application / Flights / General / Route Catalog colour**. The default is white.

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35 Traffic Volume Activation

35.1 Function Presentation

(1) This **Traffic Volume Activation** function lists the activation status of a traffic volume on the selected date.

(2) Data retrieved by the **Traffic Volume Activation** query are extracted from the ETFMS Environment database and Tactical updates.

35.2 Opening the Window

(1) From the Application / ATFCM menu, select the option **Environment / Traffic Volume Activation**.

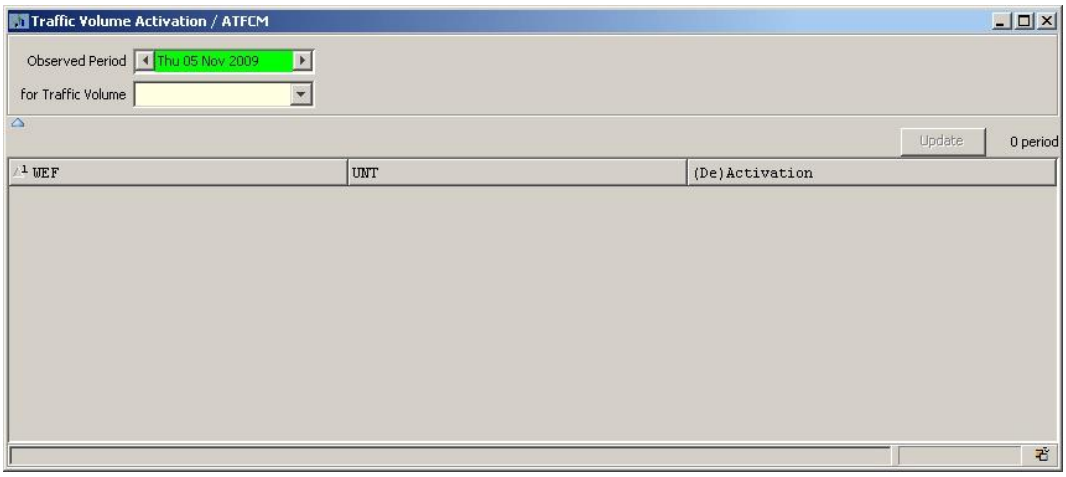


Figure Chmi-111: Query Traffic Volume Activation

(2) A new empty **Traffic Capacity** window will be opened with the following defaults:

a) **Observed period**: current or last queried traffic date is proposed (relative to current UTC date) depending on selected user preference (CHMI / ATFCM application / General / Dates and times / Default date proposed for query).

b) **for Traffic Volume**: the drop-down box will propose the most recent query data elements you have queried before with success.

(3) On initial display, the focus is set on the first mandatory query control with no default value: the **for Traffic Volume** input control.

35.3 Preparing to query

(1) The date of the **Observed Period** may be adjusted to one of the proposed values. Common ATFCM **Tactical / Pre-Tactical dates** behaviour is implemented.

(2) The **Send** button will send the query to the Network Operations system.

(3) This window may be closed before any query is issued.

(4) The **Reset** action common functionality is implemented for this function. All query criteria are considered for activating the Reset action.

35.4 Displaying the query results

(1) The window title contains the following information:

a) The selected Traffic Volume

b) The time stamp when the information was retrieved from the Network Operations system.

(2) The header box contains the selected query criteria:

a) The **Observed Period**

b) The selected **Traffic Volume** Id.

(3) At the top of the Traffic Volume Activation List a counter indicates the number of activations/deactivations retrieved.

(4) This counter may be affected by filtering (fff / rrr Activations where fff is the filtered number and rrr is the retrieved number).

(5) On query results display, the focus is set on the list of activations after selecting the first activation in the list.

(6) For each traffic volume activation/de-activation, the list display contains the following columns:

a) **WEF**: the start time of the activation/de-activation.

b) **UNT**: the end time of the activation/de-activation.

c) **(De)Activation**: whether this period is an Activation or a DeActivation.

(7) The data source for a (de)activation period can be either from the ETFMS Environment database or from a tactical update performed by an authorised user.-Tactical updates of **(De)Activation** are shown in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour**.

(8) The initial display sort sequence for the Traffic Volume Activation List is increasing order of WEF (i.e. oldest WEF at the top of the list).

(9) The Traffic Volume Activation List allows only single selection.

(10) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

(11) Access to update the Traffic Volume Activation status functionality via an **Update** button is restricted by security profile.

35.5 User interactions

35.5.1 General

(1) This window allows to modify a previous query selection and to issue a new request.

(2) On query modification, the **Reset action** is enabled.

(3) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(4) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will issue exactly the same query to obtain an up to date list.

(5) The Invalidate function may be activated for this window. Just check or uncheck the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality**.

(6) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(7) To modify the **Sort sequence** of the flights, click on column headers corresponding to your desired sort criteria.

(8) To print the Traffic Volume Activation List, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(9) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(10) Increasing / reducing the window height or width will impact the size of the main Traffic Volume Activation List control.

(11) Click the **Update** button to open the editing area of the **Traffic Volume Activation** window (see below).

(12) The **Update** button is disabled.

35.5.2 Editing Area of the Traffic Volume Activation Window

35.5.2.1 Displaying the Window

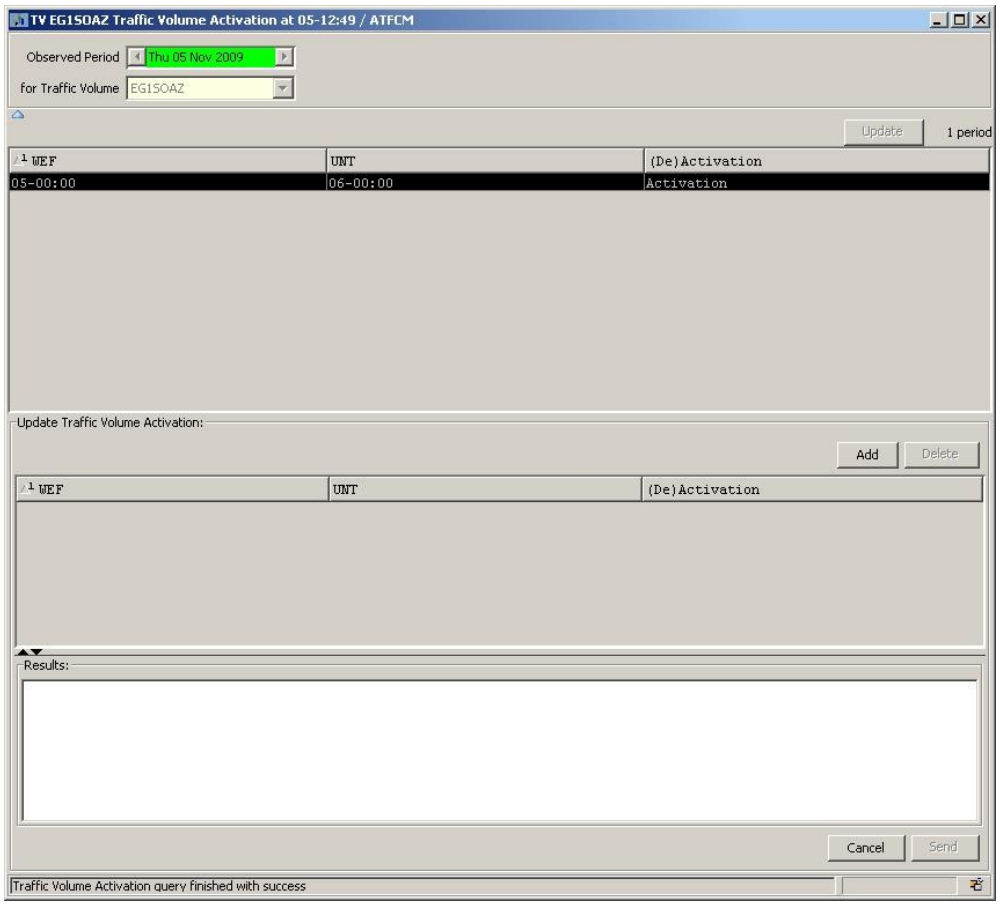


Figure Chmi-112: Update Traffic Volume Activation

(1) The editing area of the **Traffic Volume Activation** window will be opened with the following defaults:

a) The main listbox of the editing area of the window only lists the periods whose (de)activation has been updated tactically (i.e. periods containing data shown in user adjustable preference **CHMI / ATFCM application / General / Updates / Tactical Update Colour**). If no tactical updates have been performed for the **Observed Period**, the listbox will be empty.

b) The listbox is sorted by **WEF** (earliest at the top of the list).

c) By default, the first period in the list that has not started (i.e. whose **WEF** is after the current UTC) is selected.

d) The **Cancel** button is enabled.

e) The **Send** button is disabled.

(2) For each traffic volume activation/de-activation, the list display contains the following columns:

a) **WEF**: the start time of the activation/de-activation.

b) **UNT**: the end time of the activation/de-activation.

c) **(De)Activation**: whether this period is an Activation or a DeActivation.

(3) If a period has expired (i.e. the **UNT** of the period is before the current UTC time), all the fields of the period are shown as read-only.

(4) If a period has started but not yet expired, (i.e. current UTC time is between the **WEF** and **UNT**), except for the **UNT** field, all the other fields of the period are shown as read-only.

(5) If a period has not yet started (i.e. the **WEF** is after the current UTC time), all the fields of the period are editable.

35.5.2.2 User Interactions

35.5.2.2.1 To Select a Period

(1) To select a period, click on any field of the period. The period is shown selected and the previously selected period, if any, is shown deselected.

35.5.2.2.2 To Add a Period

(1) To add a new period, click the **Add Period** button.

(2) A new period is inserted in the list – below the previously selected period, if present.

(3) The new period is automatically set to be the selected period.

(4) The **Send** button is enabled.

(5) The default values for the new period are:

a) **WEF**: To value of:

i) the previously selected period, if present, otherwise

ii) the current UTC time plus 10 minutes.

b) **UNT**: From of the new period plus 3 hours limited to the end of the Observed Period.

c) **(De)Activation**: set to the value defined in the main **Traffic Volume Activation** window (i.e. merged view) at the start time of the new period.

35.5.2.2.3 To Delete a Period

(1) Select the period you wish to delete, if not already currently selected.

(2) The **Delete Period** button is only enabled when either:

a) the **WEF** of the currently selected period is later than the current UTC time or

b) the selected period has been added in the current window session (i.e. not already submitted into the system).

(3) Click the **Delete** button if enabled.

(4) The selected period is deleted from the list.

(5) The **Send** button is enabled.

35.5.2.2.4 To Modify a Period

(1) The **WEF** field, when editable, shall allow the user to change the start time of the period.

(2) The **UNT** field, when editable, shall allow the user to change the end time of the period.

(3) The **(De)Activation** field, when editable, shall allow the user to set it to either **Activation** or **DeActivation**.

(4) If any of the editable fields have been modified, the **Send** button is enabled.

35.5.2.2.5 To Send the Updates

(1) Click the **Send** button, if enabled.

(2) The results area is cleared and both the **Cancel** and the **Send** buttons are disabled.

(3) The following validation checks are performed and reported in the results area:

a) If update period was already submitted (i.e. from previous update) and is expired (**WEF** & **UNT** are before current UTC time), no validations error should be reported.

b) If update period was already submitted but only **WEF** is expired (**WEF** before current UTC time), **UNT** must be later than the current UTC time and within the **Observed Period**.

c) If a new update period is added or for an already submitted period that has not started, both **WEF** & **UNT** have to be later than the current UTC time and within the **Observed Period**. **UNT** has also to be later than the **WEF**.

d) There are no overlaps between the periods.

(4) Upon reporting of any validation errors, the **Cancel** button is re-enabled.

(5) The user shall be able to correct any validation errors.

(6) On any subsequent edit(s), the results area is cleared.

(7) If there are no validation errors:

a) The **Update** button is enabled,

b) the non-editing part of the window (i.e. the initial query/reply part) is automatically refreshed with the latest system known situation,

c) the editing part of the window becomes read-only,

d) the **Send** button is renamed to **Close** and is enabled,

e) the results area displays a message indicating the status of the last submission:

i) if no errors are returned, the message "**Update successful**" is displayed,

ii) if the update was successfully merged with parallel updates from another user, the message "**Update successfully merged with updates from other users**" is displayed,

iii) if the update was performed on the Ops (ETFMS) system for the pre-tactical day and the plan had not yet been transferred from Predict and applied, the message "**Updates for tomorrow could not be applied – the plan from Predict has not been transferred to Ops (ETFMS) and applied**" is displayed,

iv) if the update was performed on the Predict system for the pre-tactical day after the "cut-off" time – a duration parameter (TBD) related to a specific UTC time, the message "**Updates could not be applied due to late submission – plan being finalized in preparation for transfer to the Ops (ETFMS) system**" is displayed,

v) if the system is not able to merge the submitted update with an update made by another user, the message "**The updates conflict with updates received from other users. The results of the query is refreshed with the latest information. Please review your updates and re-submit if necessary.**" is displayed.

35.5.2.2.6 To Cancel the Updates

(1) Click the **Cancel** button, when available.

a) All modifications, if any, are discarded

b) The editing area of the **Traffic Volume Activation** window is closed and

c) The **Update** button is re-enabled.

35.5.2.2.7 To Close the Editing Area

(1) Click the **Close** button, when available.

(2) The editing area of the **Traffic Volume Activation** window is closed.

(3) The **Update** button is re-enabled.

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36 READY to Depart Message

36.1 Function presentation

(1) This window is used to send a READY to Depart message (**REA**) for a selected flight.

36.2 Opening the window

(1) This window may only be activated when a flight is selected in a list or from the flight data window.

(2) The REA message can only be sent for flights with a published slot in the future, relative to the ETFMS time stamp associated to the flight reply message (i.e. CTOT > ETFMS flight reply stamp). The ETFMS time stamp is the timeStamp of the ICD entity DbrFlightGeneral.

(3) Messages are not authorised for flights in following states: Proposal, ATC activated, PFD and terminated.

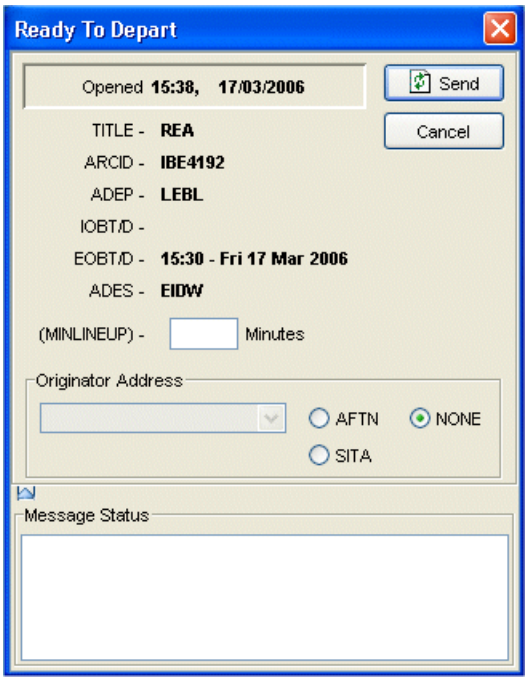


Figure Chmi-113: READY to Depart message

(4) When such a window is opened, no other function may be invoked.

(5) This display shows the READY to Depart Message to be sent. This message is sent when the flight is ready to depart before its EOBT.

(6) There is no reply to such a message.

(7) The **Opened** box at the top indicates the UTC stamp when this dialog was invoked. It is displayed there because the timing in sending messages is important.

(8) The different data elements to be included in the message are presented on display:

a) **TITLE - REA** naming the message.

b) **ARCID** for the callsign of the flight.

c) **ADEP** aerodrome of departure.

d) **IOBT/D** Initial Estimated Off Block Time and Date.

e) **EOBT/D** Last Estimated Off Block Time and Date.

f) **ADES** aerodrome of destination.

g) **MINLINEUP** Minimum time needed for the flight to get from its present holding position to airborne (if not provided, the taxi time is used). Accepted values are between 00 and the maximum identified by the system property **CHMI / ATFCM application / Flights / General / Maximum MINLINEUP**.

h) **AFTN** or **SITA** address for the reply.

(9) The field IOBT/D for the Initial OBT is only displayed if it is different from the last OBT known.

(10) The message is already prepared and only the MINLINEUP field can be supplied.

(11) If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preference (Preferences / CHMI / ATFCM application / General / Origin / Default Adress Id).

(12) To modify the proposed address, click on the **AFTN**, **SITA** or **None** radio button to possibly modify the type and then select an address in the drop-down control or input one manually. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(13) If no address is supplied on display, none will be included in the message.

(14) The system will process the message and transmit a Slot Revision Message, if a new slot is available, to all concerned including the address you have provided with the message.

(15) To send the message, click on **Send** button.

(16) To close the window without sending the message, click on **Cancel** button.

36.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) There will be no reply on the screen once the message has been transmitted. To acknowledge a good transmission, the transmit stamp is returned and displayed.

(3) The processing of the message being asynchronous, be aware this is only an acknowledgement message.

(4) If a problem occurs in the transmission, a descriptive error message is displayed in the bottom message box.

(5) All controls are disabled while processing the message.

(6) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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37 Request For direct Improvement Message

37.1 Function presentation

(1) This window is used to send a "Request For direct Improvement" message (**RFI**) for a selected flight.

37.2 Opening the window

(1) The window may only be activated when a flight is selected in a list or from the flight data window.

(2) The RFI message can be sent only for flights with an IFPL flight type which are not yet in RFI state nor in READY state.

(3) Messages are not authorised for flights in following states: Proposal, ATC activated, PFD and terminated.

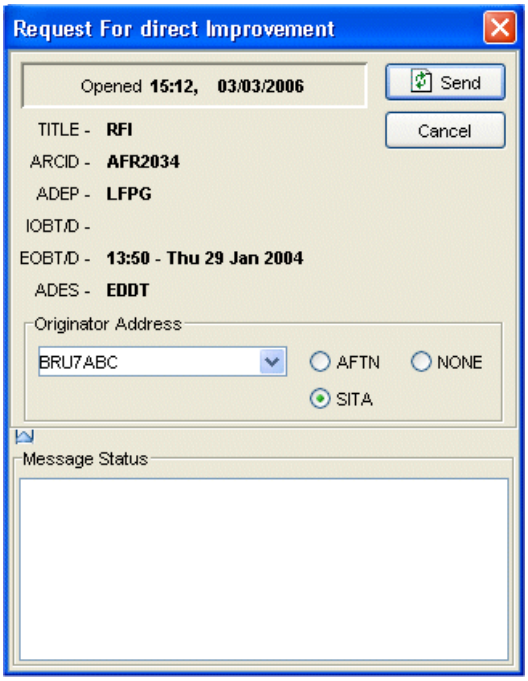


Figure Chmi-114: Request For direct Improvement message

(4) When such a window is opened, no other function may be invoked.

(5) This display shows the Request For direct Improvement Message to be sent. This message is sent when the flight can accept an SRM when an improvement is possible.

(6) There is no reply to such a message.

(7) The **Opened** box at the top indicates the UTC stamp when this dialog was invoked. It is displayed there because the timing in sending messages is important.

(8) The different data elements to be included in the message are presented on display:

a) **TITLE - RFI** naming the message.

b) **ARCID** for the callsign of the flight.

c) **ADEP** aerodrome of departure.

d) **IOBT/D** Initial Estimated Off Block Time and Date.

e) **EOBT/D** Last Estimated Off Block Time and Date.

f) **ADES** aerodrome of destination.

g) **AFTN** or **SITA** address for the reply.

(9) The field IOBT/D for the Initial OBT is only displayed if it is different from the last OBT known.

(10) If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preferences (Preferences / CHMI / ATFCM application / General / Origin).

(11) To modify the proposed address, click on the **AFTN**, **SITA** or **None** radio button to possibly modify the type and then select an address in the drop-down control or input one manually. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(12) If no address is supplied on display, none will be included in the message.

(13) To send the message, click on **Send** button.

(14) To close the window without sending the message, click on **Cancel** button.

37.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) There will be no reply on the screen once the message has been transmitted. To acknowledge a good transmission, the transmit stamp is returned and displayed.

(3) The processing of the message being asynchronous, be aware this is only an acknowledgement message.

(4) If a problem occurs in the transmission, a descriptive error message is displayed in the bottom message box.

(5) All controls are disabled while processing the message.

(6) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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38 SIP Wanted Message

38.1 Function presentation

(1) This window is used to send a "SIP Wanted Message" (**SWM**) for a selected flight.

38.2 Opening the window

(1) This window may only be activated when a flight is selected in a list or from the flight data window.

(2) The SWM message can be sent only for flights with an IFPL flight type which are not yet in SWM state nor in READY state.

(3) Messages are not authorised for flights in following states: Proposal, ATC activated, PFD and terminated.

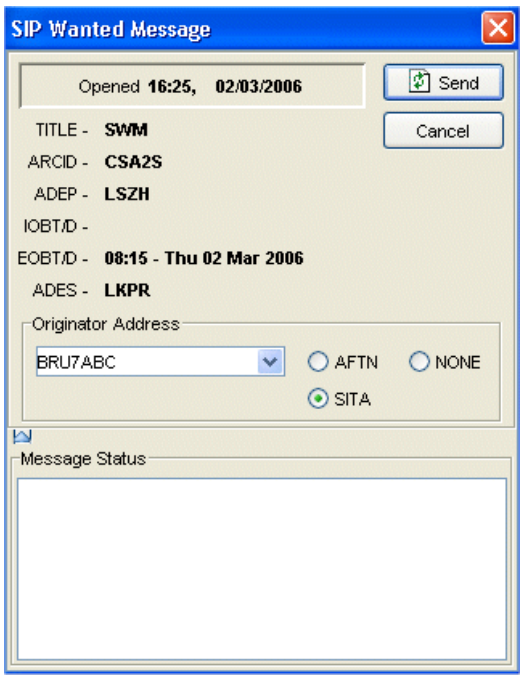


Figure Chmi-115: SIP Wanted Message

(4) When such a window is opened, no other function may be invoked.

(5) This display shows the SIP Wanted Message to be sent. This message is sent when the flight cannot accept an SRM as improvement.

(6) There is no reply to such a message.

(7) The **Opened** box at the top indicates the UTC stamp when this dialog was invoked. It is displayed there because the timing in sending messages is important.

(8) The different data elements to be included in the message are presented on display:

a) **TITLE - SWM** naming the message.

b) **ARCID** for the callsign of the flight.

c) **ADEP** aerodrome of departure.

d) **IOBT/D** Initial Estimated Off Block Time and Date.

e) **EOBT/D** Last Estimated Off Block Time and Date.

f) **ADES** aerodrome of destination.

g) **AFTN** or **SITA** address for the reply.

(9) The field IOBT/D for the Initial OBT is only displayed if it is different from the last OBT known.

(10) If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preferences (Preferences / CHMI / ATFCM application / General / Origin).

(11) To modify the proposed address, click on the **AFTN**, **SITA** or **None** radio button to possibly modify the type and then select an address in the drop-down control or input one manually. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(12) If no address is supplied on display, none will be included in the message.

(13) To send the message, click on **Send** button.

(14) To close the window without sending the message, click on **Cancel** button.

38.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) There will be no reply on the screen once the message has been transmitted. To acknowledge a good transmission, the transmit stamp is returned and displayed.

(3) The processing of the message being asynchronous, be aware this is only an acknowledgement message.

(4) If a problem occurs in the transmission, a descriptive error message is displayed in the bottom message box.

(5) All controls are disabled while processing the message.

(6) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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39 Slot Proposal Acceptance

39.1 Function presentation

(1) This window is used to send a CASA message to accept a slot proposal (**SPA**) for a selected flight.

39.2 Opening the window

(1) This window may only be activated when a flight is selected in a list or from the flight data window.

(2) The SPA message can be sent only for flights with a published slot and a valid proposed new CTOT (SIP).

(3) Messages are not authorised for flights in following states: Proposal, ATC activated, PFD and terminated.

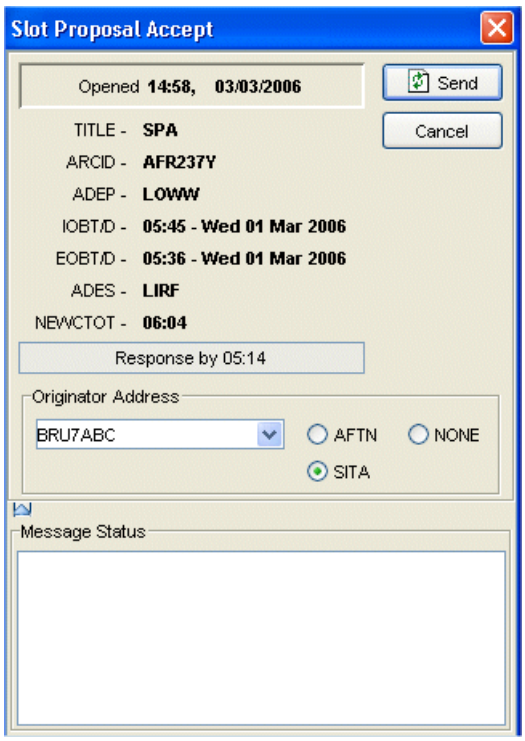


Figure Chmi-116: Slot Proposal Acceptance message

(4) When such a window is opened, no other function may be invoked.

(5) This display shows the Slot Proposal Acceptance message (SPA) to be sent. This message is sent as a positive reply to a Slot Improvement Proposal (SIP).

(6) There is no reply to such a message.

(7) The **Opened** box at the top indicates the UTC stamp when this dialog was invoked. It is displayed there because the timing in sending messages is important.

(8) The different data elements to be included in the message are presented on display:

a) **TITLE - SPA** naming the message.

b) **ARCID** for the callsign of the flight,

c) **ADEP** aerodrome of departure,

d) **IOBT/D** Initial Estimated Off Block Time and Date,

e) **EOBT/D** Last Estimated Off Block Time and Date,

f) **ADES** aerodrome of destination,

g) **NEWCTOT** is the proposed new slot to be accepted,

h) **Response by** time.

i) **AFTN** or **SITA** address for the reply.

(9) The field IOBT/D for the Initial OBT is only displayed if it is different from the last OBT known.

(10) If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preferences (Preferences / CHMI / ATFCM application / General / Origin).

(11) To modify the proposed address, click on the **AFTN**, **SITA** or **None** radio button to possibly modify the type and then select an address in the drop-down control or input one manually. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(12) If no address is supplied on display, none will be included in the message.

(13) The system will process the message and transmit a Slot Revision Message to all concerned including the address you have provided with the message.

(14) To send the message, click on **Send** button.

(15) To close the window without sending the message, click on **Cancel** button.

39.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) There will be no reply on the screen once the message has been transmitted. To acknowledge a good transmission, the transmit stamp is returned and displayed.

(3) The processing of the message being asynchronous, be aware this is only an acknowledgement message.

(4) If a problem occurs in the transmission, a descriptive error message is displayed in the bottom message box.

(5) All controls are disabled while processing the message.

(6) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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40 Slot Proposal Rejection

40.1 Function presentation

(1) This window is used to send a CASA message to reject a slot proposal (**SRJ**) for a selected flight.

40.2 Opening the window

(1) This window may only be activated when a flight is selected in a list or from the flight data window.

(2) The SRJ message can be sent only for a flight with a published slot and a valid proposed new CTOT (SIP).

(3) Messages are not authorised for flights in following states: Proposal, ATC activated, PFD and terminated.

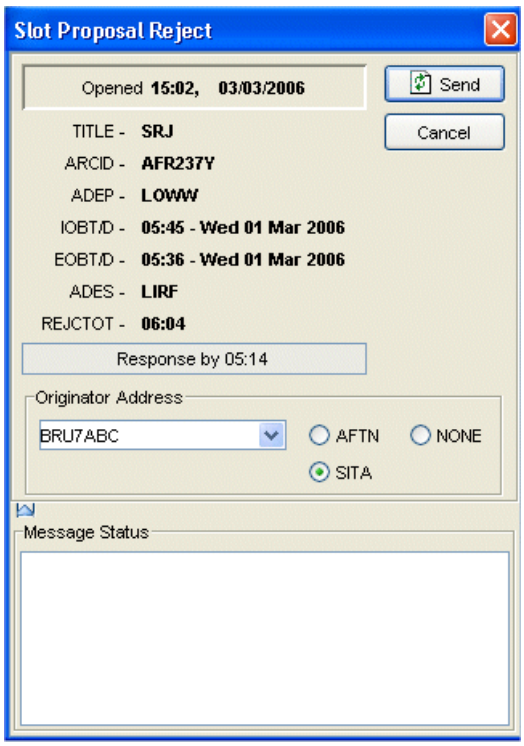


Figure Chmi-117: Slot Proposal Rejection message

(4) When such a window is opened, no other function may be invoked.

(5) This display shows the Slot Proposal Rejection message to be sent. This message is sent as a negative reply to a Slot Improvement Proposal (SIP).

(6) There is no reply to such a message.

(7) The **Opened** box at the top indicates the UTC stamp when this dialog was invoked. It is displayed there because the timing in sending messages is important.

(8) The different data elements to be included in the message are presented on display:

a) **TITLE - SRJ** naming the message.

b) **ARCID** for the callsign of the flight,

c) **ADEP** aerodrome of departure,

d) **IOBT/D** Initial Estimated Off Block Time and Date,

e) **EOBT/D** Last Estimated Off Block Time and Date,

f) **ADES** aerodrome of destination,

g) **REJCTOT** is the proposed new slot to be rejected,

h) **Response by** time.

i) **AFTN** or **SITA** address for the reply.

(9) The field IOBT/D for the Initial OBT is only displayed if it is different from the last OBT known.

(10) If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preferences (Preferences / CHMI / ATFCM application / General / Origin).

(11) To modify the proposed address, click on the **AFTN**, **SITA** or **None** radio button to possibly modify the type and then select an address in the drop-down control or input one manually. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(12) If no address is supplied on display, none will be included in the message.

(13) The system will process the message and will give no further reply.

(14) To send the message, click on **Send** button.

(15) To close the window without sending the message, click on **Cancel** button.

40.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) There will be no reply on the screen once the message has been transmitted. To acknowledge a good transmission, the transmit stamp is returned and displayed.

(3) The processing of the message being asynchronous, be aware this is only an acknowledgement message.

(4) If a problem occurs in the transmission, a descriptive error message is displayed in the bottom message box.

(5) All controls are disabled while processing the message.

(6) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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41 Slot Missed Message

41.1 Function presentation

(1) This window is used to send a slot missed message (**SMM**) for a selected flight.

41.2 Opening the window

(1) This window may only be activated when a flight is selected in a list or from the flight data window.

(2) The SMM message can be sent only for a flight with a published slot.

(3) Messages are not authorised for flights in following states: Proposal, ATC activated, PFD and terminated.

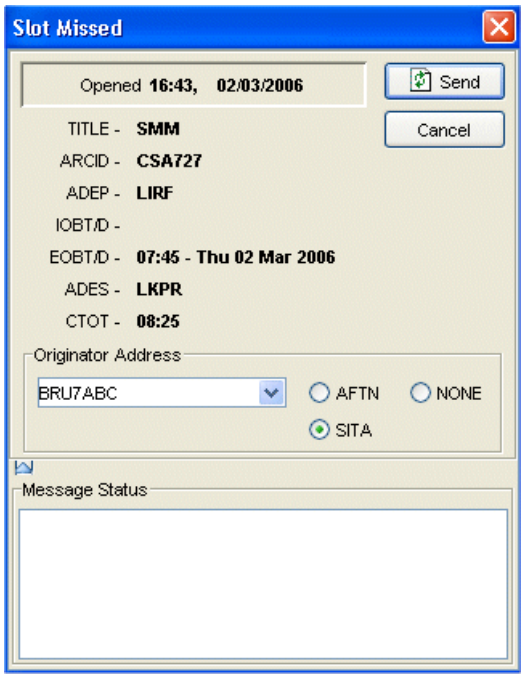


Figure Chmi-118: Slot Missed Message

(4) When such a window is opened, no other function may be invoked.

(5) This display shows the Slot Missed Message to be sent. This message is sent when the flight is not able to cope with the slot, which has been allocated, and that its new departure time is not known. The released slot will be given to another flight.

(6) There is no reply to such a message. When the departure time is known, a DLA message must be sent to IFPS.

(7) The **Opened** box at the top indicates the UTC stamp when this dialog was invoked. It is displayed there because the timing in sending messages is important.

(8) The different data elements to be included in the message are presented on display:

a) **TITLE - SMM** naming the message.

b) **ARCID** for the callsign of the flight,

c) **ADEP** aerodrome of departure,

d) **IOBT/D** Initial Estimated Off Block Time and Date,

e) **EOBT/D** Last Estimated Off Block Time and Date,

f) **ADES** aerodrome of destination,

g) **CTOT** Slot not used,

h) **AFTN** or **SITA** address for the reply.

(9) The field IOBT/D for the Initial OBT is only displayed if it is different from the last OBT known.

(10) If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preferences (Preferences / CHMI / ATFCM application / General / Origin).

(11) To modify the proposed address, click on the **AFTN**, **SITA** or **None** radio button to possibly modify the type and then select an address in the drop-down control or input one manually. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(12) If no address is supplied on display, none will be included in the message.

(13) The system will process the message and transmit a Flight Suspension Message to all concerned including the address you have provided with the message.

(14) To send the message, click on **Send** button.

(15) To close the window without sending the message, click on **Cancel** button.

41.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) There will be no reply on the screen once the message has been transmitted. To acknowledge a good transmission, the transmit stamp is returned and displayed.

(3) The processing of the message being asynchronous, be aware this is only an acknowledgement message.

(4) If a problem occurs in the transmission, a descriptive error message is displayed in the bottom message box.

(5) All controls are disabled while processing the message.

(6) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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42 Rerouting Rejection

42.1 Function presentation

(1) This window is used to send a CASA message to reject a rerouting proposal (**RJT**) for a selected flight.

42.2 Opening the window

(1) The window may only be activated from the **Flight Data** window.

(2) The RJT message can be sent only for a flight that has received a rerouting proposal.

(3) Messages are not authorised for flights in following states: Proposal, ATC activated, PFD and terminated.

(4) This rerouting reference and the corresponding limit in time for the reply are only retrieved by the **Flight Data** window.

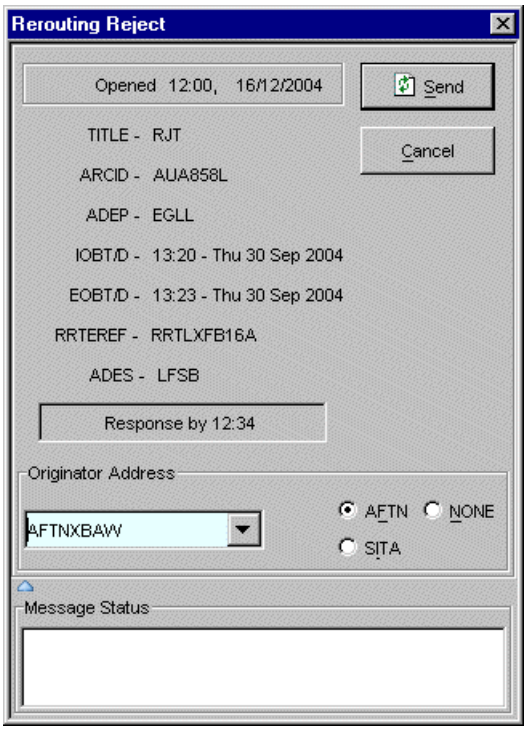


Figure Chmi-119: Rerouting Rejection message

(5) When such a window is opened, no other function may be invoked.

(6) This display shows the Rerouting Rejection message to be sent. This message is sent as a negative reply to a Rerouting Proposal (RRP). It allows to release back a slot into the system for potential use elsewhere.

(7) There is no reply to such a message.

(8) The **Opened** box at the top indicates the UTC stamp when this dialog was invoked. It is displayed there because the timing in sending messages is important.

(9) The different data elements to be included in the message are presented on display:

a) **TITLE - RJT** naming the message.

b) **ARCID** for the callsign of the flight,

c) **ADEP** aerodrome of departure,

d) **IOBT/D** Initial Estimated Off Block Time and Date,

e) **EOBT/D** Last Estimated Off Block Time and Date,

f) **ADES** aerodrome of destination,

g) **RRTEREF** is the proposed rerouting reference code,

h) **AFTN** or **SITA** address for the reply.

(10) The field IOBT/D for the Initial OBT is only displayed if it is different from the last OBT known.

(11) If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preferences (Preferences / CHMI / ATFCM application / General / Origin).

(12) To modify the proposed address, click on the **AFTN**, **SITA** or **None** radio button to possibly modify the type and then select an address in the drop-down control or input one manually. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(13) If no address is supplied on display, none will be included in the message.

(14) The system will process the message and the current CTOT will be kept.

(15) To send the message, click on **Send** button.

(16) To close the window without sending the message, click on **Cancel** button.

42.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) There will be no reply on the screen once the message has been transmitted. To acknowledge a good transmission, the transmit stamp is returned and displayed.

(3) The processing of the message being asynchronous, be aware this is only an acknowledgement message.

(4) If a problem occurs in the transmission, a descriptive error message is displayed in the bottom message box.

(5) All controls are disabled while processing the message.

(6) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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43 Flight Confirmation Message

43.1 Function presentation

(1) This window is used to send a flight confirmation message (**FCM**) for a selected flight.

43.2 Opening the window

(1) This window may only be activated when a flight is selected in a list or from the flight data window.

(2) The FCM message can be sent for any flight with a state IFPL.

(3) Messages are not authorised for flights in following states: Proposal, ATC activated, PFD and terminated.

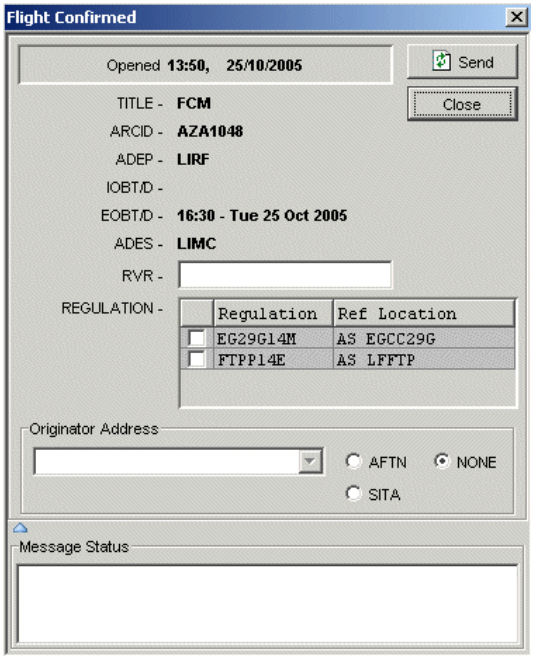


Figure Chmi-120: Flight Confirmation Message

(4) When such a window is opened, no other function may be invoked.

(5) This display shows the Flight Confirmation Message to be sent. This message is to be sent following a request made by the NM (for example during fog situations or closure of aerodromes).

(6) There is no reply to such a message.

(7) The **Opened** box at the top indicates the UTC stamp when this dialog was invoked. It is displayed there because the timing in sending messages is important.

(8) The different data elements to be included in the message are presented on display:

a) **TITLE - FCM** naming the message.

b) **ARCID** for the callsign of the flight.

c) **ADEP** aerodrome of departure.

d) **IOBT/D** Initial Estimated Off Block Time and Date.

e) **EOBT/D** Last Estimated Off Block Time and Date.

f) **ADES** aerodrome of destination.

g) **RVR** is the minimum Runway Visual Range capability for the flight.

h) **REGULATION** list for which a Confirmation Message (FCM) is still required (see later).

i) **FLS Response by** is the limit in time for the FCM message with the missing RVR.

j) **AFTN** or **SITA** address for the reply.

(9) The **REGULATION** list contains the following information:

a) The identifier of the **Regulation**.

b) The related reference location (**Ref Location**).

c) *Checkbox*: To identify the regulations to be confirmed. By default, all checkboxes are unchecked (empty).

(10) The check box in front of the regulation shall be checked to include this regulation in the confirmation message (FCM).

(11) The field IOBT/D for the Initial OBT is only displayed if it is different from the last OBT known.

(12) The field RVR must be filled if it is a weather situation. The format is XXX, i.e. 300.

(13) The confirmation message (FCM) may only be issued if either the RVR field is supplied and/or one or more regulations are checked in the list.

(14) If a **default AFTN or SITA address** is defined, it is presented in the **Originator Address** drop-down box. To have a default address defined, you need to update the **Message Originator** preferences (Preferences / CHMI / ATFCM application / General / Origin).

(15) To modify the proposed address, click on the **AFTN**, **SITA** or **None** radio button to possibly modify the type and then select an address in the drop-down control or input one manually. Any address will be checked for a valid syntax (See Default Message Originator in Common Functionalities).

(16) If no address is supplied on display, none will be included in the message.

(17) The system will process the message and transmit a slot related message to all concerned including the address you have provided with the message.

(18) To send the message, click on **Send** button.

(19) To close the window without sending the message, click on **Cancel** button.

43.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) There will be no reply on the screen once the message has been transmitted. To acknowledge a good transmission, the transmit stamp is returned and displayed.

(3) The processing of the message being asynchronous, be aware this is only an acknowledgement message.

(4) If a problem occurs in the transmission, a descriptive error message is displayed in the bottom message box.

(5) All controls are disabled while processing the message.

(6) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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44 Swap Slot

44.1 Function presentation

(1) This window is used to send a request to exchange slots in a Regulation flight list.

(2) The slot swapping function allows to issue requests:

a) to exchange a selected flight with an identified free slot.

b) to exchange the published slots of 2 flights identified in the same Regulation flight list.

44.2 Opening the window

(1) The window may only be activated when a flight is selected in a Regulation flight list.

(2) The swap request message can only be sent for flights with a published slot. The flight may already be in TACT Activated state.

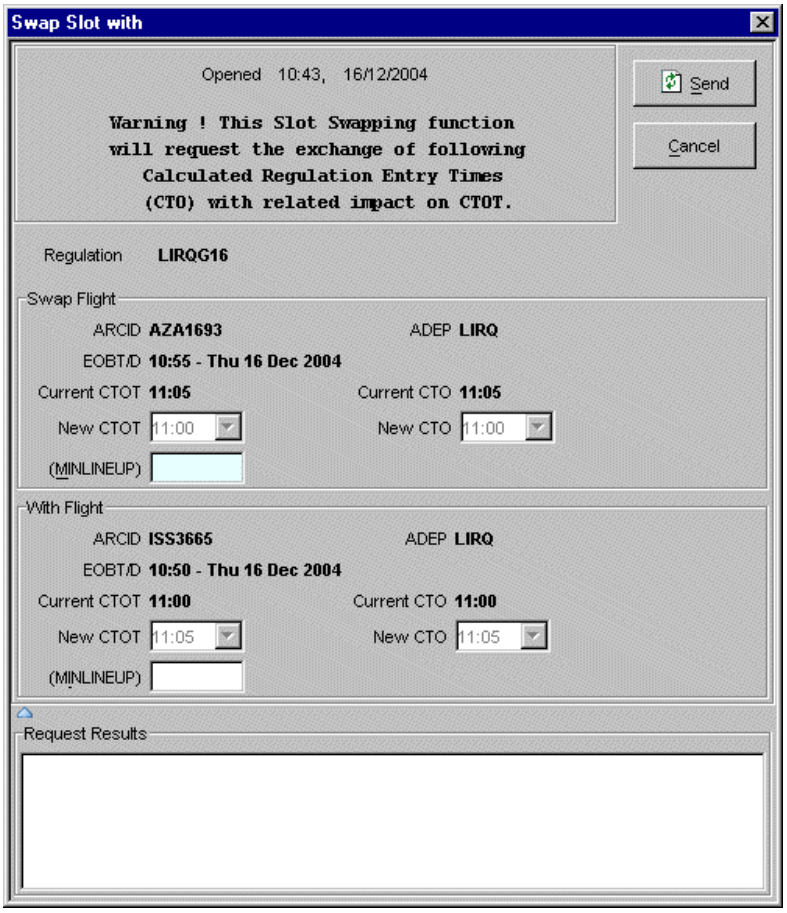


Figure Chmi-122: Swap Slot

(3) From a Regulation flight list,

a) use the option **Swap 1 with a free Slot** to open the Swap Slot dialog with a single flight selected for swapping.

b) use the option **Swap 2 Slots (new/reset)** to select/unselect a first flight followed by the option **Swap Slot with xxx** to open the Swap Slot dialog with 2 flights selected for swapping.

(4) When such a window is opened, no other function may be invoked.

(5) This display shows the slot swapping request to be sent. This message is sent when a flight shall get a better slot position inside a Regulation Flight List.

(6) The reply to such a message is either the confirmation that the requested swapping has been performed or an error message justifying why the swapping could not be performed.

(7) Here are the different data elements presented in the opened dialog:

a) **Opened**: local stamp when the dialog was opened.

b) **Warning** message.

c) **Regulation**: identifies the Regulation affecting the selected flight(s).

(8) The **Swap Flight** group box lists data elements related to the first flight to be swapped:

a) **ARCID**: callsign of this flight.

b) **ADEP**: aerodrome of departure.

c) **EOBT/D**: Last Estimated Off Block Time and Date.

d) **Current CTOT**: the current Calculated Take Off Time.

e) **Current CTO**: the current Calculated Time Over the reference of the regulation (this time might be identical to the CTOT in case the regulation is on the Departure Aerodrome).

f) **New CTOT**: desired Calculated Take Off Time resulting from the slot swapping operation.

g) **New CTO**: desired Calculated Time Over resulting from the slot swapping operation.

h) **MINLINEUP** Minimum time needed for the flight to get from its present holding position to airborne (if not provided, the taxi time is used). Accepted values are between 00 and the maximum identified by the system property **CHMI / ATFCM application / Flights / General / Maximum MINLINEUP**.

(9) The **With Flight** group box lists the same data elements on any second flight to be swapped with.

(10) The **With Flight** group box is hidden when the first flight shall be swapped with a free slot (option **Swap 1 with a free Slot** in the menu).

(11) The swap request is already prepared and only the MINLINEUP field(s) can be supplied. If not provided, the taxi time is used.

(12) In case of a swap with a free slot, either the **New CTOT** or **New CTO** shall be provided. The other value is automatically calculated by the application. These **New CTOT** and **New CTO** fields are protected in case of 2 slots swapping.

(13) **New CTOT** and **New CTO** values are considered as desired values with some tolerance.

(14) When ready, the Send button shall be clicked.

(15) If the slot swapping is successfully performed, a message is displayed with the newly allocated slot(s).

(16) In case of rejection, the reason is displayed in a message box.

(17) The new slot position of the flight that has improved is forced and will no more be modified by the slot allocation system (CASA).

(18) To close the window without sending the message, click on **Cancel**.

44.3 Displaying the query results

(1) After clicking the **Send** button, the message is issued to the Network Operations system.

(2) The result of the swap request is displayed in the bottom message box when received.

(3) It includes:

a) The processing stamp of the Network Operations system.

b) The result of the swapping operation, if successfull (for example: New CTOT for flight ... is ...)

c) The error message in case of failure.

(4) All controls are disabled while processing the message.

(5) Once the message is transmitted (successfully or not), the Cancel button becomes **Close** button and allows to terminate the dialog properly.

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45 Query Replay

45.1 Function presentation

(1) The Query Replay allows:

a) Displaying query list history data for selected user, period and additional optional selection criteria. History shall be available as early as possible but at least at the beginning of the next day after operation.

b) Requesting to replay the exact query / reply image for a CHMI ATFCM query selected in the list history.

45.2 Opening the window

(1) From the Application menu, select the option **Query Replay**.

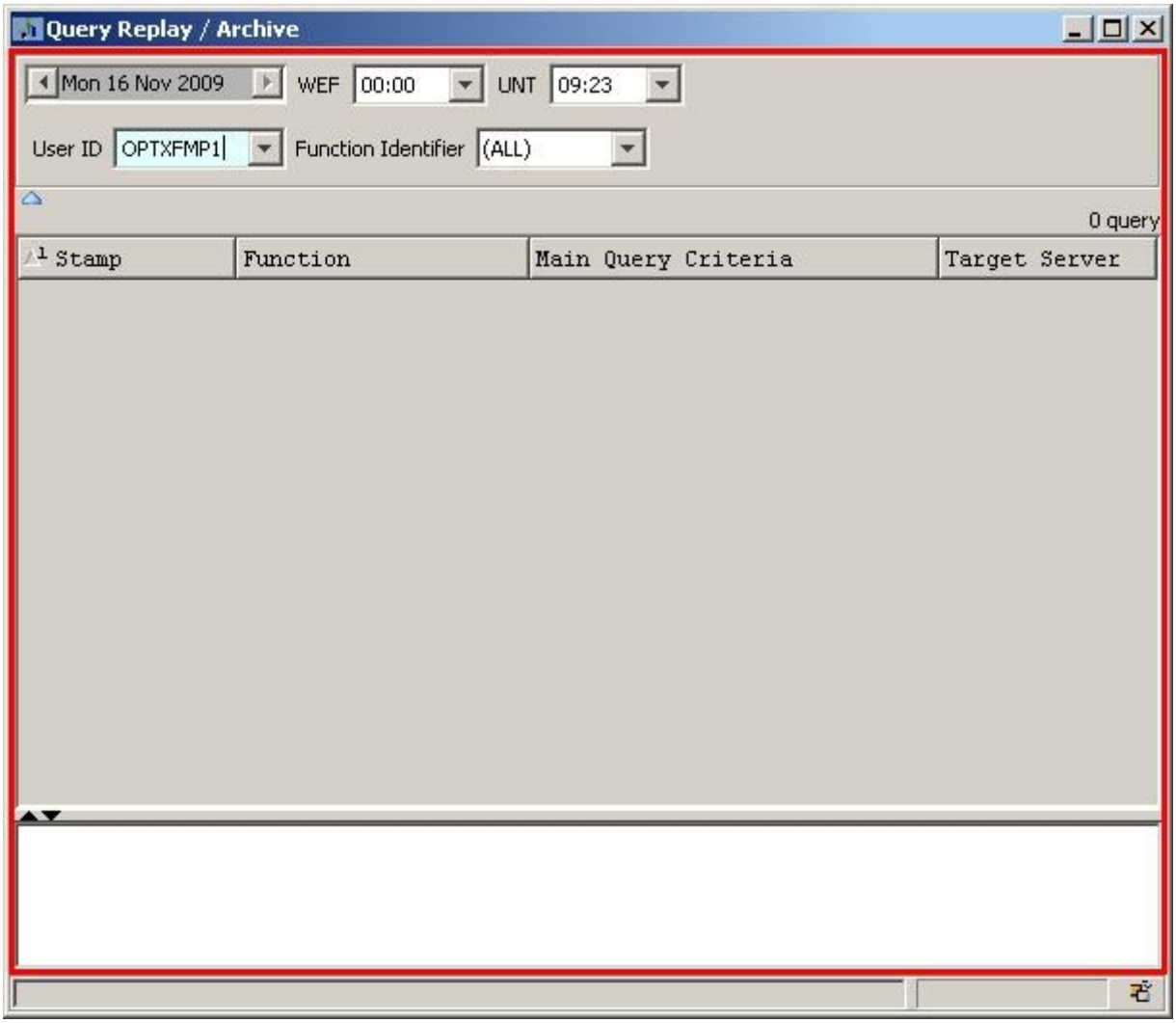


Figure Chmi-123: "Query Replay" Query

(2) A new empty **Query Replay** window will be opened with the following defaults:

a) Date: current traffic date is proposed (relative to current UTC date).

b) **WEF** time: will be intialised with 00:00 to default to the start of the selected day.

c) **UNT** time: will be intialised with 00:00 to default to the end of the selected day.

d) **User ID**: the User ID used to display this Query Replay window is proposed by default. The drop-down box will propose the most recent User ID query data elements you have queried before with success.

e) **Function Identifier**: the entry **(all)** is proposed by default. The drop-down box will propose the list of unique CHMI ATFCM function identifiers.

(3) On initial display, the focus is set on the first query control: the Traffic Date input control.

45.3 Preparing to query

(1) The Traffic Date may be adjusted to one of the proposed values.

(2) The date control shall allow selecting any date between current UTC up to 15 months in the past.

(3) The proposed default **WEF** and **UNT** times may be overwritten to better focus on the period to be analysed.

(4) Any valid time value from 00:00 to 23:59 included may be entered in the **WEF** and/or the **UNT** entry boxes. This input must respect the HH:MM format.

(5) Since the query is always limited to a single calendar day, the **WEF** time must be smaller than the **UNT** time, except if both times are set to 00:00, which corresponds to a full day query.

(6) If the default User Identifier (**User ID**) needs to be overwritten, it shall be filled in manually or selected from the drop-down list (when a previous query was successful).

(7) The possibility to overwrite the default User Identifier (**User ID**) shall be controlled via the user security profile.

(8) If the list of queries shall be displayed for a specific **Function Identifier**, it shall be selected from the drop-down list.

(9) All query data elements are mandatory but the **Function Identifier** may be set to entry value **(all)** which implies no selection on function identifier.

(10) The **Send** button will send the query to the Network Operations system.

(11) This window may be closed before any query is issued.

(12) The **Reset** action common functionality is implemented for this function. All query criteria are considered for activating the Reset action.

45.4 Displaying the query results

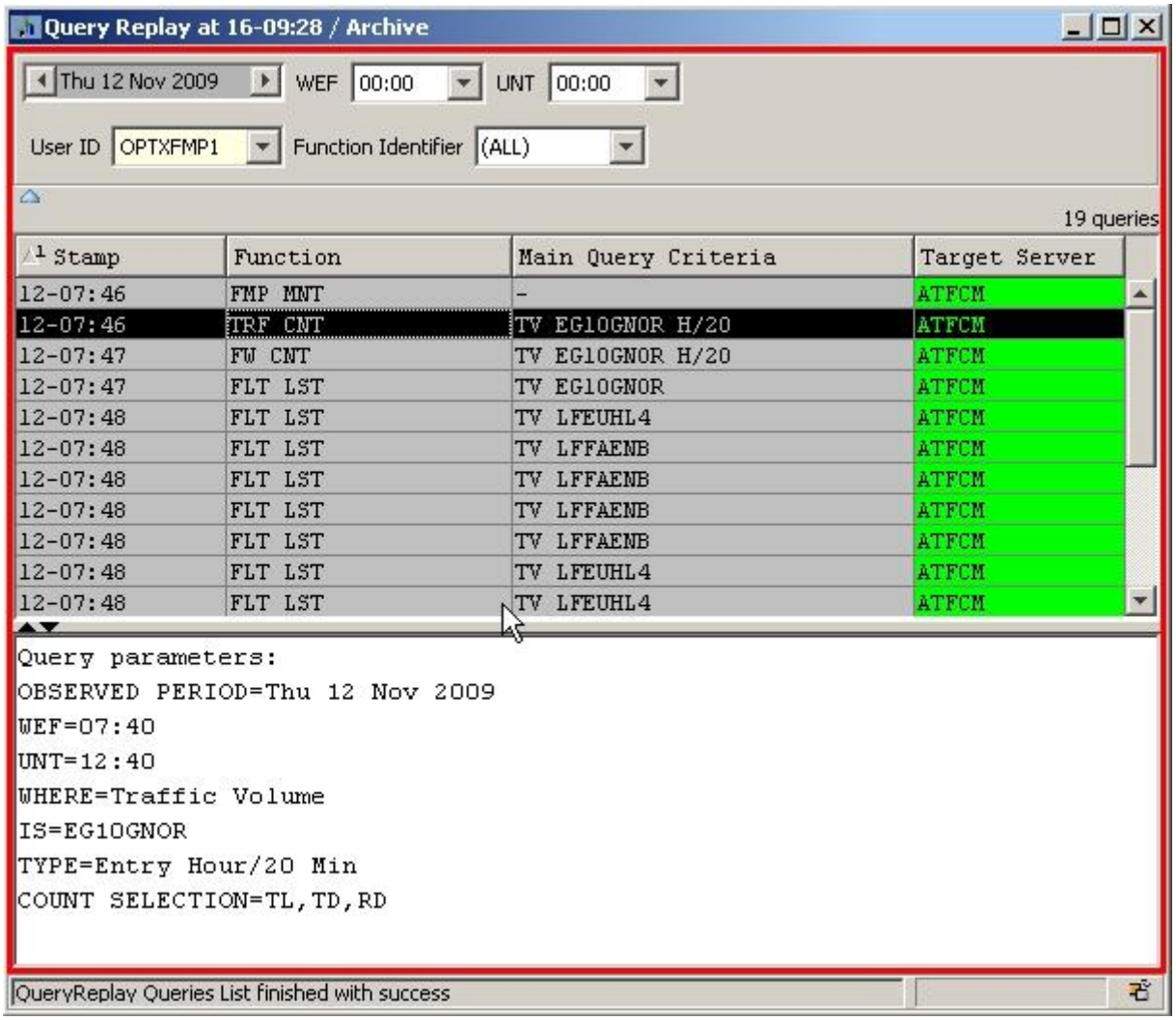


Figure Chmi-124: "Query Replay"

(1) This display shows the list of queries from the history that corresponds to the selected query criteria (Traffic Date, WEF, UNT, User ID and Function Identifier).

(2) The header section of the window contains the different **query data elements** used to retrieve from the history, the queries currently listed by the Query Replay function:

a) The Traffic Date scanned for the associated query criteria.

b) **WEF / UNT**: the period of the query history list scanned for the associated query criteria.

c) **User ID**: the User ID that have initially issued the queries currently listed from the history.

d) **Function Identifier**: either **(all)** or one specific function identifier that corresponds to the queries currently listed from the history..

(3) A **counter** identifies the number of queries retrieved from the history for the specified criteria.

(4) This counter may be affected by **filtering** (fff / rrr queries where fff is the filtered number and rrr is the retrieved number).

(5) The list shows all queries that correspond to the selected query criteria.

(6) The following information is shown for each query entry in the list:

a) The **Stamp** when the query was initially received by the CUA server (ex: 31-14:36)

b) The identification of the ATFCM **Function** that initiated the query (ex: TFC CNT)

c) The type and the identifier of the **Main Query Criteria** (ex: AD EBBR D)

d) The target server (i.e. ATFCM, Archive or Predict) possibly with the window frame colour coding associated with Archive and Predict.

(7) These data elements shall be listed in separate columns to allow standard list sorting and filtering.

(8) If the **Query Replay** is able to list ATFCM queries issued by different HMIs (i.e. CHMI ATFCM or NOP Portal), the origin of the query shall be indicated somehow and it shall be possible to filter on this criteria.

(9) When a query line is selected in the list, a detail box at the bottom shall display the complete query details in readable format.

(10) For each data element in the query: *"xxx = zzzzz"* on each sub-sequent lines, where *xxx* is the data element identifier and *zzzzz* is the associated value.

(11) The Invalidate function works for this window, if the user preference **CHMI / ATFCM application / General / Dates and times / Activate invalidate functionality** is checked.

(12) On initial display, the list is sorted in ascending sequence on the **Stamp** column (i.e. when the query was received).

(13) Even if not completely displayed, the sort algorithm of the **Stamp** column shall take into account the complete date value.

(14) On query results display, the focus is set on the query list after selecting the first query entry in the list.

45.5 User interactions

(1) Click on a line in the list to **select the query entry** and have more detailed information displayed below the list.

(2) This window allows to modify a previous query selection and to issue a new request.

(3) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option.

(4) On query modification, the **Reset action** is enabled.

(5) To display / hide the query part, click respectively the Down / Up arrow on the left underneath the split bar.

(6) To print the Query Replay list, first retrieve, sort and filter the data in the window. When ready, click on the **Print** button or select the **File / Print** menu option.

(7) To display a sub-selection of the retrieved data, click on the **Filter** button on the button bar or select the **Action / Filter** menu option and adjust the desired filtering criteria.

(8) To modify the **Sort sequence** of the queries in the list, click on column headers corresponding to your desired sort criteria.

(9) To export the displayed details, click on the **Export** button or select the **Action / Export** menu option.

(10) Increasing / reducing the window height and width will impact the size of the main query list control.

45.6 Contextual actions

(1) When a query line is selected in the list, the right mouse button may pop up a **contextual menu**.

(2) When available, this contextual menu shall allow requesting the **Replay** action of the query / reply corresponding to the query entry selected in the list.

(3) Based on initial user requirements, the **Replay** functionality shall be available for the following main traffic monitoring and flight display functionalities:

a) FMP Monitor

b) Airspace Monitor

c) Status Monitor

d) Traffic Counts

e) Flow Counts

f) Departure Delays

g) Regulation Delays

h) Flight List

i) ARCID List

j) Suspended Flight List

k) Flight Data

(4) If standard implementation implies low additional cost for all simple query / reply windows (i.e. function display based on single data exchange), the **Replay** could also be implemented for the following functions:

a) Flight History

b) Flight Operational Log

c) CASA Regulations

d) Regulation Description

e) Rerouting Measures

f) ATC Configuration

g) Traffic Capacity

h) Occupancy Traffic Monitoring Values

i) ATC Airspace OTMV List

j) Aerodrome Details

k) Route Catalogue

l) Traffic Volume Activation

(5) The following functions have been explicitly excluded of the **Replay** initial implementation because it involves either complex multiple query / reply exchanges or specific treatment with limited added value:

a) Tower Departure List

b) Alternate Routes

c) ANM Display

d) AIM Display

e) ADP List

f) ADP Current

g) ADP History

h) ATM Messages Release Notifier

i) FMP Selection

j) Query Replay

(6) If the **Query Replay** is able to list ATFCM queries issued by different HMIs (i.e. CHMI ATFCM or NOP Portal), the **Replay** function shall only be enabled in the popup if the CHMI ATFCM is technically able to replay a query / reply image exchanged initially with another HMI (i.e. the NOP Portal).

(7) When the **Replay** function is invoked, it shall display the exact image of the corresponding ATFCM function window as it was at the time when the selected query / reply was initially processed.

(8) The only differences that may appear between the initial window and the replayed image should be related to updated local preferences or PC settings (ex: different colour preferences, different PC / UTC reference time…)

(9) Because the **Replay** function is displaying historical data, it is desynchronised from current traffic. Therefore the query panel and any contextual action shall be disabled. All other local actions shall be available (i.e. Sort, Filter, Print, Export…)

(10) For traffic counts, the count sub-total "RPL / PFD / IFPL" has been split into their individual counts. Consequently the Count piling order is also revised. **Query Replay** shall be able to re-display the results as they were presented at the initial query time.

(11) The **Replay** function shall display the original **OTMV thresholds and alerts** features as described in the **Occupancy Traffic Monitor Values** function (i.e. based on the same values received for the original Query Reply).

(12) The popup display of the High Occupancy counts warning dialog, if any, does not have to be reproduced.

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46 Simulations Selection

46.1 Function presentation

(1) The **Simulations Selection** functionality shall give access to the list of Published Simulation systems and shall allow issuing further ATFCM queries against a selected Simulation System.

(2) While the main **ATFCM** Application menu option gives access to the Tactical (today) and the Pre-Tactical (tomorrow) operational data, the **Simulations** option shall give access to simulation data that will cover the **Period** identified for the SIMULATION **Hosted Dataset**.

(3) A SIMULATION **Hosted Dataset** is built based on the **Reference Dataset** where it is hosted.

(4) A **Reference Dataset**

a) Is of either type: OPERATIONAL, FORECAST or SIMULATION

b) May host several SIMULATION **Hosted Dataset**

46.2 Opening the window

(1) Selecting the **Simulations** option from the Application menu of the CHMI ATFCM, shall open a **Simulations** selection window, if authorised by your user profile.

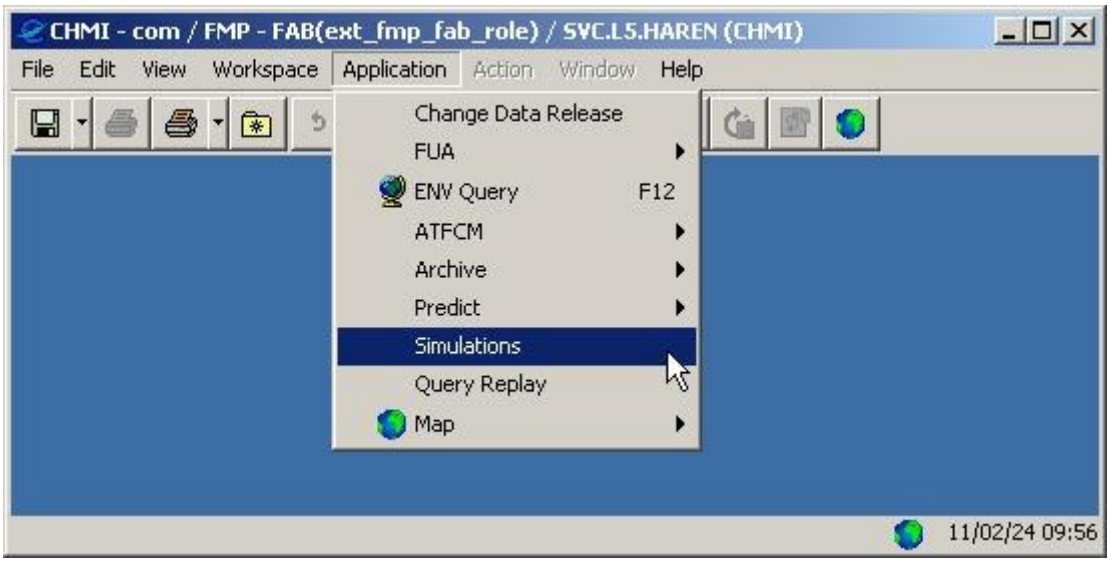


Figure Chmi-125: Simulations Menu

(2) When the **Simulations** selection window opens, it shall automatically retrieve and display the list of published simulation Datasets.

46.3 Displaying the query results

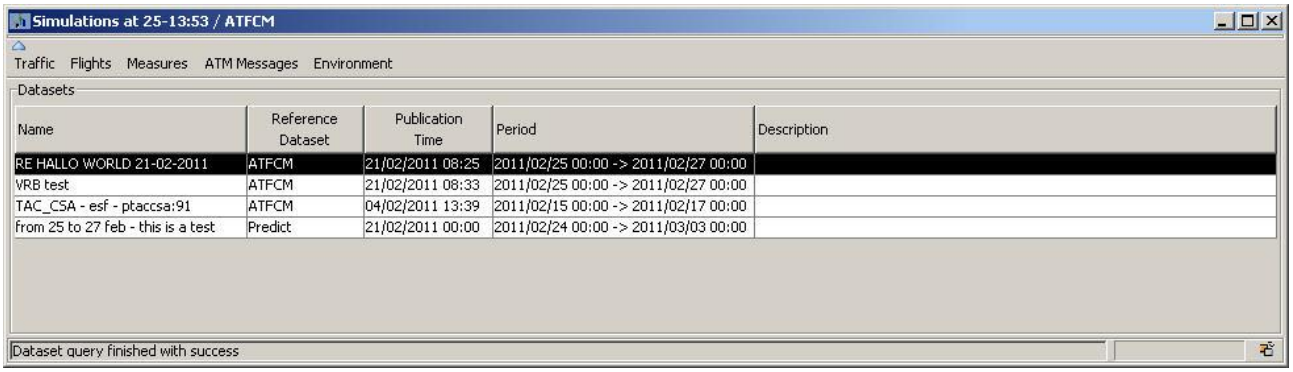


Figure Chmi-126: Simulations list

(1) The title contains the **time stamp** when the Simulations list details were retrieved from the Network Operations system.

(2) On initial display and any subsequent refresh, the focus is set on the list of published simulation Datasets after selecting the first Simulation in the list.

(3) For each simulation, the **Simulations** selection window displays the following information:

a) **Name**: The name associated to the published simulation Dataset

b) **Reference Dataset**: The type of Reference Dataset on which the simulation is built

i) **ATFCM** when host is of type OPERATIONAL

ii) **Predict** when host is of type FORECAST

iii) **Simulation** when host is of type SIMULATION

c) **Publication Time**: The date and time stamp when the Simulation was published

d) **Period**: The period covered by the Simulation Hosted Dataset

e) **Description**: The descriptive text associated to the published simulation Dataset, if any

(4) The Simulations list allows single selection to invoke contextual ATFCM functionalities for one simulation.

46.4 User interactions

(1) To update the display, click on the **Send** button on the button bar or select the **Action / Send** menu option. This will refresh the displayed list of published simulation Datasets based on the latest data available from the servers and the latest refresh **time stamp** in the title.

(2) To modify the **Sort sequence** of the simulations, click on the column headers corresponding to your desired sort criteria.

(3) When one of the published datasets is selected in the list, the menu at the top of the Simulations list display provides a contextual access to the ATFCM functionalities (i.e. usual CHMI ATFCM functionalities against the selected simulation dataset) authorised for the corresponding type of Reference Dataset by the user profile (ATFCM or Predict).

a) **ATFCM** when host of selected simulation is of type OPERATIONAL

b) **Predict** when host of selected simulation is of type FORECAST or SIMULATION

(4) Each windows' title bar includes the selected target simulation name as target platform.

(5) Any standard CHMI ATFCM window (i.e. Flight List, Counts …) opened from the Simulations list to display related simulation data is surrounded by a specific coloured frame.

(6) Any CHMI ATFCM window (i.e. Flight List, Counts …) opened from the Simulations list shall propose query dates based on the **Period** covered by the Simulation Hosted Dataset. To avoid unnecessary complexity just for this specific usage, this has not been reflected in the detailed description text of each function.

(7) Any CHMI ATFCM window (i.e. Flight List, Counts …) opened from the Simulations list shall be customised and behave based on the type of Reference Dataset on which the simulation is built:

a) As an **ATFCM** window when host is of type OPERATIONAL

b) As a **Predict** window when host is of type FORECAST or SIMULATION

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47 ATFCM Collaboration Human Machine Interface application - Archive Access

47.1 Introduction

(1) This document contains :

a) The identification of some specific behaviour related to the Archive system access.

b) The list of functionalities available in the **Archive access** of the Collaboration Human Machine Interface application.

c) A link to either the detailed description of the specific Archive implementation or to the equivalent ATFCM function description when there is no differences in the Archive implementation.

(2) The full **detailed ATFCM access** being described in the basic documentation, differences identified in this document are always described based on the comparison with the ATFCM access.

47.2 Main Menu

47.2.1 Description

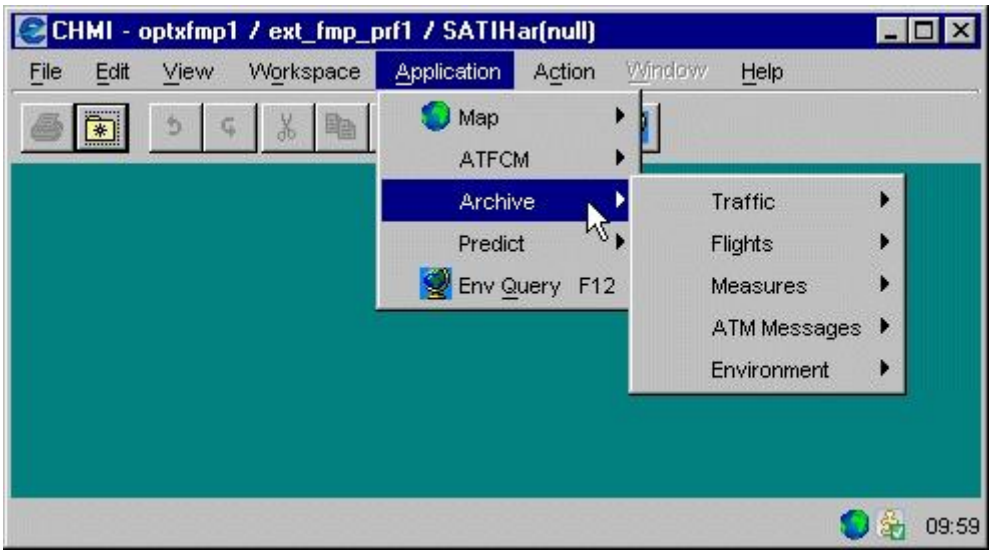


Figure ArchiveChmi-1: Main Menu

(1) From the CHMI **Application / Archive** main menu, you can select the function you want to execute.

(2) Functions are only displayed in the menu if your profile is authorising its usage.

(3) It is only when a function has been selected that the corresponding dedicated Window appears.

(4) This document only contains the functions implemented for Archive access. Look at ATFCM main page for a full list of the ATFCM menu.

47.2.2 Menu items

(1) **Traffic**

a) Monitor / FMP Monitor: displays a graph with the load states, regulation information and rerouting periods for the traffic volumes belonging to a selected FMP. Differences with ATFCM access are highlighted in the attached help page.

b) Monitor / ATC Airspace Monitor: displays a graph with the load states, regulation information and rerouting periods for the airspaces included in a selected configuration of an ATC Unit Airspace. Currently not available from the Archive server while writing this help text.

c) A separator line shall be added after the last Monitor option present in the Monitor group ("Monitor / FMP Monitor" or "Monitor / ATC Airspace Monitor") and before the following options in the Monitor group, if any is authorised.

d) Monitor / ATC Configuration: gives the ATC configuration and the activation table for one ATC Unit Airspace.

e) A separator line shall be added after the last ATC Configuration option present in the Monitor group ("Monitor / ATC Configuration") and before the following option in the Monitor group, if any is authorised.

f) Monitor / Occupancy Traffic Monitoring Values: Displays detailed Occupancy Traffic Monitor Values definition for a selected Traffic Volume and Duration.

g) Traffic Counts: displays in a graph or a table the number of flights entering in a traffic volume or over a location. Related Regulated Rates are also available. Differences with ATFCM access are highlighted in the attached help page.

h) Flow Counts: displays in a graph or a table the number of flights per flow entering in a traffic volume. Differences with ATFCM access are highlighted in the attached help page. Currently not available from the Archive server while writing this help text.

i) Delays: displays in a graph or a table the delays for a Departure Aerodrome or a Regulation. Departure Aerodrome Delays are currently not available from the Archive server while writing this help text.

(2) **Flights**

a) Flight List: lists all the flights for a given traffic type and the associated selection criteria (traffic volume, aerodrome, point, ...). Differences with ATFCM access are highlighted in the attached help page.

b) ARCID List: list of flights corresponding to a specific aircraft identifier and associated optional selection criteria. Differences with ATFCM access are highlighted in the attached help page.

c) Flight Data: gives the detailed information, route and airspace profile for one flight.

d) Point Profile: gives all details for one flight, pointing directly at the detailed route / point profile.

e) Airspace Profile: gives all details for one flight, pointing directly at the detailed airspace profile.

f) Reroute Violation: gives all details for one flight, pointing directly at all reroute violations.

g) Flight History: displays Flight Event History list for one flight.

h) Flight Operational Log: lists Tactical Operational Log entries for one flight.

(3) **Measures**

a) ATFCM Monitor: displays a graph with the regulation information for all FMPs.

b) CASA Regulations: lists the regulations for selected FMP(s) or all regulations. Differences with ATFCM access are highlighted in the attached help page.

c) Regulation Description: gives details for a selected Regulation including the Sub and Supplementary Periods.

d) Reroutings: lists the reroutings for selected or all FMP(s). Currently not available from the Archive server while writing this help text.

(4) **ATM Messages**

a) CRAM: lists details over the Conditional Route Availability Message (CRAM) for a selected date.

b) AIM: lists all AIMs valid for a selected date and displays additional details for the selected AIM.

c) ANM: lists the active ANM for a selected date and time.

d) ADP List: lists the latest ADP regulation list for a selected date.

e) ADP Current: gives details for a selected ADP Regulation including the Sub Periods and date.

f) ADP History: lists the history log for a selected ADP Regulation and date.

(5) **Environment**

a) Capacity Values : draws a graph with the capacity values for one location.

b) Traffic Volume Activation: lists the activation status for a selected traffic volume.

47.3 Common functionalities

47.3.1 Archive menu access

(1) Each function detailed description identifies the menu path to be used to start the corresponding window.

(2) For the Archive access, the Application / ATFCM part of the path shall be replaced by **Application / Archive**.

(3) Immediate shortcuts are not applicable to the Archive access.

47.3.2 Archive dates

(1) While the ATFCM option gives access to the Tactical (today) and the Pre-Tactical (tomorrow) operational data, the **Archive** gives access back to 24 months in the past (i.e. 741 days is the value used by DWH).

(2) Tactical operational data are archived daily.

(3) Once started, the content of the title bar allows identifying the start up mode.

(4) Each windows' title bar includes the selected target application environment (ATFCM, Archive or Predict Access).

(5) When talking about dates, the main help text refers to Tactical (today) and Pre-Tactical (tomorrow) dates. This is in fact the original purpose of this application.

(6) When a window is opened from the Archive Access menu, you must be aware that the proposed dates are within 24 months in the past based on your preferences (today-1, today-2, ...). To avoid unnecessary complexity just for this specific usage, this has not been reflected in the detailed description text of each ATFCM function.

(7) The list of selectable dates and the date proposed by default shall be adjusted in the Preference dialog.

(8) Following user preference shall be adjusted to modify the list of proposed dates:

a) CHMI / ATFCM application / General / Archive Dates and times / First default date.

b) CHMI / ATFCM application / General / Archive Dates and times / Last default date.

c) CHMI / ATFCM application / General / Archive Dates and times / Default date proposed for query. The default archive dates proposed are always relative to the current date.

(9) For example, if you adjust your preferences to display last week only, even on the following days, the dates proposed for last week will always be relative to the current date on opening a new window.

(10) When **Default date proposed for query** is set to **Last used**, the date proposed by default is always the same relative date as the one you have used for your last query. (for example : if you always query a date one month in the past, you will always be prompted with the date corresponding to one month in the past.)

(11) When **Default date proposed for query** is set to **Current**, the date proposed by default is always the most recent date in the list of available dates.

(12) The Archive date selection is highlighted by a specific **background color**.

(13) This color shall be identified using the following user preference :

a) CHMI / ATFCM application / General / Archive Dates and times / Archive date background color

47.3.3 Functions identical to ATFCM access

(1) Following common functionalities are implemented in the same way as in the ATFCM access.

(2) We refer you to the ATFCM access detailed description.

a) The **Time adjustment** function.

b) The **Default query period** function.

c) The **Send action**.

d) The **Reset action**.

e) The **Sort action**.

f) The **Filter action**.

g) The **Print / Export** function.

h) The **Previous / Next period** function.

47.3.4 Unavailable functions

(1) The Archive data being static, following common functions are not implemented for Archive access:

a) The **Period Shifting** function.

b) The **Polling** function.

c) The **Normal / Proposal** flights selection.

d) The **Invalidate** function.

(2) The **plot** action is implemented in the archive access but is limited to the availability of past AIRAC data (currently 3 AIRACs i.e. approximately 3 months in the past).

(3) Click here to return to the main CHMI help page

(4) Issuing a query over midnight in the archive access will trigger an application error message.

a) For code simplicity and to preserve future possible evolution, the common functionality to allow a query over midnight is enabled in the Archive Access of the ATFCM Collaboration Human Machine Interface application.

b) To the contrary, the implementation of such functionality on the Archive server could not be justified yet by an approved cost / benefit analysis. As a consequence, the Archive server will return an application error message to such query until further notice.

48 Monitor graph - Archive access

48.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Monitor graph** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

48.2 Function presentation

(1) In the Archive access, the monitoring may be requested:

a) For the Traffic Volumes belonging to 1 FMP at a time.

(2) The ATC Airspace Monitor and the ATFCM Monitor functions are currently not available from the Archive server while writing this help text.

48.3 Contextual actions

(1) Options described in the ATFCM access contextual popup menu are only displayed if the target function is available in Archive access.

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49 Traffic Counts - Archive access

49.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Traffic Counts** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

49.2 Function presentation

(1) In the Archive access, only two Traffic Types are available:

a) **TL**: Traffic Load.

b) **TD**: Traffic Demand.

49.3 Opening the window

(1) A new empty **Traffic Counts** window will be opened with the same default as in ATFCM access but :

a) Traffic Type (**TL - TD**): last selection used with your last similar query.

49.4 Preparing to query

(1) The different Traffic Types can be checked or unchecked:

a) **TL**: Traffic Load.

b) **TD**: Traffic Demand.

The "Show VFR/OAT at Aerodrome" checkbox shall be visible and checked but not editable.

The counts from Archive always include flights that are in a non-extracted portion of the route (VFR/OAT) at the Aerodrom.

For users that have access to the "Show Forecast" functionality, the "Show Forecast" checkbox shall be visible and unchecked but not editable.

The counts from Archive never include forecast flights.

For users that do not have access to the "Show Forecast" functionality, the "Show Forecast" checkbox shall be hidden.

49.5 Displaying the query results

49.5.1 Common Display

(1) The header box contains the selected query criteria as in ATFCM access, except for:

a) The combination of Traffic **Types** retrieved locally (**TL**, **TD**).

49.5.2 Graph display

(1) On the top of the graph, a group of three toggle buttons allows to select the traffic type(s) to be included in the graph :

a) **TL**: Traffic Load.

b) **TD**: Traffic Demand.

49.6 Contextual actions

(1) Options described in the ATFCM access contextual popup menu are only displayed if the target function is available in Archive access.

50 Flow Counts - Archive access

50.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Flow Counts** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

50.2 Function presentation

(1) In the Archive access, only two Traffic Types are available:

a) **TL** : Traffic Load.

b) **TD** : Traffic Demand.

50.3 Contextual actions

(1) Options described in the ATFCM access contextual popup menu are only displayed if the target function is available in Archive access.

50.4 Preparing the query

The "Show VFR/OAT at Aerodrome" checkbox shall be visible and checked but not editable.

The counts from Archive always include flights that are in a non-extracted portion of the route (VFR/OAT) at the Aerodrome.

For users that have access to the "Show Forecast" functionality, the "Show Forecast" checkbox shall be visible and unchecked but not editable.

The counts from Archive never include forecast flights.

For users that do not have access to the "Show Forecast" functionality, the "Show Forecast" checkbox shall be hidden.

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51 Flight List - Archive access

51.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Flight List** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

51.2 Function presentation

(1) In the Archive access, only two Traffic Types are available:

a) **TL**: Traffic Load.

b) **TD**: Traffic Demand.

51.3 Opening the window

(1) A new empty **Flight List** window will be opened with the same default as in ATFCM.

(2) When Traffic Volume is selected, no **Flow** may be associated.

(3) On initial selection of a **Where** type, the following **Traffic Type** comes pre-selected:

a) **Traffic Load** for Traffic Volume, Regulation, Point, Airspace, Aerodrome, Aerodrome Set.

b) **Traffic Demand** for Aircraft Operator.

(4) The Compare checkbox is unchecked.

51.4 Preparing to query

(1) When Traffic Volume is selected, no **Flow** may be associated.

(2) The Regulation is currently not available in the **Where** parameter from the Archive server while writing this help text.

(3) The traffic type (**TFC Type**) is automatically adjusted when the **Where** is changed.

(4) In the Archive access, only two Traffic Types are available :

a) **TL** : Traffic Load.

b) **TD** : Traffic Demand.

The "Show VFR/OAT at Aerodrome" checkbox shall be visible and checked but not editable.

The result of the flight list on aerodrome and aerodrome sets from archive always contains the flights that are in a non-extracted portion of the route (VFR/OAT) at the Aerodrome.

For users that have access to the "Show Forecast" functionality, the "Show Forecast" checkbox shall be visible and unchecked but not editable.

The flight lists from Archive never include forecast flights.

For users that do not have access to the "Show Forecast" functionality, the "Show Forecast" checkbox shall be hidden.

51.5 Displaying the query results

(1) All CHMI flight list columns are also available for the archive flight list. There some exceptions (see following requirements).

(2) The **READY Flights only** checkbox is not available in the archive access.

(3) The **Show Predicted Flights** functionality is not available in the archive access.

(4) Following flight list columns are NOT available from archive access:

a) **TW**: The Tolerance Window

b) **CCAMS**: The SSR code assigned by CCAMS

(5) The yoyo flight list column shows slightly different information wrt the CHMI ATFCM.

51.6 Contextual actions

(1) Options described in the ATFCM access contextual popup menu are only displayed if the target function is available in Archive access.

52 ARCID List - Archive access

52.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **ARCID List** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

52.2 Function presentation

(1) In the Archive access, the query will always include a given day.

52.3 Opening the window

(1) In the Archive access, the **Select on IOBD/T** check box is not available.

(2) The query will always include the day identified by IOBD.

52.4 Preparing to query

(1) In the Archive access, the **Select on IOBD/T** check box is not available.

52.5 Displaying the query results

(1) IOBD and IOBT controls are never disabled since **Select on IOBD/T** is not available.

(2) The **READY Flights only** checkbox is not available in the archive access.

(3) All CHMI flight list columns are also available for the archive flight list. There some exceptions (see following requirements).

(4) Following ARCID flight list columns are NOT available from archive access:

a) **TW**: The Tolerance Window

b) **CCAMS**: The SSR code assigned by CCAMS

(5) The yoyo flight list column shows slightly different information wrt the CHMI ATFCM.

52.6 Contextual actions

(1) Options described in the ATFCM access contextual popup menu are only displayed if the target function is available in Archive access.

53 Flight Data - Archive Access

53.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Flight Data** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

53.2 Displaying the query results

(1) Following features implemented for the ATFCM access will not be made available in the archive access:

a) Following data elements are not displayed by the Archive access as they are not made available by the DWH ICD back-end service:

i) Proposed Calculated Take Off Time (**Prop CTOT**) with the latest time to reply (**Resp By**)

ii) Slot Tolerance Window for a regulated flight or the Departure Tolerance Window for a non-regulated flight (**TW**)

iii) Code of the cause for the most penalising Regulation (**Regcause**)

iv) Reference of the Rerouting measure affecting the flight (**Rerouting Ref**) with related time limit to reply to the proposed Rerouting (**RRP Resp By**)

v) **Suspension Reason** with the time limit for the Confirmation message (FCM) after the issuance of a Flight Suspension (**FLS Resp By**)

vi) **CCAMS Code**: SSR code assigned by CCAMS to a flight.

b) The colouring of the Actual Point Profile and the Actual Airspace Profile based on CPR confirmation.

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54 Flight Operational Log - Archive access

54.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Flight Operational Log** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

54.2 Opening the window

(1) When opening a new empty flight **Operational Log** window, following defaults are specific to the **Archive Access**:

a) From / at : **From** date control will be initialised with the date before the one selected in IOBD. Related **at** time control will be initialised with 00:00. The date drop-down list will propose the same dates as in the IOBD list plus one day before. This is needed to query early history of the flight in the Operational Log.

b) Until / at : **Until** date control will be initialised with the date after the one selected in IOBD. Related **at** time control will be initialised with 00:00. The date drop-down list will propose the same dates as in the IOBD list plus one day before.

54.3 Preparing to query

(1) When the IOBD control is adjusted, the From and Until date controls are automatically adjusted in order to cover two full days elapse in the Operational Log (From IOBD-1 00:00 - Until IOBD+1 00:00).

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55 CASA Regulations - Archive access

55.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Casa Regulations** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

55.2 Function presentation

(1) In the Archive access, the **Active only** filter is not available since out of context.

55.3 Contextual actions

(1) Options described in the ATFCM access contextual popup menu are only displayed if the target function is available in Archive access.

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56 Traffic Capacity - Archive access

56.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Traffic Capacity** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

56.2 Function presentation

(1) Data retrieved by this query are extracted from the Archived Environment database (DWH) including possible Tactical updates.

56.3 User interactions

(1) Traffic Capacity update functionalities are not applicable to the Archive access (i.e. no modifications allowed on past data definitions).

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57 Occupancy Traffic Monitoring Values - Archive access

57.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Occupancy Traffic Monitoring Values** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

57.2 Function presentation

(1) Data retrieved by this query are extracted from the Archived Environment database (DWH) for the Daily OTMV Applicability definitions.

57.3 Displaying the query results

(1) The detailed OTMV definition is displayed on the right side for the Duration currently selected in the list.

(2) It displays for the selected Duration, the **Daily OTMV applicability for *dd/mm/yyyy*** obtained from the merge of the possible Temporary OTMV overwrites, if any, over the Permanent OTMV definition for the date selected in the **Observed period** (i.e. ***dd/mm/yyyy***).

(3) The list of **Permanent OTMV definitions** described in the ATFCM accesses is not available from Archive access.

57.4 User interactions

(1) OTMV update functionalities are not applicable to the Archive access (i.e. no modifications allowed on past data definitions).

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58 ATC Configuration - Archive access

58.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **ATC Configuration** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

58.2 Function presentation

(1) Data retrieved by this query are extracted from the Archived Environment database (DWH) including possible Tactical updates.

58.3 User interactions

(1) ATC Configuration update functionalities are not applicable to the Archive access (i.e. no modifications allowed on past data definitions).

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59 Traffic Volume Activation - Archive access

59.1 Highlighting differences with ATFCM

(1) The objective of the following paragraphs is to highlight the differences between the ATFCM and the Archive accesses.

(2) Please refer to the **Traffic Volume Activation** for a full description of the ATFCM access implemented functionality.

(3) Global differences between the ATFCM and the Archive accesses are described in the **ATFCM Collaboration Human Machine Interface application - Archive Access** page.

59.2 Function presentation

(1) Data retrieved by this query are extracted from the Archived Environment database (DWH) including possible Tactical updates.

59.3 User interactions

(1) Traffic Volume Activation update functionalities are not applicable to the Archive access (i.e. no modifications allowed on past data definitions).

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Abbreviations

Abbreviations and acronyms used in this document are available in the EUROCONTROL Air Navigation Inter-site Acronym List (AIRIAL) which may be found online:

http://www.eurocontrol.int/airial/definitionListInit.do?skipLogon=true&glossaryUid=AIRIAL.

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